

Lecture 1: Introduction to Macroeconomics

Overview

This lecture introduces the field of macroeconomics, distinguishing it from microeconomics, and provides a roadmap for the course. We examine the current economic environment (circa 2023): low unemployment, high inflation, rising interest rates, and volatile asset markets. The lecture demonstrates how macroeconomic forces create paradoxes like “good news is bad news” for financial markets.

1. What is Macroeconomics?

Macro vs. Micro: A Fundamental Distinction

- **Microeconomics:** Studies individual agents (households, firms, industries).
 - Example: Why does a particular worker choose to work at a specific firm?
 - Example: How does Apple set prices for iPhones?
- **Macroeconomics:** Studies the economy as a whole (countries, regions, global).
 - Example: Why is the US unemployment rate at 3.5%?
 - Example: Why is inflation at 8% instead of 2%?

Key Variables in Macroeconomics: - **Inflation:** Rate of change of the general price level (not individual prices). - **Unemployment Rate:** Share of labor force without jobs (not individual employment status). - **Exchange Rates:** Relative price of currencies (not individual goods across countries). - **Aggregate Output (GDP):** Total production in the economy.

Why Macro Sum of Micros

- **Fallacy of Composition:** What is true for an individual may not be true for all individuals together.
- **Interaction Effects:** In macroeconomics, individual behaviors interact to create aggregate outcomes that cannot be understood by studying individuals in isolation.
- **Example (Preview):** If one person saves more, they become wealthier. But if everyone tries to save more simultaneously, aggregate demand falls, causing a recession. Total savings may not increase (Paradox of Savings—covered in Lecture 3).

Analogy from Physics: Studying the motion of planets (macro) requires different tools than studying subatomic particles (micro). The aggregate behavior emerges from, but is not simply the sum of, individual parts.

The Art of Macroeconomics

- **Shortcuts and Simplifications:** Macro requires simplifying complex systems to capture their essence.
- **Models over Math:** This course emphasizes intuition and applied understanding, not advanced mathematics.
 - Goal: You should be able to read and critically analyze the **IMF World Economic Outlook**, **Wall Street Journal**, **Financial Times**, and **The Economist**.
 - Goal: Understand what traders and policymakers discuss, even if they sometimes get it wrong.

Course Philosophy: “If it gets very complicated, I’m failing.”

2. Current Economic Context (2023)

A. The Labor Market: Historically Tight

Unemployment at Record Lows

- **Current US Unemployment Rate:** ~3.5% (historically low).
- **Historical Context:**
 - **Great Recession (2007-2009):** Unemployment peaked at 10%.
 - **COVID-19 Recession (2020):** Unemployment spiked to 14.7% (unprecedented peacetime level).
 - **Normal Times:** Unemployment typically around 5-6%.
 - **Current:** 3.5% is far below normal, indicating an “overheating” labor market.

Key Insight: Low unemployment sounds great (and it is for workers), but in macro, it signals potential problems—namely, inflation pressures.

Wage Growth: Surging Across Sectors

- **Aggregate Wage Growth:** Rising steadily and rapidly across all sectors.
- **Service Sectors Hit Hardest:** Accommodation and food services experiencing especially sharp wage increases.
 - **Reason:** “Labor supply” problems—many workers left these sectors during COVID (early retirement, career changes, childcare responsibilities) and have not returned.
 - **Result:** Severe labor shortages driving up wages to attract workers.

Data: Wage growth in service sectors exceeding 6-8% annually (well above the 2-3% typical in normal times).

B. Inflation: Far Above Target

The 2% Target

- **Normal Inflation:** Central banks (Federal Reserve, European Central Bank) target **~2% inflation**.
 - Why 2%? Provides a buffer against deflation, allows for relative price adjustments, consistent with stable growth (explained later in course).
- **Current Inflation:** 6.5% - 8% depending on the measure (CPI, Core CPI, GDP Deflator—defined in Lecture 2).

Wage-Inflation Correlation

- **High Correlation:** Wage growth and inflation move together (correlation ~0.8+).
- **Mechanism:**
 1. Tight labor market → Firms compete for scarce workers → Wages rise.
 2. Higher wages → Higher production costs → Firms raise prices.
 3. Higher prices → Workers demand higher wages → **Wage-price spiral**.

Key Insight: This is why low unemployment + high wage growth is a macro problem, even though it sounds like good news for individuals.

C. Monetary Policy: Fighting Inflation with Interest Rates

The Central Bank's Main Tool: Interest Rates

- **Expansionary Policy** ($i \downarrow$): Lower rates → Cheaper borrowing → More investment and consumption → Higher aggregate demand → Higher output.
- **Contractionary Policy** ($i \uparrow$): Higher rates → Expensive borrowing → Less investment and consumption → Lower aggregate demand → Lower output (and lower inflation).

The Fed's Recent Path

1. **Pre-COVID (2015-2019):** Interest rates around 1.5-2.5%.
2. **COVID Shock (March 2020):** Fed cut rates to near zero (0-0.25%) to support the economy.
 - **Zero Lower Bound:** Cannot cut rates much below 0% (explained in Lecture 9).
3. **2021-2022:** Inflation surged, but Fed initially slow to respond (“behind the curve”).
4. **2022-2023:** Fed hiking rapidly (fastest pace since the 1980s) to combat inflation.
 - Rates increased from 0% to 4.5%+ in less than a year.

Key Insight: The Fed faces a trade-off—hiking rates fights inflation but also slows the economy, risking recession.

D. Asset Prices: The Wealth Channel

Equity Markets and Monetary Policy

- **Index:** S&P 500 (main US equity index, tracking 500 largest companies).

Timeline: 1. **COVID Crash (March 2020):** S&P 500 fell 30% as pandemic hit. - **Why?:** Forward-looking investors anticipated economic devastation. 2. **COVID Boom (2020-2021):** S&P 500 surged 114% (end-2020 to end-2021), exceeding pre-pandemic levels even as the economy was still recovering. - **Main Driver:** Ultra-low interest rates (not just “Amazon and Tesla stories”). - **Mechanism:** Low $i \rightarrow$ Less discounting of future profits $\rightarrow$ Higher stock valuations (explained in Lecture 22-23). 3. **2022 Decline:** S&P 500 fell 20%+ as Fed hiked rates. - **Main Driver:** Rising interest rates (can explain 100% of the decline).

Wealth Channel of Monetary Policy: - Low rates $\rightarrow$ Asset prices rise $\rightarrow$ Households feel richer $\rightarrow$ Consumption increases $\rightarrow$ Aggregate demand rises. - High rates $\rightarrow$ Asset prices fall $\rightarrow$ Households feel poorer $\rightarrow$ Consumption decreases $\rightarrow$ Aggregate demand falls.

Key Insight: Monetary policy affects the real economy significantly through asset prices, not just through borrowing costs.

3. “Good News is Bad News”: The Friday Jobs Report Paradox

The Event (January 2023)

- **Expected:** Nonfarm payroll to increase by 190,000 jobs.
- **Actual:** Nonfarm payroll increased by **500,000+ jobs** (massive beat).
- **Stock Market Response:** S&P 500 futures crashed 2% in minutes after the 8:30 AM release.

The Paradox

- **Real Economy News:** Wonderful—lots of jobs created, low unemployment.
- **Financial Market Reaction:** Catastrophic—equity prices plummeted.

Why?

Mechanism

1. **Strong Jobs Report** $\rightarrow$ Labor market still overheating.
2. **Overheating Labor Market** $\rightarrow$ Wage pressures remain high (recall wage-inflation correlation).
3. **High Wage Growth** $\rightarrow$ Inflation will stay elevated.
4. **Persistent Inflation** $\rightarrow$ **Fed must keep hiking rates** (or hike even more than previously expected).

5. **Higher Interest Rates** → Lower asset valuations (discounting effect) → **Stocks fall**.

Forward-Looking Markets

- **Financial Markets** anticipate future policy changes.
- Before the jobs report, markets expected the Fed to **start cutting rates in 4 months** (mid-2023).
- After the jobs report, markets pushed out rate cuts to **end of 2023** (or later).
- **Result:** “Good news” about jobs is “bad news” for financial assets because it implies tighter monetary policy ahead.

Key Insight: In an overheating economy, strong economic data is bad for asset prices because it signals the Fed will maintain restrictive policy longer.

4. Global Context: A Worldwide Phenomenon

Inflation is Everywhere

- **Pre-COVID (2010-2019):** 70-80% of countries had inflation at or below 2%.
- **Today (2023):** Nearly **zero countries** have inflation below 2%.
 - Even **Japan**, which struggled with deflation for decades, now has inflation above 2%.

Regional Variations

1. **United States:** Inflation driven by **aggregate demand** (fiscal stimulus, monetary easing, pent-up savings).
2. **Europe:** Similar demand-driven inflation, plus **supply shocks** from Ukraine war.
 - **Energy Prices:** War disrupted natural gas supplies (Russia was major supplier) → Energy prices surged → Inflation spiked.
3. **Latin America:** Commodity exporters benefiting from high prices but also experiencing imported inflation.

Common Thread: All regions “overheated” coming out of COVID due to massive fiscal and monetary stimulus.

5. China: The Major Exception

Zero-COVID Policy and Economic Slowdown

- **Typical Growth Rate:** China grew at 5.5-6% annually (pre-COVID); much higher (~10%+) 15 years ago.
- **2022 Growth:** ~3% (or less)—significant slowdown.
- **Cause:** Aggressive **Zero-COVID policy** shut down large parts of the economy.

- Factories closed, supply chains disrupted, domestic demand collapsed.

The Reopening (Late 2022/Early 2023)

- **Policy Shift:** China abandoned Zero-COVID policy.
- **Expected Boom:** Massive pent-up demand (same dynamic as US post-COVID).
 - Consumers have **forced savings** (couldn't spend during lockdowns).
 - Desire to spend on services (restaurants, travel, entertainment).

Global Spillovers: China's Growth Matters

Impact of 1% Increase in China's Growth Rate (by region): - **East Asia:** +0.3-0.4% growth (neighbors benefit from trade). - **Latin America:** +0.4-0.5% growth (**largest impact**). - **Why?:** Latin America exports commodities (copper, soybeans, oil); China consumes these when building/expanding.

The Double-Edged Sword

- **Good News:** China's reopening boosts global growth, particularly for commodity exporters.
- **Bad News:** China's boom adds to global **aggregate demand** at a time when other countries are trying to **reduce demand** to fight inflation.
 - **Result:** China's reopening could fuel **more global inflation**, complicating the Fed's (and other central banks') job.

Key Insight: In an interconnected world, “what happens in China doesn't stay in China.” Spillovers are large and asymmetric.

6. Recession Risk: The Cost of Disinflation

Forecasts for 2023

- **Survey of Professional Forecasters:** Median expectation is **65% probability of recession** in 2023.
- **Reason:** The Fed (and other central banks) must hike rates aggressively to fight inflation.
 - High rates slow investment, consumption, and hiring.
 - At some point, the Fed may “break something”—cause a recession.

Key Trade-Off: - Don't hike enough → Inflation persists (or worsens). - Hike too much → Severe recession.

Historical Precedent: Every time the Fed has hiked rates rapidly to fight inflation (1980s, early 2000s), a recession followed shortly after.

7. Course Structure and Goals

What You Will Learn

1. **Short Run:** Output determined by aggregate demand (goods market, financial markets, IS-LM model).
2. **Medium Run:** Prices adjust, unemployment dynamics, inflation-unemployment trade-offs.
3. **Long Run:** Growth theory—what determines living standards over decades (capital, technology).
4. **Open Economy:** Trade, exchange rates, international spillovers (Mundell-Fleming model).
5. **Financial Markets:** Asset pricing, role of expectations, monetary policy transmission.

Lecture Format

- **First 5-10 minutes:** Discuss current events (once you have the tools to understand them).
 - Repetition: Same topics revisited as your knowledge deepens.
- **Main Lecture:** Build models to explain the phenomena.
- **Applications:** Connect models to real-world policy debates.

Course Philosophy

- **Simple Models:** Capture the essence of complex phenomena without complicated math.
- **Critical Thinking:** Understand what's in the news, spot mistakes in analysis.
- **Practical Skills:** Useful for internships (Wall Street, policy work), reading economic reports (IMF, WSJ).

Not a Goal: Make you a PhD researcher in macro (that requires much more math).

8. Summary

Key Takeaways

1. **Macroeconomics** studies the economy as a whole—inflation, unemployment, aggregate output, exchange rates.
2. **Current Environment (2023):** Low unemployment, high inflation, rising interest rates, volatile asset prices.
3. **Central Bank Tool:** Interest rates are the main instrument to manage aggregate demand and inflation.
4. **Good News Can Be Bad News:** Strong economic data (jobs) can hurt asset prices because it signals tighter monetary policy.

5. **Global Interconnections:** China’s reopening will boost global growth but may complicate inflation-fighting efforts elsewhere.
6. **Recession Risk:** The Fed’s aggressive rate hikes to fight inflation raise the probability of a recession in 2023.

Coming Up

- **Lecture 2:** Definitions—GDP, inflation, unemployment (the “boring” lecture, but essential foundation).
- **Lecture 3+:** Models to explain the relationships discussed today.

Key Insight for the Course: Macroeconomics is about understanding **interactions and equilibrium effects**, not just summing up individual behaviors. The models are simple, but the insights are powerful.

Next: Definitions of key macroeconomic variables.

Lecture 2: Definitions - GDP, Unemployment, and Inflation

Overview

This lecture establishes the formal definitions of key macroeconomic variables: GDP (Gross Domestic Product), unemployment, and inflation. Unlike microeconomics, where concepts like “output” and “price” are straightforward, macroeconomics requires careful aggregation across millions of goods and services. We explore three equivalent methods of measuring GDP, distinguish nominal from real GDP, and examine current labor market and inflation data.

Context: Current environment features high inflation, supply chain disruptions (COVID, China reopening), and the war in Ukraine affecting energy prices.

1. Why Definitions Matter in Macro

The Aggregation Problem

- **Microeconomics:** Definitions are obvious.
 - Output of a car factory = Number of cars.
 - Price of a car = Dollar price of one car.
- **Macroeconomics:** Millions of goods and services produced simultaneously.
 - **Question:** What is “the output” of the US economy?
 - **Challenge:** How do we add apples, oranges, health services, financial services, and entertainment into a single number?

Key Insight: Macro definitions require sophisticated aggregation techniques. The US National Income and Product Accounts (NIPA) were only developed

in the late 1940s (post-WWII). Before that, proxies like industrial production were used.

2. Gross Domestic Product (GDP)

Definition

GDP (Gross Domestic Product): The total market value of all **final goods and services** produced within a country during a given period (typically one year or one quarter).

Key Points: - **Flow Variable:** GDP measures production over a period of time (e.g., “US GDP in 2022 was \$23 trillion”). - **Domestic:** Produced within the country’s borders (regardless of ownership nationality). - **Final Goods:** Excludes intermediate goods to avoid double counting.

Why “Gross” and not “Net”? Gross means we don’t subtract depreciation of capital. Net Domestic Product (NDP) would subtract depreciation, but we typically use GDP.

Three Equivalent Methods of Measuring GDP

We use a **simple two-firm economy** to illustrate: - **Firm 1 (Steel Company):** Revenue from sales = \$100. - **Firm 2 (Car Company):** Buys steel for \$100, sells cars for \$200.

Question: What is GDP? Is it \$300 (sum of all revenues) or \$200 (final goods only)?

Answer: $GDP = \$200$ (final goods only). Here’s why, shown three ways:

Method 1: Final Goods Approach

$GDP = \text{Value of final goods and services produced}$

- **Final Goods:** Goods purchased by end-users (consumers, firms for investment, government, foreigners).
- **Intermediate Goods:** Goods used as inputs in production (e.g., steel used to make cars).

In our example: - Cars are final goods (sold to consumers) = \$200. - Steel is an intermediate good (used to make cars), so **excluded**.

Robustness Test: Suppose the car company merges with the steel company (vertical integration). - Now steel production is “in-house” (not sold externally). - $GDP = \$200$ (only final output: cars). - **Result:** GDP unchanged by ownership structure. This confirms we should only count final goods.

Key Insight: Counting only final goods ensures GDP is invariant to organizational structure (vertical integration, outsourcing, etc.).

Method 2: Value Added Approach

$$\text{GDP} = \sum \text{Value Added by all firms}$$

Value Added: Revenue minus cost of intermediate inputs. - **Steel Company:** Value Added = \$100 - \$0 = \$100 (no intermediate inputs). - **Car Company:** Value Added = \$200 - \$100 = \$100 (revenue minus steel purchases). - **Total GDP** = \$100 + \$100 = \$200.

Key Insight: Value Added approach avoids double counting by subtracting intermediate inputs at each stage.

Method 3: Income Approach

$$\text{GDP} = \text{Sum of all incomes earned in the economy}$$

Income Categories: 1. **Labor Income (Wages):** Payments to workers. 2. **Capital Income (Profits):** Payments to owners of firms and capital. 3. **Taxes** (in realistic economies).

In our example: - **Wages:** Steel workers (\$80) + Car workers (\$70) = \$150. - **Profits:** Steel company (\$20) + Car company (\$30) = \$50. - **Total GDP** = \$150 + \$50 = \$200.

Fundamental Identity (Closed Economy):

$$\text{Production} = \text{Income}$$

Why This Matters for Macro (but not for Micro): - In **micro**, a car company's income can be spent on anything (food, entertainment, housing). - In **macro** (closed economy), aggregate income **must** be spent on the aggregate good produced (no other economy to buy from). - **Implication:** In macro, what is produced = what is earned = what is spent (circular flow).

Key Insight: This production-income-expenditure equivalence is fundamental to macroeconomic analysis. It's a distinctive feature of macro absent in micro.

Summary: All Three Methods Yield GDP = \$200

- **Final Goods Approach:** Sum only final goods (\$200).
- **Value Added Approach:** Sum value added across all firms (\$200).
- **Income Approach:** Sum all incomes (wages + profits = \$200).

Why Use Multiple Methods? Cross-validation—discrepancies indicate measurement errors.

3. Nominal vs. Real GDP

The Problem: Separating Quantity from Prices

- **Nominal GDP** can grow for two reasons:
 1. **Real Growth**: More goods/services produced.
 2. **Inflation**: Higher prices for the same goods/services.
- **Goal**: Isolate real production growth by removing the effect of price changes.

Definitions

Nominal GDP (\$Y)

$$\text{Nominal GDP}_t = \sum_i P_{it} \times Q_{it}$$

- P_{it} : Current price of good i in year t .
- Q_{it} : Quantity of good i produced in year t .
- **Example**: US Nominal GDP in 2023 = \$24 trillion.

Real GDP (Y)

$$\text{Real GDP}_t = \sum_i P_{i,\text{base}} \times Q_{it}$$

- $P_{i,\text{base}}$: Constant price from a **base year** (e.g., 2012).
- Q_{it} : Quantity of good i produced in year t (changes over time).

Key Feature: Real GDP uses constant prices → removes effect of inflation → measures true production growth.

Numerical Example

Simple Economy (produces only cars):

Year	Quantity (Cars)	Price (\$/car)	Nominal GDP	Real GDP (Base 2012)
2011	10	\$20,000	\$200,000	\$240,000
2012	12	\$24,000	\$288,000	\$288,000
2013	13	\$26,000	\$338,000	\$312,000

Calculations: - **Nominal GDP (2011)**: $10 \times 20,000 = \$200,000$. - **Real GDP (2011)**: $10 \times 24,000 = \$240,000$ (using 2012 prices). - **Nominal GDP (2012)**: $12 \times 24,000 = \$288,000$. - **Real GDP (2012)**: $12 \times 24,000 = \$288,000$ (equal because 2012 is the base year). - **Nominal GDP (2013)**: $13 \times 26,000 = \$338,000$. - **Real GDP (2013)**: $13 \times 24,000 = \$312,000$ (using 2012 prices).

Observations: 1. **Base Year Property**: Nominal GDP = Real GDP in the base year (2012). 2. **Growth Rates Differ**: Nominal GDP growing faster than Real GDP (due to inflation).

US Example (1960-2018)

- **Nominal GDP:** Increased by factor of **38x**.
- **Real GDP:** Increased by factor of **5.7x**.
- **Implication:** Most nominal GDP growth was due to inflation, not real production growth.

Key Insight: Always use **Real GDP** when analyzing economic growth. Nominal GDP is misleading, especially over long periods or in high-inflation countries (e.g., Argentina: chronic recession but nominal GDP “exploding” due to 10,000%+ inflation).

Convention in This Course

- **\$Y:** Nominal GDP (with dollar sign).
- **Y:** Real GDP (no dollar sign).
- **First Part of Course (up to Quiz 1):** Prices fixed → focus on Real GDP.

4. GDP Growth and Recessions

US GDP Growth (Historical Context)

- **Typical Growth Rate:** ~2% per year (real GDP).
- **Recessions** (shaded areas): Periods of negative or very low growth.
 - **Great Recession (2007-2009):** GDP contracted significantly; unemployment peaked at 10%.
 - **COVID-19 Recession (2020):** Massive contraction (~30% annualized in Q2 2020), but rapid recovery in 2021 as economy reopened.

Informal Recession Definition: Two consecutive quarters of negative GDP growth (not the official NBER definition, but close).

Key Insight: GDP growth is the primary indicator of economic health. Recessions = periods of falling output and rising unemployment.

5. Unemployment

Definitions

Labor Market Categories

1. **Employed:** Persons with jobs.
2. **Unemployed:** Persons who:
 - Do **not** have a job.
 - **Are actively looking** for a job (searched in the last 2 weeks).
3. **Not in Labor Force:** Persons who do not have a job and are **not** looking.
 - Examples: Students, retirees, discouraged workers, stay-at-home parents.

Key Concepts

- **Labor Force** (L): $L = \text{Employed} + \text{Unemployed}$.
- **Unemployment Rate** (u):

$$u = \frac{\text{Unemployed}}{\text{Labor Force}} = \frac{\text{Unemployed}}{\text{Employed} + \text{Unemployed}}$$

- **Participation Rate:**

$$\text{Participation Rate} = \frac{\text{Labor Force}}{\text{Working-Age Population}}$$

Important: Unemployment rate is unemployment divided by **labor force**, not total population.

Measurement: Current Population Survey (CPS)

- **Method:** Monthly survey of ~60,000 households.
- **Questions:** Employment status, job search activity over past two weeks.

Discouraged Workers

- **Definition:** People who want a job but have stopped searching (e.g., after repeated rejections).
- **Classification:** Counted as **not in labor force** (not unemployed).
- **Issue:** During recessions, discouraged workers spike → unemployment rate **understates** labor market weakness.
 - **Solution:** Use broader measures (e.g., U-6, which includes discouraged workers and part-time workers seeking full-time work).

Key Insight: The unemployment rate can be misleading during severe recessions if many workers become discouraged and drop out of the labor force.

6. Current US Labor Market (2023)

Unemployment Rate: Historically Low

- **Current:** ~3.5% (lowest since early 1960s).
- **Great Recession Peak:** 10%.
- **COVID Peak:** 14.7%.
- **Normal:** 5-6%.

Paradox: Why is such low unemployment a “problem”? - **Answer:** Tight labor market → Wage pressures → Inflation (as discussed in Lecture 1).

Participation Rate: The Missing Workers

- **Trend (1960s-1990s):** Sharp rise in participation rate.
 - **Cause:** Women entering the workforce (major social/economic shift).
- **Trend (2000-2020):** Gradual decline (aging population, earlier retirements).
- **COVID Shock:** Participation rate dropped sharply in 2020.
 - **Reasons:** Childcare/eldercare responsibilities, early retirements, sector-specific (e.g., restaurants) job losses.
- **Current Issue:** Participation rate **has not recovered** to pre-COVID levels.
 - **Implication:** Labor force smaller than expected → Severe labor shortages → Upward wage pressure.

Federal Reserve Miscalculation: Fed expected participation rate to bounce back fully. It didn't → Fed underestimated inflationary pressures from tight labor market.

Key Insight: Low unemployment rate in 2023 partly reflects strong demand, but also reflects **reduced labor supply** (low participation). This combination fuels wage inflation.

7. Inflation

Definition

Inflation (π): A sustained rise in the **general price level**. - Not about relative prices (e.g., cars becoming more expensive than hotels). - About **all prices** rising on average.

Deflation: Negative inflation (prices falling). Example: Japan (1990s-2010s).

Measuring Inflation

The Price Level (P_t)

- **Challenge:** How do we define “the price” when there are millions of goods?
- **Solution:** Construct a price index (weighted average of many prices).

Method 1: GDP Deflator

$$\text{GDP Deflator}_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t}$$

- **Interpretation:** Ratio of nominal to real GDP.
- **Inflation Rate** (using GDP Deflator):

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100\%$$

- **Usage:** Economists prefer this, but rarely mentioned in media.

Method 2: Consumer Price Index (CPI)

- **Construction:** Tracks the cost of a **fixed basket** of goods and services consumed by typical households.
 - Basket includes: Food, housing, transportation, medical care, entertainment, etc.
- **Inflation Rate** (using CPI):

$$\pi_t^{\text{CPI}} = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100\%$$

- **Usage:** Most commonly cited in media; directly relevant to consumers.

Core CPI

- **Definition:** CPI excluding **food and energy** (volatile components).
- **Rationale:** Food and energy prices fluctuate wildly due to weather, geopolitics → not indicative of underlying inflation trends.
- **Current:** Even Core CPI is elevated (~6%+), so inflation problem is **not** just volatile components.

Recent Debate: Some argue to also exclude **shelter** (housing costs), which is sticky and lagged. But even with these exclusions, inflation remains high.

Key Insight: No matter which price index you use, the US (and most of the world) has an inflation problem in 2023—far above the 2% target.

GDP Deflator vs. CPI: Key Differences

1. **Basket:** GDP Deflator covers all goods produced (including investment goods); CPI covers only consumer goods.
2. **Weights:** GDP Deflator uses current weights (Paasche index); CPI uses fixed weights (Laspeyres index).
3. **Imports:** CPI includes imported goods; GDP Deflator does not (GDP is domestic production).

Practical Implication: The two move closely together in most periods, so the choice rarely matters for broad analysis.

8. International Comparisons

China: Catch-Up Growth

Historical Growth Rates

- **1980s-2000s:** ~10% annual real GDP growth.

- **2010s:** ~6-7% (slowing).
- **2022:** ~3% (COVID-related slowdown).

Why So Fast (Historically)? - Catch-Up Growth: Starting from a very low income level → high marginal returns to capital and technology adoption.
- Convergence: Poor countries can grow faster than rich countries (explained in Lecture 14-16 on growth theory).

Current Concerns: - **Slowing Growth:** China worried about “getting old before getting rich.” - **Demographic Challenges:** One-child policy → aging population → shrinking labor force.

COVID-19 Impact

- **Zero-COVID Policy:** Extremely strict lockdowns → significant economic contraction.
- **Reopening (Late 2022/Early 2023):** Expected boom (similar to US post-COVID bounce in 2021).

Key Insight: China’s slowdown is partly structural (convergence, demographics) and partly COVID-related. Reopening will boost growth temporarily, but long-run trend is declining growth rates.

Japan: The Deflationary Trap

Post-War Boom (1950s-1980s)

- **1960s:** Very high growth (~10%), similar to China’s recent experience.
- **Reason:** Catch-up growth after WWII devastation.

The Bubble and Crash (Late 1980s-Early 1990s)

- **Late 1980s:** Massive asset price bubble (equities, real estate).
 - **Example:** Imperial Palace grounds in Tokyo worth as much as entire state of California (absurd valuation).
- **Early 1990s:** Bubble burst → decades of stagnation (“Lost Decades”).
- **Growth Since 1990:** Very low (~0-1% annually).

The Deflation Problem

- **Inflation Rate:** Near zero or negative (deflation) for most of 1990s-2010s.
- **Problem:** Deflation makes monetary policy ineffective.
 - **Mechanism:** Real interest rate = Nominal rate - Inflation. If inflation is negative, real rates are high even when nominal rates are zero → difficult to stimulate the economy.
- **Current (2023):** Japan finally has inflation above 2% (for the first time in decades).

China's Fear: China worries about following Japan's path—slowdown happening before reaching high income levels.

Key Insight: Japan's experience shows that financial bubbles followed by deflation can trap an economy in prolonged stagnation. Demographics (aging) exacerbate the problem.

9. Summary

Key Definitions

1. **GDP:** Total value of final goods and services produced in an economy during a period.
 - **Three Methods:** Final goods, value added, income (all equivalent).
2. **Nominal GDP:** Measured using current prices (\$Y).
3. **Real GDP:** Measured using constant (base year) prices (Y).
 - **Use Real GDP** for growth analysis.
4. **Unemployment Rate:** $u = \frac{\text{Unemployed}}{\text{Labor Force}}$.
 - **Unemployed:** No job + actively searching.
5. **Participation Rate:** $\frac{\text{Labor Force}}{\text{Working-Age Population}}$.
6. **Inflation:** Sustained rise in the general price level.
 - **GDP Deflator:** $\frac{\text{Nominal GDP}}{\text{Real GDP}}$.
 - **CPI:** Cost of a fixed basket of consumer goods.
 - **Core CPI:** Excludes food and energy.

Current Economic Conditions (2023)

- **US:** Low unemployment (~3.5%), high inflation (~6-8%), rising interest rates.
- **Participation Rate:** Below pre-COVID levels → labor supply constrained → wage pressures.
- **Global:** Nearly all countries have inflation above 2%.
- **China:** Reopening from Zero-COVID → expected boom, but long-run growth slowing.
- **Japan:** Finally escaping deflation after decades of stagnation.

Key Insights

1. **Production = Income** in a closed economy (distinctive macro feature).
2. **Always use Real GDP** to analyze economic growth (remove inflation effects).
3. **Unemployment rate can be misleading** if participation rate shifts (discouraged workers).
4. **Inflation is a macro problem** even though low unemployment + high wages sound good for individuals.

Coming Up

- **Lecture 3:** First model—how is equilibrium GDP determined in the short run? (The goods market and the multiplier).

Next: We begin building models to understand how GDP, unemployment, and inflation interact.

Lecture 3: The Goods Market and Aggregate Demand

Overview

This lecture introduces the first model of output determination in the short run. We analyze how equilibrium GDP is determined by **aggregate demand** in a closed economy. The key mechanism—the **multiplier effect**—shows how an initial change in demand (e.g., from government spending or consumer confidence) generates larger changes in equilibrium output through feedback loops between production, income, and consumption. We also explore fiscal policy and the **Paradox of Savings**, a counterintuitive result where attempts to save more can reduce aggregate output.

Context: Economists forecast a 65% probability of recession within 12 months (as of 2023). Understanding output determination is essential to forecasting recessions.

1. Forecasting Recessions: Why Do We Need Models?

Survey of Professional Economists

- **Question:** What is the probability of a recession within the next 12 months?
- **Current Answer (2023):** ~65% probability—historically very high.
- **Typical Periods:** Recession probability usually <10%; spikes near/during recessions.

How Do Economists Forecast Recessions? - Explicitly or implicitly, they use **models of equilibrium output determination**. - They observe factors (consumer confidence, interest rates, fiscal policy) and run them through a model to predict whether output will decline (recession) or grow (expansion).

Goal of This Course: Build progressively richer models of output determination.

2. Three Time Horizons for Output Determination

Short Run (Focus of Lectures 3-10)

- **Time Frame:** Within a year or so.

- **Mechanism:** Output determined primarily by **aggregate demand**.
- **Assumption:** Prices are sticky (fixed)—little movement in goods prices (asset prices may move).
- **Application:** Business cycle analysis (recessions, booms, expansions).

Medium Run (Lectures 11-13)

- **Time Frame:** Prices begin to adjust sufficiently.
- **Mechanism:** Interaction between demand and supply; wage-price dynamics.
- **Application:** Inflation-unemployment trade-offs (Phillips Curve).

Long Run (Lectures 14-17)

- **Time Frame:** Decades.
- **Mechanism:** Output determined by **supply factors**—capital, labor, technology.
- **Application:** Growth theory (Why does China grow faster than the US?).

Key Distinction: - **Short/Medium Run** → Business cycle (demand-driven).
 - **Long Run** → Growth (supply-driven: capital accumulation, technological progress).

3. The Core Short-Run Mechanism: Demand → Production → Income → Demand

The Feedback Loop

1. **Change in Demand** → Change in Production.
2. **Change in Production** → Change in Income (Production = Income, from national accounts).
3. **Change in Income** → Change in Demand (consumers spend part of extra income).
4. **Repeat:** This creates a **multiplier effect**.

Key Insight: In the short run, output is demand-determined. This is why **consumer confidence** matters so much—if consumers are depressed, demand falls, triggering a recession.

Why Focus on Demand? - Short-run assumption: Firms have **unused capacity** (idle factories, unemployed workers). - Firms can adjust output quickly in response to demand changes. - Contrast with long run: Output constrained by capital, labor, technology.

4. Components of Aggregate Demand

Definitions

1. Consumption (C)

- **Definition:** Goods and services purchased by households.
- **Share of GDP:** ~70% (largest component).
- **Why It Matters:** University of Michigan Consumer Sentiment Index closely watched by economists/financial markets.

2. Investment (I)

- **Definition:** $I = \text{Nonresidential Investment} + \text{Residential Investment}$.
 - **Nonresidential:** Equipment, factories, machinery.
 - **Residential:** Houses, apartment buildings.
- **Share of GDP:** ~18% (substantial, but smaller than consumption).

3. Government Spending (G)

- **Definition:** Purchases of goods/services by federal, state, local government.
- **Important Exclusion:** Government **transfers** (e.g., COVID stimulus checks) are **not** part of G .
 - Transfers are like **negative taxes** (enter through T , not G).
- **Share of GDP:** ~17%.

4. Net Exports ($X - IM$)

- **Exports (X):** Purchases of US goods by foreigners.
- **Imports (IM):** Purchases of foreign goods by US residents.
- **For Now:** Assume closed economy $\rightarrow X = IM = 0$ (we open the economy in Lecture 19).

5. Inventory Investment

- **Definition:** Difference between production and sales (unintended accumulation/decumulation of inventories).
- **Importance:** Small (volatile but not systematically important).
- **For This Course:** Assume inventory investment = 0 (unless explicitly stated).

Aggregate Demand Formula (Closed Economy)

$$Z = C + I + G$$

Interpretation: Demand for domestically-produced goods. - Consumption (C): Households demand domestic goods. - Investment (I): Firms demand capital goods (domestic). - Government (G): Government demands domestic goods/services. - (Later: Add $X - IM$ for open economy.)

US Composition of GDP (2018 Example)

- **Consumption:** 68% of GDP.
- **Investment:** 18% (13% nonresidential, 5% residential).
- **Government:** 17%.
- **Net Exports:** Small (~-3%, trade deficit).
- **Inventory Investment:** ~0.1% (volatile but tiny on average).

5. Modeling Consumption: The Behavioral Assumption

Consumption Function

$$C = C(Y_D)$$

Assumption: Consumption is an **increasing function** of disposable income (Y_D). - If disposable income rises, households consume more.

Disposable Income (Y_D)

$$Y_D = Y - T$$

- Y : Income (= Output = GDP).
- T : Taxes (net of transfers).
- Y_D : What households have available to consume or save.

Linear Consumption Function

$$C = c_0 + c_1(Y - T)$$

Parameters: 1. c_0 : **Autonomous Consumption** (Intercept). - Consumption independent of current income. - Captures: - **Consumer confidence:** If optimistic about future, consume more today even with same current income. - **Wealth effects:** If stock market doubles, consume more (not modeled explicitly here). - **Example:** If expect economy to boom next year, increase consumption today (captured by higher c_0).

2. c_1 : **Marginal Propensity to Consume (MPC)** (Slope).

- **Definition:** Share of an extra dollar of disposable income that is consumed.
- **Range:** $0 < c_1 < 1$.
- **Example:** If you get \$1 extra income and spend \$0.60 on consumption, then $c_1 = 0.6$.
- **Interpretation:** $c_1 = 0.6$ means you consume 60¢ and save 40¢ of each extra dollar.

Graphical Representation

- **Axes:** Disposable income (Y_D) on x-axis, Consumption (C) on y-axis.
- **Intercept:** c_0 (autonomous consumption).
- **Slope:** c_1 (MPC).
- **Shape:** Upward-sloping line with slope between 0 and 1.

Key Insight: The consumption function is the **only behavioral assumption** in this model. Everything else follows from definitions and equilibrium conditions.

6. Determining Equilibrium Output

Step 1: Aggregate Demand Function

$$Z = C + I + G$$

Substitute the consumption function:

$$Z = c_0 + c_1(Y - T) + I + G$$

Rearrange:

$$Z = [c_0 + I + G - c_1T] + c_1Y$$

Key Feature: Aggregate demand is an **increasing function of output** (Y). - **Slope:** c_1 (the MPC). - **Intercept:** $c_0 + I + G - c_1T$ (autonomous components).

Step 2: Equilibrium Condition

$$Y = Z$$

Interpretation: In equilibrium, production (output, Y) equals demand (Z).

Important Distinction: - $Z = c_0 + c_1(Y - T) + I + G$ is a **function** (holds everywhere). - $Y = Z$ is an **equilibrium condition** (holds only at equilibrium).

Common Mistake: Treating $Y = Z$ as a function. It's not—it's a condition that pins down equilibrium.

Step 3: Solve for Equilibrium Output

Substitute aggregate demand into equilibrium condition:

$$Y = c_0 + c_1(Y - T) + I + G$$

Rearrange:

$$Y - c_1Y = c_0 + I + G - c_1T$$

$$Y(1 - c_1) = c_0 + I + G - c_1T$$

$$Y = \frac{1}{1 - c_1} [c_0 + I + G - c_1T]$$

Equilibrium Output Formula:

$$Y = \frac{1}{1 - c_1} [c_0 + I + G - c_1T]$$

7. The Multiplier

Definition

$$\text{Multiplier} = \frac{1}{1 - c_1}$$

Why “Multiplier”? - Since $0 < c_1 < 1$, we have $\frac{1}{1 - c_1} > 1$. - Example: If $c_1 = 0.6$, then $\text{Multiplier} = \frac{1}{1 - 0.6} = \frac{1}{0.4} = 2.5$.

Interpretation

An **autonomous** increase in spending (e.g., $\Delta G = \$1$ billion) leads to a **larger** increase in equilibrium output:

$$\Delta Y = \text{Multiplier} \times \Delta G = \frac{1}{1 - c_1} \times \$1 \text{ billion}$$

Example: If $c_1 = 0.6$ and $\Delta G = \$1$ billion:

$$\Delta Y = 2.5 \times \$1 \text{ billion} = \$2.5 \text{ billion}$$

Why Does the Multiplier Exist?

Mechanism (for $\Delta G = \$1$ billion, $c_1 = 0.6$):

1. **Round 1:** Government spends \$1 billion $\rightarrow$ Output $\uparrow$ \$1 billion $\rightarrow$ Income $\uparrow$ \$1 billion.
2. **Round 2:** Households receive \$1 billion extra income $\rightarrow$ Consume $c_1 \times \$1 \text{ billion} = \$0.6 \text{ billion} \rightarrow$ Demand $\uparrow$ \$0.6 billion $\rightarrow$ Output $\uparrow$ \$0.6 billion $\rightarrow$ Income $\uparrow$ \$0.6 billion.
3. **Round 3:** Households receive \$0.6 billion extra income $\rightarrow$ Consume $c_1 \times \$0.6 \text{ billion} = \$0.36 \text{ billion} \rightarrow$ Output $\uparrow$ \$0.36 billion.
4. **Continue:** \$0.216 billion, \$0.13 billion, ...

Total Effect (geometric series):

$$\Delta Y = \$1\text{B} \times (1 + 0.6 + 0.36 + 0.216 + \dots) = \$1\text{B} \times \frac{1}{1 - 0.6} = \$2.5\text{B}$$

Key Insight: The multiplier is **larger** when c_1 (MPC) is larger. If consumers spend a high share of extra income, the feedback loop is stronger.

Policy Relevance: Size of the Multiplier

- **Large Multiplier** (high c_1): Small fiscal stimulus generates large output increase.
- **Small Multiplier** (low c_1): Large fiscal stimulus needed to boost output significantly.

Debate During Great Recession (2008-2009): - Economists disagreed about the size of the multiplier. - Estimates ranged from 0.5 to 2.5, with major policy implications (how much stimulus to inject?).

8. The Keynesian Cross Diagram

Setup

- **Horizontal Axis:** Output/Income (Y).
- **Vertical Axis:** Aggregate Demand (Z).

Two Lines

1. **Aggregate Demand (ZZ):** $Z = c_0 + c_1(Y - T) + I + G$.
 - **Intercept:** $c_0 + I + G - c_1T$.
 - **Slope:** c_1 (MPC, between 0 and 1).
 - **Why Flatter Than 45°?** Because $c_1 < 1$ (households don't spend all extra income).
2. **45-Degree Line:** $Z = Y$ (equilibrium condition).
 - **Why 45°?** Any point on this line has $Z = Y$ (vertical distance = horizontal distance).
 - This is **not a function**—it's the locus of points satisfying the equilibrium condition.

Equilibrium Point

- **Intersection** of ZZ curve and 45° line.
- At this point: Aggregate demand = Output.
- **Off equilibrium** (e.g., to the left of intersection): $Z > Y$ (demand exceeds output).
- **Off equilibrium** (to the right): $Z < Y$ (output exceeds demand).

Key Insight

- **Only one point** (the intersection) satisfies $Y = Z$.
- The ZZ curve can be evaluated everywhere (it's a function), but equilibrium holds only at the intersection.

9. Policy Experiments

Experiment 1: Increase in Autonomous Consumption ($\Delta c_0 = +\$1$ billion)

Interpretation: Consumer confidence improves—households consume more at any given income level.

Algebraic Effect:

$$\Delta Y = \frac{1}{1 - c_1} \times \Delta c_0 = \frac{1}{1 - c_1} \times \$1\text{B}$$

Example ($c_1 = 0.5$):

$$\Delta Y = \frac{1}{0.5} \times \$1\text{B} = \$2\text{B}$$

Graphical Analysis (Keynesian Cross):

1. **Step A→B:** ZZ curve shifts **up** by \$1B (parallel shift).
 - Distance A→B = \$1B (initial increase in autonomous consumption).
2. **Step B→C:** Output immediately adjusts to meet demand → Output ↑ \$1B.
 - Distance B→C = \$1B (vertical distance to 45° line).
3. **Step C→D:** Income ↑ \$1B → Consumption ↑ $c_1 \times \$1\text{B} = \0.5B (if $c_1 = 0.5$).
 - Distance C→D = \$0.5B (movement along new ZZ curve).
4. **Step D→E:** Output adjusts again → Output ↑ \$0.5B → Income ↑ \$0.5B.
5. **Continue:** \$0.25B, \$0.125B, ... → Converge to new equilibrium at $Y' = Y + \$2\text{B}$.

Key Takeaway: A \$1B increase in autonomous spending leads to \$2B increase in output (if $c_1 = 0.5$). The multiplier amplifies the initial shock.

Experiment 2: Expansionary Fiscal Policy ($\Delta G = +\$1$ billion)

Mechanism: Identical to increase in c_0 . - ZZ curve shifts up by \$1B. - Multiplier effect: $\Delta Y = \frac{1}{1 - c_1} \times \1B .

Fiscal Multiplier: A \$1 increase in government spending raises output by more than \$1.

Experiment 3: Tax Increase ($\Delta T = +\$1$ billion)

Effect on Aggregate Demand:

$$Z = c_0 + c_1(Y - T) + I + G$$

Increase in T reduces disposable income:

$$\Delta Z = -c_1 \Delta T = -c_1 \times \$1\text{B}$$

Effect on Equilibrium Output:

$$\Delta Y = \frac{1}{1 - c_1} \times (-c_1 \Delta T) = -\frac{c_1}{1 - c_1} \times \$1\text{B}$$

Example ($c_1 = 0.6$, $\Delta T = +\$1\text{B}$):

$$\Delta Y = -\frac{0.6}{0.4} \times \$1\text{B} = -\$1.5\text{B}$$

Key Insight: Tax increases are **contractionary** (reduce output), but less contractionary per dollar than government spending is expansionary. - **Why?** Consumers only reduce spending by c_1 (save the rest), so initial shock smaller.

10. Alternative Derivation: Investment = Saving

Definitions

- **Private Saving** (S): $S = Y - T - C$ (disposable income minus consumption).
- **Public Saving** ($T - G$): Government budget balance (surplus if positive, deficit if negative).

Equilibrium Condition (Alternative Form)

$$I = S + (T - G)$$

Interpretation: Investment = Private Saving + Public Saving.

Equivalence to $Y = Z$:

$$I = (Y - T - C) + (T - G)$$

$$I = Y - C - G$$

$$Y = C + I + G = Z$$

Why This Matters: The curve we'll study later—**IS curve**—stands for “Investment = Saving.”

Graphical Representation

- **Horizontal Axis:** Output (Y).
- **Vertical Axis:** Investment (I) and Saving ($S + T - G$).
- **Investment:** Horizontal line (assumed constant at $\bar{I}$).
- **Saving:** Upward-sloping line with slope $1 - c_1$ (marginal propensity to save).
- **Equilibrium:** Where Investment line intersects Saving line.

11. The Paradox of Savings

The Question

- **Individual Level:** Saving more is good (increases wealth).
- **Aggregate Level:** Is saving more good for the economy?

Answer (Short Run): No—attempts to save more can **reduce** equilibrium output and may not even increase total saving.

Mechanism (Using IS Derivation)

Scenario: Households decide to save more for any given level of income. - **Effect:** Private saving function shifts **up**. - **Problem:** Now $S > I$ (more saving than investment) → **Not an equilibrium**.

How Does Equilibrium Restore? - Investment (I) is fixed (exogenous). - Public saving ($T - G$) is fixed (exogenous). - **Only endogenous variable:** Output (Y).

Result: Output must **fall** to reduce saving back to I . - Lower Y → Lower income → Lower saving → Equilibrium restored at lower output.

Paradox: Attempts to save more lead to a **recession** (lower Y). In equilibrium, total saving may be **unchanged** (still equals I) or even lower if recession is severe.

Mechanism (Using Keynesian Cross)

Scenario: Households save more → c_0 falls (autonomous consumption declines). - **Effect:** ZZ curve shifts **down**. - **Result:** Equilibrium output falls. - **Multiplier Effect:** Output falls by **more** than the initial decline in consumption (multiplier works in reverse).

Key Insight: What is true for an individual (saving more makes you richer) is **not** true for the economy as a whole (everyone saving more causes a recession). This is **Fallacy of Composition**.

When Does the Paradox Matter?

- **Normal/Recession Times:** Paradox holds—attempts to save more are contractionary.
- **Overheated Economy** (like 2023): Higher saving would be **beneficial** (reduce demand $\rightarrow$ lower inflation).

Policy Implication: During recessions, governments try to **boost confidence** and **discourage excessive saving** to prevent spiraling declines in output.

12. Fiscal Policy: Expansionary vs. Contractionary

Definitions

Expansionary Fiscal Policy

- **Increase G** (government spending) or **Decrease T** (taxes).
- **Effect:** Shifts ZZ curve **up** $\rightarrow$ Output $\uparrow$ (boom).
- **Use:** Fight recessions.

Contractionary Fiscal Policy

- **Decrease G** or **Increase T** .
- **Effect:** Shifts ZZ curve **down** $\rightarrow$ Output $\downarrow$ (slowdown/recession).
- **Use:** Cool overheated economy (reduce inflation pressure).

Current Policy Stance (2023)

- **Fed's Goal:** Cool the economy (contractionary).
- **Method:** Raise interest rates (monetary policy—covered in Lecture 5+).
- **Communication Strategy:** Fed officials make speeches emphasizing tightening to **reduce confidence** and dampen consumption/investment.

Effect of Fed Speeches: - Financial markets react immediately (equity prices fall). - Consumers eventually respond (reduce spending). - **Goal:** Induce a mild recession to bring down inflation.

13. Summary

Key Concepts

1. **Short-Run Output Determination:** Output determined by aggregate demand ($Y = Z$).
2. **Aggregate Demand:** $Z = C + I + G$ (closed economy).
3. **Consumption Function:** $C = c_0 + c_1(Y - T)$.
 - c_0 : Autonomous consumption.
 - c_1 : Marginal propensity to consume (MPC).

4. Equilibrium Output:

$$Y = \frac{1}{1 - c_1} [c_0 + I + G - c_1 T]$$

5. Multiplier: $\frac{1}{1 - c_1} > 1$.

- Amplifies shocks to autonomous spending.
- Larger when c_1 is larger (consumers spend more of extra income).

Key Diagrams

1. **Keynesian Cross:** Intersection of ZZ curve (aggregate demand) and 45° line ($Y = Z$).
2. **Investment = Saving:** Intersection of horizontal I line and upward-sloping S curve.

Policy Implications

1. **Fiscal Policy:**
 - $\uparrow G$ or $\downarrow T \rightarrow$ Expansionary (output $\uparrow$).
 - $\downarrow G$ or $\uparrow T \rightarrow$ Contractionary (output $\downarrow$).
2. **Multiplier Effect:** Government spending increases output by **more** than the initial spending (due to feedback loops).
3. **Paradox of Savings:** Attempts to save more can backfire, causing recessions in the short run.

Key Insights

1. **Demand Drives Output** (short run): Firms produce whatever demand requires (unused capacity).
2. **Multiplier Mechanism:** Initial demand shock $\rightarrow$ Output change $\rightarrow$ Income change $\rightarrow$ Consumption change $\rightarrow$ Further output change $\rightarrow \dots$
3. **Macro Micro:** What's good for individuals (saving more) may harm the aggregate economy (Paradox of Savings).
4. **Consumer Confidence Matters:** Changes in c_0 (sentiment) can drive recessions or booms.

Coming Up

- **Lecture 4:** Introduce financial markets—interest rates, money demand/supply, central bank policy.
- **Lecture 5:** Integrate goods market (this lecture) with financial markets $\rightarrow$ **IS-LM model**.

Next: We add financial markets and monetary policy to understand how interest rates affect output.

Lecture 4: Financial Markets and Interest Rate Determination

Overview

This lecture introduces financial markets and explains how the central bank (Federal Reserve) determines interest rates. We develop a simple model with two assets—money and bonds—to understand the portfolio decision households face. The equilibrium interest rate is determined by money supply (controlled by the Fed) and money demand (determined by nominal income and interest rates). We explore **open market operations** (buying/selling bonds) as the mechanism through which the Fed controls interest rates, and introduce the role of banks and the **Federal Funds Rate**.

Context: The Fed is aggressively hiking interest rates (2022-2023) to fight inflation. Understanding how interest rates are determined is essential for understanding monetary policy.

1. Why Study Financial Markets?

Monetary Policy: The Fastest Policy Tool

Two Main Short-Run Policy Tools: 1. **Fiscal Policy** (Lecture 3): Government spending (G) and taxes (T). - **Direct Effect:** Government buys goods $\rightarrow$ Increases aggregate demand directly. - **Drawback:** Slow—requires Congressional approval. 2. **Monetary Policy** (This Lecture): Interest rates controlled by the central bank. - **Indirect Effect:** Central bank buys/sells bonds $\rightarrow$ Changes interest rates $\rightarrow$ Affects investment and consumption. - **Advantage:** Fast and nimble—Federal Open Market Committee (FOMC) can meet and change rates quickly.

Key Insight: Monetary policy is **faster** than fiscal policy, but **more indirect** (must go through financial markets).

Monetary Policy Works Through Financial Markets

The Central Bank's Challenge: - **Mandate:** Stabilize output, avoid recessions, maintain price stability (goods market objectives). - **Tools:** Can only buy/sell financial assets (bonds), not goods. - **Cannot** go out and buy hamburgers to increase demand (fiscal policy can). - **Can** buy bonds, which affects interest rates, which then affects investment and consumption.

Implication: Understanding monetary policy requires understanding **financial markets**.

2. The Federal Reserve System

Key Figures

Jerome Powell

- **Role:** Chair of the Federal Reserve (2018-present).
- **Significance:** Most watched person in financial markets—his decisions determine interest rates.

Katsuo Ueda

- **Role:** Governor of the Bank of Japan (appointed 2023).
- **Background:** MIT PhD alumnus (Economics Department has produced many central bankers).

MIT Econ PhD Alumni in Central Banking: - **Ben Bernanke:** Fed Chair during 2008 Financial Crisis. - **Mario Draghi:** ECB President during European debt crisis (“Whatever it takes” speech). - **Stanley Fischer:** Governor of Bank of Israel. - **Philip Lowe:** Governor of Reserve Bank of Australia.

Institutional Structure

Board of Governors

- **Location:** Washington, D.C.
- **Composition:** 7 members.
- **Appointment:** Nominated by the President, confirmed by the Senate.
- **Chair:** One of the 7 governors (currently Jerome Powell).

Regional Federal Reserve Banks

- **Number:** 12 banks across the US (e.g., Boston, New York, San Francisco).
- **Boston Fed:** Distinctive building made of recycled aluminum near the waterfront.

Federal Open Market Committee (FOMC)

- **Role:** Sets the **policy interest rate** (Federal Funds Rate).
- **Voting Members:**
 - **7 Governors** (permanent voters).
 - **President of New York Fed** (permanent voter, due to NY’s role as financial center).
 - **4 Regional Bank Presidents** (rotating voters).

Why NY Fed President Always Votes: New York is the financial heart of the US—the NY Fed communicates with financial markets and executes open market operations.

3. A Simple Model of Financial Markets

Two Assets Only

We simplify the financial system to **two assets**:

1. Money (M)

- **Characteristics:**
 - Used for **transactions** (buying goods/services).
 - Pays **zero interest** ($i_M = 0$).
- **Examples:** Cash (currency), checkable deposits (checking accounts).
- **Advantage:** Liquidity (can buy anything immediately).
- **Disadvantage:** No return.

2. Bonds (B)

- **Characteristics:**
 - Pay **positive interest** ($i > 0$).
 - **Cannot** be used directly for transactions.
- **Example:** One-year US Treasury bill (buy for \$95, receive \$100 in one year $\rightarrow$ 5.3% return).
- **Advantage:** Earn return.
- **Disadvantage:** Illiquid (must sell bond to get cash for transactions).

The Portfolio Decision: How much wealth to hold in money vs. bonds?

The Trade-Off

Trade-Off: - **High interest rates** $\rightarrow$ High opportunity cost of holding money $\rightarrow$ Bias portfolio toward bonds. - **Low interest rates** $\rightarrow$ Low opportunity cost of holding money $\rightarrow$ Bias portfolio toward money (liquidity convenience).

Historical Context: - **2008-2020:** Interest rates near zero $\rightarrow$ Portfolio decision not very important (everyone held cash). - **2022-2023:** Interest rates $\sim$ 5% $\rightarrow$ Portfolio decision matters a lot (giving up 5% return by holding cash).

Key Insight: The portfolio decision creates a **demand for money** that depends on the interest rate.

4. Money Demand

Money Demand Function

$$M^d = L(\$Y, i)$$

Notation: M^d = Money demand, L = Liquidity preference function (Keynes's term).

Determinants of Money Demand

1. Interest Rate (i): Movement Along Curve

$$\frac{\partial M^d}{\partial i} < 0$$

Interpretation: Higher interest rates $\rightarrow$ Lower money demand.

Why? - Higher $i \rightarrow$ Higher opportunity cost of holding money $\rightarrow$ Hold less money, more bonds. - Lower $i \rightarrow$ Lower opportunity cost $\rightarrow$ Hold more money.

Graphical Representation: Downward-sloping money demand curve in (M, i) space.

2. Nominal Income ($\$Y = P \times Y$): Shifts Curve

$$\frac{\partial M^d}{\partial \$Y} > 0$$

Interpretation: Higher nominal income $\rightarrow$ Higher money demand (curve shifts right).

Why Use Nominal (Not Real) Income?

Example 1 (Real output increases, prices fixed): - Real GDP $\uparrow \rightarrow$ More transactions $\rightarrow$ Need more money.

Example 2 (Prices double, real output fixed): - Same real transactions, but prices 2x higher $\rightarrow$ Need 2x more dollars to conduct same transactions.

Conclusion: Money demand depends on **nominal** income ($\$Y = P \times Y$), not just real income (Y), because money is denominated in **dollars**.

Money Demand Diagram

- **Axes:** Money (M) on horizontal, Interest rate (i) on vertical.
- **Curve:** Downward-sloping (higher $i \rightarrow$ lower M^d).
- **Shifts:** $\uparrow \$Y$ shifts curve **right** (for any given i , demand more money).

5. Equilibrium Interest Rate (Simple Model: No Banks)

Money Supply

Assumption: Central bank controls money supply exogenously.

$$M^s = \bar{M}$$

Key Point: Only the central bank can produce currency (printing money is illegal for anyone else).

Equilibrium Condition

$$M^s = M^d$$

Graphical Representation: - **Money Supply:** Vertical line at $M = \bar{M}$ (independent of i). - **Money Demand:** Downward-sloping curve. - **Equilibrium:** Intersection determines the interest rate i^* .

Key Insight: The central bank sets M , and the market determines i to equate supply and demand.

6. Monetary Policy: Changing the Interest Rate

Expansionary Monetary Policy ($\uparrow M$)

Mechanism: 1. Central bank **increases** money supply: M shifts **right**. 2. At the old interest rate, **excess supply** of money (people holding more money than they want). 3. To restore equilibrium, people must **demand** more money. 4. When do people demand more money? When $i \downarrow$ (lower opportunity cost). 5. **Result:** Interest rate **falls** to new equilibrium.

Summary: $\uparrow M \rightarrow \downarrow i$ (Expansionary).

Contractionary Monetary Policy ($\downarrow M$)

Mechanism: 1. Central bank **decreases** money supply: M shifts **left**. 2. At the old interest rate, **excess demand** for money. 3. Interest rate must **rise** to reduce money demand back to the (now lower) supply.

Summary: $\downarrow M \rightarrow \uparrow i$ (Contractionary).

Current Policy (2022-2023): Hiking Rates

- **Context:** Inflation at 8% (far above 2% target).
- **Policy Response:** Fed raising interest rates aggressively (0% $\rightarrow$ 5% in ~18 months).
- **Mechanism:** Fed **reducing** money supply $\rightarrow$ Interest rates **rising**.

7. Why Central Banks Don't Target Money Supply Any-more

Historical Practice (Pre-1980s)

Old Approach: Central banks targeted **monetary aggregates** (M). - Set $M = \bar{M}$. - Let market determine i .

The Problem: Volatile Interest Rates

Issue: Money demand shifts frequently due to: - **Predictable Factors:** Holidays, weekends, Super Bowl (people buy more goods $\rightarrow$ need more cash). -

Unpredictable Factors: Financial innovations, payment technology changes.
- **Example:** Y2K panic (1999-2000)—people withdrew cash fearing ATMs would fail → Huge spike in money demand.

Result: If M is fixed and M^d shifts frequently → i fluctuates wildly. - **Example:** Super Bowl weekend → $M^d \uparrow$ → At fixed M , i spikes.

Key Insight: Volatile interest rates are disruptive (firms can't plan investment, households can't plan borrowing).

Modern Practice (Post-1980s)

New Approach: Central banks target **interest rates** (i), not money supply.
- Fed announces: “We’re setting the interest rate at 5%.” - Fed adjusts M continuously to keep i at target (even as M^d shifts).

Advantage: Stable interest rates, even though money demand fluctuates.

Key Insight: When the Fed says “We’re cutting rates by 50 basis points,” they’re implicitly increasing the money supply, but they don’t announce the money supply change.

8. Shifts in Money Demand: Why Central Banks Must Be Active

Nominal Income Growth → Rightward Shift in M^d

Typical Year: - **Inflation:** $\sim 2\% \rightarrow P \uparrow \rightarrow \$Y \uparrow$. - **Real Growth:** $\sim 2\% \rightarrow Y \uparrow \rightarrow \$Y \uparrow$. - **Combined:** Nominal income grows $\sim 4\% \rightarrow M^d$ shifts **right** by $\sim 4\%$.

Implication: If Fed wants to **maintain** the interest rate at i^* : - Must **increase** M by $\sim 4\%$ to accommodate higher money demand. - **Failure to do so** → Interest rate rises unintentionally (contractionary).

Key Insight: Even to maintain a constant interest rate, the central bank must continuously expand the money supply to match growth in nominal income and money demand.

9. Open Market Operations: How the Fed Changes Money Supply

What Are Open Market Operations?

Definition: Purchases or sales of bonds by the central bank in the **open market** (public bond market, not directly from the Treasury).

Why “Open Market”? Operations occur in public markets where anyone can trade (transparency).

Expansionary Open Market Operation

Goal: Increase money supply $\rightarrow$ Lower interest rate.

Mechanism: 1. Fed **buys bonds** from the public (financial institutions, households). 2. Fed pays for bonds by **creating money** (printing currency or crediting bank reserves). 3. Public now holds **more money, fewer bonds**. 4. Money supply $\uparrow \rightarrow$ Interest rate $\downarrow$.

Analogy: Fed goes to the bond market with a “big bag of cash” and buys bonds.

Contractionary Open Market Operation

Goal: Decrease money supply $\rightarrow$ Raise interest rate.

Mechanism: 1. Fed **sells bonds** to the public. 2. Public pays for bonds with money. 3. Fed **removes money** from circulation (destroys it). 4. Money supply $\downarrow \rightarrow$ Interest rate $\uparrow$.

Central Bank Balance Sheet

Assets: Bonds (Treasury securities).

Liabilities: Money (currency issued).

Expansionary Operation (\$1 million example): - **Assets:** Bonds $\uparrow$ by \$1 million. - **Liabilities:** Money $\uparrow$ by \$1 million. - **Result:** Balance sheet **expands** (both sides grow by \$1 million).

Historical Context: - **Pre-2008:** Fed balance sheet $\sim$ \$1 trillion (small). - **Post-2008:** Global Financial Crisis + COVID-19 $\rightarrow$ Massive expansionary operations $\rightarrow$ Balance sheet $\sim$ \$9 trillion (2022).

Key Insight: The size of the central bank’s balance sheet reflects cumulative expansionary policy (buying bonds to fight recessions).

10. Bond Prices and Interest Rates: The Inverse Relationship

Bond Pricing Formula

Setup: One-year zero-coupon bond. - **Today:** Buy bond for price P_B . - **One year from now:** Receive \$100 (face value).

Interest Rate (return on bond):

$$i = \frac{100 - P_B}{P_B}$$

Example 1 (Price = 95) : $i = \frac{100-95}{95} = \frac{5}{95} \approx 5.3\%$

Example 2 (Price = 100) : $i = \frac{100-100}{100} = 0\%\$$

Inverse Relationship:

$$\frac{\partial i}{\partial P_B} < 0$$

- **Higher bond price** → **Lower interest rate**.
- **Lower bond price** → **Higher interest rate**.

Open Market Operations and Bond Prices

Expansionary Policy (Fed buys bonds): 1. Fed enters bond market with huge demand for bonds. 2. Demand for bonds $\uparrow \rightarrow$ Bond prices $\uparrow$ (basic supply-demand). 3. Bond prices $\uparrow \rightarrow$ Interest rates $\downarrow$ (inverse relationship).

Contractionary Policy (Fed sells bonds): 1. Fed floods bond market with supply of bonds. 2. Supply of bonds $\uparrow \rightarrow$ Bond prices $\downarrow$. 3. Bond prices $\downarrow \rightarrow$ Interest rates $\uparrow$.

Key Insight: Another way to understand open market operations—Fed directly affects bond prices, which mechanically affects interest rates.

11. The Role of Banks: Introducing Realism

Why Add Banks to the Model?

In Reality: Financial system has intermediaries (banks) between households and the central bank.

Two Key Roles of Banks: 1. **Create money:** Checkable deposits (checking accounts) are money—you can write checks, use debit cards. 2. **Hold reserves:** Banks deposit a fraction of their deposits at the central bank (called **reserves**).

Bank Balance Sheets

Assets: - **Loans:** Money lent to borrowers. - **Bonds:** Government securities purchased. - **Reserves:** Deposits at the central bank.

Liabilities: - **Deposits:** Money owed to depositors (checkable accounts).

Key Point: Deposits are **money** for households (can use for transactions), but they're **liabilities** for banks.

Central Bank Balance Sheet (With Banks)

Assets: Bonds (Treasury securities).

Liabilities: - **Currency:** Cash held by the public. - **Reserves:** Deposits by banks at the central bank.

Central Bank Money (also called **High-Powered Money** or **Monetary Base**):

$$H = \text{Currency} + \text{Reserves}$$

Key Distinction: - **Central Bank Money** (H): Currency + Reserves—directly controlled by the Fed. - **Total Money** (M): Currency + Checkable Deposits—includes money created by banks.

Relationship: $M > H$ (banks create additional money through deposits).

Reserve Requirements

Definition: Banks must hold a **fraction** (θ) of their deposits as reserves at the central bank. - **Example:** $\theta = 0.1$ (10%) $\rightarrow$ For every \$100 in deposits, bank must hold \$10 in reserves.

Purpose: Regulatory requirement—ensures banks have liquidity to meet withdrawal demands.

Money Demand and Reserve Demand

Assumption (Simplified): No one holds currency—all money is in checking accounts.

Money Demand: $M^d = L(\$Y, i)$ (same as before).

Reserve Demand: Banks demand reserves to back deposits.

$$R^d = \theta \times M^d$$

Example: If $M^d = \$1,000$ billion and $\theta = 0.1$:

$$R^d = 0.1 \times \$1,000\text{B} = \$100 \text{ billion}$$

Demand for Central Bank Money:

$$H^d = \theta \times M^d = \theta \times L(\$Y, i)$$

Equilibrium with Banks

Money Market Equilibrium (as before):

$$M^s = M^d$$

Reserve Market Equilibrium (new):

$$H^s = H^d = \theta \times M^d$$

Central Bank Control: Fed controls H^s (high-powered money supply).

Solving for Interest Rate:

$$H^s = \theta \times L(\$Y, i)$$

Rearrange:

$$\frac{H^s}{\theta} = L(\$Y, i)$$

Result: Same downward-sloping money demand curve, but scaled by θ .

Key Insight: The mechanism is **identical** to the simple model—equilibrium in the **reserve market** (not the full money market) determines the interest rate.

12. The Federal Funds Rate

Definition

Federal Funds Rate: The interest rate in the **market for reserves** (market where banks lend reserves to each other overnight).

Why This Market Matters: - Banks that are **short on reserves** (below required $\theta \times$ Deposits) borrow from banks with **excess reserves**. - This overnight lending market is active every night (huge volume of transactions).

Example: - Bank A has \$10 million in deposits, needs 1 *million in reserves* (= 0.1\$), but only has \$0.8 million → **Borrows** \$0.2 million overnight. - Bank B has excess reserves → **Lends** \$0.2 million to Bank A at the Federal Funds Rate.

Fed's Control Over the Federal Funds Rate

Mechanism: 1. Fed **targets** a specific Federal Funds Rate (e.g., 5.00%-5.25% range). 2. **If rate rises above target: Banks need more reserves** → Fed **injects reserves** (buys bonds) → Rate falls back to target. 3. If **rate falls below target:** Too many reserves in system → Fed **drains reserves** (sells bonds) → Rate rises back to target.

Key Insight: The Fed doesn't directly "set" the rate—it **targets** the rate and adjusts H^s (reserve supply) continuously to hit the target.

Federal Funds Rate vs. Other Interest Rates

Direct Control: Fed controls the **Federal Funds Rate** (overnight rate in reserve market).

Indirect Effect: Federal Funds Rate affects **other interest rates** in the economy (transmission mechanism): - **Treasury Bills** (3-month, 1-year): Move closely with Fed Funds Rate. - **Corporate Bonds:** Higher rates, but move in tandem. - **Mortgage Rates:** Influenced by longer-term Treasury rates.

Key Insight: By controlling the Federal Funds Rate, the Fed indirectly influences the entire **term structure** of interest rates.

13. Recent History: Federal Funds Rate (2019-2023)

Pre-COVID (2019)

- **Rate:** ~2.5%.
- **Context:** Fed had been hiking rates (2015-2018), then began cutting as economy slowed.

COVID Shock (March 2020)

- **Action:** Fed cut rates to **0-0.25%** (effective zero).
- **Reason:** Massive economic shock—output collapsed, unemployment spiked to 14.7%.
- **Duration:** Rates remained at zero for ~2 years (2020-2022).

Zero Lower Bound (ZLB)

Why Can't Rates Go Below Zero? - If $i < 0$, bonds pay negative return.
- **Alternative:** Hold cash (pays 0%) under the mattress. - **Result:** No one would hold bonds at $i < 0 \rightarrow$ Demand for money infinite. - **Implication: Zero is the floor** for nominal interest rates (ZLB).

Policy Constraint: At ZLB, conventional monetary policy is exhausted (cannot cut rates further).

Post-COVID Inflation (2021-2023)

- **Problem:** Inflation surged to 8%+ (far above 2% target).
- **Fed Response:** Aggressive rate hikes—**fastest pace since the 1980s**.
 - 2022: 0% $\rightarrow$ 2% $\rightarrow$ 3% $\rightarrow$ 4% $\rightarrow$ 5% (seven consecutive hikes of 50-75 basis points).
- **Current Rate** (as of early 2023): ~5%.

Key Insight: Fed is “catching up” after being slow to respond to inflation (was “behind the curve” in 2021).

14. Summary

Key Concepts

1. **Two Assets:** Money (liquid, zero return) vs. Bonds (illiquid, positive return).
2. **Money Demand:** $M^d = L(\$Y, i)$.
 - $\frac{\partial M^d}{\partial i} < 0$ (downward-sloping curve).
 - $\frac{\partial M^d}{\partial \$Y} > 0$ (shifts curve right).

3. **Equilibrium Interest Rate:** Determined by $M^s = M^d$ (or $H^s = H^d$ with banks).
4. **Monetary Policy:**
 - **Expansionary:** $\uparrow M$ (or $\uparrow H$) $\rightarrow \downarrow i$.
 - **Contractionary:** $\downarrow M$ (or $\downarrow H$) $\rightarrow \uparrow i$.
5. **Open Market Operations:** Fed buys/sells bonds to change money supply.
 - **Buy bonds** $\rightarrow$ Inject money $\rightarrow$ Lower interest rates.
 - **Sell bonds** $\rightarrow$ Remove money $\rightarrow$ Raise interest rates.
6. **Bond Prices and Interest Rates:** Inverse relationship.
 - $P_B \uparrow \rightarrow i \downarrow$.
 - Fed buying bonds raises bond prices $\rightarrow$ lowers interest rates.

Institutional Details

1. **Federal Reserve System:**
 - **7 Governors + 12 Regional Banks.**
 - **FOMC** sets policy (11 voting members).
2. **Federal Funds Rate:** Interest rate in the overnight reserve market—the rate the Fed targets.
3. **Central Bank Money (H):** Currency + Reserves—directly controlled by the Fed.
4. **Banks Create Money:** Checkable deposits expand money supply beyond H .
5. **Reserve Requirements:** Banks hold fraction θ of deposits as reserves.

Key Insights

1. **Modern Monetary Policy:** Central banks target interest rates (not money supply) because money demand is too volatile.
2. **Zero Lower Bound:** Nominal interest rates cannot go below zero—major constraint during deep recessions.
3. **Transmission Through Financial Markets:** Monetary policy affects the real economy indirectly (through interest rates $\rightarrow$ investment/consumption).
4. **Fed's Current Challenge (2023):** Hiking rates aggressively to fight inflation, risking recession.

Coming Up

- **Lecture 5:** Integrate financial markets (this lecture) with goods market (Lecture 3) $\rightarrow$ **IS-LM Model**.
- **Understand:** How does a change in interest rates (monetary policy) affect equilibrium output?

Next: The IS-LM model—the workhorse of short-run macroeconomics.

Lecture 5: The IS-LM Model (Part I)

Overview

The IS-LM model is the workhorse model for understanding the **joint determination of output (Y) and the interest rate (i)**. It combines: - **IS Curve**: Equilibrium in the **goods market** (Investment-Savings) - **LM Curve**: Equilibrium in the **financial market** (Liquidity-Money)

This model is fundamental for analyzing monetary and fiscal policy, and forms the basis for modern central bank thinking (including the Federal Reserve).

1. Current Economic Context (2022-2023)

Wealth Increase During COVID Recovery

- **Dramatic Rise in Net Worth**: US household wealth increased significantly during and after the COVID recovery.
- **Sources of Wealth Increase**:
 1. **Asset Price Appreciation** (Main Driver):
 - Massive equity market rallies
 - Skyrocketing house prices
 - Despite 2022 decline, wealth remains elevated relative to pre-COVID trend
 2. **Excess Savings** (Important for Lower-Income Segments):
 - Incomes remained stable or increased (due to government transfers)
 - Consumption opportunities limited during lockdowns
 - Resulted in \$2.7-2.8 trillion in excess savings

Implications for Aggregate Demand

- **Wealth Effect**: Higher wealth → higher consumption → increased aggregate demand
- **Desaving Phase**: Consumers now spending accumulated savings
 - Saving rate dropped below historical averages
 - Formula: $\text{Saving Rate} = \text{Income} - \text{Consumption}$
 - Lower saving rate → higher consumption relative to income
- **Credit Expansion**: Lower-income segments borrowing (credit cards) to fund consumption even after depleting excess savings

Consumption Patterns Post-COVID

- **Services Consumption** (~2/3 of consumption):
 - Collapsed during COVID
 - Fully recovered and now **above trend**

- Examples: Travel, restaurants, hotels
- **Goods Consumption** (~1/3 of consumption):
 - Initially collapsed, then spiked during COVID (gadgets, home equipment)
 - Slowing down but still **above trend**

The Overheating Problem

- **Current Situation:** Very high aggregate demand relative to productive capacity
 - **Consequence:** High inflation (mechanism to be studied in ~6 lectures)
 - **Policy Response:** Fed implementing **contractionary monetary policy** (raising interest rates) to cool the economy
 - **Relevance:** The IS-LM model explains exactly how this works
-

2. The IS Curve (Goods Market Equilibrium)

Building on Lecture 3

The IS curve extends the goods market model from Lecture 3 by making **investment endogenous**.

Lecture 3 Model:

$$Y = C(Y - T) + I + G$$

- Only consumption was endogenous - Investment (I) was exogenous (constant)

Lecture 5 Enhancement:

$$Y = C(Y - T) + I(Y, i) + G$$

- Investment is now a **function** of output and interest rate

Investment Function: $I = I(Y, i)$

Dependence on Output (Y)

- **Positive Relationship:** $\frac{\partial I}{\partial Y} > 0$
- **Mechanism:** Higher sales $\rightarrow$ firms invest more to expand capacity
- **Interpretation:** Investment has a “marginal propensity to invest” similar to consumption’s marginal propensity to consume

Dependence on Interest Rate (i)

- **Negative Relationship:** $\frac{\partial I}{\partial i} < 0$
- **Mechanisms:**
 1. **Borrowing Cost:** Most investment is debt-financed. Higher $i \rightarrow$ more expensive to borrow $\rightarrow$ less investment

2. **Opportunity Cost:** Even without borrowing, higher i means higher opportunity cost of using funds for physical investment versus financial investment

Important Note: This is **real investment** (purchase of capital goods: equipment, structures, machinery), **NOT** financial investment (bonds, stocks). Financial investments belong to the financial market, not goods market.

The IS Relation

Definition: The IS curve represents all combinations of (Y, i) consistent with equilibrium in the goods market.

Equilibrium Condition:

$$Y = C(Y - T) + I(Y, i) + G$$

This defines a **relationship** between Y and i , not just a single point.

Deriving the IS Curve Graphically

Step 1: Start with the Goods Market Diagram (ZZ curve) - Horizontal axis: Output (Y) - Vertical axis: Aggregate Demand (Z) - 45° line: $Y = Z$ (equilibrium condition) - ZZ curve: Aggregate demand as function of output

Key Difference from Lecture 3: - ZZ curve is **steeper** than before - **Reason:** Both consumption AND investment now increase with Y - Slope = MPC + MPI (marginal propensity to consume + marginal propensity to invest) - **New Parameter:** Interest rate (i) is now a parameter of the ZZ curve

Step 2: Trace the IS Curve 1. **Point A:** Start with interest rate $i_0 \rightarrow$ ZZ curve $\rightarrow$ equilibrium output Y_0 - This gives us one point: (Y_0, i_0)

2. **Point A':** Increase interest rate to $i_1 > i_0$
 - **Effect:** ZZ curve shifts **down**
 - **Reason:** Higher $i \rightarrow$ lower investment $\rightarrow$ lower aggregate demand for any given Y
 - **Multiplier Effect:** Output falls by more than the initial decline in investment
 - New equilibrium: (Y_1, i_1) where $Y_1 < Y_0$
3. **Repeat:** Continue varying i to trace entire curve

Result: The IS curve is **downward sloping** in (Y, i) space.

Why Is the IS Curve Downward Sloping?

Logic: - Suppose we increase i but keep Y constant - Higher $i \rightarrow$ lower investment $\rightarrow$ aggregate demand falls below output - **Excess Supply** in goods market (demand insufficient to support that output) - To restore equilibrium, output

must fall - Therefore: Higher i requires lower Y for equilibrium $\rightarrow$ downward sloping

Mathematical Intuition:

$$\left. \frac{dY}{di} \right|_{IS} = \frac{\partial I / \partial i}{1 - (\partial C / \partial Y + \partial I / \partial Y)} < 0$$

- Numerator: Negative (investment falls with i) - Denominator: Positive but less than 1 (1 minus total marginal propensity to spend) - Result: Negative slope

3. Movements Along vs. Shifts of the IS Curve

Movement Along the IS Curve

- **Cause:** Change in interest rate only
- **Interpretation:** Moving from one equilibrium point to another on the same IS curve
- **Example:** Fed raises interest rate $\rightarrow$ move up along IS $\rightarrow$ lower equilibrium output

Shifts of the IS Curve

- **Cause:** Change in **exogenous** demand factors (anything in the model except i)
- **Interpretation:** Entire curve moves to a new position

What Shifts IS to the LEFT (Contractionary)?

1. **Tax Increase** ($T \uparrow$):
 - Lower disposable income $\rightarrow$ lower consumption $\rightarrow$ lower aggregate demand $\rightarrow$ lower output
 - Magnitude: $\Delta Y = -\frac{c_1}{1-c_1-i_1} \Delta T$ (where $i_1 = \partial I / \partial Y$)
2. **Government Spending Decrease** ($G \downarrow$):
 - Direct reduction in aggregate demand
 - Multiplier effect amplifies the decline
3. **Consumer Confidence Decline** ($c_0 \downarrow$):
 - Autonomous consumption falls
 - Examples: Wealth decline, increased uncertainty, pessimism
 - This captures the wealth effects discussed at the beginning
4. **Investment Sentiment Decline** ($i_0 \downarrow$):
 - Autonomous investment falls (e.g., due to uncertainty about future)

What Shifts IS to the RIGHT (Expansionary)?

- Opposite of above: $T \downarrow$, $G \uparrow$, $c_0 \uparrow$, $i_0 \uparrow$

How to Verify a Shift: - Go back to the ZZ diagram - For a given interest rate, see if aggregate demand shifts - If yes → IS curve shifts - Example: $T \uparrow \rightarrow ZZ$ shifts down → for same i , equilibrium Y is lower → IS shifts left

4. The LM Curve (Financial Market Equilibrium)

Traditional LM Derivation

Equilibrium Condition: Real money supply = Real money demand

$$\frac{M}{P} = Y \cdot L(i)$$

Where: - M/P : Real money supply - $Y \cdot L(i)$: Real money demand - Increases with Y (transaction demand) - Decreases with i (opportunity cost of holding money)

Traditional LM (if central bank targets M): - Fix $M \rightarrow$ Trace combinations of (Y, i) that satisfy equilibrium - **Shape:** Upward sloping - **Logic:** $Y \uparrow \rightarrow$ money demand $\uparrow \rightarrow$ if M fixed, then i must $\uparrow$ to reduce money demand and restore equilibrium

Modern LM Curve: Interest Rate Targeting

Modern Practice: Central banks (Fed, ECB, etc.) target the **interest rate** directly, not money supply.

How It Works: 1. Central bank announces target rate: $i = \bar{i}$ 2. Central bank supplies whatever M is needed to maintain that rate 3. If output changes and central bank doesn't want to change i : - Money demand changes: $\Delta(Y \cdot L(i))$ - Central bank adjusts M to accommodate: $\Delta M = P \cdot \Delta(Y \cdot L(i))$

Example: - Fed sets $i = \bar{i}$ - Output rises unexpectedly: $Y \uparrow$ - If Fed doesn't react: Money demand $\uparrow \rightarrow$ upward pressure on i - Fed response: Provide more $M \rightarrow$ keeps $i = \bar{i}$

Result: Modern LM is **HORIZONTAL** at the target rate $\bar{i}$

$$\text{Modern LM: } i = \bar{i}$$

Interpretation: The central bank accommodates any level of output at the target interest rate by adjusting money supply.

What Shifts the Modern LM?

Answer: ONLY changes in the central bank's target interest rate.

- **Contractionary Monetary Policy:** Fed raises target rate → LM shifts up
- **Expansionary Monetary Policy:** Fed lowers target rate → LM shifts down

Life Is Simpler Now: - In the past: Changes in money demand (L) or money supply (M) would shift LM - Now: Only the Fed's decision matters

Why Fed Changes Rates (Fed is not just “moody”): - Responding to inflation - Responding to unemployment - Responding to economic overheating (current situation) - Forced by circumstances (e.g., COVID, financial crises)

Risks of Rapid Rate Changes: - Can “break” something in financial system - Example: Highly leveraged institutions (banks, hedge funds) can fail - 2022-2023: Fed raised rates very fast to combat inflation → risky but nothing major broke (lucky) - UK: Near-crisis with pension funds during rapid rate increases

5. IS-LM Equilibrium

Finding Equilibrium

Setup: Two curves in (Y, i) space, two unknowns (Y, i)

$$\begin{cases} \text{IS : } Y = C(Y - T) + I(Y, i) + G \\ \text{LM : } i = \bar{i} \end{cases}$$

Solution: Point A where IS and LM intersect - At Point A: **Both** goods market and financial markets are in equilibrium - This is the **only** point consistent with both equilibria

Understanding Disequilibrium Points

Points on LM but Right of IS

- **Financial markets:** In equilibrium (on LM)
- **Goods market:** NOT in equilibrium
- **Problem: Excess Supply** of goods
 - Output too high for the level of demand
 - Firms cannot sell all they produce
 - Need to reduce output to restore equilibrium

Points on LM but Left of IS

- **Financial markets:** In equilibrium (on LM)
- **Goods market:** NOT in equilibrium
- **Problem: Excess Demand** for goods
 - Demand exceeds output

- Need to increase output to restore equilibrium

Points on IS but Above LM

- **Goods market:** In equilibrium (on IS)
- **Financial markets:** NOT in equilibrium
- **Problem:** Interest rate too high
 - **Excess Demand for Money** (equivalently: excess supply of bonds)
 - Money demand > money supply at that (Y, i) combination

Points on IS but Below LM

- **Goods market:** In equilibrium (on IS)
 - **Financial markets:** NOT in equilibrium
 - **Problem:** Interest rate too low
 - **Excess Supply of Money** (equivalently: excess demand for bonds)
-

6. Policy Analysis with IS-LM

Fiscal Contraction ($T \uparrow$ or $G \downarrow$)

Which Curve Moves? IS (goods market affected)

Direction: IS shifts **left**

Mechanism: 1. Higher taxes (or lower G) $\rightarrow$ disposable income falls (or aggregate demand falls directly) 2. Consumption declines 3. Aggregate demand falls $\rightarrow$ ZZ shifts down 4. Multiplier effect $\rightarrow$ output falls by more than initial decline in demand 5. New IS curve to the left of original

New Equilibrium (assuming Fed doesn't react): - Fed keeps LM at same position: $i = \bar{i}$ - New equilibrium: Point A' where new IS intersects unchanged LM - **Result:** - Output falls: $Y' < Y$ - Interest rate unchanged: $i' = i = \bar{i}$

Graphical Summary:

Initial: Point A (Y, i)

IS shifts left

New equilibrium: Point A' (Y', i) where $Y' < Y$

Key Insight: With horizontal LM (modern central banking), fiscal policy directly affects output without changing interest rates (unless the Fed responds).

Fiscal Expansion ($G \uparrow$ or $T \downarrow$)

Which Curve Moves? IS

Direction: IS shifts **right**

Example: COVID-19 fiscal stimulus - Large transfers to households (especially low-income) - G increased dramatically - IS shifted significantly right

New Equilibrium (Fed accommodates): - Output increases: $Y' > Y$ - Interest rate unchanged if Fed accommodates

Historical Context: - COVID-19 recession saw the **largest combined monetary and fiscal policy package in history** - Massive fiscal expansion + Fed keeping rates at zero - Result: Strong recovery but eventual overheating $\rightarrow$ inflation

7. Key Takeaways

1. **IS Curve:**
 - Represents goods market equilibrium
 - Downward sloping: higher $i \rightarrow$ lower $I \rightarrow$ lower Y
 - Shifts: Fiscal policy, consumer/business confidence, wealth effects
2. **Modern LM Curve:**
 - Horizontal at central bank's target rate
 - Represents central bank's policy stance
 - Shifts: Only when central bank changes target rate
3. **Equilibrium:**
 - Unique point where both markets clear
 - Determined by intersection of IS and LM
4. **Policy Framework:**
 - This is how central banks (including the Fed) think
 - IS-LM provides the foundation; real models add more equations
 - Essential for understanding monetary and fiscal policy interactions
5. **Current Application:**
 - Overheating economy (high aggregate demand)
 - Fed raising rates (LM shifts up) to cool economy
 - Understanding requires IS-LM framework

Next Lecture: More policy experiments, combinations of fiscal and monetary policy, and applications to recent economic events.

Important Questions to Consider

1. **Always ask first:** Which curve moves? (IS or LM?)
 - Fiscal/demand shocks $\rightarrow$ IS
 - Monetary policy $\rightarrow$ LM
2. What happens to points off equilibrium?
 - Identify which market is in disequilibrium
 - Identify direction of imbalance (excess supply vs. excess demand)

3. How do multiplier effects work in the IS-LM framework?
 - Initial shock to aggregate demand
 - Multiplier amplifies through ZZ curve
 - Final change in output determined by IS-LM equilibrium

Lecture 6: The IS-LM Model (Part II) - Applications and Policy Analysis

Overview

This lecture applies the IS-LM framework to analyze various policy combinations and real-world scenarios, with special focus on the COVID-19 policy response. We examine how monetary and fiscal policies work together (or in conflict), explore the zero lower bound problem, and discuss unconventional monetary policy.

1. IS-LM Model Review

The IS Curve (Goods Market Equilibrium)

Key Equation:

$$Y = C(Y - T) + I(Y, i) + G$$

Properties: - **Downward sloping** in (Y, i) space - **Steeper than Lecture 3** because investment now depends on Y - Slope reflects: MPC + MPI (marginal propensity to consume + invest) - **Interest rate is a parameter** of the ZZ curve

Derivation Logic: 1. Fix interest rate at $i_0 \rightarrow$ Find equilibrium output Y_0 using ZZ diagram 2. Raise interest rate to $i_1 \rightarrow$ Investment falls $\rightarrow$ ZZ shifts down $\rightarrow$ Output falls to Y_1 3. Repeat to trace entire IS curve

Key Mechanism: - $i \uparrow \rightarrow I \downarrow$ (investment falls) - Aggregate demand shifts down - **Multiplier effect:** Output falls by MORE than initial investment decline - Final decline in Y is amplified through consumption and investment feedback

The Modern LM Curve (Financial Market Equilibrium)

Traditional Formulation:

$$\frac{M}{P} = Y \cdot L(i)$$

- Would give upward-sloping LM if central bank targets M

Modern Practice: Central banks target interest rate directly

Modern LM: $i = \bar{i}$

- **Horizontal** at the target rate
- Central bank adjusts M to accommodate any output level
- Result: LM is a **flat line** at the Fed's chosen rate

Why Modern LM is Simpler: - Old days: Changes in M or money demand would shift LM - Now: Only Fed's policy decision shifts LM - Fed announces rate target, then supplies whatever M is needed

2. Movements Along vs. Shifts: Critical Distinction

Movement Along the IS Curve

- **Cause:** Interest rate changes (only)
- **What's held constant:** Taxes, government spending, consumer confidence, autonomous consumption
- **Interpretation:** Tracing different equilibrium points on the **same** IS
- **Example:** Fed raises $i \rightarrow$ move up along IS $\rightarrow$ lower Y

Shift of the IS Curve

- **Cause:** Change in any **exogenous** demand factor except interest rate
- **What's held constant:** The structural parameters of the economy
- **Interpretation:** Moving to a **different** IS curve
- **Examples:**
 - $T \uparrow \rightarrow$ IS shifts **left** (contractionary)
 - $G \uparrow \rightarrow$ IS shifts **right** (expansionary)
 - Consumer confidence $\downarrow \rightarrow$ IS shifts **left**
 - Wealth $\downarrow$ (affects c_0) $\rightarrow$ IS shifts **left**

How to Verify: Go back to the ZZ diagram - For a **given** interest rate, does aggregate demand change? - If yes $\rightarrow$ IS shifts - Example: $T \uparrow \rightarrow$ Disposable income $\downarrow \rightarrow$ Consumption $\downarrow \rightarrow$ ZZ shifts down $\rightarrow$ For same i , equilibrium Y is lower $\rightarrow$ IS shifts left

3. Fiscal Policy in IS-LM

Contractionary Fiscal Policy ($T \uparrow$ or $G \downarrow$)

Which Curve Moves? IS (goods market policy)

Direction: IS shifts **left**

Mechanism: 1. Higher taxes $\rightarrow$ Lower disposable income $\rightarrow$ Consumption falls - OR: Lower $G \rightarrow$ Aggregate demand falls directly 2. Aggregate demand

curve (ZZ) shifts down 3. Multiplier effect amplifies the decline 4. New IS curve to the left of original

Equilibrium Impact (assuming Fed doesn't respond): - Output **falls**: $Y' < Y$
- Interest rate **unchanged**: $i' = i = \bar{i}$ (Fed maintains target)

Non-Policy Equivalents: - Consumer confidence decline - Wealth shock (negative wealth effect) - Business investment sentiment decline - These shift IS left just like contractionary fiscal policy

Expansionary Fiscal Policy ($T \downarrow$ or $G \uparrow$)

Direction: IS shifts **right**

Equilibrium Impact: - Output **rises**: $Y' > Y$ - Interest rate unchanged (if Fed accommodates)

Use Cases: - Recessions (stimulate demand) - Can target specific groups (unlike monetary policy) - COVID stimulus: Large transfers, especially to low-income households

4. Monetary Policy in IS-LM

Expansionary Monetary Policy (Rate Cut: $i \downarrow$)

Which Curve Moves? LM (financial market policy)

Direction: LM shifts **down**

Why Is It Expansionary?

Step 1: Fed cuts interest rate from i_0 to i_1 - Suppose output stayed at Y_0 - Is there equilibrium in goods market? **NO** - **Problem**: At lower i , investment increases $\rightarrow$ Aggregate demand $>$ Output - **Excess demand** for goods

Step 2: Output must rise to restore equilibrium - Lower $i \rightarrow$ Higher $I \rightarrow$ Higher aggregate demand - Multiplier kicks in: Consumption also rises with income - New equilibrium: Higher output $Y_1 > Y_0$

Implementation: Open Market Operations 1. Fed buys bonds from banks $\rightarrow$ Injects money into system 2. Initially: Money supply $\uparrow \rightarrow$ Interest rate $\downarrow$ (overnight) 3. Communication: Fed announces target rate (e.g., "50 basis points cut") 4. **But Fed's job isn't done**: As output expands, money demand increases 5. Fed must **continue** providing more money to maintain the lower rate 6. Final money supply increase is **larger** than initial injection

Key Insight: Modern LM means Fed accommodates increased money demand as economy expands

Contractionary Monetary Policy (Rate Hike: $i \uparrow$)

Direction: LM shifts **up**

Effect: Output falls

Current Example (2022-2023): - Fed raised rates aggressively to combat inflation - Cooling overheated economy

5. Policy Combinations and Scenarios

Scenario 1: “All In” - Both Policies Expansionary

Policy Mix: - Expansionary fiscal policy: IS shifts **right** - Expansionary monetary policy: LM shifts **down**

Combined Effect: - **Large increase in output:** Both policies reinforce each other - Maximum stimulative impact

When Used?: - **Deep recessions** (most important use case) - Monetary policy first (faster, overnight decision) - Add fiscal when monetary alone isn't enough

- **Advantages of using both:** 1. Fiscal can **target** specific groups (monetary is “blunt”) - Example: COVID transfers to service sector workers who lost jobs
2. Fiscal can address structural issues monetary policy cannot 3. Combined firepower for severe crises

Historical Example: COVID-19 recession (discussed below)

Scenario 2: The Zero Lower Bound and Liquidity Trap

The Problem Zero Lower Bound (ZLB): - Nominal interest rates cannot go (much) below zero - When $i = 0$, conventional monetary policy becomes ineffective

Why ZLB Binds: - At $i = 0$, no opportunity cost of holding money vs. bonds
- People indifferent between money and bonds - Central bank can inject more money, but **cannot push rate lower** - **Liquidity Trap:** Economy is “trapped”
- more liquidity doesn't stimulate

Graphical Representation: - Money demand becomes **perfectly elastic** (horizontal) at $i = 0$ - No matter how much money central bank injects, rate stays at zero

Historical Context: Japan Japan's Lost Decades: - Financial bubble burst: Late 1980s / Early 1990s - Interest rates hit zero and stayed there for **decades** - Chronic recessions with no conventional monetary policy tool available - **Solution:** Massive fiscal expansions (since monetary policy exhausted)

COVID-19 and the Zero Lower Bound Timeline: - COVID shock hits: Early 2020 - Fed's immediate response: Cut rates **aggressively to zero** - US stuck at ZLB during entire COVID period - No more conventional monetary policy available

Implication: Heavy reliance on fiscal policy (shown below)

Scenario 3: Unconventional Monetary Policy (QE)

Background Problem: What can central banks do at the zero lower bound?

Solution: Unconventional monetary policy - **Not** standard short-term interest rate policy - Interventions in other financial markets

What Is Quantitative Easing (QE)? Definition: Central bank purchases of longer-term and riskier assets

Standard Monetary Policy (Conventional): - Fed buys/sells **short-term Treasury bonds** - Affects **overnight federal funds rate**

Quantitative Easing (Unconventional): Central bank expands beyond short-term Treasuries to: 1. **Long-term Treasury bonds** - Reduces term premium (long-term vs. short-term rate spread) 2. **Mortgage-Backed Securities (MBS)** - Fannie Mae and Freddie Mac securities - Supports housing market 3. **Corporate bonds** - Investment-grade corporate debt - Directly lowers business borrowing costs 4. **"Fallen Angels"** - Companies that were investment-grade before crisis but downgraded - Examples: Airlines, cruises, hotels (during COVID)

Why It Works: - Multiple interest rates in economy (not just one like in our model) - Risk spreads between different assets - Central bank can target these spreads even when short-term rate is zero - Operates "like" monetary policy but through different channels

Balance Sheet Expansion Traditional Monetary Policy:

Before: Assets: Bonds	After: Assets: More Bonds
Liabilities: Money	Liabilities: More Money

- Balance sheet expands, but modestly

Quantitative Easing: - **Massive** balance sheet expansion - Fed buys huge quantities of diverse assets - Both assets and liabilities (money) expand dramatically

Fed's Balance Sheet Evolution: - **Pre-2008:** ~\$1 trillion (stable, unremarkable) - **Global Financial Crisis (2008-2009):** - First time US hit ZLB - Massive expansion via QE - **Recovery (2010-2019):** - Continued QE during

slow recovery - Financial sector was compromised, took years to heal - Then began unwinding (reducing balance sheet) - **COVID-19 (2020)**: - Interest rates to zero immediately - **Enormous QE**: Balance sheet to ~\$9 trillion - Largest expansion in Fed history - **Current (2022-2023)**: - Unwinding QE to fight inflation - Gradually reducing balance sheet

International Context: - European Central Bank (ECB): Similar pattern - Bank of Japan: Even more extreme, started earlier (1990s) - Japan has been doing this for decades - Balance sheet looks like continuous expansion

Scenario 4: Fiscal Consolidation with Monetary Accommodation

Policy Mix: - Contractionary fiscal policy: IS shifts **left** (reduce deficit) - Expansionary monetary policy: LM shifts **down** (offset recession risk)

Net Effect: - Output roughly **unchanged** (policies offset each other) - Government deficit **improves** (fiscal consolidation achieved) - Interest rate **lower**

When Used?: - **Fiscal deficit is unsustainable** (debt accumulation problematic) - **Economy is NOT overheating** (no inflation problem) - Government wants to “fix fiscal accounts” without causing recession

Mechanism: 1. Treasury announces fiscal contraction (higher T or lower G) 2. Central bank recognizes this would cause recession 3. Central bank has mandate for price stability and output near potential 4. Central bank cuts rates to offset contractionary fiscal impact 5. (Often implicit coordination, not explicit)

Political Economy: - Government may complain: “You’re fighting our policy!” - But that’s the point of **central bank independence** - Central bank must offset inappropriate fiscal stimulus/contraction

Counterexample - Current US (2022-2023): - If US announced 5% fiscal contraction today - Fed would likely **not** cut rates - Why? Economy is **overheating** - Fiscal contraction would be welcome to cool economy

Scenario 5: Fiscal Expansion with Monetary Tightening (Conflict)

Policy Mix: - Expansionary fiscal policy: IS shifts **right** - Contractionary monetary policy: LM shifts **up**

Net Effect: - Output impact **ambiguous** (depends on relative magnitudes) - Interest rate **rises** - Government likely **unhappy** with central bank

When This Occurs: - **Uncoordinated policies** (conflict between Treasury and Fed) - Government spends, but central bank thinks economy is overheating - Central bank offsets fiscal stimulus to prevent inflation

Recent Example: United States (2021)

Background: - Early 2021: Large fiscal expansion - Transfers to households - Infrastructure spending - **Problem:** Economy was already near full employment - Supply constrained - Not much spare capacity

What Should Have Happened: - Fed should have raised rates immediately to offset - This would be the “fiscal expansion + monetary tightening” scenario

What Actually Happened: - Fed did **not** respond for a long time - Fed was “behind the curve” - Result: **Overheating economy** → Inflation

Why Fed Didn’t Respond: - Thought inflation would be “transitory” - Expected supply side to recover faster - Hoped dis-inflationary dynamics would dominate - Also: Russian invasion of Ukraine (2022) → Oil price spike → More inflation - By the time Fed responded (late 2021/2022), inflation was entrenched

Assessment: - Fed made a mistake in not responding - Very uncertain environment (COVID unprecedented) - May have been right over 3-year horizon - But everything “compressed into 3 months” → problem

Lesson: This scenario shows importance of central bank independence and coordination

6. COVID-19 Policy Response: A Case Study

The Shock

Nature of Crisis: - **Unprecedented:** Economy “imploding” into severe recession - **Unique:** Health crisis requiring lockdowns - **Sectoral:** Some sectors devastated (services), others booming (goods)

Monetary Policy Response

Immediate Action: - Fed cut interest rates **aggressively** to zero - Happened very quickly (March 2020)

Problem: Now at zero lower bound - Conventional monetary policy exhausted - Economy still in free fall

Unconventional Response: - Massive QE program - Purchased: - Long-term Treasuries - Mortgage-backed securities - Corporate bonds (investment grade) - “Fallen angels” bonds - Balance sheet expansion: ~\$4 trillion → ~\$9 trillion

Fiscal Policy Response

Scale: Unprecedented in peacetime - Total fiscal expansion: ~**20% of GDP** (combining all packages) - Comparable only to **wartime** spending - Multiple rounds of stimulus

Components: 1. **Direct transfers** to households - Stimulus checks - Enhanced unemployment benefits 2. **Small business support** - Paycheck Protection Program (PPP) 3. **State and local government aid** 4. **Infrastructure spending** (later)

International Context: - Similar massive fiscal responses **worldwide** - Not unique to US - Global coordination of fiscal expansion - Exception: China (different approach, different reasons)

The “All In” Policy Package

Visualization:

IS: Massive shift RIGHT (fiscal expansion)

LM: Massive shift DOWN (monetary expansion, then stuck at zero + QE)

Result: Large increase in output (recovery from deep recession)

Outcome: - **Short-term:** Successful at preventing depression - **Medium-term:** Economy recovered strongly - **Long-term problem:** Overheating → Inflation (by 2021-2022)

Hindsight Assessment: - Policy response may have been **too large** or **too prolonged** - Difficult to calibrate in real time during crisis - Better to err on side of too much stimulus than too little? - Debate ongoing

7. Empirical Evidence: How Monetary Policy Actually Works

Time Lags in Monetary Policy

Key Challenge: Monetary policy works with “long and variable lags”

Empirical Evidence (From VAR studies):

Effect on Output/Sales

- **Immediate:** Almost no effect (takes time for impact)
- **Peak effect:** 5-6 quarters (1.5 years) after rate change
- **Long-lasting:** Effects persist for 8+ quarters

Effect on Employment

- **Pattern:** Similar to output - slow build-up
- **Peak:** 5-6 quarters after shock
- **Problem:** Hard to know real-time impact

Effect on Unemployment

- **Mirror image** of employment
- Slow rise over 6+ quarters
- **Current concern (2022-2023):**
 - Unemployment very low now
 - But Fed has raised rates aggressively
 - Full impact not yet felt
 - Where will unemployment end up?

Effect on Prices/Inflation

- **Slowest response** of all variables
- Takes **very long time** to see full impact on inflation
- **Policy Challenge:**
 - Inflation still high now (2022-2023)
 - But Fed has done a lot
 - Need to wait for effects
 - Risk of over-tightening

Quote: *“Monetary policy works with long and variable lags”* (Famous statement)

Current Policy Dilemma (2022-2023)

The Tension: 1. **Inflation:** Still around 6% (unacceptably high) 2. **Fed Action:** Raised rates rapidly over 8 months 3. **Lag Problem:** Full effects won't be felt for 6+ more quarters 4. **Risk:** Over-tightening could cause deep recession

Questions: - Has Fed done enough? - Should Fed do more now? - Or wait to see effects of past tightening? - Will consumers and markets have patience?

Historical Pattern: When Fed raises rates quickly, something often “breaks” - Highly leveraged institutions fail - Financial instability - 2022-2023: So far lucky, nothing major has broken

Time Scales: Financial vs. Real Economy

Financial Markets: - React **instantly** to policy announcements - Asset prices adjust within minutes/hours - Interest rates adjust overnight

Real Economy (Output, Employment): - React with **long lags** - Begin to see effects: 1 quarter later - Peak effects: 5-6 quarters later - Continued effects: 8+ quarters

Implication for Analysis: - When studying impact on asset prices: Look at small windows (minutes around announcement) - When studying impact on real activity: Look over quarters

8. Key Principles for IS-LM Analysis

Always Ask: Which Curve Moves?

Fiscal Policy / Demand Shocks: - Moves **IS curve** (goods market) - Examples: T , G , c_0 , i_0 , confidence shocks - Does **not** move LM (unless Fed responds)

Monetary Policy: - Moves **LM curve** (financial market) - Central bank's interest rate target - Does **not** move IS (unless it affects expectations/confidence)

Common Mistake: Thinking fiscal policy moves LM or monetary policy moves IS - Fiscal and monetary policies operate in **different markets** - They can **respond** to each other, but that's a separate decision

Movements Along vs. Shifts

Critical for exam questions: - **Movement along** = Only interest rate changes - **Shift** = Other exogenous factors change

Understanding Disequilibrium

Points off the IS-LM intersection: - Not both markets in equilibrium - Identify which market has excess supply/demand - Understand adjustment mechanism

9. Summary

Core IS-LM Results

1. **Fiscal Policy:**
 - Shifts IS curve
 - With horizontal LM: Direct impact on output, no interest rate change (if Fed accommodates)
2. **Monetary Policy:**
 - Shifts LM curve
 - Works through investment channel
 - Long lags (6+ quarters for full effect)
3. **Policy Combinations:**
 - “All in” for deep recessions
 - Consolidation with accommodation for deficit reduction
 - Conflicts when policies misaligned

Real-World Applications

1. **Zero Lower Bound:**

- Major constraint on conventional monetary policy
 - Requires fiscal policy and/or unconventional monetary policy
 - Japan (decades), US/Europe (2008-2015, 2020-2021)
2. **Unconventional Monetary Policy:**
 - QE and other balance sheet tools
 - Effective at ZLB
 - Massive scale during crises
 3. **COVID-19 Response:**
 - Largest policy response in peacetime history
 - Both fiscal (~20% GDP) and monetary (rates to zero + huge QE)
 - Successful at preventing depression
 - Created overheating problem
 4. **Current Challenges:**
 - Fighting inflation with monetary tightening
 - Uncertainty about lags
 - Risk of over-correction

Key Takeaway

The IS-LM model is: - **Fundamental** to central bank thinking (Fed, ECB, etc.)
 - **Practical** for policy analysis - **Flexible** enough to analyze diverse scenarios -
Foundation for understanding macro policy debates

For Quiz: Master which curve moves, understand mechanisms, be able to analyze policy combinations.

Important Concepts Checklist

- ☐ Understand movements along vs. shifts of IS
- ☐ Understand movements along vs. shifts of LM
- ☐ Know what shifts IS (all exogenous demand factors except i)
- ☐ Know what shifts LM (only central bank's policy decision)
- ☐ Understand multiplier effects through ZZ curve
- ☐ Understand zero lower bound and liquidity trap
- ☐ Understand policy combinations (all in, consolidation, conflicts)
- ☐ Understand monetary policy lags
- ☐ Can analyze any policy scenario by asking "which curve moves?"

Lecture 7: Extensions to IS-LM - Nominal vs. Real Interest Rates and Credit Spreads

Overview

This lecture extends the basic IS-LM model along two realistic dimensions that significantly enhance its explanatory power for real-world macroeconomic phenomena: 1. **Nominal vs. Real Interest Rates**: Distinguishing between interest rates in dollar terms versus goods terms, and the role of expected inflation. 2. **Credit Spreads (Risk Premia)**: Introducing the wedge between safe government borrowing rates and the risky rates at which corporations actually borrow.

These extensions are critical for understanding modern macroeconomic crises, particularly the Global Financial Crisis (2008-2009) and the COVID-19 pandemic response. They explain why conventional monetary policy can become ineffective and what unconventional policy tools central banks might deploy.

Key Insight: Even when the central bank controls the nominal policy rate, the **real borrowing cost** for firms—which determines investment decisions—depends on both expected inflation and credit market conditions. These two channels can move independently from monetary policy, creating powerful shocks to the economy.

1. Nominal vs. Real Interest Rates

1.1 Definitions

Nominal Interest Rate (i)

- **Definition**: Interest rate expressed in dollar (or currency) terms.
- **Example**: If you invest \$100 in a one-year bond with nominal rate $i = 10\%$, you receive \$110 in one year.
- **Formula**: $\text{Return} = \text{Principal} \times (1 + i)$

Characteristics: - Most financial instruments in the US are denominated in nominal terms. - The Federal Reserve directly sets short-term nominal interest rates (Federal Funds Rate). - Directly observable in financial markets.

Real Interest Rate (r)

- **Definition**: Interest rate expressed in terms of goods (a basket like CPI).
- **Example**: If you invest 1 unit of goods today at real rate $r = 3\%$, you receive 1.03 units of goods next year.
- **Interpretation**: Measures the true purchasing power gain from saving.

Characteristics: - Less commonly traded directly (though TIPS—Treasury Inflation-Protected Securities—exist in US). - What matters for real economic

decisions (investment, consumption of durables). - Must be calculated from nominal rates and inflation expectations.

1.2 The Fisher Equation: Linking Nominal and Real Rates

Derivation Consider two investment strategies starting with 1 unit of goods today:

Strategy 1: Real Bond (Direct) - Invest 1 unit of goods in a real bond. - Receive $(1 + r_t)$ units of goods at time $t + 1$.

Strategy 2: Nominal Bond (Indirect) - Convert 1 unit of goods to P_t dollars (today's price level). - Invest P_t dollars in a nominal bond. - Receive $P_t(1 + i_t)$ dollars at time $t + 1$. - Convert back to goods: $\frac{P_t(1+i_t)}{P_{t+1}^e}$ units of goods (expected).

Arbitrage Condition: In equilibrium, both strategies must yield the same expected return:

$$(1 + r_t) = (1 + i_t) \frac{P_t}{P_{t+1}^e}$$

Expected Inflation Define **expected inflation** between t and $t + 1$:

$$\pi_{t+1}^e = \frac{P_{t+1}^e - P_t}{P_t}$$

This implies:

$$\frac{P_t}{P_{t+1}^e} = \frac{1}{1 + \pi_{t+1}^e}$$

Substituting into the arbitrage condition:

$$(1 + r_t) = \frac{1 + i_t}{1 + \pi_{t+1}^e}$$

Linear Approximation (Fisher Equation) For small interest rates and inflation (typical in developed economies):

$$r_t \approx i_t - \pi_{t+1}^e$$

Interpretation: The real interest rate equals the nominal rate minus expected inflation.

Example: - Nominal interest rate: $i = 6\%$ - Expected inflation: $\pi^e = 3\%$ - Real interest rate: $r \approx 6\% - 3\% = 3\%$

Even though you receive 6% more dollars, goods are 3% more expensive, so your purchasing power only increases by 3%.

1.3 Why Real Rates Matter for Investment

Question: Why do firms care about real rates, not nominal rates, when making investment decisions?

Answer: Firms invest to produce and sell goods in the future. Their decision depends on comparing: - **Cost of Borrowing:** The interest rate they must pay. - **Revenue from Investment:** The value of goods they'll produce and sell.

Thought Experiment Scenario 1: Zero Inflation - Nominal rate: $i = 5\%$ - Real rate: $r = 5\%$ - A project yielding 7% real return is profitable ($7\% > 5\%$).

Scenario 2: 10% Inflation - Nominal rate: $i = 15\%$ (Fed raised rates to offset inflation) - Expected inflation: $\pi^e = 10\%$ - Real rate: $r = 15\% - 10\% = 5\%$ - **The same project** is still profitable!

Why?: The goods you produce will sell for 10% more dollars next year. Your nominal revenue increases by 10%, exactly offsetting the higher nominal interest rate. The **real profitability** (in goods terms) is unchanged.

Conclusion: Investment depends on r , not i . If r remains constant, doubling both i and π^e doesn't change investment incentives.

1.4 Historical Evidence: Great Recession (2008-2009)

The Setup The Global Financial Crisis created a severe recession. The Fed responded by cutting the nominal interest rate to near zero.

Data (approximate from lecture figures): - **Pre-Crisis (2007):** - Nominal rate (i): $\sim 5\%$ - Real rate (r): $\sim 2\%$ - Expected inflation (π^e): $\sim 2-3\%$

- **Crisis Peak (2009):**
 - Nominal rate (i): $\sim 0\%$ (zero lower bound)
 - Real rate (r): $\sim 4\%$
 - Expected inflation (π^e): $\sim -4\%$ (deflationary expectations!)

Key Observations 1. Real Rates Rose Despite Fed Easing Even though the Fed cut nominal rates aggressively, **real rates increased**. This perverse outcome occurred because expected inflation collapsed faster than nominal rates.

2. Deflation Fears Expected inflation turned **negative** (deflation), reaching approximately -4% . Deflation expectations arose from: - Massive demand collapse (unemployment, wealth destruction). - Precedent from Japan's "lost decade." - Financial market panic and deleveraging.

3. Policy Implication The Fed "lost control" of the real interest rate. Even at $i = 0\%$:

$$r = i - \pi^e = 0\% - (-4\%) = 4\%$$

A **4% real rate** during a severe recession is highly contractionary, explaining the depth and persistence of the downturn.

4. The Zero Lower Bound Problem Nominal rates cannot go significantly below zero (people would hoard cash). This constraint becomes binding when:

- Economy needs very low real rates (recession).
- Expected inflation is low or negative (deflation).
- Result: $r = i - \pi^e$ cannot be lowered enough via conventional monetary policy.

1.5 COVID-19 Episode: Negative Real Rates (2020-2021)

The COVID pandemic created a different dynamic—a supply-side shock with unusual inflation behavior.

Initial Shock (March-April 2020)

- **Recession fears:** Expected inflation initially fell (like 2008).
- **Policy response:** Fed cut i to 0% immediately.
- **Result:** Brief spike in real rates (similar to 2008 pattern).

Recovery Phase (2021) Unlike the Great Recession, **inflation surged** due to:

- **Supply chain disruptions:** Bottlenecks in production and shipping.
- **Fiscal stimulus:** Massive government spending (\$3+ trillion).
- **Pent-up demand:** Reopening after lockdowns.
- **Labor shortages:** Workers leaving workforce (“Great Resignation”).

Data (2021): - Nominal rate: $i \approx 0\%$ (Fed kept rates at zero) - Expected inflation: $\pi^e \approx 4 - 5\%$ - Real rate: $r \approx 0\% - 4\% = -4\%$

Implications of Negative Real Rates **Highly Stimulative:** - Borrowing costs effectively negative in real terms. - Strong incentive for investment, asset purchases, speculation.

Asset Price Boom: - Equity markets soared (S&P 500, NASDAQ). - Real estate prices surged. - Cryptocurrencies and “meme stocks” flourished.

Fed’s Dilemma (2022): Eventually, the Fed decided inflation was not “transitory” and began raising rates aggressively. But initial rate hikes didn’t immediately raise real rates because expected inflation kept rising.

The Tightening Problem (2022-2023) **Challenge:** The Fed wants to tighten financial conditions (raise r), but:

- Raising i is straightforward.
- But if π^e rises alongside i , then $r = i - \pi^e$ may not increase much.

Breakthrough (August 2022): - **Jackson Hole Speech:** Fed Chair Powell gave a “hawkish” speech committing to do “whatever it takes” to reduce inflation. - **Effect:** Convinced markets that inflation would come down (π^e fell).

- **Result:** Real rates rose sharply, triggering stock market decline (especially tech stocks).

Current Challenge (2023, as of lecture): - Fed continues raising i , but π^e has stopped declining. - Real rates have plateaued or even declined slightly. -

Problem: Financial conditions not tightening as intended.

2. Credit Spreads (Risk Premia)

2.1 Definitions and Concepts

Riskless Interest Rate

- **Definition:** Interest rate on safe assets with essentially zero default risk.
- **Examples:** US Treasury bonds, German Bunds, Swiss government bonds.
- **Notation:** r (real) or i (nominal) when referring to Treasuries.

Risky Borrowing Rate Most corporations and households cannot borrow at the Treasury rate. They face a **risk premium**:

$$r^f = r + x$$

Where: - r^f : Real interest rate firms actually pay (“ f ” for firms). - r : Safe real interest rate (Treasury rate). - x : **Risk premium** or **credit spread**.

Credit Spread (x)

- **Definition:** The extra interest rate charged on risky debt above the safe rate.
- **Measured as:** Yield on corporate bonds minus yield on Treasuries of same maturity.
- **Example:** If 10-year Treasury yields 4% and Boeing’s 10-year bond yields 7%, then $x = 3\%$.

2.2 Determinants of Credit Spreads

The risk premium x compensates lenders for two factors:

1. Probability of Default (p)

- **Definition:** Likelihood that the borrower will fail to repay the loan.
- **Sources:**
 - Credit rating agencies (Moody’s, S&P, Fitch).
 - Historical default rates by rating category.
 - Market-implied probabilities from Credit Default Swaps (CDS).

Example: - AAA-rated bonds: $p \approx 0.01\%$ (virtually no default risk). - BBB-rated bonds: $p \approx 0.5\%$ (investment grade, moderate risk). - High-yield bonds (“junk”): $p \approx 5 - 10\%$ (substantial default risk).

2. Risk Aversion of Investors

- **Definition:** Investors' unwillingness to bear uncertainty.
- **Effect:** Even if expected default probability is p , investors may demand compensation exceeding p because they dislike risk.
- **Variation:** Risk aversion fluctuates dramatically over the business cycle.

Key Point: In practice, separating these two effects is difficult. We can think of the “perceived probability of default” as including both the actuarial probability and an extra premium due to risk aversion.

2.3 Deriving the Credit Spread Formula

Setup Consider a risk-neutral investor choosing between: - **Safe bond:** Pays $(1 + r)$ with certainty. - **Risky bond:** Pays $(1 + r^f)$ with probability $(1 - p)$, and 0 with probability p (default, zero recovery).

Indifference Condition Expected return on risky bond must equal return on safe bond:

$$(1 - p)(1 + r^f) + p \cdot 0 = (1 + r)$$

Simplify:

$$(1 - p)(1 + r^f) = (1 + r)$$

Solve for r^f :

$$1 + r^f = \frac{1 + r}{1 - p}$$

$$r^f = \frac{1 + r}{1 - p} - 1 = \frac{1 + r - (1 - p)}{1 - p} = \frac{r + p}{1 - p}$$

For small p (approximation):

$$r^f \approx r + \frac{p}{1 - p} \approx r + p + p^2 + \dots \approx r + p$$

Thus, the **credit spread**:

$$x \equiv r^f - r \approx p$$

More precisely:

$$x = \frac{p(1 + r)}{1 - p}$$

Interpretation

- Credit spread is **increasing** in default probability p .
- For small p , spread approximately equals p .
- For large p , spread grows faster than p (convex relationship).

Example: - $r = 3\%$, $p = 5\%$ (high-yield bond):

$$x = \frac{0.05 \times 1.03}{1 - 0.05} = \frac{0.0515}{0.95} \approx 5.4\%$$

So the firm must pay $r^f \approx 3\% + 5.4\% = 8.4\%$.

2.4 Cyclical Behavior of Credit Spreads

Credit spreads are **strongly countercyclical**: - **Recessions**: Spreads widen dramatically ($x \uparrow$). - **Expansions**: Spreads narrow ($x \downarrow$).

Why Spreads Rise in Recessions 1. Higher Actual Default Rates

- Firms' revenues fall, making debt service harder. - Weaker balance sheets increase bankruptcy risk. - Objectively, p rises.

2. Increased Risk Aversion - Investors become more fearful, demanding higher compensation for any risk. - "Flight to safety": Money flows from corporate bonds to Treasuries. - Even if p doesn't change much, perceived p spikes.

3. Liquidity Concerns - Corporate bond markets can become illiquid during crises. - Harder to sell bonds quickly without taking a loss. - Investors demand extra premium for illiquidity.

2.5 Historical Evidence: Great Recession

Credit Spread Data (High-Yield Bonds) Pre-Crisis (2006-2007): - Credit spread: $x \approx 3 - 4\%$ (typical for high-yield). - Interpretation: Markets relatively calm, low perceived default risk.

Crisis Peak (2008): - Credit spread: $x \approx 20\%$ (spiked to unprecedented levels). - Interpretation: Massive fear, perceived default probabilities extremely high. - **Impact**: Corporate borrowing costs exploded, even for healthy firms.

Recovery (2009-2010): - Spreads gradually declined as financial panic subsided. - By 2010: $x \approx 5 - 7\%$ (still elevated but improving).

Policy Implication The credit spread shock was a **major contractionary force** independent of monetary policy: - Fed cut the safe rate (r) to zero. - But corporate borrowing costs ($r + x$) remained very high due to massive x . - Investment collapsed not just from low aggregate demand, but from inability to borrow affordably.

2.6 Historical Evidence: COVID-19

Initial Spike (March 2020)

- Credit spread: Jumped from ~4% to ~11% (high-yield).
- **Trigger:** Uncertainty about pandemic duration, lockdowns, corporate survival.
- **Specific sectors hit hard:** Airlines, hospitality, retail.

Policy Response Unlike 2008, the Fed acted **immediately and forcefully**:
- Cut policy rate to zero (within weeks). - Announced **corporate bond purchase programs** (unprecedented for Fed). - Primary Market Corporate Credit Facility (PMCCF). - Secondary Market Corporate Credit Facility (SMCCF). - **Goal:** Directly reduce x by buying corporate bonds.

Effect

- Credit spreads quickly fell back to ~4-5% by summer 2020.
- **Key insight:** Mere announcement of Fed support was enough (Fed barely used the facilities).
- Markets regained confidence that Fed would backstop corporate credit.

Recent Period (2022-2023)

- Spreads remained relatively low (~4-5%) despite Fed tightening.
- **Fed's frustration:** Wanted to tighten financial conditions, but x not rising.
- Only very recently (late 2023) have spreads begun to widen modestly.

3. The Extended IS-LM Model

3.1 Modifications to the Standard Model

LM Curve: No Change

$$i = \bar{i}$$

The central bank still sets the **nominal** interest rate. This is the Fed's direct policy instrument (Federal Funds Rate).

Key Point: The Fed controls i , but not r , and definitely not $(r + x)$.

IS Curve: Modified Investment Function

 In the standard model:

$$Y = C(Y - T) + I(Y, i) + G$$

Problem: This assumes investment depends on the nominal rate i . But we now know investment depends on the **real borrowing cost**.

Corrected Model:

$$Y = C(Y - T) + I(Y, r^f) + G$$

Where the **real borrowing cost**:

$$r^f = i - \pi^e + x$$

Full IS Equation:

$$Y = C(Y - T) + I(Y, i - \pi^e + x) + G$$

New Parameters in IS Curve The IS curve now depends on: - G : Government spending (as before). - T : Taxes (as before). - i : Nominal interest rate set by Fed (as before). - π^e : **Expected inflation** (new parameter). - x : **Credit spread** (new parameter).

Interpretation: π^e and x are exogenous variables (parameters) in the short-run IS-LM model. Later in the course, we'll endogenize inflation. For now, we treat them as given and analyze their effects.

3.2 How π^e and x Shift the IS Curve

Effect of Expected Inflation (π^e) Increase in Expected Inflation ($\pi^e \uparrow$):
- Real borrowing cost: $r^f = i - \pi^e + x \downarrow$ (falls). - Investment: $I \uparrow$ (cheaper to borrow in real terms). - Aggregate demand: $Z \uparrow$. - **IS curve shifts RIGHT** (expansionary).

Mechanism: For a given nominal rate i , higher expected inflation means goods will be more expensive in the future, reducing the real cost of borrowing today. Firms find more projects profitable.

Graphical Representation: - In (Y, i) space: IS shifts right. - In ZZ-curve diagram (Lecture 3): ZZ shifts up for any given i .

Effect of Credit Spread (x) Increase in Credit Spread ($x \uparrow$): - Real borrowing cost: $r^f = i - \pi^e + x \uparrow$ (rises). - Investment: $I \downarrow$ (more expensive to borrow). - Aggregate demand: $Z \downarrow$. - **IS curve shifts LEFT** (contractionary).

Mechanism: Even if the Fed keeps i constant and inflation expectations are stable, a rise in credit spreads (due to financial crisis, increased risk aversion, etc.) raises the effective cost of borrowing for firms. Investment falls, contracting the economy.

Graphical Representation: - In (Y, i) space: IS shifts left. - In ZZ-curve diagram: ZZ shifts down for any given i .

3.3 The IS-LM Diagram with Extensions

Setup

- **Vertical axis:** Nominal interest rate i .
- **Horizontal axis:** Output Y .
- **LM curve:** Horizontal line at $i = \bar{i}$ (Fed's target rate).
- **IS curve:** Downward sloping, but depends on π^e and x .

Equilibrium Intersection of IS and LM determines: - Output: Y^* - Interest rate: $i^* = \bar{i}$ (set by Fed)

Important: The **real borrowing cost** determining investment is:

$$r^f = \bar{i} - \pi^e + x$$

This is **not** shown directly on the diagram but is implicit in the IS curve position.

3.4 Policy Experiments

Experiment 1: Increase in Expected Inflation ($\pi^e \uparrow$) **Scenario:** Markets become more optimistic about future growth, or Fed successfully raises inflation expectations.

Effects: 1. Real borrowing cost falls: $r^f = i - \pi^e + x \downarrow$. 2. IS curve shifts **RIGHT**. 3. Output increases: $Y \uparrow$. 4. (If Fed accommodates, i unchanged.)

Intuition: This acts like an **expansionary monetary policy**, even though the Fed didn't change i . The real rate fell automatically because inflation expectations rose.

Historical Example: - **2021:** Supply shocks raised inflation and inflation expectations. Combined with $i = 0\%$, this created very negative real rates, fueling economic boom and asset price surge.

Experiment 2: Increase in Credit Spread ($x \uparrow$) **Scenario:** Financial crisis, bank failures, or panic increases perceived default risk.

Effects: 1. Real borrowing cost rises: $r^f = i - \pi^e + x \uparrow$. 2. IS curve shifts **LEFT**. 3. Output falls: $Y \downarrow$. 4. (If Fed accommodates, i unchanged, but this isn't enough.)

Intuition: This acts like a **contractionary monetary policy shock**, even though the Fed didn't raise rates. The effective borrowing cost for firms increased due to credit market disruption.

Historical Example: - **2008:** Credit spreads exploded ($x \uparrow$ by ~20 percentage points for high-yield). This was a massive contractionary shock, overwhelming the Fed's attempts to cut i .

Experiment 3: Fed Response to Credit Spread Shock Scenario: Credit spread rises ($x \uparrow$), and Fed attempts to offset by cutting i .

Initial Shock: - $x \uparrow$ shifts IS left. - Output would fall significantly.

Policy Response: - Fed cuts i (LM shifts down). - This shifts IS back to the right (partially offsetting the x shock).

Outcome: - If Fed cuts i by exactly Δx , then $r^f = (i - \Delta x) - \pi^e + (x + \Delta x) = i - \pi^e + x$ (unchanged). - Investment unchanged, output unchanged. - **Full offset is possible** if Fed can cut rates enough.

Problem: Zero Lower Bound: - If i is already near zero, Fed cannot cut further. - x shock cannot be fully offset. - Economy falls into deep recession.

Historical Example: - 2008-2009: x rose ~20 points, but Fed could only cut i by ~5 points (from 5% to 0%). Net effect: Real borrowing costs rose, economy collapsed.

Experiment 4: Combined Shock (Great Recession) Scenario: Simultaneous credit crisis and deflation fears.

Shocks: 1. Credit spread rises: $x \uparrow$ (say, +15 points). 2. Expected inflation falls: $\pi^e \downarrow$ (say, -6 points, from +2% to -4%).

Effect on Real Borrowing Cost:

$$\Delta r^f = \Delta i - \Delta \pi^e + \Delta x = 0 - (-6) + 15 = +21 \text{ percentage points}$$

Even with i cut to zero, real borrowing costs increased by **21 percentage points**.

IS-LM Diagram: - IS shifts massively **LEFT** (both $x \uparrow$ and $\pi^e \downarrow$ are contractionary). - LM shifts down to zero (Fed's maximum response). - New equilibrium: Y much lower (deep recession).

Policy Implication: Conventional monetary policy (cutting i) is **insufficient**. Fed "out of ammunition" in conventional sense.

4. Unconventional Monetary Policy

4.1 The Problem

When i hits the zero lower bound but the economy needs further stimulus, conventional monetary policy is exhausted. The Fed cannot cut i below zero (or only marginally, due to cash hoarding).

Constraint:

$$i \geq 0$$

But the economy needs:

$$r^f = i - \pi^e + x \text{ to be very low}$$

Options: 1. Raise π^e (inflation expectations). 2. Lower x (credit spreads).

4.2 Large-Scale Asset Purchases (QE)

Quantitative Easing Definition: Central bank purchases long-term government bonds and other assets (mortgages, corporate bonds) to directly influence long-term interest rates and risk premia.

Traditional Monetary Policy: - Target: Very short-term Treasury rates (overnight Federal Funds Rate). - Assets: Minimal holdings of short-term Treasuries.

Quantitative Easing: - Target: Long-term Treasury yields, mortgage rates, corporate bond yields. - Assets: Massive holdings of 10-year Treasuries, mortgage-backed securities (MBS), even corporate bonds.

Mechanism: - Fed buys long-term bonds $\rightarrow$ Demand for bonds rises $\rightarrow$ Bond prices rise $\rightarrow$ Yields fall. - For corporate bonds: Direct reduction in x by increasing demand for risky assets.

Applications Great Recession (2008-2009): - **QE1, QE2, QE3:** Fed purchased \$3.5+ trillion in long-term Treasuries and MBS. - **Effect:** Long-term Treasury yields fell from ~4% to ~2%. Mortgage rates fell from ~6% to ~3.5%. - **Limited corporate bond purchases:** Focused on Treasuries and MBS.

COVID-19 (2020): - **Innovation:** Fed created facilities to purchase **corporate bonds** directly. - Primary Market Corporate Credit Facility (PMCCF). - Secondary Market Corporate Credit Facility (SMCCF). - **Goal:** Directly target x (credit spreads). - **Effect:** Credit spreads fell sharply just from announcement (Fed barely used facilities).

International Examples: - **Japan:** Bank of Japan purchases equities (ETFs) directly. - **European Central Bank:** Bought corporate bonds, covered bonds. - **Hong Kong (1997):** Government directly purchased equities during Asian Financial Crisis.

4.3 Forward Guidance

Definition Forward Guidance: Central bank communicates future policy intentions to influence expectations.

Goal: Raise expected inflation (π^e) or lower expected future interest rates.

Mechanism If Fed commits to keeping $i = 0$ for an extended period (e.g., “through 2025”), then: - Markets expect low rates to persist. - Long-term rates fall (average of expected future short rates). - Expected inflation may rise (persistent stimulus). - Real rates fall: $r = i - \pi^e \downarrow$.

Challenges

- **Credibility:** Will Fed actually follow through, or change policy if conditions improve?
- **Time-inconsistency:** Fed may want to raise rates sooner if inflation rises unexpectedly.
- **Communication:** Markets must understand and believe the commitment.

Historical Example: - **2012-2015:** Fed pledged to keep rates near zero until unemployment fell below 6.5% or inflation exceeded 2.5%. This helped keep long-term rates low and supported recovery.

4.4 Fiscal Policy as Complement

When monetary policy is constrained by the zero lower bound, **fiscal policy** becomes more important:

Automatic Stabilizers

- Unemployment insurance, progressive taxes, welfare programs.
- These cushion the fall in aggregate demand automatically.

Discretionary Fiscal Stimulus

- Increased government spending: Infrastructure, defense, transfers.
- Tax cuts: Rebates, payroll tax holidays.

Example: - **2009:** American Recovery and Reinvestment Act (\$800 billion stimulus). - **2020-2021:** CARES Act, American Rescue Plan (\$3+ trillion combined).

Effectiveness: Fiscal multiplier is **larger** at the zero lower bound because: - Fed won't raise rates to offset (since already at zero). - No “crowding out” via interest rate channel.

5. Summary and Key Takeaways

Main Concepts

1. Fisher Equation

$$r \approx i - \pi^e$$

- Real rate = Nominal rate - Expected inflation. - Investment depends on r , not i .

2. Credit Spreads

$$r^f = r + x$$

- Firms borrow at risky rate r^f , not safe rate r . - Spread x is countercyclical (high in recessions).

3. Extended IS-LM

$$Y = C(Y - T) + I(Y, i - \pi^e + x) + G$$

- IS depends on i , π^e , and x . - $\pi^e \uparrow$ or $x \downarrow$ shifts IS right (expansionary). - $\pi^e \downarrow$ or $x \uparrow$ shifts IS left (contractionary).

Historical Episodes

Great Recession (2008-2009)

- **Shocks:** $x \uparrow \uparrow$ (credit crisis), $\pi^e \downarrow \downarrow$ (deflation fears).
- **Policy:** Fed cut i to zero, but $r^f = i - \pi^e + x$ remained high.
- **Result:** Deep recession, slow recovery.
- **Lesson:** Conventional monetary policy can become ineffective.

COVID-19 (2020-2021)

- **Initial shock:** $x \uparrow$ (panic), $\pi^e \downarrow$ (uncertainty).
- **Policy:** Fed cut i to zero, launched corporate bond purchases.
- **Supply shock:** $\pi^e \uparrow \uparrow$ (bottlenecks, stimulus).
- **Result:** Real rates turned deeply negative ($r \approx -4\%$), fueling recovery and asset boom.

Recent Challenge (2022-2023)

- **Fed goal:** Raise r to cool economy, fight inflation.
- **Problem:** Raising i doesn't raise r if π^e rises simultaneously.
- **Breakthrough:** Jackson Hole speech (August 2022) lowered π^e , finally raising r .
- **Ongoing challenge:** Credit spreads remain low, limiting financial tightening.

Policy Implications

For Monetary Policy

1. **Control nominal rate (i), not real rate (r):** Real rate depends on inflation expectations (partially outside Fed control).
2. **Zero lower bound matters:** When $i \approx 0$, further easing requires unconventional tools.
3. **Credit conditions matter:** Fed may need to intervene in credit markets directly during financial crises.

For Financial Stability

1. **Monitor credit spreads:** Sharp widening signals financial stress, requires policy response.
2. **Prevent deflation:** Negative expected inflation amplifies recessions even if nominal rates are zero.
3. **Communication is powerful:** Fed's ability to shape expectations (π^e) is a key tool at zero bound.

Looking Ahead

In upcoming lectures, we will: - **Endogenize inflation:** How is π (and π^e) determined? (Phillips Curve, expectations formation) - **Long-run vs. short-run:** Reconciling price flexibility in long run with stickiness in short run. - **Open economy:** How do these concepts extend to exchange rates and international capital flows?

6. Mathematical Appendix

A. Exact Fisher Equation

Starting from:

$$(1 + r_t) = (1 + i_t) \frac{P_t}{P_{t+1}^e}$$

Define:

$$\pi_{t+1}^e = \frac{P_{t+1}^e - P_t}{P_t}$$

Then:

$$\frac{P_t}{P_{t+1}^e} = \frac{1}{1 + \pi_{t+1}^e}$$

Substitute:

$$(1 + r_t) = \frac{1 + i_t}{1 + \pi_{t+1}^e}$$

Expand:

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}^e} \approx (1 + i_t)(1 - \pi_{t+1}^e) = 1 + i_t - \pi_{t+1}^e - i_t \pi_{t+1}^e$$

For small i_t and π_{t+1}^e , drop the product term $i_t \pi_{t+1}^e$:

$$r_t \approx i_t - \pi_{t+1}^e$$

Error: The approximation error is $i_t \pi_{t+1}^e$. For $i_t = 5\%$ and $\pi_{t+1}^e = 3\%$, the error is $0.05 \times 0.03 = 0.15\%$ (negligible).

B. Credit Spread Derivation (With Recovery)

More generally, assume recovery rate θ (fraction of principal recovered in default).

Expected return on risky bond:

$$(1-p)(1+r^f) + p(1+\theta) = (1+r)$$

Solve for r^f :

$$(1-p)(1+r^f) = (1+r) - p(1+\theta)$$

$$1+r^f = \frac{(1+r) - p(1+\theta)}{1-p}$$

$$r^f = \frac{(1+r) - p(1+\theta) - (1-p)}{1-p} = \frac{r - p\theta}{1-p}$$

Credit spread:

$$x = r^f - r = \frac{r - p\theta}{1-p} - r = \frac{r - p\theta - r(1-p)}{1-p} = \frac{rp - p\theta}{1-p} = \frac{p(r-\theta)}{1-p}$$

Special cases: - $\theta = 0$ (zero recovery): $x = \frac{pr}{1-p} \approx pr \approx p$ (for small p). - $\theta = -1$ (lose principal): $x = \frac{p(r+1)}{1-p}$ (larger spread).

C. IS Curve with Extensions (Formal)

Goods market equilibrium:

$$Y = C(Y - T) + I(Y, r^f) + G$$

Investment function (linear approximation):

$$I = \bar{I} - b(r^f - \bar{r})$$

where $b > 0$ is the interest sensitivity of investment.

Real borrowing cost:

$$r^f = i - \pi^e + x$$

Substitute:

$$I = \bar{I} - b(i - \pi^e + x - \bar{r})$$

Full equilibrium:

$$Y = C(Y - T) + \bar{I} - b(i - \pi^e + x - \bar{r}) + G$$

Solve for IS curve:

$$Y = \frac{1}{1 - c_1} [\bar{C} - c_1 T + \bar{I} - b(i - \pi^e + x - \bar{r}) + G]$$

Comparative statics: - $\frac{dY}{di} = -\frac{b}{1-c_1} < 0$ (downward sloping IS). - $\frac{dY}{d\pi^e} = \frac{b}{1-c_1} > 0$ (higher expected inflation shifts IS right). - $\frac{dY}{dx} = -\frac{b}{1-c_1} < 0$ (higher credit spread shifts IS left).

Note: Effects of i , π^e , and x on output are symmetric (enter as $i - \pi^e + x$). A 1 percentage point increase in x has the same effect as a 1 percentage point decrease in π^e or a 1 percentage point increase in i .

Lecture 8: The Labor Market

Overview

This lecture introduces the labor market as the foundation for understanding inflation dynamics. We transition from the IS-LM model (fixed prices, demand-determined output) to the medium-run model where supply constraints and labor market tightness affect prices. The labor market is crucial because: (1) unemployment is a key macroeconomic health indicator, and (2) labor market conditions are a primary driver of inflation.

1. Motivation: Why Labor Markets Matter for Macro

Two Key Reasons

1. **Unemployment as Economic Indicator:** The unemployment rate signals the overall health of the economy.
2. **Labor Market and Inflation:** Tight labor markets are a main driver of inflation.

Current Context (2023)

- **US Inflation:** ~6% (down from peaks but still well above the Fed's 2% target)
- **Labor Market:** Extremely tight
 - Unemployment rate: 3.4% (lowest since the 1960s)
 - Labor force participation: Not fully recovered to pre-COVID trend
- **Fed Policy:** High interest rates maintained because “labor market conditions are very tight,” suggesting persistent inflation pressure

The Central Question

Why does a tight labor market create inflation? This lecture begins to answer this by examining how labor market conditions affect wages, which in turn affect prices.

2. Transition: From Short Run to Medium Run

IS-LM Limitations

The IS-LM model (Lectures 3-7) made two key assumptions: 1. **Fixed Prices:** $P = \bar{P}$ (completely sticky) 2. **Demand-Determined Output:** Whatever aggregate demand wanted, firms produced at constant prices

These assumptions work well for: - **Recessions:** Plenty of slack, firms can expand without hitting constraints - **Short-run analysis:** Prices adjust slowly

These assumptions fail when: - **Supply constraints bind:** Firms struggle to find workers - **Labor markets are tight:** Difficult to expand production - **Inflationary pressures emerge:** Firms raise prices because they can't meet demand at old prices

The New Framework

We will now **endogenize inflation** by: 1. Understanding wage determination (today's lecture) 2. Connecting wages to prices (price-setting behavior) 3. Deriving the Phillips Curve (next lecture) 4. Integrating supply and demand (IS-LM-PC model)

3. Labor Market Structure and Definitions

Population Breakdown (US, circa 2018)

Starting from total population and narrowing to those actively working:

Total Population (330M)

↓

Civilian Noninstitutional Population (260M)

[Excludes: under 16, incarcerated, armed forces]

↓

Labor Force (162M) = Employed + Unemployed

Employed (156M)

Unemployed (6M)

Out of Labor Force (98M)

[Retirees, students, discouraged workers, etc.]

Key Definitions

- **Labor Force (L):** $L = N + U$ (employed + unemployed)
- **Unemployment Rate:**

$$u = \frac{U}{L} = \frac{\text{Unemployed}}{\text{Labor Force}}$$

Note: NOT unemployment divided by total population

- **Labor Force Participation Rate:**

$$\text{LFPR} = \frac{L}{\text{Civilian Noninstitutional Population}}$$

- **Employment-to-Population Ratio:**

$$\text{EPOP} = \frac{N}{\text{Civilian Noninstitutional Population}}$$

Recent Data (January 2023)

- **Unemployment Rate:** 3.4% (historically very low)
- **Change in Unemployment:** -28,000 workers
 - **But:** This is net of two large offsetting flows:
 - * Employment increased: +894,000
 - * Labor force increased: +866,000
 - **Interpretation:** Strong job creation, but also people entering labor force (out of labor force → employed/unemployed)
- **LFPR:** 62.5% (still below pre-COVID trend)
- **EPOP:** 60% (recovering but off trend)

Historical Patterns (1950s-Present)

- **Unemployment Rate:**
 - Countercyclical: Rises sharply in recessions (peaks ~10% in 2009), falls in expansions
 - Pre-COVID low: ~3.5% (similar to today)
 - COVID spike: Brief surge to ~15% (unprecedented speed)
 - Current: 3.4% (record low, matching 1960s levels)
- **Employment Rate (EPOP):**
 - Upward trend (1950s-2000): Driven by rising female labor force participation
 - Decline post-2000: Demographics (aging), discouraged workers, structural changes
 - COVID collapse: Sharp drop as workers left labor force
 - Incomplete recovery: Currently at pre-COVID levels but **off trend** (missing ~2-3 percentage points)

4. Labor Market Dynamics: Flows Not Stocks

Why Flows Matter

The labor market is not static. Each month, millions of workers change status.

Monthly Labor Flows (Typical Month, 2018)

Employment (132M)

↓↑ 3M (Job-to-Job)
 ↓ 1.8M → Unemployment
 ↑ 2M ← Unemployment
 ↓ 3.7M → Out of Labor Force
 ↑ 3.4M ← Out of Labor Force

Unemployment (8.6M)
 ↓ 2M → Employment
 ↑ 1.8M ← Employment
 ↓ 1.5M → Out of Labor Force
 ↑ 1.3M ← Out of Labor Force

Out of Labor Force (79M)
 ↓ 3.4M → Employment
 ↑ 3.7M ← Employment
 ↓ 1.3M → Unemployment
 ↑ 1.5M ← Unemployment

Key Insights from Flows

1. **Large Gross Flows:** Even when unemployment changes by small amounts (e.g., -28K), the underlying flows are enormous (millions)
2. **Job-to-Job Transitions:** ~3M workers switch jobs each month without unemployment
3. **Out of Labor Force Matters:** Many workers bypass unemployment entirely (especially during COVID)

The Job-Finding Rate

Definition: Percentage of unemployed workers who find a job each month.

Relationship to Unemployment: - **High Unemployment** ($u \uparrow$) $\Rightarrow$ **Low Job-Finding Rate** ($\downarrow$) - **Why?** Firms aren't hiring aggressively; more competition for fewer jobs - **Result:** Unemployed workers stay unemployed longer (duration $\uparrow$)

- **Low Unemployment** ($u \downarrow$) $\Rightarrow$ **High Job-Finding Rate** ($\uparrow$)
 - **Why?** Firms hiring aggressively; plentiful vacancies
 - **Result:** Unemployed workers find jobs quickly (duration $\downarrow$)

Empirical Relationship: Nearly perfect negative correlation (inverted scale shows unemployment and job-finding rate move together)

Implications for Workers

When unemployment is **high** (recessions), workers face two risks: 1. **Employed Workers:** Higher probability of job loss (firms laying off) 2. **Unemployed Workers:** Lower probability of finding job (longer spells)

Combined Effect: High unemployment is **scary** for workers. This fear weakens their bargaining power.

When unemployment is **low** (today), workers have: 1. **Job Security:** Low probability of layoff 2. **Job Opportunities:** Easy to find new job if desired

Combined Effect: Low unemployment strengthens workers' bargaining power. They can credibly threaten to leave.

5. Wage Determination (Wage-Setting Relation)

Institutional Background

Collective Bargaining

- **Unions:** Set wages through collective negotiation
 - **US:** Union density low (~10%) and declining (much higher in 1950s-1970s)
 - **Europe/Japan:** Much higher union density (30-50%+ in some countries)
 - **Levels:** Firm-level, industry-level, or national-level bargaining

Individual Bargaining

- **High-skill workers:** More likely to negotiate individually (idiosyncratic, specialized skills)
- **Low-skill workers:** Often face posted wages (less negotiation)

Key Point: Macro Drivers Are Similar

Regardless of whether wages are set by unions or individuals, the **macroeconomic determinants** are the same: 1. Labor market tightness (unemployment rate) 2. Expected future prices (inflation) 3. Institutional factors (bargaining power)

The Reservation Wage

Definition: The wage at which a worker is **indifferent** between being employed vs. unemployed.

Fact of Life: Actual wages typically **exceed** reservation wages. - Workers prefer employment but have limited bargaining power - Firms pay more than minimum to retain workers, avoid turnover, elicit effort

Wage-Setting Equation (Aggregated)

$$W = P^e F(u, z)$$

Where: - W : **Nominal wage** (dollars per hour) - P^e : **Expected price level** (what workers expect prices to be during wage contract) - u : **Unemployment**

rate (measures labor market tightness) - z : **Institutional factors** (unemployment benefits, firing costs, minimum wage, union strength, etc.)

Understanding Each Component

1. Expected Price Level (P^e): Why It Matters **Question:** Why do wages depend on expected prices, not just current prices?

Answer: Wages are sticky. Once set, they last for a period (typically 1 year or more). - If $P^e = P$ (no expected inflation): Workers demand wage sufficient for current purchasing power - If $P^e > P$ (expect inflation): Workers demand **higher nominal wage** to maintain real purchasing power

Example: - Current price level: $P = 100$ - Expected price level next year: $P^e = 110$ (10% inflation expected) - Workers won't accept $W = \$20/\text{hour}$ if they expect prices to rise 10% - They'll demand $W = \$22/\text{hour}$ to maintain constant real wage ($W/P^e = 22/110 = 0.20$)

Why $\partial W / \partial P^e > 0$:

$$\frac{\partial W}{\partial P^e} > 0$$

Higher expected prices $\Rightarrow$ Workers demand higher nominal wages to protect real purchasing power.

2. Unemployment Rate (u): Bargaining Power Channel

$$\frac{\partial F}{\partial u} < 0 \quad \Rightarrow \quad \frac{\partial W}{\partial u} < 0$$

Why is wage decreasing in unemployment?

Worker Perspective (high unemployment): 1. **Job Loss Risk:** "If I push too hard, I might get fired, and it's hard to find another job" 2. **Low Outside Options:** "Even if I quit, I'll struggle to find work elsewhere" 3. **Weak Bargaining Position:** "The firm knows I'm replaceable"

Firm Perspective (high unemployment): 1. **Easy to Replace Workers:** "Plenty of qualified candidates in the market" 2. **Less Fear of Quits:** "Workers won't leave because alternatives are scarce" 3. **Strong Bargaining Position:** "We can offer lower wages and still attract talent"

Result: High $u \Rightarrow$ Low W (workers accept lower wages)

Opposite for Low Unemployment (today's situation): - **Worker:** "I can easily find another job if you don't pay me well" - **Firm:** "We desperately need to retain workers; competitors are poaching" - **Result:** Low $u \Rightarrow$ High W (wage pressures)

Current Example (2023): Restaurant industry - Unemployment extremely low - Restaurants struggling to hire - Wages rising rapidly (10%+ in some cases) - Firms very afraid of losing current workers

3. Institutional Factors (z): Structural Bargaining Power

$$\frac{\partial F}{\partial z} > 0 \quad \Rightarrow \quad \frac{\partial W}{\partial z} > 0$$

What is z ? A catch-all variable representing factors that **structurally strengthen workers' bargaining power**.

Examples of High z : 1. **Generous Unemployment Benefits:** Workers less desperate (can afford to wait for better offer) 2. **High Firing Costs:** Firms hesitant to fire (workers have job security) 3. **Strong Employment Protection Laws:** Difficult to lay off workers (European labor markets) 4. **High Minimum Wage:** Floor on wages 5. **Strong Unions:** Collective bargaining power 6. **Tight Labor Regulations:** Restrictions on hours, conditions, etc.

Cross-Country Variation: - **US:** Relatively low z (at-will employment, weak unions, modest unemployment insurance) - **France/Germany:** High z (strong employment protection, generous benefits, powerful unions) - **Nordics:** High z but combined with active labor market policies (training, reemployment support)

Wage-Setting as Real Wage Relation

Divide both sides by P^e :

$$\frac{W}{P^e} = F(u, z)$$

Interpretation: For a given expected price level, workers demand a real wage that: - **Decreases** with unemployment (weaker bargaining power) - **Increases** with institutional strength (z)

Graphical Representation: Downward-sloping curve in $(u, W/P)$ space (holding z constant)

6. Price Determination (Price-Setting Relation)

Production Function (Simplified)

$$Y = AN$$

Where: - Y : Aggregate output (GDP) - N : Employment (number of workers) - A : Labor productivity (output per worker)

Further Simplification: Set $A = 1$

$$Y = N$$

Interpretation: To produce one more unit of output, need one more worker.

Marginal Cost of Production

Question: What does it cost the firm to produce one additional unit of output?

Answer: 1. Need one more worker ($\Delta N = 1$ for $\Delta Y = 1$) 2. Worker costs W (the wage) 3. Therefore, **Marginal Cost** = W

$$MC = W$$

Price-Setting Equation: Markup Pricing

Firms set prices as a markup over marginal cost:

$$P = (1 + m)W$$

Where: - m : **Markup** (e.g., $m = 0.2$ means 20% markup) - $m > 0$: Positive due to imperfect competition, market power

Example: - Wage: $W = \$20/\text{hour}$ - Markup: $m = 0.25$ (25%) - Price: $P = 1.25 \times 20 = \$25$ per unit

What Determines m ? 1. **Market Structure:** Monopolistic competition, oligopoly (higher m), vs. perfect competition ($m \rightarrow 0$) 2. **Demand Elasticity:** If consumers sensitive to price, firms keep m low 3. **Entry Barriers:** Harder to enter market $\Rightarrow$ Higher m 4. **Input Costs:** If non-labor costs (energy, materials) rise but not well-measured in value-added, effective m increases

Price-Setting as Real Wage Relation

Divide both sides by P and rearrange:

$$P = (1 + m)W \quad \Rightarrow \quad \frac{W}{P} = \frac{1}{1 + m}$$

Critical Insight: The real wage is pinned down by the markup. - Firms are willing to pay real wage of $\frac{1}{1+m}$ - **Not a function of unemployment** or any other variable (in this simple model) - Horizontal line in $(u, W/P)$ space

Intuition: Why Is Real Wage Fixed by Markup?

1. Firms set $P = (1 + m)W$ regardless of economic conditions (taking m as given)
2. Rearranging: $W/P = 1/(1 + m)$
3. The real wage firms offer is purely determined by their desired markup

Comparative Statics: - $m \uparrow$ (markup increases) $\Rightarrow W/P \downarrow$ (real wage falls) - Firms take a larger share of revenue as profits, leaving less for workers

7. Equilibrium: The Natural Rate of Unemployment

Definition of Natural Rate (u_n)

The Natural Rate of Unemployment is the unemployment rate when **expected prices equal actual prices**:

$$P^e = P$$

Key Point: “Natural” is a misnomer. It’s not: - Optimal - Unchangeable - Determined by nature/God - Necessarily desirable

It’s simply the **medium-run equilibrium** unemployment rate when expectations are correct.

Why Is It Called “Natural”?

Historical naming (Milton Friedman, Edmund Phelps, 1960s). Better terms:
- **Structural unemployment rate** - **Equilibrium unemployment rate** - **Medium-run unemployment rate**

Why Does It Approximate the Long-Run Average?

If people aren’t systematically wrong about prices: - Sometimes $P^e > P$ (overestimate inflation) $\Rightarrow u < u_n$ - Sometimes $P^e < P$ (underestimate inflation) $\Rightarrow u > u_n$ - **On average:** $P^e = P \Rightarrow u = u_n$

Therefore, u_n is a good proxy for the **average unemployment rate** over longer periods (when forecast errors cancel out).

Deriving the Natural Rate

Step 1: Wage-Setting (with $P^e = P$)

$$W = P^e F(u, z) \quad \Rightarrow \quad W = P F(u, z)$$

Divide by P :

$$\frac{W}{P} = F(u, z)$$

When we solve for equilibrium assuming $P^e = P$, the resulting unemployment rate is u_n :

$$\frac{W}{P} = F(u_n, z)$$

Step 2: Price-Setting

$$\frac{W}{P} = \frac{1}{1 + m}$$

Step 3: Equilibrium Condition Set wage-setting equal to price-setting:

$$F(u_n, z) = \frac{1}{1+m}$$

Solve for u_n :

$$u_n = F^{-1}\left(\frac{1}{1+m}, z\right)$$

Interpretation: The natural rate is determined by: 1. **Markup** (m): Higher markup $\Rightarrow$ Higher u_n 2. **Institutions** (z): Stronger worker bargaining power $\Rightarrow$ Higher u_n (paradoxically)

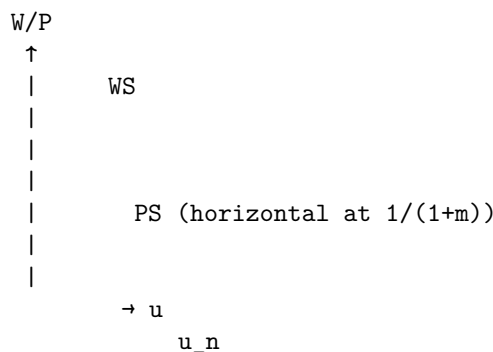
Graphical Determination

Axes: - Horizontal: Unemployment rate (u) - Vertical: Real wage (W/P)

Wage-Setting (WS) Curve: - **Slope:** Downward (higher $u \Rightarrow$ lower W/P) - **Position:** Shifts up if $z \uparrow$ (stronger bargaining power)

Price-Setting (PS) Curve: - **Slope:** Horizontal (real wage independent of u) - **Position:** $W/P = \frac{1}{1+m}$ - Shifts down if $m \uparrow$ (higher markup)

Equilibrium: Intersection of WS and PS determines u_n and equilibrium real wage.



Stability of Equilibrium

To the Right of u_n (unemployment too high): - Workers demand: $W/P = F(u, z)$ (low, because high u) - Firms offer: $W/P = \frac{1}{1+m}$ (same as always) - **Result:** Firms offering **more** than workers demand $\Rightarrow$ Firms hire $\Rightarrow u \downarrow$ (back toward u_n)

To the Left of u_n (unemployment too low): - Workers demand: $W/P = F(u, z)$ (high, because low u) - Firms offer: $W/P = \frac{1}{1+m}$ (same as always) - **Result:** Workers demand **more** than firms offer $\Rightarrow$ Firms reluctant to hire, may fire $\Rightarrow u \uparrow$ (back toward u_n)

Conclusion: u_n is stable equilibrium.

8. Comparative Statics: What Shifts the Natural Rate?

1. Increase in Bargaining Power ($z \uparrow$)

Examples: - Increase in unemployment benefits - Stricter employment protection laws - Stronger unions - Higher minimum wage

Effect on Wage-Setting: - WS curve shifts **up/right** - For any u , workers demand higher W/P

Effect on Price-Setting: - PS curve unchanged (firms' markup unaffected)

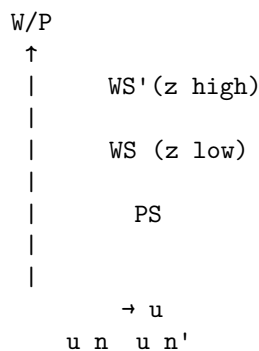
New Equilibrium: - WS shifts up, but PS is horizontal (real wage can't change in equilibrium) - **Only way to restore equilibrium:** $u_n \uparrow$ - Higher unemployment reduces workers' bargaining power back down until W/P demand falls to $\frac{1}{1+m}$

Mathematical Intuition:

$$F(u_n, z) = \frac{1}{1+m}$$

- $z \uparrow \Rightarrow F(\cdot)$ increases (for given u_n) - Right-hand side unchanged - **Restore equality:** Need $u_n \uparrow$ to bring $F(\cdot)$ back down (since $\frac{\partial F}{\partial u} < 0$)

Graphical:



Real-World Example: France (1980s) - Policy: Major labor market reforms in early 1980s under Mitterrand government - Increased employment protection (harder to fire) - Expanded unemployment benefits - Stronger union rights - Higher minimum wage - **Short-Run:** Real wages rose, workers celebrated - **Medium-Run:** Unemployment rate soared from ~6% to 12-15% by late 1980s - **Mechanism:** 1. $z \uparrow \Rightarrow$ Workers demanded higher wages 2. Firms couldn't sustain higher real wages (PS curve pinned by markup) 3. Firms reduced hiring, increased layoffs 4. Unemployment rose until workers' demands came back down - **Long-Run:** Decades of high unemployment, slow growth - **Recent:** France has been gradually unwinding some of these policies (Macron reforms)

2. Increase in Markup ($m \uparrow$)

Examples: - Reduced competition (mergers, lax antitrust enforcement) - Oil price shocks (increases effective markup if not well-measured in value-added) - Productivity decline (if firms maintain prices, effective markup rises) - Increased market power (tech monopolies, network effects)

Effect on Price-Setting: - PS curve shifts **down** - Firms now offer lower real wage: $\frac{W}{P} = \frac{1}{1+m} \downarrow$

Effect on Wage-Setting: - WS curve unchanged (workers' bargaining power unaffected)

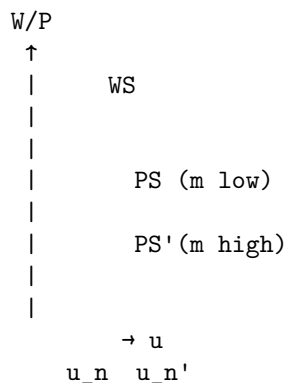
New Equilibrium: - At old u_n , workers demand more than firms offer (gap opens) - **Only way to restore equilibrium:** $u_n \uparrow$ - Higher unemployment reduces workers' wage demands until they accept lower real wage $\frac{1}{1+m}$

Mathematical Intuition:

$$F(u_n, z) = \frac{1}{1+m}$$

- $m \uparrow \Rightarrow$ Right-hand side $\downarrow$ - Left-hand side unchanged (for given u_n) - **Restore equality:** Need $u_n \uparrow$ to bring $F(u_n, z)$ down (since $\frac{\partial F}{\partial u} < 0$)

Graphical:



Real-World Example: Oil Shocks (1970s) - Event: OPEC oil embargo (1973) and Iranian Revolution (1979) $\Rightarrow$ Oil prices quadrupled - **Effect on Markup:** - Firms' costs rose (energy is input) - If value-added accounting doesn't fully adjust, effective m rises - Real wage firms can afford to pay fell - **Result:** - u_n increased in most advanced economies - US unemployment: Rose from 4-5% (1960s) to 7-10% (1970s-1980s) - Stagflation: High unemployment + High inflation (next lecture explains inflation part via Phillips Curve) - **Recovery:** As oil prices stabilized (1980s-1990s), u_n gradually declined

3. Decrease in Markup ($m \downarrow$)

Examples: - Increased competition (deregulation, trade liberalization, antitrust enforcement) - Positive productivity shock ($A \uparrow$ in full model) - Technological innovation reducing costs

Effect: Opposite of markup increase - PS curve shifts **up** - Firms can afford to pay higher real wage - At old u_n , firms offer more than workers demand - **Result:** $u_n \downarrow$ (lower equilibrium unemployment)

Real-World Example: US (1990s) - Productivity boom: Tech revolution (internet, computers) - **Effect:** $A \uparrow$ and/or $m \downarrow$ - **Result:** - Unemployment fell to ~4% (below previous estimates of $u_n \approx 6\%$) - Real wages rose - Low inflation despite tight labor market (productivity gains offset wage pressures)

4. Decrease in Bargaining Power ($z \downarrow$)

Examples: - Declining unionization (US since 1970s) - Reduced unemployment benefits - Weakening of employment protection

Effect: Opposite of bargaining power increase - WS curve shifts **down/left** - At old u_n , workers demand less than firms offer - **Result:** $u_n \downarrow$ (lower equilibrium unemployment)

Real-World Example: US (1980s-2000s) - Union Decline: Membership fell from ~25% (1970s) to ~10% (2000s) - **Effect:** $z \downarrow$ - **Result:** - u_n declined from ~6-7% (1970s) to ~5% (1990s) to ~4-5% (2000s) - Real wage growth moderated (but unemployment lower on average)

9. Key Takeaways

1. Labor Market Dynamics

- Unemployment rate is not just a stock; massive flows underneath (millions moving each month)
- Job-finding rate strongly countercyclical (hard to find jobs in recessions)
- High unemployment scares workers (bargaining power falls)

2. Wage Determination (WS)

$$\frac{W}{P} = F(u, z)$$

- Real wage **decreases** with unemployment (bargaining power channel) - Real wage **increases** with institutional strength (z) - Downward-sloping curve in $(u, W/P)$ space

3. Price Determination (PS)

$$\frac{W}{P} = \frac{1}{1+m}$$

- Real wage determined by markup (firms' pricing power) - **Independent** of unemployment (horizontal curve) - Higher markup $\Rightarrow$ Lower real wage

4. Natural Rate of Unemployment (u_n)

$$F(u_n, z) = \frac{1}{1 + m}$$

- Equilibrium when $P^e = P$ (correct expectations) - “Natural” is misleading—it's not optimal or fixed - Depends on structural factors (m, z) - Approximates medium-run average unemployment

5. Comparative Statics

- $z \uparrow$ (stronger unions, higher benefits) $\Rightarrow u_n \uparrow$ (paradoxically hurts workers in medium run)
- $m \uparrow$ (less competition, higher markups) $\Rightarrow u_n \uparrow$ (lower real wages or higher unemployment)
- Both shifts lead to higher unemployment (the “escape valve” for restoring equilibrium)

6. Why This Matters for Inflation (Next Lecture)

- Today's model determines u_n (medium-run equilibrium)
- But what if $P^e \neq P$? (Expectations wrong)
- **Phillips Curve** will show:
 - When $u < u_n$: Upward pressure on wages/prices (inflation accelerates)
 - When $u > u_n$: Downward pressure on wages/prices (inflation decelerates)
- This connects labor market to inflation dynamics

10. Current Policy Relevance (2023)

Why Is the Fed Keeping Rates High?

1. **Unemployment = 3.4%**: Well below most estimates of u_n (consensus ~4-5%)
2. **Implication**: $u < u_n \Rightarrow$ Wage pressures $\Rightarrow$ Inflation pressures (via Phillips Curve)
3. **Fed Goal**: Raise rates $\Rightarrow$ Slow economy $\Rightarrow$ Raise u toward $u_n \Rightarrow$ Reduce wage/inflation pressures

Labor Market “Tightness” Indicators

1. **Unemployment Rate**: 3.4% (record low)
2. **EPOP**: Still below trend (missing workers due to early retirements, COVID, etc.)

3. **Wage Growth:** 5-6% year-over-year (above inflation target + productivity growth)
4. **Job Openings:** Near record highs (JOLTS data)
5. **Quits Rate:** High (workers confident they can find better jobs)

All point to **very tight** labor market $\Rightarrow$ Sustained wage/inflation pressures unless unemployment rises.

Open Questions

1. **Has u_n Changed?**
 - Pre-COVID: Some argued u_n fell to $\sim 3.5\%$ (we sustained it without high inflation)
 - Post-COVID: Demographics, long COVID, early retirements may have raised u_n
2. **Will Wage Growth Slow?**
 - If u_n unchanged ($\sim 4-5\%$), need u to rise to cool wages
 - If u_n has fallen ($\sim 3.5\%$), current unemployment consistent with stable inflation
3. **Soft Landing vs. Recession?**
 - Fed hopes to raise u gently (to $\sim 4.5\%$) without recession
 - History suggests difficult: Unemployment usually rises >2 percentage points in recessions

11. Limitations of This Model

What We've Assumed Away

1. **Constant Markup:** In reality, m varies with demand (procyclical?)
2. **Constant Productivity:** $A = 1$ is extreme simplification
3. **One Labor Market:** In reality, heterogeneous (skilled vs. unskilled, sectors, regions)
4. **No Capital:** Full model includes investment, capital accumulation
5. **Static Expectations:** Next lecture relaxes $P^e = P$

What's Next

- **Lecture 9:** Phillips Curve (what happens when $P^e \neq P$? Inflation dynamics)
- **Lecture 10:** IS-LM-PC model (integrating demand, supply, and inflation)
- **Later:** Growth theory (endogenizing A , capital, long-run trends)

12. Historical Context: The Natural Rate Debate

Origins (1960s)

- **Phillips Curve Discovery (1958):** Negative correlation between unemployment and wage growth

- **Keynesian View:** Permanent tradeoff (can choose low unemployment at cost of higher inflation)
- **Friedman-Phelps Critique (1968):**
 - No long-run tradeoff
 - Only temporary deviations from “natural rate” u_n
 - Attempting to keep $u < u_n$ causes accelerating inflation

Stagflation (1970s)

- High unemployment + High inflation (contradicted simple Phillips Curve)
- Vindicated natural rate view (oil shocks raised u_n , attempts to lower u below new u_n caused inflation)

Modern Consensus

- Natural rate exists but:
 - Not constant (varies over time with m, z)
 - Not precisely known (Fed estimates $u_n \approx 4.5\%$ but uncertain)
 - Not normative (not “optimal,” just equilibrium)

Recent Puzzles

- **Pre-COVID (2018-2019):** $u = 3.5\%$, yet inflation stayed near 2% (was u_n lower than thought?)
- **Post-COVID (2022-2023):** Rapid inflation despite unemployment only modestly below historical u_n (were there additional shocks? Supply chains? Expectations?)

These puzzles motivate ongoing research in labor macroeconomics.

Next Lecture: The Phillips Curve (connecting unemployment to inflation dynamics)

Lecture 9: The Phillips Curve

Overview

This lecture introduces one of the most important relationships in macroeconomics: the Phillips Curve, which describes the trade-off between unemployment and inflation. We derive this relationship from the wage-setting and price-setting equations, examine its evolution over time, and explore its critical implications for monetary policy and expectations formation.

1. Historical Origins of the Phillips Curve

A.W. Phillips (1958)

- **Discovery:** A.W. Phillips, an economist at the London School of Economics (LSE), documented an empirical relationship using historical data for the UK (1861-1957).
- **Finding:** A stable **negative correlation** between the unemployment rate and the rate of wage inflation.
- **Key Observation:** At very low levels of unemployment, economies typically experienced very high levels of inflation; conversely, at very high levels of unemployment, they experienced low inflation or even deflation.

Samuelson & Solow (1960)

- **Extension:** Paul Samuelson and Robert Solow replicated Phillips' work using US data (1900-1960).
- **Naming:** They named this empirical relationship the "Phillips Curve" in honor of A.W. Phillips.
- **Significance:** The relationship became a central concept in macroeconomic policy, suggesting a menu of choices between unemployment and inflation.

Early Data (1900-1960)

- The data showed a clear downward-sloping relationship.
- **Historical Context:** This period included the Great Depression (high unemployment, deflation) and various expansions (low unemployment, higher inflation).
- **Initial Interpretation:** Policymakers believed they could exploit this trade-off—accepting slightly higher inflation to achieve lower unemployment.

2. Theoretical Foundation: Deriving the Phillips Curve

Starting Ingredients (From Lectures 7-8)

Wage Setting Equation (WS)

$$W = P^e \cdot F(u, z)$$

- W : Nominal wage
- P^e : Expected price level
- u : Unemployment rate
- z : Catchall variable for labor market institutions (unemployment benefits, bargaining power, etc.)
- $F(u, z)$: **Decreasing in u , Increasing in z**

Economic Intuition: - **Higher unemployment** ($u \uparrow$): Workers are “scared”—more likely to lose jobs and harder to find new ones if unemployed. Firms have more bargaining power (easier to replace workers). Result: Lower wage demands ($W \downarrow$). - **Higher institutional support** ($z \uparrow$): Stronger unemployment benefits, unions, etc., embolden workers to demand higher wages ($W \uparrow$).

Price Setting Equation (PS)

$$P = (1 + m)W$$

- P : Price level
- m : Markup over marginal cost (wage)
- Firms set prices by marking up wages (assuming one worker produces one unit).

Economic Intuition: - Firms need to cover costs and earn profits. - Higher wages $\Rightarrow$ Higher costs $\Rightarrow$ Higher prices (given constant markup).

Combining WS and PS

Substitute the wage-setting equation into the price-setting equation:

$$P = P^e(1 + m)F(u, z)$$

This gives us a relationship between the **price level** and unemployment (conditional on expected prices).

Functional Form Assumption

For tractability, assume $F(u, z)$ is locally linear:

$$F(u, z) = 1 - \alpha u + z$$

where $\alpha > 0$ captures the sensitivity of wages to unemployment.

Substituting:

$$P = P^e(1 + m)(1 - \alpha u + z)$$

Interpretation: - For given expected prices P^e , markups m , and institutions z : - Higher unemployment $\Rightarrow$ Lower wages $\Rightarrow$ Lower prices.

From Price Levels to Inflation Rates

The Phillips Curve relates **inflation** (not price levels) to unemployment. To get there:

Step 1: Divide both sides by P_{t-1} (previous period's price level):

$$\frac{P_t}{P_{t-1}} = \frac{P_t^e}{P_{t-1}}(1+m)(1-\alpha u_t + z)$$

Step 2: Recognize that:

$$\frac{P_t}{P_{t-1}} = 1 + \pi_t$$

where π_t is the inflation rate.

Step 3: For expected inflation:

$$\pi_t^e = \frac{P_t^e - P_{t-1}}{P_{t-1}} \implies \frac{P_t^e}{P_{t-1}} = 1 + \pi_t^e$$

Note: We use P_{t-1} (not P_{t-1}^e) in the denominator because at time t , agents **know** what happened at $t-1$ —there's no expectation needed for past data.

Step 4: Substitute into the equation:

$$(1 + \pi_t) = (1 + \pi_t^e)(1+m)(1-\alpha u_t + z)$$

Step 5: Apply the log approximation. For small x , $\log(1+x) \approx x$:

$$\log(1 + \pi_t) \approx \pi_t$$

$$\log(1 + \pi_t^e) \approx \pi_t^e$$

$$\log(1 + m) \approx m$$

$$\log(1 - \alpha u_t + z) \approx -\alpha u_t + z$$

Taking logs of both sides:

$$\pi_t \approx \pi_t^e + m + z - \alpha u_t$$

The Basic Phillips Curve

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

Interpretation: - **Negative relationship:** Higher unemployment $\Rightarrow$ Lower inflation (holding expectations, markups, and institutions constant). - **Economics:** Same intuition as before—higher unemployment lowers wages, which lowers prices, which lowers inflation. - **Three components:** 1. **Expected inflation** (π_t^e): Baseline expectation. 2. **Supply shocks** ($m + z$): Changes in markups (e.g., oil shocks) or labor market institutions. 3. **Demand pressure** ($-\alpha u_t$): Tightness of the labor market.

3. Evolution of the Phillips Curve Over Time

Period 1: The Original Phillips Curve (Pre-1970s)

Context

- **Low, stable inflation:** Inflation in the US and UK was generally low and stable (0-2%).
- **Well-anchored expectations:** People expected inflation to remain around historical averages.

Expectations Assumption

$$\pi_t^e = \bar{\pi} \quad (\text{constant})$$

This means expected inflation is “anchored” at some long-run average $\bar{\pi}$ (e.g., 2%).

Simplified Phillips Curve

$$\pi_t = \bar{\pi} + (m + z) - \alpha u_t$$

or equivalently:

$$\pi_t = \text{constant} - \alpha u_t$$

Result: A **stable, downward-sloping line** relating inflation to unemployment.

Policy Implications (1960s)

- **Perceived Trade-Off:** Policymakers (including advisors Samuelson and Solow) believed they could choose a point on this curve.
- **Exploitation:** The US government pursued expansionary policies to lower unemployment, accepting slightly higher inflation.
- **Initial Success:** The curve was relatively flat, so unemployment fell significantly without much inflation (early 1960s).
- **Diminishing Returns:** As unemployment fell further, the curve became steeper (convex), and inflation accelerated (late 1960s).

Period 2: The Breakdown (1970s)

What Happened? If you plot the same data for **1970-1995**, the negative relationship **disappears**. The data is scattered all over the place—no clear pattern.

Question: Had Mr. Phillips been born later and estimated his regression in the 1970s, would there be a curve named after him? **Answer:** Probably not. He would have found no relationship.

Why Did the Relationship Break Down? Two Main Causes:

1. Oil Shocks (Supply Shocks):

- 1973: OPEC oil embargo.
- 1979: Iranian Revolution.
- **Effect on m :** Firms faced higher energy costs $\Rightarrow$ Needed to increase markups to cover costs $\Rightarrow m \uparrow$.
- **Result:** For any given level of unemployment, inflation was higher.
- **Graphically:** The Phillips Curve **shifted up**.

2. De-Anchoring of Inflation Expectations:

- As inflation persisted at high levels, people **stopped believing** it would return to 2%.
- **Adaptive Expectations:** Instead of expecting $\bar{\pi}$, people began forming expectations based on recent inflation:

$$\pi_t^e = \pi_{t-1}$$

- **Implication:** Expected inflation became **endogenous** (moving with actual inflation), rather than fixed.

Period 3: The Accelerationist Phillips Curve (1970s-1990s)

General Expectations Model A more flexible model for expectations:

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

- θ : Weight on recent inflation (persistence parameter).
- $\theta = 0$: Anchored expectations ($\pi_t^e = \bar{\pi}$).
- $\theta = 1$: Fully adaptive expectations ($\pi_t^e = \pi_{t-1}$).

The Accelerationist Case ($\theta = 1$) During the 1970s-1980s, $\theta \approx 1$ (empirical estimates).

Expectations:

$$\pi_t^e = \pi_{t-1}$$

Phillips Curve:

$$\pi_t = \pi_{t-1} + (m + z) - \alpha u_t$$

Rearrange:

$$\pi_t - \pi_{t-1} = (m + z) - \alpha u_t$$

Interpretation: - Now the Phillips Curve relates the **change in inflation** to unemployment. - Low unemployment $\Rightarrow$ Inflation **accelerates** ($\pi_t > \pi_{t-1}$). - High unemployment $\Rightarrow$ Inflation **decelerates** ($\pi_t < \pi_{t-1}$).

Why “Accelerationist”? - Because it describes inflation **acceleration** (or deceleration), not just the level. - **Implication:** Very low unemployment doesn't just produce high inflation—it produces **rising** inflation, which can spiral out of control.

Empirical Fit When you estimate the accelerationist Phillips Curve on 1970s-1990s data, you recover a **negative relationship** (between $\Delta\pi$ and u).

Conclusion: The theory still works—you just need the right specification for expectations.

Period 4: Re-Anchoring (Mid-1990s - 2019)

The Volcker Disinflation

- **Paul Volcker** (Fed Chair, 1979-1987): Aggressively raised interest rates to break the back of inflation.
- **Cost:** Deep recession (1981-1982), unemployment peaked at ~10.8%.
- **Benefit:** Inflation fell from ~14% (1980) to ~3% (1983).
- **Key Achievement:** Re-anchored inflation expectations around 2%.

Global Adoption

- Other central banks (e.g., ECB, Bank of England, Latin American central banks) followed suit.
- **Inflation targeting regimes:** Explicit 2% targets became the norm.

Return to the Original Phillips Curve By the mid-1990s: - $\theta \approx 0$ again (expectations re-anchored). - $\pi_t^e \approx 2\%$ (stable, independent of recent inflation).
- **Result:** The original downward-sloping relationship between **inflation** and **unemployment** returned.

Data (1995-2019) Plotting inflation vs. unemployment for this period shows a **clear negative relationship** again (similar to the 1960s).

Success of Monetary Policy: Central banks successfully maintained low, stable inflation for ~25 years (“Great Moderation”).

4. The Natural Rate of Unemployment (u_n)

Definition

The **natural rate of unemployment** (u_n) is the unemployment rate at which:

$$\pi_t = \pi_t^e$$

Meaning: Actual inflation equals expected inflation—there are no “surprises.”

Why “Natural”?

- **Not divinely ordained:** It’s not a fixed number given by nature or God.
- **Structural:** It’s determined by labor market institutions (z), markups (m), and wage-setting behavior.

- **Equilibrium:** It's the unemployment rate at which the wage-setting and price-setting equations are consistent.

Derivation

From the Phillips Curve:

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

Set $\pi_t = \pi_t^e$:

$$0 = (m + z) - \alpha u_n$$

Solve for u_n :

$$u_n = \frac{m + z}{\alpha}$$

Determinants of u_n

1. Markups (m)

- **Higher m :** Firms want to pay lower real wages (to maintain profits).
- Workers (at the original unemployment level) won't accept lower real wages.
- **Equilibrium:** Unemployment must rise to weaken workers' bargaining power.
- **Result:** $u_n \uparrow$ when $m \uparrow$.

2. Labor Market Institutions (z)

- **Higher z :** Workers demand higher real wages (e.g., stronger unions, generous unemployment benefits).
- Firms aren't willing to pay higher real wages.
- **Equilibrium:** Unemployment must rise to reduce wage demands.
- **Result:** $u_n \uparrow$ when $z \uparrow$.

3. Wage Sensitivity (α)

- **Higher α :** Wages are more responsive to unemployment.
- **Result:** A smaller increase in unemployment is needed to restore equilibrium $\Rightarrow u_n \downarrow$ when $\alpha \uparrow$.

Phillips Curve in Terms of the Unemployment Gap

From the Phillips Curve:

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

We know that:

$$m + z = \alpha u_n$$

Substitute:

$$\pi_t = \pi_t^e + \alpha u_n - \alpha u_t$$

Rearrange:

$$\pi_t - \pi_t^e = -\alpha(u_t - u_n)$$

Interpretation: - **Unemployment below natural rate** ($u_t < u_n$): “Tight” labor market $\Rightarrow$ **Upward pressure on inflation** ($\pi_t > \pi_t^e$). - **Unemployment above natural rate** ($u_t > u_n$): “Slack” labor market $\Rightarrow$ **Downward pressure on inflation** ($\pi_t < \pi_t^e$). - **Unemployment at natural rate** ($u_t = u_n$): No pressure $\Rightarrow$ Inflation equals expectations.

This is the form most commonly used by central banks.

5. Policy Implications and Current Context (2020s)

The Challenge of Estimating u_n

- **Problem:** The natural rate is **not directly observable**.
- **Estimates:** Economists use statistical methods, but estimates are uncertain and vary.
- **Real-time challenge:** When unemployment is falling, policymakers must guess whether they’re approaching u_n (and risking inflation) or if there’s still slack.

Example Estimate: US Natural Rate Over Time

A typical estimate shows: - **Blue Line:** Estimated natural rate of unemployment. - **Red Line:** Actual unemployment rate.

Episode 1: Great Recession (2008-2013)

- $u_t \gg u_n$: Unemployment far above the natural rate (peaked at 10%).
- **Prediction:** Strong **downward pressure** on inflation.
- **Reality:** Inflation fell below 2%, even briefly negative (deflation).
- **Implication:** The gap correctly predicted disinflationary pressure.

Episode 2: Pre-COVID Expansion (2016-2019)

- $u_t < u_n$: Unemployment fell below most estimates of the natural rate (reached ~3.5%).
- **Expected:** Upward pressure on inflation.
- **Reality:** Inflation remained stubbornly around 1.5-2% (didn’t accelerate much).

- **Mystery:** Suggested either:
 1. The natural rate had declined (not captured by models), or
 2. The Phillips Curve had “flattened” (inflation less responsive to unemployment).

Episode 3: Post-COVID (2021-2023)

- $u_t \ll u_n$: Unemployment fell rapidly (reached ~3.4% in 2023).
- **Supply shocks:** Pandemic disruptions, supply chain bottlenecks, energy prices.
- **Result:** Inflation surged to ~9% (June 2022).
- **Implication:** Negative unemployment gap + supply shocks = very high inflation.

Current Policy Debate (2022-2023)

Question: Is a recession necessary to bring inflation back to 2%?

View 1: Yes, Recession is Necessary

- **Logic:** The Phillips Curve says $u_t > u_n$ is needed to bring $\pi_t < \pi_t^e$.
- If $\pi_t^e \approx 5 - 6\%$ (unanchored), then we need $u_t \gg u_n$ to bring inflation back to 2%.
- **Implication:** The Fed must induce a recession (raise unemployment significantly).

View 2: Maybe Not (Fed’s Initial Hope)

- **Alternative indicators:** Other measures of labor market tightness (job openings, quits rate, hiring flows) are improving.
- **Supply-side healing:** Bottlenecks resolving, energy prices moderating $\Rightarrow m \downarrow$.
- **If expectations stay anchored** ($\pi_t^e \approx 2 - 3\%$): Only a modest rise in unemployment is needed.
- **Implication:** A “soft landing” (disinflation without recession) is possible.

The Central Role of Expectations **Critical Variable:** π_t^e (expected inflation).

Scenario A: Expectations Stay Anchored ($\pi_t^e \approx 2\%$)

- **Phillips Curve:**

$$\pi_t = 2\% - \alpha(u_t - u_n)$$

- **To bring π_t to 2%:** Just need $u_t = u_n$ (close the gap).
- **Cost:** Modest rise in unemployment (not a deep recession).

Scenario B: Expectations Become Unanchored ($\pi_t^e \approx 6\%$)

- **Phillips Curve:**

$$\pi_t = 6\% - \alpha(u_t - u_n)$$

- **To bring π_t to 2%:** Need $\pi_t = 2\% = 6\% - \alpha(u_t - u_n)$

$$\alpha(u_t - u_n) = 4\%$$

$$u_t - u_n = \frac{4\%}{\alpha}$$

- **Implication:** Unemployment must rise **far above** u_n (deep recession).
- **Cost:** Severe recession to “break” expectations and bring them back down.

What Happened in 2022-2023?

Summer 2022: Near-Miss

- **One-year expected inflation** crept up to $\sim 6\%$ (based on surveys and market measures).
- **Fed response:** Very hawkish speeches (Powell at Jackson Hole), aggressive rate hikes.
- **Result:** Expectations quickly fell back to $\sim 2.5\text{-}3\%$.
- **Success:** Avoided full de-anchoring.

2023-2024: Continued Uncertainty

- Expected inflation fluctuates between $2.5\text{-}3.5\%$.
- **Fed’s goal:** Keep expectations anchored to avoid needing a deep recession.
- **Risk:** If inflation lingers at $5\text{-}6\%$ and expectations rise again, a recession may become unavoidable.

The Role of Credibility

- **Volcker’s lesson:** Aggressive action + clear communication can re-anchor expectations.
- **Fed’s challenge:** Balance tightening enough to control inflation without triggering a severe recession.
- **Anchor or spiral:** The difference between a soft landing and a hard landing depends on θ (persistence of expectations).

6. Additional Factors and Extensions

Supply Shocks (Oil, Pandemics, Wars)

- **Effect on m :** Higher energy costs $\Rightarrow$ Firms mark up more $\Rightarrow m \uparrow$.
- **Phillips Curve shifts:** For given u , inflation is higher.

- **Historical examples:**
 - 1973-1974: OPEC embargo.
 - 1979-1980: Iranian Revolution.
 - 2021-2022: Pandemic supply chain disruptions, Russia-Ukraine war.

Labor Market Institutions (z)

- **Unemployment benefits:** More generous benefits $\Rightarrow$ Higher reservation wages $\Rightarrow z \uparrow \Rightarrow u_n \uparrow$.
- **Unions:** Stronger unions $\Rightarrow$ Higher wage demands $\Rightarrow z \uparrow \Rightarrow u_n \uparrow$.
- **COVID example:** Emergency pandemic benefits (2020-2021) likely raised z temporarily, contributing to wage growth and inflation.

Labor Force Participation

- **Definition:** Share of working-age population in the labor force (employed or actively seeking work).
- **Post-COVID decline:** Many workers retired early or dropped out due to health concerns.
- **Effect:** Tighter labor market (lower unemployment) for given employment $\Rightarrow$ More wage pressure.
- **Policy relevance:** Increasing participation could ease inflation without raising unemployment.

Immigration

- **US context:** Immigration flows declined during COVID (~500,000 workers/year lost).
- **Effect:** Reduced labor supply $\Rightarrow$ Tighter labor market $\Rightarrow$ Upward wage pressure.
- **Policy tool:** Increasing immigration could help cool wage growth without requiring higher unemployment.

7. Summary

Historical Evolution

1. **1900-1960s:** Stable negative relationship (anchored expectations).
2. **1970s-1980s:** Breakdown due to oil shocks and de-anchored expectations (accelerationist Phillips Curve).
3. **1990s-2019:** Re-anchoring and return to original relationship (successful monetary policy).
4. **2020s:** New challenges from pandemic shocks and inflation surge.

Key Equations

Basic Phillips Curve

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

Natural Rate of Unemployment

$$u_n = \frac{m + z}{\alpha}$$

Phillips Curve (Unemployment Gap Form)

$$\pi_t - \pi_t^e = -\alpha(u_t - u_n)$$

Accelerationist Phillips Curve ($\theta = 1$)

$$\pi_t - \pi_{t-1} = (m + z) - \alpha u_t$$

Policy Takeaways

- **Expectations are critical:** Anchored expectations allow disinflation without severe recessions.
- **Natural rate is structural:** Changes in m and z shift u_n (not just cyclical).
- **Trade-offs exist:** In the short run, there's tension between unemployment and inflation goals.
- **Long run:** No permanent trade-off if expectations adjust (accelerationist view).

Current Relevance

- **Fed's challenge:** Bring inflation from ~6% to 2% without triggering a deep recession.
- **Battle on expectations:** Success depends on keeping π_t^e anchored near 2%.
- **Uncertainty:** Estimating u_n in real time is difficult; policy risks missing on either side (too tight or too loose).

Next Lectures: We'll integrate the Phillips Curve into the IS-LM framework to build a complete model of output, unemployment, and inflation dynamics (the IS-LM-PC model).

Lecture 10: Comprehensive Review for Quiz 1

Overview

This lecture provides a comprehensive review of material from Lectures 1-8, covering basic definitions, the goods market, financial markets, the IS-LM model,

extensions incorporating risk and inflation, and the labor market. This review emphasizes key concepts, equations, graphical analysis, and policy applications. Approximately two-thirds of Quiz 1 will focus on IS-LM analysis.

1. Measuring Aggregate Output (GDP)

The Measurement Problem

At the individual firm level, measuring output is straightforward. At the aggregate level, we must avoid double-counting intermediate goods.

Example: Two-Firm Economy

- **Steel Company:** Produces \$100 of steel (all sold to car company as intermediate input)
- **Car Company:** Produces \$200 of cars (final goods)
- **Naive Sum:** $\$100 + \$200 = \$300$ (WRONG - double counts steel)
- **Correct GDP:** \$200

Three Equivalent Methods for Measuring GDP

Method 1: Final Goods Approach

- **Definition:** $\text{GDP} = \text{Sum of value of all final goods only}$
- **Final Goods:** Goods sold to end users, not as inputs to other firms
- **Example:** Only cars (\$200) are final goods, steel is intermediate
- **Result:** $\text{GDP} = \$200$

Method 2: Value Added Approach

- **Definition:** $\text{Value Added} = \text{Revenue} - \text{Cost of Intermediate Inputs}$
- **Steel Company:**
 - Revenue = \$100
 - Intermediate inputs = \$0
 - Value added = \$100
- **Car Company:**
 - Revenue = \$200
 - Intermediate inputs = \$100 (steel)
 - Value added = $\$200 - \$100 = \$100$
- **Total Value Added:** $\$100 + \$100 = \$200$
- **Result:** $\text{GDP} = \$200$

Method 3: Income Approach

- **Definition:** $\text{GDP} = \text{Sum of all incomes (wages + profits)}$
- **Steel Company:**

- Wages = \$80
- Profits = \$20
- Total income = \$100
- **Car Company:**
 - Wages = \$70
 - Profits = \$30
 - Total income = \$100
- **Total Income:** \$100 + \$100 = \$200
- **Result:** GDP = \$200

Key Properties

1. **All three methods yield identical results**
2. **Immune to organizational structure changes:** If the two companies merged, GDP would still be \$200
3. **Avoids double counting:** Unlike naive summation, which would incorrectly change from \$300 to \$200 upon merger

Nominal vs. Real GDP

Definitions

- **Nominal GDP:** $GDP_{\text{nominal}} = P_t \times Q_t$ (measured at current prices)
- **Real GDP:** $GDP_{\text{real}} = P_{\text{base}} \times Q_t$ (measured at base year prices)

Example: Single Good Economy (Cars)

Year	Quantity	Price	Nominal GDP	Real GDP (base=2012)
2012	10	\$24,000	\$240,000	\$240,000
2013	12	\$25,000	\$300,000	\$288,000
2014	13	\$26,000	\$338,000	\$312,000

- **Real GDP in 2012:** $\$24,000 \times 10 = \$240,000$
- **Real GDP in 2013:** $\$24,000 \times 12 = \$288,000$ (NOT $\$25,000 \times 12$)
- **Real GDP in 2014:** $\$24,000 \times 13 = \$312,000$

Note on Multiple Goods

- With one good, choice of base year doesn't affect growth rate
- With multiple goods, base year matters because relative prices change over time

Other Key Definitions

Unemployment Rate

$$u = \frac{\text{Number of Unemployed}}{\text{Labor Force}}$$

- **Note:** Denominator is labor force, NOT total population - Labor Force = Employed + Unemployed (actively seeking work)

Inflation Rate

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{\Delta P}{P}$$

- Multiple price indices exist: GDP deflator, CPI, etc.

2. The Goods Market: Keynesian Cross Model

Components of Aggregate Demand (Closed Economy)

$$Z = C + I + G$$

- **Closed Economy Assumption:** No exports or imports (not on Quiz 1) - C : Consumption - I : Investment - G : Government expenditure

The Consumption Function

$$C = c_0 + c_1(Y - T)$$

Parameters

- c_0 : Autonomous consumption (consumption when disposable income = 0)
- c_1 : Marginal Propensity to Consume (MPC)
 - $0 < c_1 < 1$
 - Interpretation: Out of an additional dollar of disposable income, c_1 is consumed
- $Y - T$: Disposable income (income minus taxes)

Properties

- **Upward sloping** in disposable income
- **Slope** = $c_1 < 1$

Equilibrium Condition

$$Y = Z$$

Output equals aggregate demand (demand-determined output).

Solving for Equilibrium Output

Starting from:

$$Y = c_0 + c_1(Y - T) + I + G$$

Rearranging:

$$Y - c_1 Y = c_0 + I + G - c_1 T$$
$$Y(1 - c_1) = c_0 + I + G - c_1 T$$

Equilibrium Output:

$$Y = \frac{1}{1 - c_1} [c_0 + I + G - c_1 T]$$

The Multiplier

Definition

$$\text{Multiplier} = \frac{1}{1 - c_1} > 1$$

Interpretation

- An increase in autonomous spending (e.g., $\Delta G = 1$) leads to a larger increase in equilibrium output
- **Why?:** Initial spending increase generates income, which is partly consumed, generating more income, etc.
- **Example:** If $c_1 = 0.5$, then multiplier = $\frac{1}{1-0.5} = 2$

Multiplier Effect in Action If $\Delta c_0 = 1$: 1. Initial increase in demand: $\Delta Z = 1$ 2. Output rises by 1 (to meet demand) 3. Income rises by 1, consumption rises by c_1 4. Demand rises further by c_1 , output rises by c_1 5. Process continues... 6. **Final change in output:** $\Delta Y = \frac{1}{1-c_1} > 1$

Graphical Representation

Aggregate Demand Curve

- **Equation:** $Z = c_0 + c_1(Y - T) + I + G$
- **Vertical intercept:** $c_0 + I + G - c_1 T$
- **Slope:** c_1 (less than 1)

Equilibrium

- **45-degree line:** Points where $Y = Z$
- **Intersection:** Determines equilibrium output Y^*
- At equilibrium: Aggregate demand = Aggregate supply

Shifts in Aggregate Demand

Upward Shifts (Expansionary)

- Increase in c_0 (consumer confidence)
- Increase in I (investment optimism)

- Increase in G (government spending)
- Decrease in T (tax cuts)

Example: Increase in c_0

- Aggregate demand shifts up by Δc_0
- Equilibrium output increases by $\frac{1}{1-c_1}\Delta c_0$ (multiplier effect)

Alternative Equilibrium Condition: Saving = Investment

Derivation Starting from $Y = C + I + G$:

$$Y - C - G = I$$

$$(Y - T - C) + (T - G) = I$$

$$S_{\text{private}} + S_{\text{government}} = I$$

Where: - **Private Saving:** $S_{\text{private}} = Y - T - C$ - **Government Saving:**
 $S_{\text{government}} = T - G$ (budget surplus)

The Paradox of Thrift

Scenario Suppose consumers decide to save more: $c_0 \downarrow$

What Happens?

1. **Aggregate demand decreases:** Z shifts down
2. **Output falls:** $Y \downarrow$ (through multiplier)
3. **Income falls:** Since $Y = \text{Income}$
4. **Actual saving:** $S = Y - T - C$ may fall (or stay constant) because income fell

Interpretation

- Individual attempts to save more lead to lower aggregate income
- In equilibrium, total saving doesn't increase (or even decreases)
- **Paradox:** Virtue at individual level (thrift) becomes vice at aggregate level

Alternative Explanation (Saving-Investment Framework)

- If, for any given Y , saving increases, then $S > I$ (imbalance)
- To restore equilibrium ($S = I$), output must fall
- Lower Y reduces saving, restoring $S = I$

3. Financial Markets

Simplifying Assumptions

- Only two assets: **Money** (cash) and **Bonds**
- Focus on money market equilibrium

Money Demand

$$M^d = \$Y \cdot L(i)$$

Determinants

1. **Nominal Income (\$Y)**:
 - M^d increasing in $\$Y$
 - Interpretation: Higher income means more transactions, need more money
2. **Interest Rate (i)**:
 - M^d decreasing in i
 - Interpretation: i is opportunity cost of holding cash (return on bonds)
 - Example: When $i = 5\%$, holding cash costs you 5% per year

Functional Form

$$L(i)$$

is a decreasing function of i

Money Supply

- **Controlled by the Central Bank**
- Denoted M^s or simply M

Equilibrium in Money Market

$$M^s = M^d$$

At equilibrium interest rate i^* , money supply equals money demand.

Modern Monetary Policy Implementation

Interest Rate Targeting

- **Modern approach**: Central bank sets target interest rate i^*
- Then supplies whatever money (M) is needed to achieve that rate
- **Contrast**: Old approach was to set M and let i adjust

Why Interest Rate Targeting?

- Interest rates directly affect investment decisions
- More predictable and transparent
- Better communication tool

Open Market Operations (OMO)

Expansionary Monetary Policy (Lowering i)

1. **Action:** Fed buys bonds from private sector
2. **Payment:** Fed gives cash (money) in exchange
3. **Effect on Balance Sheet:**
 - **Assets:** More bonds
 - **Liabilities:** More currency in circulation (cash is a Fed liability)
4. **Market Effect:**
 - Money supply increases: $M^s \uparrow$
 - Interest rate decreases: $i \downarrow$

Contractionary Monetary Policy (Raising i)

1. **Action:** Fed sells bonds to private sector
2. **Payment:** Fed receives cash in exchange
3. **Effect:** Money supply decreases, interest rate increases

Bond Prices and Interest Rates

The Inverse Relationship For a one-year bond with face value F :

$$i = \frac{F - P_B}{P_B}$$

Where: - P_B : Current price of bond - F : Face value (payment at maturity)

Example

- Face value: $F = \$100$
- Current price: $P_B = \$95$
- Interest rate: $i = \frac{100-95}{95} = \frac{5}{95} \approx 5.26\%$

Intuition for OMO Effect

1. Fed buys bonds (expansionary policy)
2. High demand for bonds pushes $P_B \uparrow$
3. Since $i = \frac{F - P_B}{P_B}$, when $P_B \uparrow$, then $i \downarrow$

Extreme Case If $P_B = 100 = F$, then $i = 0\%$

4. The IS-LM Model

The IS-LM model integrates goods and financial markets to determine equilibrium output and interest rates.

Building the IS Curve: Goods Market Equilibrium

Making Investment Endogenous In the basic model, I was exogenous (fixed). Now:

$$I = I(Y, i)$$

Investment Function Properties

1. **Increasing in Y :** $\frac{\partial I}{\partial Y} > 0$
 - Higher output/income increases investment (accelerator effect)
 - Interpretation: Firms invest more when the economy is strong
2. **Decreasing in i :** $\frac{\partial I}{\partial i} < 0$
 - Higher interest rates increase cost of borrowing
 - Interpretation: More expensive to finance investment projects

The IS Curve: Derivation

Equilibrium Condition

$$Y = C + I(Y, i) + G$$

$$Y = c_0 + c_1(Y - T) + I(Y, i) + G$$

Key Insight

- For each interest rate i , there is an equilibrium level of output Y
- The IS curve traces out these (i, Y) combinations

Slope of IS Curve

- **Downward sloping** in (Y, i) space
- **Why?:**
 - Higher i reduces investment I
 - Lower investment reduces aggregate demand
 - Lower demand requires lower Y to maintain equilibrium

Constructing the IS Curve

1. Start with initial interest rate i_0
2. Find equilibrium output Y_0 (from Keynesian cross)
3. Plot point (Y_0, i_0) on IS curve
4. Increase interest rate to $i_1 > i_0$
5. Investment falls, aggregate demand shifts down
6. New equilibrium output $Y_1 < Y_0$

7. Plot point (Y_1, i_1) on IS curve
8. Connect points to trace IS curve

Shifts in the IS Curve

Rightward Shifts (Expansionary) For any given i , output is higher.

Causes: - $G \uparrow$ (increased government spending) - $T \downarrow$ (tax cuts) - $c_0 \uparrow$ (increased consumer confidence) - Autonomous investment increases

Leftward Shifts (Contractionary) For any given i , output is lower.

Causes: - $G \downarrow$ (decreased government spending) - $T \uparrow$ (tax increases) - $c_0 \downarrow$ (decreased consumer confidence)

Important Distinction

- **Movement along IS:** Change in i (caused by monetary policy)
- **Shift of IS:** Change in parameters other than i (e.g., G , T , c_0)

The LM Curve: Financial Market Equilibrium

In Modern Policy Framework The central bank sets the interest rate i directly.

LM Curve: Horizontal line at $i = i^{\text{policy}}$

Interpretation

- Fed chooses target interest rate
- Supplies whatever money is needed to achieve that rate
- Money supply is endogenous, interest rate is exogenous

Shifts in LM Curve

- **Downward shift:** Expansionary monetary policy ($i \downarrow$)
- **Upward shift:** Contractionary monetary policy ($i \uparrow$)

IS-LM Equilibrium

Finding Equilibrium

- **Intersection of IS and LM** determines equilibrium (Y^*, i^*)
- At this point:
 - Goods market in equilibrium: $Y = Z$
 - Financial market in equilibrium: $M^s = M^d$

Policy Analysis Using IS-LM

1. Expansionary Fiscal Policy Action: Increase G or decrease T

Effects: 1. IS curve shifts right 2. For given i , output increases 3. New equilibrium: Higher Y , same i (if Fed doesn't react)

Graphical Analysis: - IS shifts right from IS_0 to IS_1 - LM remains horizontal at i^* - Equilibrium moves from (Y_0, i^*) to (Y_1, i^*) - $Y_1 > Y_0$

Multiplier Still Applies:

$$\Delta Y = \frac{1}{1 - c_1 - d_1} \Delta G$$

where $d_1 = \frac{\partial I}{\partial Y}$ captures investment response to output

2. Contractionary Fiscal Policy Action: Decrease G or increase T

Effects: 1. IS curve shifts left 2. Equilibrium output decreases

Example: Fiscal austerity, deficit reduction

3. Balanced Budget Change Question: What if $\Delta G = \Delta T$ (same magnitude)?

Answer: IS still shifts!

Why?: - ΔG affects demand one-for-one: $\Delta Z = \Delta G$ - ΔT affects demand through consumption: $\Delta Z = -c_1 \Delta T$ - Net effect: $\Delta Z = \Delta G - c_1 \Delta T = (1 - c_1) \Delta G > 0$ (if both increase)

Direction: - If $\Delta G = \Delta T > 0$: IS shifts right (expansionary) - If $\Delta G = \Delta T < 0$: IS shifts left (contractionary)

Key Insight: Balanced budget expansion is still expansionary (though less than deficit-financed)

4. Expansionary Monetary Policy Action: Fed lowers interest rate ($i \downarrow$)

Effects: 1. LM shifts down 2. Lower i stimulates investment 3. Higher investment increases aggregate demand 4. Output increases through multiplier

Mechanism:

$$i \downarrow \Rightarrow I \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow \Rightarrow C \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow \text{ (multiplier)}$$

Graphical Analysis: - LM shifts down from $LM_0 (i_0)$ to $LM_1 (i_1)$ - IS remains unchanged - Equilibrium moves from (Y_0, i_0) to (Y_1, i_1) - $Y_1 > Y_0$ and $i_1 < i_0$

Implementation: Open market purchase of bonds

5. Contractionary Monetary Policy **Action:** Fed raises interest rate ($i \uparrow$)

Effects: 1. LM shifts up 2. Higher i reduces investment 3. Output decreases

Use: Combat inflation or overheating economy

Policy Mixes

Mix 1: Fighting a Recession **Combination:** Expansionary fiscal + Expansionary monetary

Actions: - Increase G and/or decrease T (IS shifts right) - Decrease i (LM shifts down)

Effects: - Very strong increase in Y - Both policies work in same direction

Historical Examples: - COVID-19 pandemic (2020): Massive fiscal stimulus + near-zero interest rates - Global Financial Crisis (2008-09): Fiscal stimulus + aggressive rate cuts

Mix 2: Fiscal Consolidation with Monetary Offset **Combination:** Contractionary fiscal + Expansionary monetary

Goal: Reduce budget deficit without causing recession

Actions: - Decrease G or increase T (IS shifts left) - Decrease i (LM shifts down) to offset

Effects: - Output can remain constant: $\Delta Y = 0$ - Improved fiscal position (lower deficit) - Lower interest rates

Example: “Austerity with monetary accommodation”

Mix 3: Both Policies Expansionary (Strong Stimulus) **Combination:** Both expansionary

When Used: Deep recessions, crisis situations

Effects: Maximum stimulus to output

Mix 4: Both Policies Contractionary **Combination:** Both contractionary

When Used: Fighting inflation, cooling overheated economy

Effects: Strong contraction in output

The Zero Lower Bound (ZLB)

The Problem

- Nominal interest rates cannot go significantly below zero
- If $i = 0$ and recession persists, monetary policy is ineffective

- Cannot lower i further to stimulate investment

When Does This Matter?

1. Deep recessions requiring very low rates
2. Low inflation or deflation environment
3. When optimal policy rate would be negative

Historical Examples

- Japan (1990s-2000s): Lost decade with $i \approx 0$
- United States (2008-2015): Fed funds rate at 0-0.25%
- Europe (2010s): Even negative rates in some countries

Implications

- **Fiscal policy becomes more important:** When monetary policy is constrained
- **Unconventional monetary policy:** Quantitative easing, forward guidance
- **Need for higher inflation targets:** Creates more room to cut rates

5. The Extended IS-LM Model: Risk and Inflation

Motivation for Extensions

Problem 1: Real vs. Nominal Interest Rates

- Firms care about **real cost of borrowing**, not nominal rate
- With inflation, real and nominal rates differ

Problem 2: Credit Spreads

- Firms don't borrow at risk-free rate (Treasury rate)
- They pay a premium over risk-free rate due to default risk

The Real Interest Rate

Fisher Equation

$$r = i - \pi^e$$

Where: - r : Real interest rate - i : Nominal interest rate (what we observe) - π^e : Expected inflation

Interpretation

- Real interest rate = nominal rate minus inflation
- If $i = 5\%$ and $\pi^e = 2\%$, then $r = 3\%$
- Real rate is the actual purchasing power cost of borrowing

The Credit Spread (Risk Premium)

Definition

$$x = i_{\text{corporate}} - i_{\text{Treasury}}$$

Where: - x : Credit spread (risk premium) - $i_{\text{corporate}}$: Interest rate on corporate bonds - i_{Treasury} : Risk-free rate (government bonds)

Why Credit Spreads Exist

1. **Default risk:** Corporations can go bankrupt
2. **Risk aversion:** Investors require compensation for risk
3. **Varies over business cycle:**
 - Low during booms (investor optimism)
 - High during recessions (investor pessimism, “flight to quality”)

Formal Representation Expected return on risky bond = Risk-free return:

$$(1 - \theta)(1 + i_{\text{corporate}}) = 1 + i_{\text{Treasury}}$$

Where θ is probability of default.

This implies:

$$x \approx \frac{\theta}{1 - \theta}$$

Extended Investment Function

New Specification

$$I = I(Y, r + x) = I(Y, i - \pi^e + x)$$

Interpretation

- Investment depends on **effective real borrowing cost**: $r + x$
- Firms face: Nominal rate i , minus expected inflation π^e , plus credit spread x

The Extended IS Curve

Equilibrium Condition

$$Y = C + I(Y, i - \pi^e + x) + G$$

New Parameters in IS Three interest rate components enter symmetrically:
1. $+i$ (contractionary) 2. $-\pi^e$ (expansionary) 3. $+x$ (contractionary)

Key Insight: Symmetric Effects The effects are symmetric: - $\Delta i = +100$ bp has same effect as $\Delta x = +100$ bp - $\Delta i = +100$ bp has same effect as $\Delta \pi^e = -100$ bp - All work through same channel: effective cost of capital for firms

Policy Analysis with Extensions

1. Increase in Credit Spreads ($\Delta x > 0$) **Cause:** Financial crisis, increased risk aversion, “flight to quality”

Effect on Investment: - Higher effective borrowing cost: $(i - \pi^e + x) \uparrow$ - Investment decreases: $I \downarrow$

Effect on IS Curve: - IS shifts **left** (contractionary) - For any given i , output is lower

Effect on Output: - If Fed doesn’t respond: $Y \downarrow$ (recession)

Example: 2008 Financial Crisis - Credit spreads spiked (e.g., from 1% to 5%) - Even with Fed cutting i to 0%, $(i - \pi^e + x)$ remained high - Deep recession resulted

2. Increase in Expected Inflation ($\Delta \pi^e > 0$) **Effect on Investment:** - Lower real interest rate: $(i - \pi^e + x) \downarrow$ - Investment increases: $I \uparrow$

Effect on IS Curve: - IS shifts **right** (expansionary) - For any given i , output is higher

Effect on Output: - If Fed doesn’t respond: $Y \uparrow$ (expansion)

3. Fed Response to Financial Conditions **Scenario:** Credit spreads increase by 100 bp

Fed Goal: Maintain current output level

Required Action: - Decrease i by exactly 100 bp - This keeps $(i - \pi^e + x)$ constant - IS curve doesn’t move, output unchanged

Formula:

$$\Delta i = -\Delta x$$

(to offset)

Current Example (2023): - Financial conditions loosened (spreads compressed, stocks up) - Fed responded by raising rates more than expected - Goal: Offset easier financial conditions to control inflation

4. Deflation and the ZLB Scenario: $i = 0$, but $\pi^e < 0$ (deflation) and x is high

Problem: - Real borrowing cost: $(0 - \pi^e + x) = (-\pi^e) + x$ - If $\pi^e = -2\%$ and $x = 3\%$, then effective rate = $2\% + 3\% = 5\%$ - Very high real borrowing cost despite zero nominal rate!

Result: - Investment collapses - Deep recession/depression - **Liquidity trap:** Monetary policy ineffective

Historical Example: Great Depression (1930s) - Deflation: $\pi^e < 0$ - High credit spreads due to bank failures - Nominal rates near zero but real rates very high - Output collapsed

Policy Response: - Need fiscal policy (monetary policy ineffective) - Unconventional monetary policy (if possible) - Prevent deflation expectations from forming

Summary: Extended IS-LM

Key Equation:

$$I = I(Y, \underbrace{i - \pi^e + x}_{\text{effective real cost}})$$

Three channels affect investment: 1. **Nominal interest rate i :** Controlled by Fed 2. **Expected inflation π^e :** Driven by expectations 3. **Credit spread x :** Driven by financial conditions

Policy Implications: - Fed must monitor all three components - Financial shocks (Δx) can require policy response - ZLB is more severe with deflation or high credit spreads

6. The Labor Market and Natural Rate of Unemployment

This section introduces medium-run analysis where prices can adjust. Will appear on Quiz 1 but less prominently than IS-LM.

Labor Market Framework

Key Variables

- W : Nominal wage
- P : Price level
- W/P : Real wage
- u : Unemployment rate
- L : Employment
- U : Unemployment

The Production Function

Specification

$$Y = N$$

Where N is employment (number of workers).

Interpretation

- Very simple: one worker produces one unit of output
- **Marginal product of labor:** $MPL = 1$
- **Marginal cost of production:** $MC = W$ (cost of hiring one more worker)

Price Setting by Firms

Markup Pricing Firms set price as a markup over marginal cost:

$$P = (1 + m) \times MC = (1 + m)W$$

Where: - m : Markup (e.g., $m = 0.2$ means 20% markup) - Reflects market power, competition level

Implied Real Wage (Price Setting) Rearranging:

$$\frac{W}{P} = \frac{1}{1 + m}$$

Interpretation: - Maximum real wage firms are willing to pay - Determined by markup - **Horizontal line** in $(W/P, u)$ space - Does not depend on unemployment

Wage Setting by Workers

Wage Setting Equation

$$W = P^e \cdot F(u, z)$$

Where: - P^e : Expected price level - u : Unemployment rate - z : Institutional/structural factors (bargaining power, unemployment benefits, etc.)

Properties of $F(u, z)$

1. **Decreasing in u :** $\frac{\partial F}{\partial u} < 0$
 - Higher unemployment weakens workers' bargaining position
 - Lower wage demands when unemployment is high
2. **Increasing in z :** $\frac{\partial F}{\partial z} > 0$
 - Higher z means more bargaining power
 - Examples: stronger unions, higher unemployment benefits

Real Wage (Wage Setting) In real terms:

$$\frac{W}{P} = \frac{P^e}{P} \cdot F(u, z)$$

If $P = P^e$ (expectations correct):

$$\frac{W}{P} = F(u, z)$$

Graph: Downward sloping in $(W/P, u)$ space

The Natural Rate of Unemployment

Definition The natural rate u_n is the unemployment rate when **actual price equals expected price**: $P = P^e$

Finding u_n : Equilibrium Condition Set wage-setting equal to price-setting:

$$F(u_n, z) = \frac{1}{1+m}$$

This equation determines u_n .

Graphical Representation

- **Price-setting line:** Horizontal at $W/P = \frac{1}{1+m}$
- **Wage-setting curve:** Downward sloping, $W/P = F(u, z)$
- **Intersection:** Natural rate u_n

Why “Natural”?

- **Misleading term:** Nothing inherently “natural” or desirable
- Better term: “equilibrium unemployment rate”
- Reflects structural features of economy (m, z)

Comparative Statics: Changes in Natural Rate

1. Increase in Markup ($\Delta m > 0$) **Cause:** Reduced competition, increased market power, oil shocks (raising costs)

Effect on Price Setting:

$$\frac{1}{1+m} \downarrow \text{ (price-setting line shifts down)}$$

New Equilibrium: - Lower real wage offered by firms - Workers unwilling to accept lower wage at u_n - Unemployment must rise to $u'_n > u_n$ - Higher unemployment forces workers to accept lower real wage

Result: $u_n \uparrow$ (natural rate increases)

2. Increase in Worker Bargaining Power ($\Delta z > 0$) **Causes:** - Stronger unions - Higher unemployment benefits - Better worker protections - Labor market regulations

Effect on Wage Setting: - For any given u , workers demand higher W/P - Wage-setting curve shifts **up**

New Equilibrium: - At u_n , workers now demand real wage higher than $\frac{1}{1+m}$ - Firms unwilling to pay higher wage - Unemployment must rise to $u'_n > u_n$ - Higher unemployment moderates wage demands back to $\frac{1}{1+m}$

Result: $u_n \uparrow$ (natural rate increases)

Interpretation: - Benefits to workers ($\uparrow z$) are offset by higher equilibrium unemployment - Policy trade-off: worker protection vs. employment level

Cross-Country Example

- **Europe:** Higher labor protections (z higher) $\rightarrow$ typically higher u_n
- **United States:** Lower labor protections (z lower) $\rightarrow$ typically lower u_n
- Trade-off between job security/benefits and unemployment rate

Key Takeaways: Labor Market

1. **Natural rate is endogenous:** Depends on structural parameters (m, z)
 2. **Not a policy target per se:** But policies affect it through m and z
 3. **Short-run vs. medium-run:**
 - Short-run (IS-LM): Prices fixed, output demand-determined
 - Medium-run: Prices adjust, unemployment returns to u_n
 4. **Important for later:** Phillips curve, inflation dynamics (not on Quiz 1)
-

7. Summary and Quiz Preparation

What You Must Know Cold

1. GDP Measurement

- All three methods and why they're equivalent
- Nominal vs. real GDP calculations
- Base year concept

2. Goods Market

- Consumption function: $C = c_0 + c_1(Y - T)$
- Equilibrium condition: $Y = Z$
- **Solving for equilibrium output**
- Multiplier: $\frac{1}{1-c_1}$
- Graphical analysis (Keynesian cross)

- Paradox of thrift

3. Financial Markets

- Money demand: increasing in income, decreasing in i
- Equilibrium: $M^s = M^d$
- Open market operations
- Bond prices and interest rates (inverse relationship)

4. IS-LM Model (Most Important!)

- **Deriving IS curve** from goods market
- **IS slope:** downward in (Y, i) space
- **IS shifts:**
 - Right: $\uparrow G, \downarrow T, \uparrow c_0$
 - Left: $\downarrow G, \uparrow T, \downarrow c_0$
- **LM curve:** horizontal at policy rate i
- **Finding equilibrium:** IS-LM intersection
- **Policy experiments:**
 - Fiscal policy (shifts IS)
 - Monetary policy (shifts LM)
 - Policy mixes
 - Balanced budget changes
- **Graphical analysis:** Must be able to draw and shift curves correctly

5. Extended IS-LM

- Investment function: $I = I(Y, i - \pi^e + x)$
- Symmetric effects of i , π^e , and x
- Credit spread shocks
- Inflation expectation shocks
- Fed offsetting financial conditions
- Zero lower bound problems

6. Labor Market (Less Emphasis)

- Wage setting: $W/P = F(u, z)$, decreasing in u
- Price setting: $W/P = \frac{1}{1+m}$, horizontal
- Natural rate: $F(u_n, z) = \frac{1}{1+m}$
- Effects of Δm and Δz on u_n

Problem-Solving Skills

1. **Algebraic:**
 - Solve for equilibrium output given parameters
 - Calculate multiplier effects
 - Find required policy changes to achieve target output

2. **Graphical:**

- Draw and label IS-LM diagrams correctly
- Show shifts vs. movements along curves
- Identify new equilibrium after shocks

3. **Conceptual:**

- Explain mechanisms (e.g., how monetary policy affects output)
- Compare policy effectiveness
- Understand trade-offs and limitations (e.g., ZLB)

Common Mistakes to Avoid

1. **Confusing movements along vs. shifts:**

- Change in $i \rightarrow$ movement along IS
- Change in $G \rightarrow$ shift of IS

2. **Wrong multiplier calculation:**

- Remember $\frac{1}{1-c_1}$, not $\frac{1}{c_1}$

3. **Sign errors in balanced budget:**

- $\Delta G = \Delta T > 0$ is still expansionary
- Net effect: $(1 - c_1)\Delta G$

4. **Forgetting the extended IS components:**

- All three (i, π^e, x) affect investment
- They enter symmetrically

5. **Labor market:**

- Both $\Delta m \uparrow$ and $\Delta z \uparrow$ increase u_n
- Don't confuse which curve shifts

Approximate Quiz Breakdown

- **IS-LM Analysis:** ~60-70%
 - Basic IS-LM
 - Policy experiments
 - Extended IS-LM with risk and inflation
- **Goods Market / Financial Markets:** ~15-20%
- **Labor Market:** ~10-15%
- **Definitions / Measurement:** ~5-10%

Final Advice

“If you understood what I said today, you’re in good shape.” - Professor Caballero

- Practice solving problems algebraically and graphically
- Understand mechanisms, not just formulas
- Know how to offset one policy with another
- Be prepared for balanced budget scenarios
- Understand the extended IS-LM thoroughly (risk, inflation, ZLB)

Lecture 11: The IS-LM-PC Model

Overview

This lecture introduces the IS-LM-PC model, which integrates the IS-LM framework (short run) with the Phillips Curve (medium run). This unified model allows us to analyze not only the immediate impact of policies and shocks but also their dynamics over time, distinguishing between short-run and medium-run effects. It is the central workhorse model for understanding business cycle fluctuations and monetary policy.

Current Events: The Collapse of Silicon Valley Bank (SVB)

Background

On March 10, 2023, Silicon Valley Bank (SVB), the 16th largest bank in the US by asset size (\$209 billion), was shut down by the FDIC following a classic bank run. This was the largest bank failure since Washington Mutual during the 2008 Global Financial Crisis.

Causes of the Collapse

1. Tech Sector Downturn

- SVB served primarily tech startups and venture-backed companies.
- As the tech sector struggled in 2022 (difficulty raising capital, falling valuations), companies began withdrawing deposits to fund operations.
- Net deposit outflows accelerated throughout 2022.

2. Investment Strategy and Interest Rate Risk

- SVB grew rapidly during 2020-2021, doubling its asset size.
- Rather than making risky loans, SVB invested heavily in long-term (10-year) Treasury bonds.
- **Problem:** These bonds were purchased when interest rates were near zero.
- When the Fed began aggressive rate hikes in 2022, bond prices fell sharply (inverse relationship between yields and prices).
- **Unrealized Losses:** SVB held approximately \$80 billion in securities with significant unrealized losses.

3. Accounting Rules and Liquidity Crisis

- Banks need not recognize losses on securities held to maturity (unless sold).
- As deposit withdrawals continued, SVB needed liquidity.
- Forced to sell securities, SVB realized \$1.8 billion in losses.

- CEO announced need to raise \$2.25 billion in new equity to cover the losses.

4. Bank Run Dynamics

- News of capital raising spread rapidly via social media and venture capital networks.
- **Coordination Failure:** Each depositor rational to withdraw, but collective action caused failure.
- **Uninsured Deposits:** Approximately 95% of SVB deposits exceeded FDIC insurance limit (\$250,000).
 - Most banks have majority of deposits insured (small savers).
 - SVB concentrated in large business accounts.
- Attempted withdrawals exceeded \$42 billion in single day (Thursday, March 9).

5. Venture Capital Amplification

- VC firms, seeing trouble, instructed all portfolio companies to withdraw immediately.
- Created self-fulfilling prophecy: fear of failure caused the failure.

Policy Response

Weekend Emergency Measures

1. **Full Deposit Guarantee:** FDIC invoked “systemic risk exception” to guarantee all deposits (not just under \$250k).
 - Protected small businesses using SVB for payroll.
 - Prevented contagion to other regional banks.
2. **Bank Term Funding Program (BTFP):**
 - Fed created new facility allowing banks to pledge securities at par value (not market value) for loans.
 - Removed need for fire sales of depreciated bonds.
 - Addressed the specific vulnerability that brought down SVB.

Macro Connections to Course Material

Risk Premium (x) and the Expanded IS-LM Model

- **Credit Spreads Spike:** Banking sector stress increased borrowing costs for private sector.
- From Lecture 7: Increase in x shifts IS curve left (contractionary).
 - Higher $x \Rightarrow$ Higher cost of borrowing for firms $\Rightarrow$ Lower investment $\Rightarrow$ Lower aggregate demand.
- **Fed Response:** When x rises exogenously, Fed can offset by lowering policy rate.
 - Maintains total borrowing cost ($i + x$) constant.

Market Expectations of Fed Policy

- Before SVB crisis: Markets expected 50 basis point hike at March 22 meeting.
 - Economic data coming in “hot” (strong labor market, inflation picking up again).
 - Expected value of next hike exceeded 40 basis points.
- After SVB crisis: Expected value collapsed to 15 basis points.
 - About 50% probability of 25bp hike.
 - About 50% probability of pause (0bp).
- **Reasoning:** Financial stability concerns dominate inflation concerns.
 - Something “broke” from aggressive tightening.
 - Fed may need to slow or pause hiking cycle.

Regulatory Failures and Lessons

1. **Dodd-Frank Rollback (2018):**
 - Originally, banks above \$50 billion subject to strict stress tests.
 - Threshold raised to \$250 billion (SVB lobbied for this).
 - SVB fell just below new threshold (\$209B), avoiding stringent oversight.
2. **Concentration Risk:**
 - Single sector exposure (tech) in deposits.
 - Single sector exposure (long-term Treasuries) in assets.
 - No diversification in funding sources.
3. **Interest Rate Risk Management:**
 - Failure to hedge duration risk during period of rising rates.
 - Regulator (San Francisco Fed) apparently missed this vulnerability.
4. **Speed of Modern Bank Runs:**
 - Digital banking + social media = runs happen in hours, not days.
 - Traditional regulatory response too slow.

The IS-LM-PC Model: Structure and Components

1. Motivation for the Model

What IS-LM Tells Us (Short Run)

- Determines equilibrium output (Y) given monetary and fiscal policy.
- Assumes prices fixed in the short run.
- Cannot address: What happens to inflation? How do things evolve over time?

What the Phillips Curve Adds (Medium Run)

- Links output to inflation dynamics.
- Shows how economy adjusts from short-run equilibrium to medium-run equilibrium.

- Allows analysis of disinflation, overheating, and stabilization policies.

2. The Three Components

IS Curve (Goods Market Equilibrium)

- **Equation:** $Y = C(Y - T) + I(Y, r) + G$
- **Key Change:** Now expressed in terms of **real interest rate** (r), not nominal rate (i).
- **Interpretation:** Higher real rate $\Rightarrow$ Lower investment $\Rightarrow$ Lower output.
- **Graphical Representation:** Downward sloping in (r, Y) space.

LM Curve (Monetary Policy Rule)

- **Modern Specification:** $r = \bar{r}$
- **Interpretation:** Central bank sets the real interest rate directly.
 - In practice, Fed sets nominal rate (i) and adjusts for expected inflation (π^e).
 - Assuming Fed can control r : $r = i - \pi^e$.
 - Most of the time (away from zero lower bound), Fed can achieve any desired r .
- **Graphical Representation:** Horizontal line in (r, Y) space.

Phillips Curve (Inflation Dynamics)

- **Original Form** (unemployment): $\pi_t - \pi_t^e = -\alpha(u_t - u_n)$
 - Inflation exceeds expectations when unemployment below natural rate.
- **Transformed to Output Form:**
 - Start with production function: $Y = N$ (output equals employment).
 - Employment: $N = L(1 - u)$ where L is labor force.
 - Potential output: $Y_n = L(1 - u_n)$
 - **Output gap:** $Y - Y_n = -L(u - u_n)$
 - Therefore: $u - u_n = -(Y - Y_n)/L$
- **Phillips Curve in Output Space:**

$$\pi_t - \pi_t^e = \lambda(Y_t - Y_n)$$

where $\lambda = \alpha/L$.

- **Interpretation:**
 - $Y > Y_n$ (positive output gap) $\Rightarrow \pi > \pi^e$ (inflation rises).
 - $Y < Y_n$ (negative output gap) $\Rightarrow \pi < \pi^e$ (inflation falls).
 - $Y = Y_n \Rightarrow \pi = \pi^e$ (inflation stable).
- **Graphical Representation:** Upward sloping in (Y, π) space.

- Why upward? Higher output $\Rightarrow$ Lower unemployment $\Rightarrow$ Wage pressure $\Rightarrow$ Price pressure $\Rightarrow$ Higher inflation.

3. The Natural Rate of Interest (r_n or r^*)

Definition The **natural rate of interest** (also called neutral rate, Wicksellian rate, or r^*) is the real interest rate that equates output with potential output:

$$Y(r_n) = Y_n$$

- Defined **implicitly** by solving the IS equation with $Y = Y_n$.
- r_n is the rate at which goods market clears at full employment.

Interpretation

- If Fed sets $r = r_n$: Output remains at potential, inflation stable.
- If Fed sets $r < r_n$: Output exceeds potential, inflation rises (economy overheats).
- If Fed sets $r > r_n$: Output falls below potential, inflation falls (economy cools).

Determinants of r_n r_n is determined by real factors (not monetary policy):

1. **Productivity growth**: Higher trend growth $\Rightarrow$ Higher r_n (more investment opportunities).
2. **Consumer preferences**: Higher desire to save $\Rightarrow$ Lower r_n (excess saving).
3. **Fiscal policy**: Expansionary fiscal policy $\Rightarrow$ Higher r_n (competes for resources).
4. **Demographics**: Aging population $\Rightarrow$ Higher saving $\Rightarrow$ Lower r_n .
5. **Investment opportunities**: Fewer profitable projects $\Rightarrow$ Lower r_n .

Why r_n Matters for Policy

- Fed must estimate r_n to determine appropriate policy stance.
- **Tight policy**: $r > r_n$ (contractionary).
- **Loose policy**: $r < r_n$ (expansionary).
- **Problem**: r_n is unobservable and time-varying.

The IS-LM-PC Model: Dynamics and Equilibrium

1. Short-Run Equilibrium

Determining Output

- **IS-LM intersection** determines output in the short run.
- Given Fed's choice of $\bar{r}$, output determined by IS curve: $Y = Y(\bar{r})$.
- **Nothing new here**: Same logic as standard IS-LM model.

Reading Inflation Pressure

- Once we know Y , Phillips Curve tells us inflation dynamics.
- Plot (Y, π) on Phillips Curve diagram.
- **Three cases:**
 1. $Y > Y_n$: $\pi > \pi^e$ (inflation rising).
 2. $Y = Y_n$: $\pi = \pi^e$ (inflation stable).
 3. $Y < Y_n$: $\pi < \pi^e$ (inflation falling).

Example: Economy Overheating

- Suppose Fed sets r too low: $\bar{r} < r_n$.
- IS-LM: Output determined at $Y_1 > Y_n$.
- Phillips Curve: $\pi_1 - \pi^e = \lambda(Y_1 - Y_n) > 0$.
- **Change in inflation:** $\Delta\pi = \pi_1 - \pi^e > 0$ (inflation accelerating).

2. Medium-Run Adjustment

The Problem with Persistent Gaps

- If $Y \neq Y_n$, inflation either rising or falling.
- **Central bank mandate:** Maintain price stability (typically 2% inflation target).
- Fed cannot tolerate persistent inflation changes.
- **Must act:** Adjust r to bring output back to potential.

Adjustment Mechanism

1. **Initial position:** $r < r_n$, so $Y > Y_n$ and inflation rising.
2. **Fed observes** inflation exceeding target.
3. **Fed responds:** Hikes real interest rate ($\bar{r} \uparrow$).
4. **IS-LM:** Higher $r \Rightarrow$ Lower Y (movement along IS curve).
5. **Phillips Curve:** As Y declines toward Y_n , rate of inflation increase slows.
6. **Process continues** until $r = r_n$ and $Y = Y_n$.
7. **Final outcome:** $\pi = \pi^e$ (inflation stable).

Graphical Representation

- **Top panel** (IS-LM space): Fed raises $\bar{r}$ from initial level up to r_n .
 - LM curve shifts up (higher horizontal line).
 - Output declines along IS curve from Y_1 to Y_n .
- **Bottom panel** (Phillips Curve space): Economy moves along PC curve.
 - As Y falls, $\pi - \pi^e$ declines.
 - At $Y = Y_n$, $\pi = \pi^e$ (vertical line at Y_n).

3. Short Run vs. Medium Run: Key Distinctions

Short Run

- **Time horizon:** Weeks to months.
- **Determined by:** IS-LM (aggregate demand).
- **Variables determined:** Output (Y), real interest rate (r).
- **Monetary policy:** Fully effective in influencing output.
- **Prices:** Treated as sticky (not adjusting).

Medium Run

- **Time horizon:** Quarters to a few years.
- **Determined by:** Supply-side factors (labor market, potential output).
- **Variables determined:** Potential output (Y_n), natural rate of unemployment (u_n), natural real rate (r_n).
- **Monetary policy:** Neutral (affects only nominal variables).
- **Inflation:** Adjusts to bring economy to potential.

The Role of Inflation Expectations

1. Two Models of Expectations

Adaptive Expectations: $\pi_t^e = \pi_{t-1}$

- Agents use last period's inflation to forecast future inflation.
- **Backward-looking:** Based on experience, not forward-looking analysis.
- **Implication:** If inflation rises, expectations rise next period.
- **Problem for central banks:** Creates inflation inertia.

Anchored Expectations: $\pi_t^e = \bar{\pi}$

- Agents believe central bank will hit its inflation target in medium run.
- **Forward-looking:** Based on credibility of monetary authority.
- **Implication:** Even if inflation temporarily deviates, expectations remain at target.
- **Benefit for central banks:** Easier to stabilize inflation.

2. Disinflation Under Different Expectations

Scenario: Economy with High Inflation

- Suppose $\pi = 9\%$ (as US experienced in 2022).
- Central bank wants to return to 2% target.
- How does expectation formation affect required policy?

Case 1: Anchored Expectations ($\pi^e = 2\%$) **Policy Task:** - Need to bring inflation from 9% down to 2%. - Phillips Curve: $\pi - \pi^e = \lambda(Y - Y_n)$ - Currently: $9\% - 2\% = \lambda(Y - Y_n)$, so $Y > Y_n$ (positive gap). - **Solution:** Raise r until $Y = Y_n$. - At $Y = Y_n$: $\pi - \pi^e = 0$, so $\pi = 2\%$ (back to target).

Outcome: - No recession required. - Only need to eliminate output gap (slow growth to trend, not negative growth). - If economy growing over time, just need below-trend growth for a period.

Example: - Initial: $r = -1\%$, $r_n = 1\%$, $Y_1 > Y_n$, $\pi = 9\%$. - Fed action: Raise r from -1% to 1% (200bp hike). - Result: Y falls to Y_n , π stabilizes at 2% . - No period of negative growth, just slower growth during adjustment.

Case 2: Adaptive Expectations ($\pi^e = \pi_{-1}$) Policy Task: - Initial: $\pi = 9\%$, so $\pi^e = 9\%$ next period. - Phillips Curve becomes: $\pi - 9\% = \lambda(Y - Y_n)$. - If Fed only raises r to r_n : $Y = Y_n \Rightarrow \pi = 9\%$ (inflation stays high!). - **Problem:** Inflation expectations now anchored at 9% , not 2% .

Solution (Requires Recession): - Need $\pi < \pi^e$ to bring expectations down. - To get $\pi = 2\%$ when $\pi^e = 9\%$: $2\% - 9\% = \lambda(Y - Y_n)$. - Requires: $Y < Y_n$ (negative output gap of -7% !). - **Must raise r well above r_n** to create recession.

Multi-Period Process: - Period 1: Raise r well above r_n , create $Y \ll Y_n$, π falls to (say) 6% . - Period 2: π^e now 6% , continue tight policy, π falls to 4% . - Period 3: π^e now 4% , continue tight policy, π falls to 3% . - Period 4: π^e now 3% , continue tight policy, π falls to 2% . - Period 5: π^e now 2% , can finally set $r = r_n$ and $Y = Y_n$.

Outcome: - Deep, prolonged recession required. - Multiple years of $Y < Y_n$ to “wring out” inflation expectations. - High unemployment, significant output loss.

Historical Example: Volcker Disinflation (1980-82)

- Inflation peaked at 14% in 1980.
- Fed Chair Paul Volcker raised policy rate above 15% .
- Created severe recession (unemployment reached 10.8%).
- Took 3-4 years to bring inflation back to acceptable levels.
- Necessary because expectations had become unanchored in 1970s.

3. Why Central Banks Care About Credibility

The Inflation Target as Anchor

- Modern central banks announce explicit inflation targets (typically 2%).
- Constant communication about commitment to target.
- **Goal:** Keep $\pi^e = \bar{\pi} = 2\%$ at all times.

Benefits of Anchored Expectations

1. **Easier stabilization:** Can respond to shocks without causing deep recessions.
2. **Lower sacrifice ratio:** Less output lost per percentage point of disinflation.

3. **Automatic stabilizer:** If inflation rises temporarily, expected inflation doesn't follow, so real rates rise automatically (fiscal-like stabilization).
4. **Policy credibility:** Markets trust central bank, reducing need for extreme actions.

Risks of Losing Credibility

- If expectations become unanchored: $\pi^e = \pi_{-1}$.
- Creates inflation inertia, requires recessions to disinflate.
- Difficult to rebuild credibility once lost (see 1970s US experience).

Current Situation (2022-2023)

- US inflation peaked at 9% (June 2022).
- Fed raised rates from 0% to 5.25% (March 2023).
- **Key question:** Are expectations still anchored at 2%?
- **Evidence:** Survey-based expectations remain near 2% for medium term.
- **Implication:** If credibility holds, can return to 2% inflation without deep recession (requires only $Y \approx Y_n$, not $Y \ll Y_n$).

The Neutrality of Money

1. Short Run vs. Medium Run

Short Run (IS-LM): Money is Not Neutral

- Monetary policy affects real variables: output (Y), employment (N), unemployment (u).
- Mechanism: Fed changes $r \Rightarrow$ Investment changes $\Rightarrow$ Output changes.
- **Real effects:** Fed's actions have real consequences for production and employment.

Medium Run (Natural Rate): Money is Neutral

- Monetary policy does not affect real variables: Y_n , u_n , r_n .
- These determined by real factors:
 - Labor force (L), price-setting behavior, wage-setting behavior (for u_n).
 - Productivity, preferences, fiscal policy (for r_n).
- **Nominal effects only:** Fed determines average inflation ($\bar{\pi}$) and nominal interest rate ($i = r_n + \bar{\pi}$).

2. What Central Banks Can and Cannot Do

What Central Banks CAN Do (Short Run)

- Stabilize output around potential during shocks.
- Smooth business cycles (reduce volatility of Y around Y_n).
- Example: Recession $\Rightarrow$ Lower $r \Rightarrow$ Boost Y back toward Y_n .

What Central Banks CANNOT Do (Medium Run)

- Permanently raise output above potential.
- Permanently lower unemployment below natural rate.
- Attempt to do so only causes accelerating inflation.
- **Trade-off:** In medium run, can only choose inflation rate, not output level.

Example: 1960s US Policy Mistake

- Policymakers believed they could maintain $u < u_n$ permanently.
- Kept monetary policy loose for extended period.
- Result: Rising inflation throughout late 1960s and 1970s.
- Eventually forced Volcker disinflation (painful recession).

3. Implications for Policy Design

Dual Mandate (Fed's Objectives)

1. **Price stability:** Keep inflation near 2% target.
2. **Maximum employment:** Keep output near potential ($Y \approx Y_n$).

Interpretation: - Short run: Trade-off between objectives (can temporarily boost Y above Y_n at cost of higher inflation). - Medium run: No trade-off (can only achieve price stability; employment determined by supply side).

Optimal Policy Strategy

1. Identify potential output (Y_n) and natural rate (r_n).
2. Set policy to keep output near potential: $r \approx r_n$.
3. Allow automatic adjustment: If shock pushes Y away from Y_n , adjust r to bring it back.
4. Maintain credibility: Keep π^e anchored so adjustments don't require recessions.

Current Macroeconomic Environment (2023)

1. The Post-COVID Inflation Surge

Origins (2020-2021)

- **Demand shock:** Massive fiscal stimulus (checks to households, expanded unemployment insurance).
- **Monetary accommodation:** Fed kept rates near zero through 2021.
- **Supply constraints:** COVID disruptions reduced potential output ($Y_n \downarrow$).
- **Result:** Large positive output gap ($Y \gg Y_n$), inflation rose from 2% to 9%.

Was Policy “Behind the Curve”?

- Fed kept $r < r_n$ for too long (through 2021).
- **Reasons:**
 1. Expected supply recovery: Thought Y_n would bounce back quickly (didn't happen).
 2. Transitory narrative: Believed inflation was temporary (some supply-side, some base effects).
 3. Risk management: Scarred by 2010s experience of too-low inflation, wanted to “run economy hot.”
- **Result:** By early 2022, clear that inflation was persistent and expectations at risk of becoming unanchored.

2. The Tightening Cycle (2022-2023)

Fed's Response

- March 2022: First rate hike (25bp).
- Followed by multiple 75bp and 50bp hikes (unusually aggressive).
- Peak rate: 5.25% (March 2023).
- Real rate went from approximately -4% (2021) to +2% (early 2023).

The Challenge: Finding r_n

- Is r_n still around 1% (pre-COVID estimate)?
- Or has r_n risen (due to fiscal stimulus, deglobalization, energy transition investment)?
- **If r_n higher:** Current policy not as tight as it seems.
- **If r_n unchanged:** Current policy quite restrictive.

3. The SVB Crisis and Policy Implications

New Considerations (March 2023)

- Banking stress adds risk premium (x) to borrowing costs.
- Equivalent to additional monetary tightening even if Fed doesn't hike.
- **Trade-off:**
 - Inflation still elevated (need higher r).
 - Financial stability concerns (argue for lower r or pause).

Market Expectations Shift

- Pre-SVB: Expected 50bp hike at March meeting.
- Post-SVB: Expected pause or 25bp hike at most.
- **Reasoning:** Increase in x substitutes for increase in r .
 - Total borrowing cost: $r + x$.
 - If x rises, Fed need not raise r as much.

Fed's Eventual Decision (March 22, 2023)

- Fed hiked 25bp (continued tightening, but at slower pace).
- Acknowledged financial stability concerns.
- Emphasized commitment to price stability.
- **Interpretation:** Balancing act between inflation and financial stability.

Summary and Key Takeaways

The IS-LM-PC Framework

- **Integrates** short-run demand analysis (IS-LM) with medium-run supply-side constraints (Phillips Curve).
- **Three curves:**
 1. **IS:** Goods market equilibrium (output depends negatively on real rate).
 2. **LM:** Monetary policy rule (Fed sets real rate).
 3. **PC:** Inflation dynamics (inflation rises when output exceeds potential).

How the Three Curves Interact

1. **IS-LM determines short-run output:** Given Fed's choice of r , IS curve determines Y .
2. **PC determines inflation pressure:** Given Y , Phillips Curve determines whether inflation rising or falling.
3. **Fed adjusts LM over time:** If inflation deviating from target, Fed changes r (shifts LM) to move output toward potential.
4. **Medium-run equilibrium:** Process continues until $r = r_n$, $Y = Y_n$, and $\pi = \pi^e$.

The Natural Rate of Interest (r_n)

- Defined as real rate that equates output with potential.
- Determined by real factors, not monetary policy.
- Central bank must estimate r_n to calibrate policy stance.
- Setting $r < r_n$ causes overheating; setting $r > r_n$ causes slack.

Inflation Expectations are Critical

- **Anchored expectations** ($\pi^e = \bar{\pi}$): Easier to stabilize inflation, no recession needed to disinflate.
- **Adaptive expectations** ($\pi^e = \pi_{-1}$): Creates inertia, requires recession to disinflate.
- Central bank credibility essential to maintaining anchor.

Neutrality of Money

- **Short run:** Monetary policy affects real variables (output, employment).
- **Medium run:** Monetary policy neutral, affects only nominal variables (inflation, nominal interest rate).
- Real variables determined by supply-side factors in medium run.

Policy Implications

- Central bank can stabilize output around potential in short run.
- Cannot permanently alter real variables (attempting to do so causes inflation).
- Must balance short-run stabilization with medium-run price stability.
- Maintaining credibility allows less costly adjustments.

Lecture 12: The IS-LM-PC Model - Dynamics and Policy Analysis

Overview

This lecture completes the IS-LM-PC framework by analyzing its dynamic properties and policy implications. We examine how the economy adjusts from short-run to medium-run equilibrium, the role of inflation expectations, and how policymakers respond to various economic shocks. Special emphasis is placed on the ongoing banking crisis and its macroeconomic implications.

1. The Complete IS-LM-PC Framework

The Three Building Blocks

IS Curve (Goods Market)

$$Y = C(Y - T) + I(Y, r) + G$$

- **Definition:** Combinations of output (Y) and real interest rate (r) that equilibrate the goods market.
- **Key Property:** Downward sloping in (Y, r) space.
 - Higher $r \Rightarrow$ Lower investment ($I \downarrow \Rightarrow$ Lower output ($Y \downarrow$)).
- **Shifts:**
 - Fiscal expansion ($G \uparrow$ or $T \downarrow$): IS shifts **right**.
 - Financial shock ($x \uparrow$): IS shifts **left** (credit tightening).
 - Confidence shock ($c_0 \downarrow$): IS shifts **left**.

Monetary Policy Rule (LM Simplification)

$$r = \bar{r}$$

- **Assumption:** Central bank controls the **real interest rate** directly (simplification).
- **Reality:** Central bank controls **nominal** interest rate (i), and $r = i - \pi^e$.
- **Why simplify?:** Cleaner diagrams; avoids shifting curves when inflation changes.
- **Key Point:** Central bank adjusts r based on inflation outcomes.

Phillips Curve (Labor Market + Inflation)

$$\pi_t = \pi_t^e + \lambda(Y_t - Y_n)$$

Output Gap Version: - Original PC: $\pi_t - \pi_t^e = \alpha(u_n - u_t)$ - Production function: $Y = N = L(1 - u)$ - Potential output: $Y_n = L(1 - u_n)$ - Output gap: $Y - Y_n = L(u_n - u)$

Result: $\pi_t - \pi_t^e = \lambda(Y_t - Y_n)$ where $\lambda = \alpha/L$

- **Key Property:** Upward sloping in $(Y, \pi - \pi^e)$ space.
 - If $Y > Y_n$ (positive output gap): Inflation **rises** ($\pi > \pi^e$).
 - If $Y < Y_n$ (negative output gap): Inflation **falls** ($\pi < \pi^e$).
 - If $Y = Y_n$ (zero output gap): Inflation **stable** ($\pi = \pi^e$).

The Output Gap Concept

Definition: Output Gap = $Y - Y_n$

- **Positive Gap** ($Y > Y_n$):
 - Unemployment **below** natural rate ($u < u_n$).
 - Economy “running hot” / “overheating”.
 - Labor market tight $\Rightarrow$ Wage pressure $\Rightarrow$ Price pressure $\Rightarrow$ Inflation rises.
- **Negative Gap** ($Y < Y_n$):
 - Unemployment **above** natural rate ($u > u_n$).
 - Economy “running cold” / in recession.
 - Labor market slack $\Rightarrow$ Wage pressure falls $\Rightarrow$ Disinflation / deflation.
- **Zero Gap** ($Y = Y_n$):
 - Unemployment at natural rate ($u = u_n$).
 - No pressure on inflation.
 - **Medium-run equilibrium.**

Potential Output (Y_n)

Definition: Output when unemployment equals natural rate.

$$Y_n = L(1 - u_n)$$

- **Interpretation:** “Sustainable” level of output without accelerating inflation.

- **Not Fixed:** Y_n changes with:
 - Labor force size (L).
 - Natural rate of unemployment (u_n).
 - Productivity (not in this simple model).

2. Short Run vs. Medium Run Equilibrium

Short Run (IS-LM Intersection)

Determination: Output determined by intersection of IS and LM (central bank's chosen r).

$$Y^{SR} = IS(r, G, T, x, \text{confidence}, \dots)$$

Key Feature: Y^{SR} need **not** equal Y_n . - Any level of output is a valid short-run equilibrium. - IS-LM determines output; no constraint that $Y = Y_n$.

Implications: - If $Y^{SR} > Y_n$: Inflation **accelerating** (Phillips curve). - If $Y^{SR} < Y_n$: Inflation **decelerating** (Phillips curve). - If $Y^{SR} = Y_n$: Inflation **stable** (lucky case).

Medium Run (Full Adjustment)

Determination: Output returns to potential output.

$$Y^{MR} = Y_n$$

Mechanism: Central bank adjusts interest rate in response to inflation. - If inflation rising ($Y > Y_n$): CB raises $r \Rightarrow$ IS-LM intersection moves left $\Rightarrow Y \downarrow$. - If inflation falling ($Y < Y_n$): CB lowers $r \Rightarrow$ IS-LM intersection moves right $\Rightarrow Y \uparrow$. - Process continues until $Y = Y_n$ and inflation stabilizes.

Medium-Run Equilibrium Conditions: 1. $Y = Y_n$ (output at potential). 2. $\pi = \pi^e$ (inflation stable). 3. $r = r_n$ (interest rate at natural rate).

The Natural Rate of Interest (r_n or r^*)

Definition: Real interest rate that delivers $Y = Y_n$ given current IS curve.

Implicit Definition: $Y_n = C(Y_n - T) + I(Y_n, r_n) + G$

Key Properties: - Not a constant; changes with: - Fiscal policy (G, T). - Confidence (c_0). - Credit conditions (x). - Foreign demand, demographics, productivity, etc. - Central bank cannot observe r_n directly; must infer from inflation. - If CB sets $r < r_n$: Economy overheats ($Y > Y_n$), inflation rises. - If CB sets $r > r_n$: Economy cools ($Y < Y_n$), inflation falls. - If CB sets $r = r_n$: Economy at potential ($Y = Y_n$), inflation stable.

3. Inflation Expectations and Anchoring

Two Models of Expectations

Model 1: Unanchored Expectations (Adaptive)

$$\pi_t^e = \pi_{t-1}$$

- Expectations = Last period's inflation.
- **Problem:** If inflation rises, expectations follow $\Rightarrow$ Persistence.
- **Implication:** Hard to reduce inflation without recession.

Model 2: Anchored Expectations (Credible Central Bank)

$$\pi_t^e = \pi^* = 2\%$$

- Expectations pinned to central bank target.
- **Benefit:** Inflation shocks don't become persistent.
- **Implication:** Can reduce inflation without deep recession (soft landing possible).

Why Anchoring Matters

Scenario: Economy overheats; inflation rises to 9%.

With Unanchored Expectations:

1. $\pi = 9\% \Rightarrow \pi^e = 9\%$ next period.
2. To reduce inflation, need $Y < Y_n$ (recession).
3. Inflation falls slowly: $9\% \rightarrow 8\% \rightarrow 7\% \rightarrow \dots \rightarrow 2\%$.
4. Long recession required ("hard landing").

With Anchored Expectations:

1. $\pi = 9\%$ but $\pi^e = 2\%$ (people trust Fed).
2. CB raises r to cool economy ($Y \downarrow$ toward Y_n).
3. As $Y \rightarrow Y_n$, inflation falls quickly to 2%.
4. Shorter, milder recession ("soft landing").

Central Bank Credibility: The most valuable asset a central bank has. - Built over decades of hitting inflation targets. - Lost quickly if central bank "falls behind the curve". - Explains why central banks react aggressively to inflation surprises.

4. The Deflationary Trap and the Zero Lower Bound

The Problem

Scenario: Deep recession with low/negative inflation.

- Economy needs **negative real interest rate** to boost demand: $r < 0$.
- Central bank controls **nominal rate**: $i = r + \pi^e$.
- If $i = 0$ (zero lower bound) and $\pi^e < 0$ (deflation), then $r = 0 - \pi^e > 0$.

Vicious Cycle: 1. Recession $\Rightarrow$ Deflation ($\pi < 0$). 2. $\pi^e \downarrow$ (expectations turn negative). 3. $r = i - \pi^e = 0 - \pi^e > 0$ (real rate **rises** even though nominal rate at zero). 4. Higher $r \Rightarrow$ Lower $Y \Rightarrow$ More deflation. 5. Spiral deepens.

Result: Monetary policy impotent; economy stuck in deflationary trap.

Historical Example: The Great Depression (1929-1933)

Year	Unemployment	Nominal Rate (i)	Inflation (π)	Real Rate (r)
1929	3%	5%	0%	5%
1930	9%	4%	-2.5%	6.5%
1931	16%	2%	-9%	11%
1932	24%	1%	-10%	11%

What Happened: - Fed lowered nominal rates from 5% to 1%. - But deflation accelerated (0% $\rightarrow$ -10%). - Real interest rate **rose** from 5% to 11%. - Fed was “pushing on a string” - monetary policy ineffective. - Unemployment reached 25%.

Lesson: Central banks must act **aggressively** to prevent deflation. - 2008 Global Financial Crisis: Fed learned lesson, cut rates to zero **immediately**. - Avoided Great Depression 2.0.

Modern Solutions to ZLB

1. **Quantitative Easing (QE):** Buy long-term bonds $\Rightarrow$ Lower long-term rates.
2. **Forward Guidance:** Promise to keep rates low for extended period $\Rightarrow$ Raise inflation expectations.
3. **Negative Interest Rates:** Europe, Japan experimented with $i < 0$ (limited success).
4. **Fiscal Policy:** If monetary policy impotent, fiscal expansion ($G \uparrow$) becomes critical.

5. Dynamic Adjustment to Aggregate Demand Shocks

Case Study: Fiscal Consolidation (Austerity)

Shock: Government cuts deficit ($G \downarrow$ or $T \uparrow$).

Short Run (Impact Effect) IS-LM Diagram: - IS shifts **left** (lower autonomous spending). - For given r , output falls to $Y_1 < Y_n$.

Phillips Curve Diagram: - Economy now at $Y_1 < Y_n$ (negative output gap). - Inflation **declines**: $\pi_1 < \pi^e$ (disinflation).

Immediate Effects: - Output $\downarrow$ (recession). - Unemployment $\uparrow$. - Inflation $\downarrow$ (or deflation if severe).

Medium Run (Adjustment Process) Central Bank Reaction: - Observes falling inflation. - Lowers interest rate: $r \downarrow$. - IS-LM intersection moves **right**: $Y \uparrow$ toward Y_n .

Mechanism: - Lower $r \Rightarrow$ Higher investment ($I \uparrow$). - Higher I offsets lower G (crowding in). - Output gradually returns to Y_n .

Final Equilibrium (Medium Run): - Output: $Y = Y_n$ (back to potential). - Interest rate: $r < r_{\text{initial}}$ (lower than before consolidation). - Composition: More investment (I), less government spending (G). - Inflation: Stable at target (or slightly below if expectations adjusted).

Speed of Adjustment: Key Disagreements Why Don't We Adjust Quickly?

1. **Monetary Policy Lags** ("Long and Variable Lags" - Milton Friedman):
 - Recognition lag: Takes time to realize economy in recession.
 - Decision lag: Central bank meets infrequently (6-8 times/year).
 - Implementation lag: Rate cuts take 6-18 months to affect economy.
2. **Zero Lower Bound:** If rates already low, CB cannot cut much.
3. **Uncertainty About Y_n :** Central bank doesn't know potential output; must infer from inflation.

Policy Debates: - **Advocates of Slow Consolidation:** "Adjustment is slow and painful; spread cuts over many years." - **Advocates of Fast Consolidation:** "Markets reward credibility; adjustment faster than models predict."

Historical Evidence: Mixed. Greece (2010s) suffered prolonged depression; UK (1990s) recovered relatively quickly.

6. Dynamic Adjustment to Supply Shocks

Case Study: Oil Price Shock

Shock: Oil price spikes $\Rightarrow$ Higher production costs $\Rightarrow$ Markup (m) increases.

Impact on Natural Rate of Unemployment Wage-Setting Relation: $W = P^e F(u, z)$ (unchanged).

Price-Setting Relation: $P = (1 + m)W$ (markup rises).

Real Wage Implied by Price-Setting: $\frac{W}{P} = \frac{1}{1+m}$ (falls when $m \uparrow$).

New Equilibrium: Requires higher unemployment to reduce wage demands.

$$u_n \uparrow \quad \text{when} \quad m \uparrow$$

Potential Output Falls: $Y_n = L(1 - u_n) \downarrow$ when $u_n \uparrow$.

Short Run (Impact Effect) Phillips Curve Diagram: - PC shifts **left** (lower Y_n). - If output unchanged ($Y = Y_{\text{old}}$), now $Y > Y_n^{\text{new}}$. - Inflation surges: $\pi \uparrow\uparrow$.

IS-LM Diagram: - No immediate shift (supply shock doesn't affect goods market directly). - Output unchanged initially.

Immediate Effects: - Output unchanged (or slight decline if CB reacts). - Inflation $\uparrow\uparrow$ (supply-side inflation). - Unemployment unchanged initially.

Central Bank Dilemma Two Objectives: 1. Stabilize inflation (requires raising r). 2. Stabilize output (requires keeping r unchanged or lowering).

Tradeoff: Cannot achieve both.

Options: - **Accommodate shock:** Keep r unchanged. - Pro: Avoids recession. - Con: Inflation spirals (if expectations unanchored). - **Fight inflation:** Raise r aggressively. - Pro: Keeps inflation near target. - Con: Causes recession ($Y \downarrow$ to new, lower Y_n).

Medium Run (Adjustment Process) If CB Fights Inflation: - Raises $r \Rightarrow$ IS-LM intersection moves **left**. - Output falls to new, lower Y_n . - Inflation stabilizes at target.

Result: Stagflation (stagnant output + high inflation) during transition.

If CB Accommodates: - Keeps r unchanged. - Inflation stays high: $\pi > \pi^e$. - If expectations unanchored: $\pi^e \uparrow \Rightarrow$ Wage demands $\uparrow \Rightarrow$ Further inflation. - Risk of inflation spiral.

Historical Example: 1970s Oil Shocks US Experience: - 1973-74: Oil prices quadrupled. - Inflation rose from 3% to 12%. - Fed accommodated (kept rates low) $\Rightarrow$ Inflation persisted. - 1979-80: Second oil shock. - Fed (Volcker) finally raised rates aggressively (20%). - Deep recession (1981-82) but inflation fell to 3%.

Lesson: Supply shocks are difficult; require accepting either high inflation or recession.

Case Study: COVID-19 Supply Disruptions (2020-2022)

Shocks: 1. Supply chains disrupted (China lockdowns, shipping). 2. Labor force participation fell (early retirements, long COVID). 3. Energy shock (2022 Russia-Ukraine war).

Central Bank Mistake (with hindsight): - Fed thought supply shocks “transitory”. - Kept rates near zero through 2021. - Inflation rose to 9% (June 2022).

Why the Mistake?: - Some disruptions (chips, shipping) did resolve quickly. - But labor force participation stayed low $\Rightarrow Y_n \downarrow$ persistently. - By the time Fed realized, inflation already high.

Current Situation (March 2023): - Fed hiked aggressively (0% to 5% in one year). - Inflation falling (9% $\rightarrow$ 6%). - But now banking crisis emerges $\Rightarrow$ New complications.

7. Long and Variable Lags in Monetary Policy

Milton Friedman’s Insight

Quote: “Monetary policy acts on the economy with long and variable lags.”

Implications: 1. **Long:** Takes 6-18 months for rate changes to affect output. 2. **Variable:** Lag length unpredictable; depends on financial conditions, expectations, etc.

Why Lags Matter

Scenario: Fed raises rates aggressively (as in 2022-2023).

Question: Has Fed done enough? Too much? Too little?

Problem: Effects not yet visible. - Inflation still high today (lagged response). - But economy may crash in 6 months (future response). - Fed “flying blind” - can’t see the full effects of past actions.

Analogy: Driving a car with a fogged windshield and delayed steering.

Non-Linearities: When Things Break

Pattern: - Rate hikes work slowly at first. - Economy seems resilient (unemployment low, spending strong). - Suddenly, “something breaks” (bank failure, credit crunch). - Economy collapses quickly.

Why?: - Financial system amplifies shocks. - Small cracks become large fissures. - Confidence evaporates suddenly.

Current Situation (March 2023): - Fed hiked 4.5% in one year. - Economy held up... until Silicon Valley Bank failed (March 10). - Now markets fear broader credit crunch.

8. Current Events: Banking Crisis and Policy Response

The Silicon Valley Bank Collapse (March 2023)

What Happened? Timeline: - **March 8:** SVB announces large losses on bond portfolio. - **March 9:** Depositors panic; bank run begins. - **March 10:** SVB fails; FDIC takes over (second-largest bank failure in US history). - **March 12:** Government announces blanket deposit insurance.

Why Did SVB Fail?: 1. SVB held long-term bonds (10-year Treasuries). 2. Rates rose sharply (0% to 5%). 3. Bond prices fell (duration risk). 4. Mark-to-market losses $\Rightarrow$ Capital eroded. 5. Depositors lost confidence $\Rightarrow$ Run $\Rightarrow$ Failure.

Not a 2008-Style Crisis (yet): - SVB held safe assets (Treasuries), not sub-prime mortgages. - Losses due to interest rate risk, not credit risk. - Systemic risk: Other banks face similar unrealized losses.

Credit Shock: The x Effect

Recall from Lecture 10: Investment depends on borrowing costs.

$$I = I(Y, r + x)$$

- r : Risk-free rate (controlled by Fed).
- x : Credit spread / risk premium.

Banking Crisis $\Rightarrow x \uparrow$: - Banks tighten lending standards. - Small businesses can't get loans. - Commercial real estate, mortgages affected.

Effect on IS Curve: Shifts **left** (for given r , less investment).

$$Y \downarrow \quad \text{for given } r$$

Implications for Monetary Policy

Before SVB Crisis (March 1-9):

- Inflation still 6% (target: 2%).
- Labor market strong (unemployment 3.6%).
- Markets expected 50 basis point hike on March 22.

After SVB Crisis (March 10-present):

- Credit crunch fear: $x \uparrow \Rightarrow$ IS shifts left.
- Deflationary pressure emerges.
- Markets now expect **no hike** (or even cut).

Fed's Dilemma: - **Problem 1:** Inflation still too high (6%). - **Problem 2:** Financial instability / credit crunch. - **Cannot solve both:** Raising r fights inflation but worsens credit crunch.

Market Expectations (Diagram from lecture): - **Before:** Peak Fed Funds rate 5.6% (by June 2023). - **After:** Peak Fed Funds rate 5.0% (by May 2023), then cuts to 4.1% by end of year.

Interpretation: Markets believe credit crunch will do Fed's job. - Higher x acts like higher $r \Rightarrow$ Demand cools $\Rightarrow$ Inflation falls. - Fed can pause hikes (or even cut to prevent recession).

Flight to Safety

Treasury Yields Collapsed: - 1-year Treasury: 5.1% (March 9) $\rightarrow$ 4.3% (March 15). - Drop of 80 basis points in one week.

Why?: - Investors flee risky assets (bank stocks, corporate bonds). - Buy safe assets (Treasuries, gold). - Expect Fed to cut rates (lowers future yields).

Inflation Expectations Fell: - 1-year breakeven inflation: 5.0% $\rightarrow$ 3.8%. - Markets expect recession $\Rightarrow$ Disinflation.

Credit Suisse Crisis (March 15)

Second Shoe Drops: - Credit Suisse (major European bank) faces run. - Stock price falls 35% in one day. - Credit default swaps (insurance against default) spike to crisis levels.

Contagion Fears: - If Credit Suisse fails, will other European banks follow? - Will US banks face renewed pressure?

Broader Market Reaction: - European bank stocks fall 10-15%. - US bank stocks fall 5-10%. - VIX (volatility index / "fear gauge") spikes.

How to Analyze Banking Crisis in IS-LM-PC Framework

Step 1: Identify the Shock - Credit shock: $x \uparrow$ (risk premium rises). - Confidence shock: $c_0 \downarrow$ (consumers scared).

Step 2: Short-Run Effect (IS-LM) - IS shifts left: $Y \downarrow$ for given r . - If Fed doesn't react: Output falls, unemployment rises.

Step 3: Inflation Response (Phillips Curve) - Lower output: $Y < Y_n \Rightarrow$ Negative output gap. - Inflation falls: $\pi \downarrow$ (disinflationary pressure).

Step 4: Central Bank Reaction (Medium Run) - Observes falling inflation (or falling inflation expectations). - **Cuts interest rate:** $r \downarrow$. - Two channels:
1. **Direct:** Lower r boosts investment. 2. **Indirect:** Lower r partially offsets higher x ($r + x$ is what matters).

Step 5: Medium-Run Equilibrium - Output returns to Y_n (assuming credit crunch temporary). - Interest rate lower than before: $r < r_{\text{initial}}$. - Credit spread elevated but stabilized: x high but not rising.

Silver Lining of High Inflation (Current Environment)

Why High Inflation Helps Central Bank Response:

Recall: Real rate = Nominal rate - Expected inflation.

$$r = i - \pi^e$$

If Starting from High Inflation ($\pi^e = 5\%$): - Can cut nominal rate to zero: $i = 0$. - Real rate becomes **negative**: $r = 0 - 5\% = -5\%$. - Large stimulus possible.

If Starting from Low Inflation ($\pi^e = 0\%$): - Can cut nominal rate to zero: $i = 0$. - Real rate stuck at **zero**: $r = 0 - 0\% = 0\%$. - Hit zero lower bound immediately.

Current Situation: - Inflation 6%, expectations 4-5%. - Fed has room to cut nominal rates from 5% to 0%. - Real rate could fall to -5% if needed. - **Much more policy space than 2008** (when inflation already low).

Paradox: High inflation is a **problem** (Fed needs to fight it), but also a **tool** (allows aggressive cuts if crisis worsens).

9. Key Takeaways

IS-LM-PC Framework Recap

Three Equations: 1. **IS:** $Y = C(Y - T) + I(Y, r + x) + G$ (goods market). 2. **LM:** $r = \bar{r}$ (central bank sets rate). 3. **PC:** $\pi = \pi^e + \lambda(Y - Y_n)$ (inflation dynamics).

Short Run vs. Medium Run

Short Run (IS-LM alone): - Output determined by IS-LM intersection. - Can be away from potential: $Y \neq Y_n$. - Creates inflation/deflation pressure.

Medium Run (IS-LM-PC together): - Central bank reacts to inflation. - Output returns to potential: $Y = Y_n$. - Interest rate adjusts to natural rate: $r = r_n$.

Aggregate Demand vs. Supply Shocks

Demand Shocks (fiscal policy, credit crunch): - **Short run:** Output and inflation move **together**. - Expansionary: $Y \uparrow, \pi \uparrow$. - Contractionary: $Y \downarrow, \pi \downarrow$. - **Medium run:** Output returns to Y_n ; interest rate adjusts.

Supply Shocks (oil prices, productivity): - **Short run**: Output and inflation move **opposite**. - Negative supply shock: $Y \downarrow, \pi \uparrow$ (stagflation). - **Medium run**: Y_n permanently changes; central bank must accept lower output or higher inflation.

Importance of Inflation Expectations

Anchored Expectations ($\pi^e = 2\%$): - Shocks don't become persistent. - Soft landings possible. - Central bank credibility critical.

Unanchored Expectations ($\pi^e = \pi_{-1}$): - Shocks amplified. - Hard landings required. - Deep recessions needed to reduce inflation.

Deflationary Trap

Zero Lower Bound: Most dangerous scenario. - Deflation $\Rightarrow$ Real rates rise $\Rightarrow$ Deeper recession $\Rightarrow$ More deflation. - Monetary policy impotent. - Requires aggressive fiscal policy.

Long and Variable Lags

Central Banking Is Hard: - Can't observe Y_n or r_n directly. - Long lags between action and outcome. - Non-linearities: "Something breaks" suddenly. - Must balance multiple objectives (inflation, employment, financial stability).

Current Crisis in Context

March 2023 Banking Crisis: - Credit shock ($x \uparrow$) shifts IS left. - Does some of Fed's work (cools demand). - But creates financial stability risk. - Fed must balance: Fight inflation vs. Prevent credit crunch. - Markets expect fewer rate hikes, earlier cuts.

Historical Parallel: 2008 Financial Crisis (but much less severe so far).

Key Difference: Fed has policy space (inflation high, can cut rates deeply if needed).

Next Steps: Wait and see if credit crunch spreads or stabilizes.

10. Summary Table: Shock Analysis in IS-LM-PC

Shock Type	IS Shift	PC Shift	Short-Run Effect	CB Reaction	Medium-Run Effect
Fiscal Expansion ($G \uparrow$)	Right	None	$Y \uparrow, \pi \uparrow$	Raise r	$Y = Y_n, r$ higher

Shock Type	IS Shift	PC Shift	Short-Run Effect	CB Reaction	Medium-Run Effect
Fiscal Contraction ($G \downarrow$)	Left	None	$Y \downarrow, \pi \downarrow$	Lower r	$Y = Y_n, r$ lower, I higher
Credit Crunch ($x \uparrow$)	Left	None	$Y \downarrow, \pi \downarrow$	Lower r	$Y = Y_n, r$ lower (if x temporary)
Oil Shock ($m \uparrow$)	None	Left ($Y_n \downarrow$)	$\pi \uparrow, Y$ stable initially	Raise r	$Y = Y_n^{\text{new}}$ (lower), π stable
Confidence Shock ($c_0 \downarrow$)	Left	None	$Y \downarrow, \pi \downarrow$	Lower r	$Y = Y_n, r$ lower (if temporary)

11. Looking Ahead

Next Topics (subject to change if crisis escalates): - **Long-run growth:** Solow model, technological progress. - **Expectations and asset prices:** How forward-looking behavior affects macro outcomes. - **Financial crises in depth:** Leverage, bank runs, systemic risk (if current crisis worsens).

For Now: Monitor banking situation closely. The next few weeks will determine whether this is: 1. **Contained crisis:** A few bank failures, policy response stabilizes system, economy continues (base case). 2. **Spreading crisis:** Credit crunch deepens, more bank failures, recession emerges (tail risk).

“Interesting times”: We are living through real-time application of IS-LM-PC model. Pay attention to data, market reactions, and policy responses.

Lecture 13: Introduction to Economic Growth

Overview

This lecture marks the transition from short-run/medium-run analysis (business cycles) to long-run economic growth. We examine why growth matters, how to measure it across countries and time periods, and introduce the fundamental framework for understanding the sources of growth.

1. Transition from Short Run to Long Run

Important Policy Caveat: Speed Can Kill

Before leaving the short/medium run completely, a critical lesson about macroeconomic policy implementation:

In Financial Crises: Require overwhelming and rapid response. - **Why?:** Things happen so fast that even healthy corporations cannot adjust. - Prices become non-informative, fire sales occur, decision-making breaks down. - **Policy Response:** Large and fast (“overwhelming force”).

In Normal Times (e.g., Interest Rate Hiking): Require gradualism. - **Why?:** For sufficiently large adjustments, something can break. - The danger: Banks are highly leveraged (small capital relative to assets), so small asset price changes can destroy capital.

Historical Episodes of Interest Rate Hikes Breaking Things

Year	Event	What Broke
1982	Volcker Hikes	Latin American Debt Crisis (“Lost Decade”) - Massive capital flows to emerging markets reversed - US banks distressed → emerging market crisis
1989-1990	Fed Hikes	Savings & Loan Crisis - Small regional banks couldn’t withstand sharp rate rise - Similar to 2023 SVB concerns
1990-1991	Fed Hikes	Japanese Bubble Burst - Real estate collapsed in Japan - Japan hasn’t recovered growth trajectory since
1994-1995	Fed Hikes	Mexican “Tequila” Crisis - Emerging market bond markets exploded - Capital flows to emerging markets tightened
2004-2006	Fed Hikes	Great Financial Crisis (2008) - House prices steadily rising during hikes - Financial assets built on housing wealth - Hikes stopped appreciation → reversal → crisis
2022-2023	Fed Hikes	Regional Banking Stress (SVB, etc.) - Already seeing tremors - Unknown what else might blow up

Key Insight: You don't know exactly what will blow up, but history shows something typically does. This is why the Fed practices gradualism when hiking.

2. Why Growth Matters: The Big Picture

IMF World Economic Outlook (Growth Forecasts)

Region	2022 (Actual)	2023 (Forecast)	2024 (Forecast)
World	3.4%	3.0%	3.1%
Advanced Economies	2.7%	1.3%	1.4%
Emerging Markets & Developing	3.9%	4.0%	4.2%

Key Observation: Regardless of year, emerging markets grow faster than advanced economies. - **Near-term forecasts** (2023): Dominated by cyclical factors (recessions, booms, IS-LM-PC dynamics). - **Further-out forecasts** (2024): Dominated by structural trends (growth models we'll study now).

Question to Answer: Why do emerging markets systematically grow faster than advanced economies?

Growth Dominates Business Cycles Over Time

US GDP (1890-2017): - Real GDP increased **50-fold** (in constant 2012 dollars). - Compare to business cycle fluctuations: 2-3% up and down. - **Lesson:** Over longer horizons, you can almost ignore business cycles—it's all about the trend.

The Great Depression: - The only visible disruption in the 130-year trend. - Even the Great Depression looks small relative to the long-run trend.

Population Growth vs. GDP Growth

US Population Growth (1890-2017): - From 63 million → 320 million (5-fold increase).

Two Different Measures: 1. **Total GDP:** Increased 50-fold. 2. **GDP per Capita:** Increased only 10-fold (not 50-fold).

Why This Matters: - Total GDP growth captures both population growth and productivity/living standards. - **GDP per Capita** measures welfare/well-being more accurately. - The “final pie” is 50x larger, but you have 5x more people to split it among.

The Declining Role of Population Growth

Historical: Steady, high population growth was normal. - This contributed significantly to GDP growth.

Current Trend: Many major regions have zero or negative population growth:
- Japan, South Korea, China - Most of continental Europe - Parts of Latin America

Implication: Going forward, GDP growth must come from productivity/technology, not population growth.

3. Measuring Growth: GDP per Capita and PPP

Within-Country Comparisons: Real GDP per Capita

For a single country over time (e.g., 40-70 years): - **Reasonable:** Use real GDP per capita (constant prices). - **Not Perfect:** For very long periods (300+ years), price adjustments become complex.

Cross-Country Comparisons: The PPP Problem

Problem: Exchange rate-based comparisons are misleading. - Example: US GDP per capita = \$70,000; Italy GDP per capita = \$50,000. - This comparison doesn't capture true differences in living standards.

Why?: Prices differ dramatically across countries. - Poorer countries have much cheaper goods (especially food and non-tradeables). - Using market exchange rates understates living standards in poor countries.

Solution: Purchasing Power Parity (PPP) Adjustment.

PPP Adjustment: The Logic

Goal: Compare real purchasing power, not nominal dollars.

Method: Value a common basket of goods at a common set of prices (e.g., US prices).

Example: US vs. Russia

Assumptions: - Both consume cars and food. - Cars and food bundles are identical quality in both countries.

US Household: - Buys: 1 car/year (\$10,000) + 1 food bundle/year (\$10,000)
- **Total:** \$20,000/year

Russian Household: - Buys: 0.07 cars/year (40,000 rubles) + 1 food bundle/year (80,000 rubles) - **Total:** 120,000 rubles/year

Exchange Rate: 60 rubles per dollar.

Without PPP Adjustment

- Convert Russian consumption: 120,000 rubles $\div$ 60 = **\$2,000/year**
- **Ratio:** US is 10x richer than Russia ($\$20,000 \div \$2,000 = 10$).

With PPP Adjustment

- Value Russian consumption at US prices:
 - 0.07 cars $\times$ \$10,000 = \$700
 - 1 food bundle $\times$ \$10,000 = \$10,000
 - **Total:** \$10,700/year
- **Ratio:** US is only 1.87x richer than Russia ($\$20,000 \div \$10,700 = 1.87$).
- Russia is 53% as rich as the US, not 10%.

Key Insight: The Russian household consumes fewer cars (changes less frequently—once every 15 years vs. annually), but the food bundle is the same. Valuing at common prices reveals true consumption differences.

Where to Find PPP-Adjusted Data

- **Penn World Tables:** Comprehensive PPP-adjusted national accounts for most countries.
 - Problem: Not updated frequently.
- **FRED:** Has PPP-adjusted data for some countries (using one of the Penn datasets).

4. Empirical Patterns in Growth

Pattern 1: Growth Rates and Initial Income (1950-2017)

Developed Economies:

	GDP per Country Capita (1950)	GDP per Capita (2017)	Average Annual Growth	Ratio (2017/1950)
Japan	\$6,800	\$37,200	4.1%	5.5x
France	\$7,000	\$39,300	2.6%	5.6x
UK	\$9,000	\$38,500	2.1%	4.3x
US	\$15,200	\$58,000	2.0%	3.8x

Key Pattern: Richer countries tend to grow slower. - The richest country (US in 1950) had the lowest growth rate. - The poorest country (Japan in 1950) had the highest growth rate.

Pattern 2: Convergence Among Developed Countries

Broader Evidence (25 developed countries, 1950-1987): - **Scatter Plot:** GDP per capita (1950) on x-axis, Average Annual Growth Rate on y-axis. - **Clear Pattern: Downward sloping relationship.** - Countries with lower GDP in 1950 grew faster. - Countries with higher GDP in 1950 grew slower.

Visual Evidence: - 1950: Large dispersion in GDP per capita across countries. - 2018: Much smaller dispersion. - **Interpretation:** Convergence—poorer countries catching up to richer countries.

Pattern 3: Convergence Within Regions, Not Globally

By Region: - **OECD (blue squares):** Clear negative relationship (convergence). - **Asia (green triangles):** Negative relationship, slightly noisier. - **Sub-Saharan Africa (red):** No convergence pattern at all.

Why Africa Doesn't Converge: - Political conflicts, wars, institutional failures. - Factors outside standard economic models. - **Scope of This Course:** We'll focus on OECD/Asia patterns (blue/green).

Historical Acceleration: The Modern Growth Era

Very Long-Run Trends: - **Before 1850:** Flat growth rates (Malthusian era). - **1850-1950:** Rapid acceleration in Western Europe and Western Hemisphere. - **Post-1950:** Global acceleration (except Sub-Saharan Africa).

The Malthusian Era (pre-1850): - Population grew when harvests were good. - Food constraints limited population growth. - Most people worked in agriculture; little room for growth. - **Takeaway:** If you took this course in year 1000 or during the Renaissance, you wouldn't study growth—it didn't exist.

Modern Era: Growth is a very modern phenomenon (last 150-200 years).

Pattern 4: Growth Makes a Huge Difference

Comparing Countries Over 1950-2016:

Example 1: Taiwan vs. Democratic Republic of Congo - 1950: Both had GDP per capita \$1,700. - **2016:** - **Taiwan:** \$50,000 (grew by 30x). - **DRC:** \$800 (declined by 0.5x). - **Today:** Taiwan is one of the richest economies; DRC is one of the poorest.

The Asian Tigers: - Taiwan, Singapore, Hong Kong, South Korea. - Grew very fast from 1960s onwards. - Today: Among the richest economies in the world.

Lesson: Small differences in growth rates, sustained over decades, create massive differences in living standards.

5. The Production Function Framework

From Short Run to Long Run: What Changes?

Timeframe	Focus	Production Concerns
Short Run	Aggregate Demand	Supply side ignored; demand determines output.
Medium Run	Wage Bargaining	Production function: $Y = N$ (labor only). Capital ignored (doesn't change much at business cycle frequency).
Long Run	Growth	Must model capital accumulation explicitly. Capital stock changes significantly over decades.

The Aggregate Production Function

$$Y = F(K, N)$$

Inputs: - Y : Output (GDP) - K : Capital stock (machinery, buildings, equipment, infrastructure) - N : Labor (workers; assume full employment, so N = population)

Properties:

1. Constant Returns to Scale (CRS) **Definition:** If you scale all inputs by x , output scales by x .

$$F(xK, xN) = xY$$

Interpretation: - If you increase capital and labor by 10%, output increases by 10%. - The production function is scalable. - **Example:** Doubling a factory (same machines, same workers) doubles output.

2. Decreasing Returns to Capital (Diminishing Marginal Product)

Definition: As you increase capital (K) for a fixed amount of labor (N), the extra output generated decreases.

$$\frac{\partial^2 Y}{\partial K^2} < 0$$

Intuition: - Each additional unit of capital works with fewer and fewer workers.
- Capital and labor are complementary; fixing one makes the other scarce.

Example: - Start with 100 workers, 100 machines $\rightarrow$ 100 units of output. - Add 10 machines (now 110) $\rightarrow$ +7 units of output (not +10). - Add 10 more machines (now 120) $\rightarrow$ +5 units of output (not +7). - Add 10 more machines (now 130) $\rightarrow$ +3 units of output (not +5).

Why?: With fixed labor, each new machine has fewer workers to operate it effectively.

3. Decreasing Returns to Labor Symmetric Logic: If you fix capital and increase only labor, each additional worker contributes less output.

6. The Per-Worker Production Function

Deriving Per-Worker Form

Using Constant Returns to Scale, set $x = \frac{1}{N}$:

$$F\left(\frac{K}{N}, \frac{N}{N}\right) = \frac{Y}{N}$$

$$F\left(\frac{K}{N}, 1\right) = \frac{Y}{N}$$

Define $k \equiv \frac{K}{N}$ (capital per worker) and $y \equiv \frac{Y}{N}$ (output per worker):

$$y = f(k)$$

Interpretation: Output per worker depends on capital per worker.

Properties of $f(k)$

1. **Positive Slope:** $f'(k) > 0$ (more capital per worker $\rightarrow$ more output per worker).
2. **Concave:** $f''(k) < 0$ (decreasing returns to capital per worker).

Graphical Representation

Diagram: Production Function - **X-axis:** Capital per worker ($k = K/N$) - **Y-axis:** Output per worker ($y = Y/N$) - **Curve:** $y = f(k)$ is upward sloping and concave.

Key Features: - **At low k :** Steep slope (high marginal product of capital). - Small increase in $k \rightarrow$ large increase in y . - Capital is scarce; adding capital has big impact. - **At high k :** Flat slope (low marginal product of capital). - Same

increase in $k \rightarrow$ small increase in y . - Capital is abundant; adding capital has small impact.

Why Concave?: Decreasing returns to capital.

7. Sources of Growth: Two Channels

Based on $y = f(k)$, there are only **two ways** to increase output per worker:

Channel 1: Capital Accumulation (Moving Along the Curve)

Mechanism: Increase $k = \frac{K}{N}$ (capital per worker).

Effect: y increases (move up along the production function).

Limitation: Decreasing returns to capital. - As k increases, the gain in y gets smaller and smaller. - **Cannot sustain growth indefinitely** through capital accumulation alone.

Example: - Poor country (low k): Investing in machines yields large output gains. - Rich country (high k): Investing in machines yields small output gains.

This explains convergence: Poor countries can grow fast by accumulating capital, while rich countries face diminishing returns.

Channel 2: Technological Progress (Shifting the Curve Up)

Mechanism: The function $f(\cdot)$ itself shifts upward.

Effect: For the same k , you now produce more y .

Interpretation: Better technology, better management, better institutions.

Key Difference: - **Capital accumulation:** Subject to diminishing returns (cannot sustain growth). - **Technological progress:** Not subject to diminishing returns (can sustain growth indefinitely).

Example: - Same number of machines and workers, but better software $\rightarrow$ higher output.

8. Preview of Next Lectures

Lecture 14: The Solow Model (Capital Accumulation)

Focus: Channel 1—Capital Accumulation.

Questions to Answer: 1. How does capital accumulate over time? 2. Why do poor countries grow faster than rich countries? 3. What determines the

steady-state level of capital per worker? 4. Can capital accumulation alone sustain long-run growth?

Key Insight: The Solow model explains **convergence** but **not sustained growth**.

Lecture 15: Technological Progress

Focus: Channel 2—Technological Progress.

Questions to Answer: 1. How do we model shifts in the production function? 2. What drives technological progress? 3. Can technological progress sustain growth indefinitely?

Key Insight: Technological progress is the engine of **sustained long-run growth**.

9. Summary

Key Takeaways

1. **Growth Dominates:** Over long periods, trend growth (10x-50x) dwarfs business cycle fluctuations (2-3%).
2. **Population Matters:** GDP growth ≠ welfare growth. Must use GDP per capita.
3. **PPP Adjustment:** Essential for cross-country comparisons. Poorer countries have cheaper goods; market exchange rates mislead.
4. **Empirical Pattern: Convergence** among developed/emerging countries (poorer grow faster than richer).
 - Explained by decreasing returns to capital.
5. **Growth is Modern:** Didn't exist before 1850 (Malthusian era). Acceleration post-1950.
6. **Growth Matters:** Small differences in growth rates → huge differences in living standards over decades.
 - Example: Taiwan (30x growth) vs. DRC (0.5x) from same starting point in 1950.
7. **Production Function:** $Y = F(K, N)$ with:
 - Constant returns to scale (scalability).
 - Decreasing returns to capital/labor (diminishing marginal products).
8. **Per-Worker Form:** $y = f(k)$ where $y = Y/N$, $k = K/N$.
 - Concave (decreasing returns).

9. Two Sources of Growth:

- **Capital Accumulation** ($k \uparrow$): Moving along $f(k)$ (explains convergence, limited by diminishing returns).
- **Technological Progress**: Shifting $f(k)$ upward (explains sustained growth).

Looking Ahead

- **Next Lecture**: Solow Model—Capital Accumulation and Convergence.
 - **Future Lecture**: Technological Progress and Sustained Growth.
-

Appendix: Key Formulas

PPP Adjustment

$$\text{Real Consumption (PPP)} = \sum (\text{Quantities consumed}) \times (\text{Common prices})$$

Production Function Properties

- **Constant Returns to Scale**: $F(xK, xN) = xY$
- **Decreasing Returns to Capital**: $\frac{\partial^2 Y}{\partial K^2} < 0$
- **Per-Worker Form**: $y = f(k)$ where $k = K/N$, $y = Y/N$

Concavity

$$f'(k) > 0, \quad f''(k) < 0$$

- $f'(k)$: Marginal product of capital (positive).
- $f''(k)$: Diminishing marginal product (negative).

Lecture 14: The Solow Growth Model

Overview

This lecture introduces the **Solow Growth Model**, the foundational framework for understanding long-run economic growth. Developed by Robert Solow (MIT Nobel Laureate, 1987), this model explains how capital accumulation drives growth and why countries converge to steady states. We analyze the dynamics of capital accumulation, the role of savings rates, and the implications for long-run growth.

Key Context (March 2023): The Fed hiked rates by 25bps as expected, but signaled potential pauses due to banking stress (SVB crisis).

1. The Central Mechanism of Growth Theory

The Circular Flow of Capital Accumulation

The Solow model revolves around a fundamental feedback loop:

Capital Stock (K) → Output (Y) → Savings (S) → Investment (I) → Capital Accumulation (ΔK) →

Key Insight: Unlike short-run macro (Keynesian multiplier), long-run growth is driven by **capital accumulation**.

Distinction from Short-Run Models

- **Short-Run (IS-LM):** Focus on aggregate demand, prices sticky, unemployment matters.
- **Long-Run (Growth):** Focus on supply side, capital accumulation, full employment assumed.
- **Time Scale:** Capital accumulates slowly (decades), unlike short-run fluctuations (quarters).

2. The Basic Solow Model (No Population Growth, No Technology)

2.1 Assumptions

1. **Constant Population:** $N = \bar{N}$ (fixed labor supply).
2. **Closed Economy:** $I = S$ (no international capital flows).
3. **No Government:** $G = T = 0$ (simplification).
4. **Constant Returns to Scale (CRS):** $F(\lambda K, \lambda N) = \lambda F(K, N)$.

2.2 Production Function

Aggregate Form:

$$Y = F(K, N)$$

Properties: - $\frac{\partial F}{\partial K} > 0$: More capital increases output (marginal product of capital). - $\frac{\partial^2 F}{\partial K^2} < 0$: **Diminishing marginal returns** to capital. - $\frac{\partial F}{\partial N} > 0$: More labor increases output.

Per Worker Form (dividing by N):

$$y = \frac{Y}{N} = f(k)$$

where: - $y \equiv \frac{Y}{N}$: Output per worker. - $k \equiv \frac{K}{N}$: Capital per worker. - $f(k) = F(k, 1)$: Production function in intensive form.

Key Property: $f'(k) > 0$ and $f''(k) < 0$ (increasing and concave).

Intuition for Concavity: With fixed labor, adding more capital yields diminishing returns. Each additional machine has less labor to work with, so output increases at a decreasing rate.

2.3 Savings and Investment

Saving Function (proportional to income):

$$S = sY$$

where $s \in (0, 1)$ is the **saving rate** (exogenous parameter).

Why Proportional? (vs. Short-Run Model with C_0): - **Short-Run:** Consumption has autonomous component ($C = C_0 + c_1 Y_d$) capturing wealth effects, asset prices, etc. - **Long-Run:** These factors (wealth, asset prices) scale with income over time, so proportional form is reasonable approximation.

Equilibrium (closed economy):

$$I = S = sY$$

In Per Worker Terms:

$$\frac{I}{N} = s \frac{Y}{N} = sf(k)$$

2.4 Capital Accumulation Equation

Aggregate Form:

$$K_{t+1} = K_t - \delta K_t + I_t$$

where $\delta \in (0, 1)$ is the **depreciation rate** (fraction of capital that breaks down each period).

Components: - K_t : Existing capital stock. - $-\delta K_t$: Capital lost to depreciation. - I_t : New investment (gross investment).

Per Worker Form (dividing by N):

$$k_{t+1} = k_t(1 - \delta) + sf(k_t)$$

Change in Capital per Worker:

$$\Delta k = k_{t+1} - k_t = sf(k_t) - \delta k_t$$

Fundamental Equation of the Solow Model:

$$\boxed{\Delta k = sf(k) - \delta k}$$

Interpretation: - $sf(k)$: **Investment per worker** (adds to capital stock).
 - δk : **Depreciation per worker** (subtracts from capital stock). - $\Delta k > 0$:
 Capital per worker is **growing** (investment exceeds depreciation). - $\Delta k < 0$:
 Capital per worker is **shrinking** (depreciation exceeds investment). - $\Delta k = 0$:
Steady state (capital per worker constant).

3. The Solow Diagram

3.1 The Basic Diagram

Axes: - Horizontal: k (capital per worker). - Vertical: Output, investment, depreciation (all per worker).

Three Curves: 1. **Production Function** (blue): $y = f(k)$ - **Concave** (diminishing returns). - Shows output per worker for each level of capital per worker.

2. **Investment per Worker** (green): $sf(k)$
 - **Scaled-down version** of production function (multiply by $s < 1$).
 - Shows savings/investment per worker.
3. **Depreciation per Worker** (red): δk
 - **Linear** line through origin (slope = δ).
 - Shows capital lost to depreciation.

Key Point: The vertical distance between green and red lines is Δk (change in capital per worker).

3.2 Steady State (k^*)

Definition: A steady state is a point where capital per worker is constant ($\Delta k = 0$).

Condition:

$$sf(k^*) = \delta k^*$$

Graphically: Steady state occurs where investment curve intersects depreciation line.

Properties at Steady State: - $\Delta k = 0$: Capital per worker is constant. - $y^* = f(k^*)$: Output per worker is constant. - **Growth rate of output per worker = 0.**

Important: Since N is constant, if y is constant, then total output $Y = yN$ is also constant. **No growth in steady state.**

3.3 Solving for Steady State (Example)

Cobb-Douglas Production Function:

$$F(K, N) = K^{1/2}N^{1/2}$$

Per Worker Form:

$$y = f(k) = k^{1/2}$$

Steady State Condition:

$$sk^{1/2} = \delta k$$

Solving for k^* :

$$k^{1/2} = \frac{\delta}{s}k$$

$$k^{-1/2} = \frac{\delta}{s}$$

$$\boxed{k^* = \left(\frac{s}{\delta}\right)^2}$$

Output per Worker at Steady State:

$$\boxed{y^* = f(k^*) = \frac{s}{\delta}}$$

Key Result: Doubling the saving rate ($s \uparrow 2s$) doubles steady-state output per worker (y^*) but quadruples steady-state capital per worker (k^*).

Why? Diminishing returns require much more capital to achieve double the output.

4. Dynamics and Convergence

4.1 Transitional Dynamics

Starting Below Steady State ($k_0 < k^*$): - At low k , investment $\$ > \$$ depreciation ($sf(k) > \delta k$). - $\Delta k > 0$: Capital per worker **grows**. - Economy moves **rightward** along horizontal axis. - Growth rate **positive** but **declining** over time (approaching steady state).

Starting Above Steady State ($k_0 > k^*$): - At high k , investment $\$ < \$$ depreciation ($sf(k) < \delta k$). - $\Delta k < 0$: Capital per worker **shrinks**. - Economy moves **leftward**. - Eventually converges to k^* .

Key Insight: Regardless of initial conditions, economy converges to k^* (stable steady state).

4.2 Transitional Growth

Definition: Transitional growth is growth that occurs as an economy moves from initial capital stock to steady state.

Characteristics: - Growth is **temporary** (not sustained). - Growth rate is **highest** when far from steady state. - Growth rate **declines** as economy

approaches steady state. - Eventually, growth rate returns to **zero** at steady state.

Explanation for Cross-Country Growth Patterns: - **Poorer countries** (low k) tend to grow **faster** (catching up). - **Rich countries** (near k^*) grow **slower** (already at steady state). - This explains the **downward-sloping** relationship between initial income and growth rates observed in OECD countries.

Notable Exception: Africa has not exhibited catch-up growth (institutional, conflict, and governance issues).

5. Comparative Statics: Policy Experiments

5.1 Increase in Savings Rate ($s \uparrow$)

Initial Condition: Economy at steady state k^* (no growth).

Shock: Savings rate increases ($s \uparrow s'$).

Which Curve Moves? - **Red line** (δk): **No change** (unrelated to savings). - **Blue line** ($f(k)$): **No change** (technology unchanged). - **Green line** ($sf(k)$): **Shifts up** (scaled by higher s).

Graphical Analysis: 1. Investment curve shifts up. 2. At old steady state, now $s'f(k^*) > \delta k^*$. 3. $\Delta k > 0$: Capital starts accumulating. 4. Economy moves **rightward** toward new steady state k^{**} .

Dynamics Over Time: - **Short Run** (immediately after shock): - Output y **cannot jump** (capital stock is fixed, slow to adjust). - Savings rate jumps, but investment takes time to accumulate capital.

- **Transition Period:**
 - Capital accumulates ($\Delta k > 0$).
 - Output per worker **rises** (y moves up along production function).
 - Growth rate **positive** but **declining** over time.
- **New Steady State:**
 - Capital per worker reaches $k^{**} > k^*$.
 - Output per worker reaches $y^{**} > y^*$.
 - Growth rate returns to **zero**.

Conclusion: Higher savings rate leads to: - **Higher level** of output per worker (permanently). - **Temporarily higher growth rate** (during transition). - **No effect on long-run growth rate** (returns to zero).

Historical Example: Asian Miracle (1960s-1980s): - Countries like South Korea, Taiwan, Singapore dramatically increased savings rates. - Experienced rapid growth (8-10% per year) for decades. - Much of this growth was **transitional** (moving from low k to higher k^*).

5.2 Consumption and the Golden Rule

Consumption per Worker:

$$c = y - sy = (1 - s)y = (1 - s)f(k)$$

At Steady State:

$$c^* = (1 - s)f(k^*)$$

Key Question: Does higher savings rate increase or decrease consumption?

Two Opposing Forces: 1. **Higher** s directly reduces consumption share: $(1 - s) \downarrow$. 2. **Higher** s indirectly increases output: $y^* \uparrow$ (via higher k^*).

Net Effect Depends on Initial Savings Rate:

Case 1: Low Initial s - Economy at low k , so $f'(k)$ is **high** (steep part of production function). - Small increase in s leads to large increase in k^* and y^* . - **Output effect dominates:** $c^* \uparrow$.

Case 2: High Initial s - Economy at high k , so $f'(k)$ is **low** (flat part of production function). - Increase in s leads to small increase in y^* . - **Savings effect dominates:** $c^* \downarrow$.

Golden Rule Savings Rate (s_{GR}): - The savings rate that **maximizes steady-state consumption** c^* . - **Condition:** $f'(k^*_{GR}) = \delta$ (marginal product of capital equals depreciation rate). - **Intuition:** Invest just enough to maintain capital stock; consume the rest.

Example (Cobb-Douglas):

$$f(k) = k^{1/2}$$

$$f'(k) = \frac{1}{2}k^{-1/2}$$

Setting $f'(k^*) = \delta$:

$$\frac{1}{2\sqrt{k^*}} = \delta$$

$$k^* = \frac{1}{4\delta^2}$$

Using $k^* = (s/\delta)^2$:

$$s_{GR} = 0.5$$

Optimal savings rate is 50% in this example.

Dynamic Inefficiency: If $s > s_{GR}$, economy is **over-saving** (consuming less than possible). Could increase consumption by reducing savings.

Historical Context: Asian Miracle Critique: - Countries like Singapore had savings rates 40-50%. - Critics argued: “Yes, output is growing, but consumption isn’t growing as fast. Are people actually better off?” - Counterargument: High savings necessary for catch-up; can reduce later.

5.3 Numerical Example

Assume: $f(k) = k^{1/2}$, $\delta = 0.1$ (10% depreciation).

Steady State Values:

s	$k^* = (s/\delta)^2$	$y^* = s/\delta$	$c^* = (1 - s) \cdot y^*$
0	0	0	0
0.1	1	1.0	0.9
0.2	4	2.0	1.6
0.3	9	3.0	2.1
0.4	16	4.0	2.4
0.5	25	5.0	2.5 (maximum)
0.6	36	6.0	2.4
0.7	49	7.0	2.1
1.0	100	10.0	0

Observations: - Consumption peaks at $s = 0.5$ (Golden Rule). - Beyond $s = 0.5$, higher savings **reduces** consumption (dynamic inefficiency). - At $s = 1$, output is maximized but consumption is zero (everyone saves everything).

5.4 Time Path After Savings Shock

Transition Dynamics (from $s = 0.1$ to $s = 0.2$):

Time 0 (shock occurs): - $k = 1$ (old steady state). - s jumps from 0.1 to 0.2.
- Output $y = 1$ (unchanged immediately). - Investment jumps to $0.2 \times 1 = 0.2$ (was 0.1 before).

Time 1-50 (transition): - Capital accumulates: k grows from 1 toward 4. - Output rises: y grows from 1 toward 2. - Growth rate is positive but declining.

Time 50+ (new steady state): - $k = 4$, $y = 2$, $c = 1.6$. - Growth rate returns to zero.

Key Timing Result: Takes approximately **50 years** to reach new steady state (slow adjustment due to capital accumulation).

6. China: A Case Study

Context: - **1980s:** Very low capital stock (k far below k^*). - **Policy Shift:** Dramatic increase in savings rate (40-50%) and opening to foreign investment.
- **Result:** 8-15% annual growth for 30+ years.

Solow Model Interpretation: - Large gap between $sf(k)$ and δk (investment far exceeds depreciation). - Rapid capital accumulation (transitional growth).

Current Challenge (2020s): - Growth rate slowing (now 4-6%, down from 10%+). - **Why?** Approaching steady state ($k \rightarrow k^*$). - Diminishing returns kicking in (each unit of new capital adds less output).

Policy Implications: - **Cannot sustain high growth** through capital accumulation alone. - Need alternative growth sources: **technology** (next lecture), human capital, productivity improvements.

Quote from Lecture: “This is sometimes called the easy part of growth. It’s sort of running out in China.”

7. Adding Population Growth

7.1 Motivation

Historical Context: - For centuries, output growth was driven by **population growth**, not per capita growth. - **Malthusian Era:** Population growth kept per capita income constant (food scarcity constraint). - **Modern Era:** Many developed countries have low or negative population growth (Europe, Japan, South Korea, China).

Model Modification: Relax assumption of constant N .

7.2 New Assumption

Population Growth:

$$N_{t+1} = N_t(1 + g_N)$$

where g_N is the **population growth rate** (can be positive or negative).

7.3 Capital Accumulation with Population Growth

Aggregate Form (unchanged):

$$K_{t+1} = K_t(1 - \delta) + I_t$$

Per Worker Form (now tricky):

$$\frac{K_{t+1}}{N_{t+1}} = \frac{K_t(1 - \delta)}{N_{t+1}} + \frac{I_t}{N_{t+1}}$$

Conversion (multiply and divide by N_t):

$$k_{t+1} = \frac{N_t}{N_{t+1}} k_t (1 - \delta) + \frac{N_t}{N_{t+1}} \frac{I_t}{N_t}$$

Using $N_{t+1} = N_t(1 + g_N)$:

$$k_{t+1} = \frac{k_t(1 - \delta)}{1 + g_N} + \frac{sf(k_t)}{1 + g_N}$$

Approximation (g_N small):

$$\frac{1}{1 + g_N} \approx 1 - g_N$$

Change in Capital per Worker:

$$\Delta k = k_{t+1} - k_t \approx sf(k_t) - \delta k_t - g_N k_t$$

Fundamental Equation with Population Growth:

$$\Delta k = sf(k) - (\delta + g_N)k$$

7.4 Interpretation

Economic Meaning:

$$\underbrace{sf(k)}_{\text{Investment per worker}} - \underbrace{\delta k}_{\text{Depreciation}} - \underbrace{g_N k}_{\text{Capital dilution}}$$

Three Components: 1. **Investment:** Adds to capital per worker. 2. **Depreciation:** Subtracts from capital stock (machines break down). 3. **Capital Dilution:** Subtracts from capital per worker (new workers need to be equipped).

Key Insight: To maintain constant k , must: - Replace depreciated capital (δk).
- Equip new workers with capital ($g_N k$).

Example: If $g_N = 0.02$ (2% population growth) and $k = 100$: - Need to invest 2 units just to maintain $k = 100$ (equip 2 new workers). - This is **in addition** to replacing depreciated capital.

7.5 Steady State with Population Growth

Steady State Condition:

$$sf(k^*) = (\delta + g_N)k^*$$

Modified Solow Diagram: - **Green line:** $sf(k)$ (unchanged). - **Red line:** $(\delta + g_N)k$ (steeper than before, rotates upward).

Effect of Higher Population Growth ($g_N \uparrow$): - Red line rotates **upward** (steeper). - New intersection at **lower** k^* and y^* . - **Intuition:** More investment needed to equip new workers, leaving less for deepening capital per worker.

7.6 Growth in Steady State with Population Growth

At Steady State: - $\Delta k = 0$: Capital per worker is **constant** ($k = k^*$). - Output per worker is **constant** ($y = y^*$).

But Total Output is Growing:

$$Y = y \times N$$

If y is constant and N grows at g_N , then:

$$\text{Growth rate of } Y = g_N$$

Key Result: In steady state with population growth: - **Output per worker growth = 0**. - **Total output growth = g_N** (population growth rate).

Historical Significance: For most of human history, economic growth was driven entirely by population growth, not productivity improvements.

7.7 Counterintuitive Result

Question: If population growth increases ($g_N \uparrow$), what happens to: 1. Output per worker (y)? 2. Total output (Y)? 3. Growth rate of total output?

Answers: 1. $y \downarrow$ (lower steady-state capital per worker). 2. Y grows **faster** (grows at higher g_N). 3. Growth rate of Y increases (from g_N to g'_N).

Intuition: - Higher population growth **lowers living standards** (y falls). - But it **raises total output** (Y grows faster). - **Trade-off:** More people vs. higher income per person.

Modern Context: - **Aging societies** (Japan, Europe): $g_N < 0$ (shrinking labor force). - Implies **negative output growth** (unless productivity improves). - Major policy concern: How to maintain living standards with shrinking population?

8. Limitations of the Solow Model

8.1 No Sustained Per Capita Growth

Key Limitation: In steady state, output per worker is **constant** (zero growth).

Reality: Advanced economies have experienced sustained per capita growth for 200+ years (1-2% per year).

What's Missing? Technological progress (next lecture).

8.2 Exogenous Savings Rate

Model Assumption: Savings rate s is exogenous (given).

Reality: Savings decisions are **endogenous** (households choose how much to save based on interest rates, income expectations, etc.).

Extension: Ramsey-Cass-Koopmans model (optimal savings, not covered in this course).

8.3 No Role for Institutions, Policies

Model Assumption: All countries have access to same technology ($f(k)$ is universal).

Reality: Institutions, governance, property rights, education systems differ dramatically across countries.

Why Doesn't Africa Catch Up? Solow model predicts convergence, but Africa hasn't converged due to: - Weak institutions. - Political instability. - Poor infrastructure. - Low human capital.

Extension: Endogenous growth theory (technology depends on investment in R&D, education, etc.).

9. Key Takeaways

9.1 Core Insights

1. **Capital Accumulation Drives Transitional Growth:** Economies grow as they accumulate capital toward steady state.
2. **Diminishing Returns Limit Growth:** Concave production function means capital accumulation cannot sustain growth indefinitely.
3. **Steady State Has Zero Per Capita Growth:** In long run, output per worker is constant (no growth without technology).
4. **Savings Rate Affects Level, Not Long-Run Growth:** Higher savings increases y^* but not long-run growth rate.
5. **Convergence:** Poor countries (low k) grow faster than rich countries (high k), leading to catch-up (conditional convergence).
6. **Population Growth:** Increases total output growth but lowers output per worker in steady state.

9.2 Policy Implications

To Increase Living Standards (y^*): - Increase savings rate s (but beware of over-saving beyond Golden Rule). - Reduce depreciation rate δ (better infrastructure, maintenance). - Slow population growth g_N (demographic transition).

But: These policies only increase the **level** of income, not the **growth rate** (in long run).

For Sustained Growth: Need technological progress (next lecture).

9.3 The Solow Diagram: Master This

Axes: k (horizontal), y (vertical).

Curves: - **Blue:** $f(k)$ (production function, concave). - **Green:** $sf(k)$ (investment, scaled-down production function). - **Red:** $(\delta + g_N)k$ (depreciation + dilution, linear).

Steady State: Where green intersects red.

Dynamics: - Left of steady state: $\Delta k > 0$ (capital accumulates). - Right of steady state: $\Delta k < 0$ (capital declines).

Comparative Statics: Practice shifting curves for changes in s , δ , g_N , and technology (next lecture).

10. Summary Table

Variable	Meaning	Effect on Steady State
$s \uparrow$	Higher savings rate	$k^* \uparrow, y^* \uparrow, c^*$ ambiguous
$\delta \uparrow$	Higher depreciation	$k^* \downarrow, y^* \downarrow, c^* \downarrow$
$g_N \uparrow$	Higher population growth	$k^* \downarrow, y^* \downarrow$, growth of $Y \uparrow$
$f(k)$ shifts up	Technological progress	$k^* \uparrow, y^* \uparrow, c^* \uparrow$ (next lecture)

11. Key Equations Reference

Production Function (Per Worker):

$$y = f(k)$$

Capital Accumulation (No Population Growth):

$$\Delta k = sf(k) - \delta k$$

Capital Accumulation (With Population Growth):

$$\Delta k = sf(k) - (\delta + g_N)k$$

Steady State Condition:

$$sf(k^*) = (\delta + g_N)k^*$$

Consumption (Per Worker):

$$c = (1 - s)f(k)$$

Golden Rule:

$$f'(k_{GR}^*) = \delta$$

Example (Cobb-Douglas: $f(k) = k^{1/2}$):

$$k^* = \left(\frac{s}{\delta + g_N} \right)^2$$

$$y^* = \frac{s}{\delta + g_N}$$

12. Practice Problems (Quiz Preparation)

Problem 1: Increase in Depreciation Rate

Question: Starting at steady state, depreciation rate increases ($\delta \uparrow$). Using Solow diagram, analyze effects on k^* , y^* , and transition dynamics.

Answer: - Red line rotates **upward** (steeper). - New steady state has **lower** k^* and y^* . - Transition: Capital declines ($\Delta k < 0$) until reaching new steady state.

Problem 2: Comparing Countries

Question: Two countries (A and B) have same technology and depreciation, but Country A has $s_A = 0.2$ and Country B has $s_B = 0.4$. Compare steady states and growth rates.

Answer: - Country B has **higher** $k_B^* > k_A^*$ and $y_B^* > y_A^*$ (double in Cobb-Douglas example). - In steady state, **both have zero per capita growth**. - During transition, Country B grows **faster** than Country A.

Problem 3: Golden Rule

Question: For $f(k) = k^{1/2}$ and $\delta = 0.1$, find the Golden Rule savings rate and maximum steady-state consumption.

Answer: - Golden Rule: $f'(k^*) = \delta \Rightarrow \frac{1}{2\sqrt{k^*}} = 0.1 \Rightarrow k^* = 25$. - Using $k^* = (s/\delta)^2 = 25 \Rightarrow s = 0.5$. - $y^* = 5$, $c^* = 0.5 \times 5 = 2.5$.

Next Lecture: We relax the assumption of constant technology and introduce **technological progress** (the Solow model with productivity growth). This will allow us to explain sustained long-run growth in output per worker.

Lecture 15: Technological Progress and Growth

Overview

This lecture introduces technological progress into the Solow growth model, showing how it enables sustained long-run growth in living standards. We extend the model with population growth to include labor-augmenting technological change and analyze its implications for economic growth across countries.

1. Review: The Solow Model with Population Growth

Basic Setup (from Lecture 14)

Starting point: Production function with constant returns to scale in capital (K) and labor (N).

$$Y = F(K, N)$$

Note: In growth models, we use labor force and population interchangeably. Over the long run, these aggregates move in tandem, making the distinction less important for growth analysis.

Capital Accumulation Equation

$$K_{t+1} = K_t - \delta K_t + I_t$$

Where: - $K_{t+1} - K_t$: Change in capital stock - δK_t : Depreciation of existing capital - I_t : Investment

Key Assumption (Closed Economy, No Deficit):

$$I_t = S_t = sY_t$$

Investment equals saving, and saving is proportional to income (saving rate s).

Normalization by Population

Production Function (Per Capita):

$$y = f(k) \quad \text{where } y = Y/N, \quad k = K/N$$

- y : Output per worker
- k : Capital per worker
- $f(k)$: Increasing in k at a decreasing rate (diminishing returns)

Dynamics with Population Growth

With Population Growing at Rate g_N :

Starting from capital accumulation equation, divide by N_{t+1} :

$$\frac{K_{t+1}}{N_{t+1}} = \frac{K_t}{N_{t+1}}(1 - \delta) + \frac{sY_t}{N_{t+1}}$$

Key Step: Multiply and divide by N_t to express in per-worker terms:

$$\frac{K_{t+1}}{N_{t+1}} = \frac{K_t}{N_t} \frac{N_t}{N_{t+1}}(1 - \delta) + \frac{Y_t}{N_t} \frac{N_t}{N_{t+1}} s$$

Approximation: For small g_N :

$$\frac{N_t}{N_{t+1}} = \frac{1}{1 + g_N} \approx 1 - g_N$$

Result (dropping second-order terms like $\delta \cdot g_N$):

$$k_{t+1} - k_t \approx sf(k_t) - (\delta + g_N)k_t$$

Interpretation: - $sf(k_t)$: **Investment per worker** (increases capital per worker) - δk_t : **Depreciation** (reduces capital per worker as machines break down) - $g_N k_t$: **Population dilution** (reduces capital per worker as population grows)

Why Does Population Growth Reduce Capital Per Worker?

Mechanical Effect: Suppose capital stock K doesn't change, but population grows. - Denominator of $k = K/N$ is growing $\Rightarrow k$ falls (becomes negative) - **Population dilution:** Need investment just to maintain the same k as population expands

Policy Implication: Fast population growth requires more investment to maintain the same capital-labor ratio.

Steady State Analysis (With Population Growth)

Steady State Condition: $k_{t+1} = k_t$ (capital per worker constant)

$$sf(k^*) = (\delta + g_N)k^*$$

Graphical Representation: - **Blue curve:** $f(k)$ (production function per worker) - **Green curve:** $sf(k)$ (saving/investment per worker) - **Red line:** $(\delta + g_N)k$ (break-even investment line)

Steady State: Where green curve intersects red line.

Impact of Higher Population Growth

Experiment: Increase g_N from initial steady state.

Effect: - **Red line rotates up** (steeper slope) - At old steady state, now: Investment < Break-even investment - **Capital per worker declines** to new (lower) steady state

Intuition: Higher population growth “dilutes” capital faster. With unchanged saving rate, economy settles at lower k^* and y^* .

Output Growth vs. Output Per Capita: - **Total output growth rate:** Increases to higher g_N (more people $\Rightarrow$ more output) - **Output per worker:** Lower level in new steady state (each worker has less capital) - **During transition:** Output grows, but less than population (so output per capita falls)

2. The Limits of Capital Accumulation

Why Capital Accumulation Cannot Drive Long-Run Growth

Problem: Due to diminishing returns to capital, economies converge to steady state where: - Capital per worker is **constant** - Output per worker is **constant** - No growth in living standards (output per capita)

In steady state (without technological progress): - Output grows only because population grows - Living standards (output per worker) stagnate

Empirical Reality: Most countries experience sustained growth in output per worker over time (U.S., Europe, Asia).

Conclusion: Need another source of growth beyond capital accumulation.

3. Technological Progress: The Engine of Sustained Growth

What is Technological Progress?

Total Factor Productivity (TFP): Captures improvements in technology over time.

Empirical Evidence: TFP in the U.S. has been growing steadily over decades (see TFP index chart from lecture).

Forms of Technological Progress

1. **More Output from Same Inputs:**
 - 10 machines + 10 workers used to produce 12 units
 - Now produce 14, 15, 16 units with same inputs
 - **Efficiency gains**
2. **Better Products:**
 - Not more cars, but better cars

- Better computers, better medical devices
 - **Quality improvements**
3. **New Products:**
- Goods that didn't exist before (smartphones, AI tools)
 - Satisfy needs that were previously unmet
 - **Product innovation**
4. **Greater Variety:**
- More types of products within categories
 - Better matches between consumer needs and available goods
 - **Welfare gains from variety**

Modeling Technological Progress

Production Function with Labor-Augmenting Technology:

$$Y = F(K, AN)$$

Where: - A : State of technology (TFP level) - AN : **Effective labor** (labor units in efficiency terms) - g_A : Growth rate of A (rate of technological progress)

Interpretation: Technology acts as if economy has more workers. - With same number of people (N), higher A means more “effective workers” - Each worker becomes a better input (more productive)

Why Model This Way? - Simple and tractable - Allows us to use same diagrams as before - Equivalent to other specifications (e.g., output-augmenting, capital-augmenting) under certain conditions

4. The Solow Model with Technological Progress

Normalization by Effective Labor

Key Insight: To find steady state, normalize by effective labor (AN) instead of population (N).

Production Function:

$$Y = F(K, AN)$$

By constant returns to scale, set $x = 1/(AN)$:

$$\frac{Y}{AN} = F\left(\frac{K}{AN}, 1\right)$$

Define: - $\tilde{y} = Y/(AN)$: Output per effective worker - $\tilde{k} = K/(AN)$: Capital per effective worker

Result:

$$\tilde{y} = f(\tilde{k})$$

Same functional form as before, but now in terms of effective labor.

Capital Accumulation Dynamics

Start with Basic Equation:

$$K_{t+1} = (1 - \delta)K_t + sY_t$$

Divide by $A_{t+1}N_{t+1}$:

$$\frac{K_{t+1}}{A_{t+1}N_{t+1}} = \frac{K_t}{A_{t+1}N_{t+1}}(1 - \delta) + \frac{sY_t}{A_{t+1}N_{t+1}}$$

Multiply and divide by A_tN_t :

$$\tilde{k}_{t+1} = \tilde{k}_t \frac{A_tN_t}{A_{t+1}N_{t+1}}(1 - \delta) + s\tilde{y}_t \frac{A_tN_t}{A_{t+1}N_{t+1}}$$

Approximation: For small g_A and g_N :

$$\frac{A_tN_t}{A_{t+1}N_{t+1}} = \frac{1}{(1 + g_A)(1 + g_N)} \approx 1 - g_A - g_N$$

(Dropping second-order terms like $g_A \cdot g_N$)

Final Dynamics Equation:

$$\tilde{k}_{t+1} - \tilde{k}_t \approx sf(\tilde{k}_t) - (\delta + g_N + g_A)\tilde{k}_t$$

Break-Even Investment (Three Components)

Capital per effective worker declines due to:

1. **Depreciation** ($\delta\tilde{k}$): Physical wear and tear of capital
2. **Population Growth** ($g_N\tilde{k}$): More people dilute capital per person
3. **Technological Progress** ($g_A\tilde{k}$): Higher A means more effective workers, diluting capital per effective worker

Intuition for g_A Term: - If A grows but K doesn't, then $K/(AN)$ falls
- Need investment to keep up with growing productivity per worker - Faster technological progress $\Rightarrow$ Need more investment to maintain $\tilde{k}$

Steady State with Technological Progress

Steady State Condition: $\tilde{k}_{t+1} = \tilde{k}_t$ (capital per effective worker constant)

$$sf(\tilde{k}^*) = (\delta + g_N + g_A)\tilde{k}^*$$

Graphical Representation: - **Blue curve:** $f(\tilde{k})$ (output per effective worker)
- **Green curve:** $sf(\tilde{k})$ (saving per effective worker) - **Red line:** $(\delta + g_N + g_A)\tilde{k}$ (break-even investment per effective worker)

Steady State: Where green curve = red line.

In Steady State: - Capital per effective worker ($\tilde{k}$): **Constant** - Output per effective worker ($\tilde{y}$): **Constant**

5. Growth Rates in Steady State

Deriving Steady State Growth Rates

Question: If $\tilde{k}$ and $\tilde{y}$ are constant, what is growing?

Output Per Worker ($y = Y/N$) **Relationship:**

$$y = \frac{Y}{N} = \frac{Y}{AN} \cdot A = \tilde{y} \cdot A$$

Growth Rate: - $\tilde{y}$ is constant (growth rate = 0) - A grows at rate g_A - **Therefore:** y grows at rate g_A

$$g_y = g_{\tilde{y}} + g_A = 0 + g_A = g_A$$

Key Result: Output per worker grows at rate of technological progress.

Capital Per Worker ($k = K/N$) **Relationship:**

$$k = \frac{K}{N} = \frac{K}{AN} \cdot A = \tilde{k} \cdot A$$

Growth Rate:

$$g_k = g_{\tilde{k}} + g_A = 0 + g_A = g_A$$

Key Result: Capital per worker also grows at rate g_A (capital intensity increases with productivity).

Total Output (Y) **Relationship:**

$$Y = \tilde{y} \cdot AN = \tilde{y} \cdot A \cdot N$$

Growth Rate:

$$g_Y = g_{\tilde{y}} + g_A + g_N = 0 + g_A + g_N = g_A + g_N$$

Key Result: Total output grows at rate of technological progress plus population growth.

Total Capital (K) Similarly:

$$g_K = g_A + g_N$$

Capital stock must grow at same rate as output to maintain constant $\tilde{k}$.

Summary Table: Steady State Growth Rates

Variable	Symbol	Growth Rate	Interpretation
Output per effective worker	$\tilde{y} = Y/(AN)$	0	Constant (steady state)
Capital per effective worker	$\tilde{k} = K/(AN)$	0	Constant (steady state)
Output per worker	$y = Y/N$	g_A	Living standards grow with technology
Capital per worker	$k = K/N$	g_A	Capital intensity grows with technology
Total output	Y	$g_A + g_N$	Technology + population
Total capital	K	$g_A + g_N$	Technology + population
Labor/Population	N	g_N	Exogenous
Technology	A	g_A	Exogenous

Critical Insight: g_A is the **only** source of sustained growth in living standards (output per worker).

6. Policy Experiments and Implications

Increase in Saving Rate ($s \uparrow$)

Effect on Diagram: - Green curve ($sf(\tilde{k})$) shifts **up** - At old steady state, now: Investment > Break-even investment - Economy moves to **higher** steady state $\tilde{k}^*$ and $\tilde{y}^*$

Transitional Dynamics: - Capital per effective worker grows (faster than $g_A + g_N$) - Output per effective worker grows (faster than zero) - **Transitional growth:** Output per worker grows faster than g_A temporarily

Long-Run Effect: - Higher **level** of output per worker - But **growth rate** returns to g_A (unchanged)

In log space (output per worker): - Old path: Growing at slope g_A - New path: Same slope g_A , but higher level (parallel shift up) - Transition period: Growth rate > g_A (converging to new path)

Conclusion: Saving rate affects **levels**, not long-run **growth rates**.

Increase in Technological Progress ($g_A \uparrow$)

Effect on Diagram: - Red line $((\delta + g_N + g_A)\tilde{k})$ rotates **up** (steeper) - At old steady state, now: Investment < Break-even investment - Capital per effective worker **declines** to new (lower) steady state

Why Does $\tilde{k}$ Fall? - Faster technological progress means effective labor grows faster - With unchanged saving rate, cannot keep up with dilution from rapid A growth - Economy settles at lower capital per effective worker

But This is Deceptive!

What Happens to Output Per Worker? - $y = \tilde{y} \cdot A$ - Even though $\tilde{y}$ falls slightly (lower $\tilde{k}^*$), A grows much faster - **Net effect:** y grows faster (new growth rate = higher g_A)

Transitional Dynamics: - During transition, growth rate of y is **slower** than new steady state rate - Why? Capital per effective worker is declining (drag on growth) - Once reach new steady state, growth rate jumps to new (higher) g_A

In log space (output per worker): - Old path: Slope g_A - New path: Steeper slope (new g_A) - Transition: Growth rate between old and new g_A (below new steady state rate)

Key Lesson: Increase in g_A permanently raises long-run growth rate (unlike increase in s).

7. The Asian Miracle: A Growth Decomposition

Fast Growth in Southeast Asia (1960s-1990s)

Observations: - Japan, South Korea, Taiwan, Singapore: Growth rates of 8-12% per year - Later: China (1990s-2010s) with similarly high growth rates -

Question: What drove this exceptional growth?

Sources of Fast Growth

1. Low Initial Capital Stock (Convergence): - Many Asian economies had low k after WWII (Japan) or colonialism (Korea, Taiwan) - Starting far below steady state $\Rightarrow$ Fast transitional growth - Diminishing returns less binding when k is low

2. Dramatic Increase in Saving Rate: - Japan: Saving rate rose from 15% to 35% - Korea/Taiwan: Similar increases - China: Saving rate reached 40-50% - This shifts green curve up $\Rightarrow$ Rapid capital accumulation

Combination: - Low initial k + high saving rate $\Rightarrow$ Very fast transitional growth - Growth rates of 10-12% sustainable for decades during catch-up phase

The Slowdown Problem

What Happens Now? - Capital per worker has caught up (approaching world frontier) - Saving rates cannot go much higher (already very high) - **Growth rates declining** (Japan, Korea, Taiwan now ~2-3%, China slowing to 5-6%)

Fighting the Solow Model: - Policymakers concerned about slowdown (China especially) - But model predicts: Transitional growth must end - Once reach steady state, growth rate = $g_A + g_N$

Challenge: - g_A : Must innovate (harder for catch-up countries) - g_N : Declining or negative in many Asian economies (aging populations)

Only Solution: Increase g_A (become technology leaders, not just adopters).

Why China is Obsessed with Technology: - To maintain growth at 5-6%, need higher g_A (especially as $g_N \rightarrow 0$ or negative) - Massive investment in R&D, AI, green technology - Understanding: Solow model says g_A is only source of sustained per capita growth

8. Cross-Country Growth Patterns

Convergence Hypothesis

Prediction: Countries with lower k (relative to steady state) should grow faster.
- Diminishing returns less binding when k is low - More room for capital accumulation

Conditional Convergence: - Countries converge to **their own** steady state (determined by s , g_N , g_A , institutions) - Not all countries converge to same level (different steady states)

Empirical Evidence: - **Unconditional convergence:** Weak (poor countries don't automatically catch up to rich countries) - **Conditional convergence:** Strong (controlling for saving rates, institutions, education, growth rates negatively related to initial income)

Why Some Countries Stay Poor

Model Suggests: - Low steady state $\tilde{k}^*$ due to: - Low saving rate (s) - High population growth (g_N) without productivity gains - Low technological progress (g_A): Poor institutions, low education, barriers to technology adoption - High effective depreciation (conflict, poor property rights)

Policy Implications: - Increasing s alone insufficient (levels effect, not growth effect) - Must address g_A : Education, R&D, openness to technology transfer, institutions

9. The Role of Technological Progress: Summary

Why g_A is Critical

- 1. Only Source of Sustained Per Capita Growth:** - Capital accumulation has diminishing returns - Only technological progress can drive long-run increases in living standards
- 2. Amplifies Effect of Other Factors:** - Higher $g_A \Rightarrow$ More incentive to invest (returns to capital higher) - Human capital and physical capital complementary with technology
- 3. Spillovers Across Countries:** - Technology diffuses internationally (trade, FDI, imitation) - World's g_A benefits all countries (though leaders benefit most)

The Importance of MIT and Innovation

What MIT Does (and similar institutions): - Generates new technology (increases A) - Trains human capital (increases effective labor) - Contributes to g_A for the U.S. and world

Long-Run Impact: - Innovation from universities, R&D labs, firms drives g_A - This is the **only sustainable way** to increase living standards indefinitely

Contrast with Other Policies: - Infrastructure spending (increases K): Levels effect, not growth effect - Education spending (increases human capital): Partly a levels effect, partly increases g_A (if generates innovation) - R&D subsidies: Directly increase g_A (permanent growth effect)

10. Extensions and Caveats

What Determines g_A ?

In this model: g_A is exogenous (taken as given).

In reality: g_A is determined by: - R&D investment (private and public) - Education and human capital - Institutions (property rights, rule of law, openness) - Market size (larger markets incentivize more innovation)

Endogenous Growth Theory: Models where g_A responds to economic decisions (beyond this course).

Other Forms of Technological Progress

Labor-Augmenting (used here): $Y = F(K, AN)$ - Technology acts like more workers

Capital-Augmenting: $Y = F(BK, N)$ - Technology makes capital more productive

Hicks-Neutral: $Y = AF(K, N)$ - Technology scales up output for given inputs

Under certain conditions (balanced growth), all specifications equivalent in steady state.

Human Capital

Extension: Can think of AN as: - A : Technology - N : Raw labor - AN : Effective labor (includes education, skills)

Education increases effective labor, similar to technological progress.

Resource Constraints

This model ignores: Natural resources, energy, environment.

Extensions: Include resources $\Rightarrow$ Growth may be constrained unless technology can substitute (e.g., green technology, efficiency gains).

11. Key Takeaways

1. **Capital accumulation alone cannot sustain growth** in output per worker due to diminishing returns.
2. **Technological progress (g_A) is the only source of sustained growth** in living standards.
3. **In steady state** with technology:
 - Output per worker grows at rate g_A
 - Total output grows at rate $g_A + g_N$
4. **Saving rate affects levels** (transitional growth), not long-run **growth rates**.
5. **Policy implications:**
 - To grow fast: Increase saving rate, start with low capital (transitional growth)
 - To sustain growth: Increase g_A through innovation, education, institutions
6. **Asian miracle:** Combination of low initial capital + high saving rates $\Rightarrow$ Fast transitional growth.
7. **Future growth** (China, developed world): Depends critically on increasing g_A as population growth slows or reverses.
8. **The role of innovation:** Universities, R&D, and technological innovation are fundamental to long-run prosperity.

Next Steps: Growth accounting (how to measure contributions of capital, labor, and technology to growth).

Lecture 16: Growth Accounting and Conditional Convergence

Overview

This lecture completes the growth theory section by examining what our models can and cannot explain about cross-country income differences. We introduce growth accounting (the Solow residual) to decompose growth into contributions from capital, labor, and technology. We then extend the model to include human capital and explore conditional convergence - why some countries remain poor while others catch up.

1. Balanced Growth Path Revisited

Complete Model Characteristics

In the Solow model with technological progress (A grows at g_A) and population growth (N grows at g_N), balanced growth occurs when all variables grow at constant rates.

Key Normalization: Effective Workers

- **Effective Workers:** $A \times N$ (productivity times population)
- **Purpose:** Normalize variables by effective workers to achieve steady-state diagram where curves don't shift

Balanced Growth Path Properties

Variable	Growth Rate	Explanation
Output per Effective Worker (Y/AN)	0	Constant (balanced growth definition)
Capital per Effective Worker (K/AN)	0	Constant (steady state in diagram)
Output per Worker (Y/N)	g_A	Only technology drives per-capita growth
Capital per Worker (K/N)	g_A	Capital deepening at rate of tech progress
Total Output (Y)	$g_A + g_N$	Both drivers contribute to absolute growth
Total Capital (K)	$g_A + g_N$	Grows with effective workers

Critical Insight: In the long run (balanced growth), only technological

progress (g_A) drives growth in living standards (output per person).

Example: Cobb-Douglas Production Function

Production function:

$$Y = K^{1-\alpha}(AN)^\alpha$$

Properties: - Constant returns to scale: $(1-\alpha) + \alpha = 1$ - Capital share: $1-\alpha$ (approximately 0.3) - Labor share: α (approximately 0.7)

Taking logs and derivatives:

$$\frac{\dot{Y}}{Y} = (1-\alpha)\frac{\dot{K}}{K} + \alpha\left(\frac{\dot{A}}{A} + \frac{\dot{N}}{N}\right)$$

Or:

$$g_Y = (1-\alpha)g_K + \alpha(g_A + g_N)$$

In **balanced growth**, $g_K = g_A + g_N$, so:

$$g_Y = (1-\alpha)(g_A + g_N) + \alpha(g_A + g_N) = g_A + g_N$$

For **output per worker**:

$$g_{Y/N} = g_Y - g_N = g_A$$

This confirms our table entries.

2. Growth Accounting: The Solow Residual

The Fundamental Problem

We can observe: - g_Y : Output growth (from national accounts) - g_K : Capital growth (investment minus depreciation) - g_N : Labor force growth (population statistics) - α : Labor share (wage bill / total revenue 0.7)

But we cannot directly observe: - g_A : Technological progress

Goal: Measure technological progress indirectly by accounting for all other contributions.

The Solow Decomposition

From the production function, we can decompose output growth:

$$g_Y = (1-\alpha)g_K + \alpha g_N + g_A$$

Interpretation: - $(1-\alpha)g_K$: Contribution of capital accumulation - αg_N : Contribution of labor force growth - g_A : **Solow Residual** - contribution of Total Factor Productivity (TFP)

Deriving the Formula

Step 1: Production function contribution logic - If capital increases by 1%, output increases by $(1-\alpha)\%$ - Why $(1-\alpha)$? That's capital's share of income (under competitive markets) - **Competitive market assumption:** Factor payments equal marginal products - Capital receives: rK (rental rate $\times$ capital stock) - Labor receives: wN (wage $\times$ employment) - Capital share: $\frac{rK}{Y} = 1 - \alpha$ - Labor share: $\frac{wN}{Y} = \alpha$

Step 2: Measuring each contribution

$$g_{Y,K} = (1 - \alpha)g_K \quad (\text{growth due to capital})$$

$$g_{Y,N} = \alpha g_N \quad (\text{growth due to labor})$$

Step 3: Solve for the residual

$$g_A = g_Y - (1 - \alpha)g_K - \alpha g_N$$

This is the **Solow residual** - the portion of growth not explained by measurable factor accumulation.

What Does the Solow Residual Capture?

The residual g_A captures everything not explained by K and N : 1. **Technological innovation:** New inventions, better machinery 2. **Organizational improvements:** Better management, supply chains 3. **Education and skills:** Human capital improvements 4. **Reallocation:** Resources moving to more productive sectors 5. **Measurement error:** Mismeasurement of inputs or outputs

This is why it's sometimes called "Total Factor Productivity" (TFP) growth or the "measure of our ignorance."

3. Application: China's Growth (1978-2017)

The Chinese Growth Miracle

Between 1978 (economic reforms) and 2017, China experienced extraordinary growth:

Variable	Annual Growth Rate
Output (g_Y)	7.2%
Capital (g_K)	9.2%
Labor (g_N)	1.7%
TFP (g_A)	4.2%

Using growth accounting (with $\alpha \approx 0.7$):

$$7.2\% = (0.3)(9.2\%) + (0.7)(1.7\%) + 4.2\%$$

$$7.2\% = 2.76\% + 1.19\% + 4.2\%$$

Is This Balanced Growth?

Question: Does China's growth pattern look like a steady state?

Answer: No! Here's why:

Clue 1: Capital growing faster than output - $g_K = 9.2\% > g_Y = 7.2\%$ - This means K/Y is rising - In balanced growth, K and Y should grow at same rate ($g_A + g_N$)

Clue 2: Check the balanced growth prediction - Balanced growth rate should be: $g_A + g_N = 4.2\% + 1.7\% = 5.9\%$ - But capital growing at 9.2%, not 5.9%!

Conclusion: China was experiencing **transitional growth**, not balanced growth.

Transitional Growth Dynamics

What does this mean? - China's capital stock was **below its steady state level** - Investment rate exceeded steady-state requirements - Economy was in the "catching up" phase - Growth was temporarily boosted above long-run rate

Why was capital stock low? Potential reasons: 1. Command economy era (pre-1978) had low saving/investment 2. Cultural Revolution disrupted capital accumulation 3. Economic reforms suddenly increased saving rate (45%+ of GDP) 4. Opening to foreign investment

The Solow diagram interpretation: - Initial k (capital per effective worker) was below k^* (steady state) - Gap between saving curve $sf(k)$ and depreciation line $(n + \delta + g_A)k$ was positive - This gap generated high investment and transitional growth

Implications for China's Future

Steady-state growth prediction: - Long-run growth = $g_A + g_N = 4.2\% + 1.7\% = 5.9\%$

But this is likely an overestimate: 1. **Population growth turning negative:** One-child policy aftermath - g_N is declining, turning negative by 2020s - This directly reduces g_Y in steady state 2. **Transitional growth ending:** As $k \rightarrow k^*$, growth boost from capital deepening disappears 3. **TFP growth may slow:** Harder to maintain 4.2% TFP growth as you approach technology frontier

Realistic forecast: - Long-run growth likely around 3-5% (depending on TFP trends) - Lower than 7.2% average of high-growth period - Creates challenges for debt sustainability, employment

Policy options to maintain growth: 1. **Increase saving rate further:** Generates more transitional growth - Costly: Requires lower consumption - Limited upside: Already saving 45%+ of GDP 2. **Boost TFP growth:** R&D investment, education, institutional reforms - More sustainable but difficult to engineer 3. **Immigration:** Offset declining g_N - Politically challenging

4. Human Capital Extension

Motivation

Education levels vary dramatically across countries: - US, Korea, Japan: ~13 years average schooling - Sub-Saharan Africa: ~5 years average schooling

Could this explain income differences?

The Model

Modified production function:

$$Y = K^{1-\alpha}(AH)^\alpha$$

Where: - $H = N \cdot h$ (human capital = population $\times$ human capital per worker)
 - $h = e^{\psi \times s}$ (human capital as function of schooling years) - s : Years of schooling
 - ψ : Return to schooling (approximately 0.1)

Interpretation: - Each additional year of schooling raises human capital by $\approx 10\%$ - 10 years of schooling: $h = e^{0.1 \times 10} = e^1 \approx 2.72$ - Effectively multiplies labor force by 2.72

Balanced Growth with Human Capital

Key insight: Human capital affects the **level** but not the long-run **growth rate** of income per capita.

Why? - In steady state, $g_h = 0$ (can't increase schooling forever) - Average schooling might rise from 5 to 13 years (transition) - But can't rise from 13 to 100 years (physical/time constraint)

Balanced growth properties (same as before): - Output per worker grows at g_A - Total output grows at $g_A + g_N$

Capital Accumulation with Human Capital

Normalization: Divide all variables by N (workers, not effective workers)

Output per worker:

$$y = \frac{Y}{N} = \left(\frac{K}{N} \right)^{1-\alpha} (Ah)^\alpha = k^{1-\alpha} (Ah)^\alpha$$

Capital accumulation equation:

$$k_{t+1} - k_t = sy_t - (\delta + n)k_t$$

Key difference from before: - Before: Normalized by AN , so had $(\delta + n + g_A)$ term - Now: Normalized by N only, so just $(\delta + n)$ term

Steady-state condition: In balanced growth, k (capital per worker) grows at rate g_A :

$$g_A = \frac{sy_t}{k_t} - (\delta + n)$$

Solving for Steady-State Capital

From the steady-state condition:

$$g_A + \delta + n = \frac{sy_t}{k_t}$$

Substitute production function:

$$g_A + \delta + n = s \frac{k^{1-\alpha}(Ah)^\alpha}{k} = sk^{-\alpha}(Ah)^\alpha$$

Solve for k :

$$k = \left(\frac{s}{g_A + \delta + n} \right)^{1/\alpha} Ah$$

Define capital per effective worker:

$$\tilde{k} = \frac{k}{Ah} = \left(\frac{s}{g_A + \delta + n} \right)^{1/\alpha}$$

This is exactly the same as the basic Solow model!

Steady-State Output per Worker

Output per worker:

$$y = k^{1-\alpha}(Ah)^\alpha = \left[\left(\frac{s}{g_A + \delta + n} \right)^{1/\alpha} Ah \right]^{1-\alpha} (Ah)^\alpha$$

Simplifying:

$$y = A \cdot h \cdot \left(\frac{s}{g_A + \delta + n} \right)^{(1-\alpha)/\alpha}$$

Key components: 1. **Technology:** A (level of productivity) 2. **Human capital:** h (education) 3. **Saving rate:** s (positive effect on level) 4. **Population growth:** n (negative effect on level)

5. Cross-Country Income Comparisons

Setting Up the Comparison

Goal: Explain why income per capita varies so much across countries.

Method: Compare country i to the United States.

Income ratio:

$$\hat{y}_i = \frac{y_i}{y_{US}} = \frac{A_i \cdot h_i \cdot \left(\frac{s_i}{g_A + \delta + n_i}\right)^{(1-\alpha)/\alpha}}{A_{US} \cdot h_{US} \cdot \left(\frac{s_{US}}{g_A + \delta + n_{US}}\right)^{(1-\alpha)/\alpha}}$$

Assumption 1: Same technology ($A_i = A_{US}$) - Solow's original assumption
- Justification: Technology can be imported/copied - Questionable for many developing countries

Under this assumption:

$$\hat{y}_i = \frac{h_i}{h_{US}} \cdot \left[\frac{s_i}{s_{US}} \cdot \frac{g_A + \delta + n_{US}}{g_A + \delta + n_i} \right]^{(1-\alpha)/\alpha}$$

First Experiment: Capital and Labor Only

Ignore human capital: Set $h_i = h_{US}$ for all countries.

What can explain income differences? 1. **Saving rates** (s_i): Higher saving $\Rightarrow$ higher income 2. **Population growth** (n_i): Higher $n \Rightarrow$ lower income 3. **Depreciation** (δ): Assume constant across countries

Result: The world would be much more equal!

Graphical evidence: - Plot predicted $\hat{y}_i$ vs. actual y_i/L_i - Points cluster much closer to equality than reality - Cannot explain the 30-fold differences we observe

Conclusion: Capital and labor accumulation alone explain a small fraction of cross-country income differences.

Second Experiment: Add Human Capital

Now include education: Measure h_i using schooling data.

What changes? - Countries with more education (Japan, Korea) get boost
- Countries with less education (Sub-Saharan Africa) get penalty - Dispersion increases somewhat

Result: Still far from explaining observed differences.

Key finding: Even with education, saving rates, and population growth, the model predicts a much flatter world than we observe.

The puzzle deepens: What's missing?

6. The Role of Technology (TFP)

Dropping the Equal Technology Assumption

New approach: Allow $A_i \neq A_{US}$ and measure it.

Method: Use growth accounting to compute TFP levels

$$A_i = \frac{y_i}{k_i^{1-\alpha} h_i^\alpha}$$

Measuring Relative TFP

For each country, compute:

$$\hat{A}_i = \frac{A_i}{A_{US}}$$

Data: - Singapore: $\hat{A} \approx 0.9$ (close to US) - Mexico: $\hat{A} \approx 0.6$ - Kenya: $\hat{A} \approx 0.3$

Technology and Income: The Key Relationship

Plot: Relative TFP ($\hat{A}_i$) vs. Relative output per worker ($\hat{y}_i$)

Finding: Very strong positive relationship!

Quantitative result: Between 50% and 67% of cross-country income differences can be attributed to TFP differences.

Interpretation: - Poor countries are poor primarily because they are unproductive - Not just because they lack capital or education - Something about the way they organize production is inefficient

What Does TFP Capture?

TFP differences reflect: 1. **Institutions:** Property rights, rule of law, corruption 2. **Barriers to technology adoption:** Lack of infrastructure, skills 3. **Misallocation:** Resources in wrong sectors/firms 4. **Market distortions:** Monopolies, trade barriers, financial repression 5. **Political instability:** Uncertainty deters investment 6. **Geography:** Disease burden, agricultural productivity

Critical insight: Growth theory (Solow model) tells us TFP is crucial, but doesn't explain why TFP differs. That requires institutional economics, political economy, etc.

7. Convergence Among Similar Countries

Conditional Convergence Concept

Question: Do poor countries grow faster than rich countries (absolute convergence)?

Answer: Not unconditionally. But if we compare countries with similar steady states, yes!

Conditional convergence: Countries converge to their own steady states, which may differ.

Evidence: Rich Country Convergence

Sample: Major industrial economies (US, UK, France, Germany, Japan, Canada, Australia)

Period: 1870-2010

Pattern: Clear convergence - Japan was far behind in 1870, grew very fast - US/UK were ahead in 1870, grew more slowly - By 2010, all countries have similar income per capita

Why does this work? - Similar institutions (democracy, rule of law, property rights) - Similar technology adoption capability - Similar education systems - Therefore, similar A and h (similar steady states)

Graph interpretation: - Plot log income per capita (1870) vs. average growth rate (1870-2010) - Strong negative relationship: poor countries in this group grew faster - This is exactly what the Solow model predicts for countries converging to the same steady state

Time Series: Visualizing Convergence

Diagram: Log income per capita over time - **1870:** Japan far below US/UK - **1900-1945:** Japan catching up, then war - **1945-1990:** Rapid Japanese catch-up (“Japanese miracle”) - **1990-2010:** Convergence complete, growth rates equalize

Other patterns: - Germany: Behind in 1870, caught up by 1900, war setback, catch-up again - France: Steady convergence throughout - US/UK: Always at frontier, moderate steady-state growth

Why Doesn’t Global Convergence Hold?

Problem: When we plot all countries (including poor ones), the negative relationship disappears or even reverses.

Example: Some African countries had low income in 1960 and low growth 1960-2010.

Why? - They have different (lower) steady states - Low A (poor institutions, technology barriers) - Converging to a low steady state, not to US/rich country levels

8. Conditional Convergence Framework

The Conditional Convergence Hypothesis

Statement: Countries grow faster when they are further below their own steady state.

Formalization: Define distance from steady state:

$$\text{Gap}_i = \frac{y_i}{y_i^*}$$

Where: - y_i : Current output per worker - y_i^* : Steady-state output per worker (computed using country's own A_i, h_i, s_i, n_i)

Prediction: Growth rate should be decreasing in Gap_i

$$g_i = \lambda \left(1 - \frac{y_i}{y_i^*} \right)$$

Where λ is the convergence speed (typically 2-3% per year).

Computing Country-Specific Steady States

Step 1: Measure country i 's fundamentals - A_i : TFP level (from growth accounting) - h_i : Human capital (from education data) - s_i : Saving/investment rate - n_i : Population growth rate

Step 2: Compute steady-state output per worker

$$y_i^* = A_i \cdot h_i \cdot \left(\frac{s_i}{g_A + \delta + n_i} \right)^{(1-\alpha)/\alpha}$$

Step 3: Compare actual to steady state

$$\frac{y_i(1970)}{y_i^*(1970)} = ?$$

- If < 1 : Country below steady state (should grow fast)
- If $= 1$: Country at steady state (should grow at g_A)
- If > 1 : Country above steady state (should grow slowly) - rare

Evidence: Cross-Country Regression

Regression specification:

$$g_i = a + b \log \left(\frac{y_i(1970)}{y_i^*(1970)} \right) + \varepsilon_i$$

Prediction: $b < 0$ (negative relationship)

Sample: 100+ countries, 1970-2010

Result: Strong negative relationship (as predicted) - Countries far below their steady state grew faster - Countries close to their steady state grew at steady-state rate - Coefficient b implies convergence speed $\lambda \approx 2\%$ per year

Interpretation: Conditional on country-specific characteristics (TFP, education, saving), the Solow model works very well!

Visualizing Conditional Convergence

Scatter plot: - **X-axis:** $\log(y_i/y_i^*)$ in 1970 (distance from steady state) - **Y-axis:** Average growth 1970-2010

Observations: - **Japan:** Far below steady state in 1970 (war recovery), grew very fast - **Botswana:** Below steady state, grew fast (but steady state was low) - **US:** Close to steady state, grew at moderate pace - **Some African countries:** Close to (low) steady state, grew slowly

The key insight: Poor countries with low steady states (low A , low h) don't catch up to rich countries. They catch up to their own lower steady states.

9. The Great Income Divergence

Rising Global Inequality

Historical trend: The ratio of income per worker between rich and poor countries has been increasing.

Measure: Ratio of 90th percentile country to 10th percentile country - **1870:** ~5:1 - **1960:** ~15:1 - **2000:** ~25:1 - **Recent:** Slight decline due to China/India growth

What does this mean? - The world has become more unequal in terms of living standards - Not just that poor countries are poor, but the gap is widening

Why Is the Gap Growing?

Solow model explanation: Different rates of TFP growth

$$y = A \cdot h \cdot \left(\frac{s}{g_A + \delta + n} \right)^{(1-\alpha)/\alpha}$$

In balanced growth, $g_y = g_A$

If $g_{A,rich} > g_{A,poor}$, then gap widens:

$$\frac{y_{rich}(t)}{y_{poor}(t)} = \frac{y_{rich}(0)}{y_{poor}(0)} \times e^{(g_{A,rich} - g_{A,poor})t}$$

Evidence: Rich countries have maintained faster TFP growth than many poor countries.

Why? 1. **Innovation concentrated in rich countries:** R&D, universities, tech sector 2. **Technology diffusion barriers:** Poor countries can't adopt new technologies 3. **Institutional deterioration:** Some poor countries got worse institutions over time 4. **Poverty traps:** Low income $\Rightarrow$ low investment $\Rightarrow$ low TFP growth

Recent Reversal (China and India)

2000-2020: Global inequality declined slightly

Drivers: - **China:** Massive catch-up growth (7-10% per year) - **India:** Accelerated growth after 1990s reforms (6-8% per year) - Both countries have 1/3 of world population!

Future: Will this continue? - China: Likely slowdown (transitional growth ending) - India: Potential for continued catch-up - Africa: Still mostly stagnant (institutional challenges)

10. What We've Learned and What We Haven't

What the Solow Model Explains Well

1. **Balanced growth facts:**
 - Output per worker grows at rate of technological progress
 - Capital-output ratio stable over time
 - Factor shares (capital, labor) stable over time
2. **Transitional dynamics:**
 - Countries far below steady state grow fast (Japan, Korea, China)
 - Growth slows as countries approach steady state
3. **Conditional convergence:**
 - Countries with similar institutions/education converge to each other
 - Growth rates decline as countries approach their own steady states
4. **Role of saving/investment:**
 - Higher saving leads to higher income level (not higher growth rate)
 - Can generate transitional growth boost

What the Solow Model Doesn't Explain

1. **Why TFP differs across countries:**
 - Model takes A as exogenous
 - Doesn't explain institutions, barriers to technology adoption
2. **Why TFP growth differs across countries:**
 - Model assumes common g_A
 - Doesn't explain why some countries innovate/adopt faster
3. **Long-run growth:**

- Growth ultimately driven by exogenous g_A
 - Need endogenous growth theory (Romer, etc.) to explain innovation
4. **Development traps:**
- Why do some countries stay poor for centuries?
 - Need models with multiple equilibria, complementarities

Beyond Growth Accounting: Next Steps

Institutional economics (Acemoglu, Robinson): - Property rights, rule of law, political institutions - Historical origins: colonization, geography

Endogenous growth theory (Romer, Aghion): - R&D decisions, knowledge spillovers - Scale effects, variety of goods

Misallocation (Hsieh, Klenow): - Resources in wrong firms/sectors - Distortions in credit markets, regulations

Structural transformation (Kuznets): - Shift from agriculture to manufacturing to services - Productivity gaps across sectors

11. Policy Implications

For Developing Countries

What works: 1. **Increase investment:** Raise saving rate, attract foreign investment - Generates transitional growth - But not sufficient for long-run catch-up

2. **Improve education:** Invest in human capital
 - Raises income level significantly (10% per year of schooling)
 - But takes time (decades) to raise average education
3. **Technology adoption:** Remove barriers to importing technology
 - Trade openness, FDI, technology licensing
 - Most important for TFP growth
4. **Institutional reform:** Property rights, rule of law, reduce corruption
 - Hardest to implement but highest long-run payoff
 - Requires political will and stability

What doesn't work: 1. **Expecting automatic catch-up:** Convergence is conditional, not absolute 2. **Only accumulating capital:** China's growth driven by TFP, not just K 3. **Ignoring education:** Human capital essential complement to physical capital

For Advanced Countries

Maintaining growth: 1. **R&D investment:** Push technology frontier forward - Public funding for basic research - IP protection, innovation incentives

2. **Education:** Keep skill levels rising
 - Especially STEM fields

- Lifelong learning, retraining
3. **Reallocation:** Let resources flow to productive uses
- Minimize distortions, zombie firms
 - Creative destruction (Schumpeter)

Challenges: 1. **Slowing TFP growth:** Measured g_A declining in many advanced countries 2. **Aging populations:** $g_N < 0$ in many countries (Japan, Europe) 3. **Environmental constraints:** Climate change, resource limits

12. Summary

Key Concepts

Growth Accounting: - Decompose growth into contributions from K , N , and A - Solow residual measures TFP growth - Most important tool for understanding growth sources

Human Capital: - Education raises income level (10% per year of schooling) - But doesn't change long-run growth rate - Important for explaining cross-country differences

Cross-Country Income Differences: - 50-67% due to TFP differences - 20-30% due to education differences - 10-20% due to capital accumulation differences

Conditional Convergence: - Countries converge to their own steady states - Fast convergence within groups (OECD, East Asia) - Little convergence globally (different steady states) - Poor countries often converging to low steady states

The Big Mystery: - Why do TFP levels and growth rates differ so much? - Growth theory can't answer this - need institutional economics, political economy

Formulas to Remember

Growth accounting:

$$g_A = g_Y - (1 - \alpha)g_K - \alpha g_N$$

Steady-state output per worker (with human capital):

$$y^* = A \cdot h \cdot \left(\frac{s}{g_A + \delta + n} \right)^{(1-\alpha)/\alpha}$$

Convergence equation:

$$g_i \approx g_A - \lambda (\log y_i - \log y_i^*)$$

Where $\lambda \approx 2\%$ (convergence speed).

Looking Forward

Next topic: Open economy macroeconomics - How do trade and capital flows affect growth? - Exchange rates, current accounts, international financial markets - Interaction between domestic and foreign economies

Big picture: Growth theory provides the foundation for understanding long-run prosperity. But institutions, policy, and technology diffusion determine whether countries actually achieve their potential.

Lecture 17: Open Economy Macroeconomics: Introduction

Overview

This lecture transitions from closed economy analysis to open economy macroeconomics. It introduces the key concepts of openness in goods and financial markets, defines exchange rates (nominal and real), and explains why international linkages matter for macroeconomic policy and business cycles. The lecture also provides context on the current state of the US economy following rapid interest rate hikes and banking sector stress.

1. Current Economic Context (2023): Banking Crisis and Credit Crunch

Post-COVID Economic Dynamics

- **Demand-Supply Imbalance:** Following COVID reopening, demand recovered faster than supply.
 - **Strong Demand:** Accumulated savings, fiscal stimulus, pent-up consumption (people “fed up of staying at home”).
 - **Weak Supply:** Supply chain bottlenecks, labor force participation below pre-COVID levels, sectoral labor shortages.
 - **Result:** Positive output gap ($Y > Y^*$) → Inflationary pressure.

Fed’s Policy Response

- **Initial Stance (Late 2021):** Fed viewed inflation as “transitory.”
 - Expected supply to recover quickly and demand to normalize naturally.
 - Delayed raising interest rates despite rising inflation.
- **Policy Error:** Inflation proved persistent (new shocks like Ukraine war + sustained demand).
- **Rapid Tightening (2022-2023):** Fed hiked rates unusually aggressively.
 - **Gradualism Abandoned:** Typically, central banks raise rates slowly to avoid “breaking something.”
 - **Risk:** Fast rate hikes strain the financial system, especially banks.

Silicon Valley Bank (SVB) Collapse (March 2023)

- **What Broke:** Regional and small banks experienced severe deposit outflows (bank runs).
- **Mechanism:**
 1. **Deposit Outflows:** As rates rose, depositors moved funds to higher-yielding money market funds and US Treasuries.
 2. **Initially Gradual:** Large banks handled this well (diversified funding sources).
 3. **Acceleration (2023):** Small/medium banks saw sharp, sudden deposit declines.
 4. **Asset-Side Losses:** Banks held long-term bonds (purchased when rates were low); rising rates caused market value losses.
 5. **Credit Crunch:** To cover deposit withdrawals, banks had to sell assets at losses and cut lending sharply.

Impact on the Economy

- **Flight to Quality:** Deposits moved from small banks to large banks (and out of banking system entirely).
- **Differential Impact:** Small/regional banks are critical lenders to **small businesses** (no access to bond markets).
 - Large corporations have alternatives: bond issuance, commercial paper, multiple bank relationships.
 - Small businesses rely exclusively on bank loans (or family financing).
- **Sectoral Exposure:** Construction and services sectors have high concentrations of small businesses.
 - These sectors face disproportionate credit tightening.
- **Modeling Implication:** In IS-LM-PC framework, credit crunch appears as an **increase in the risk premium x** .

Policy Implications

- **Automatic Tightening:** Credit crunch does part of Fed's work (reducing aggregate demand).
- **Fed Response:** Reduced pace of rate hikes (from 50bp expected to 25bp actual in March 2023).
- **Outlook:** Economy expected to slow significantly (possibly recession avoided if financial stress contained).

Labor Market Silver Lining

- **Participation Rate Recovery:** Labor force participation gradually returning to pre-COVID trend.
 - Previously suppressed recovery was a key Fed forecasting error.
- **Employment-to-Population Ratio:** Approaching pre-COVID trend.

- **Immigration Rebound:** Immigration flows (critical for US labor supply) have normalized after COVID disruptions.
 - Particularly important for accommodation, food services, and other sectors with high immigrant workforce shares.
- **Wage Pressure Declining:** Wage inflation peaked at ~6% (incompatible with 2% inflation target) but now moderating.
 - Sustainable wage growth: ~3.5% (2% inflation + 1.5% productivity growth).
 - Current level (~4-4.5%) still too high but improving.

Key Insight: Even with low unemployment, expanding labor supply (via participation and immigration) can reduce inflationary pressure without requiring recession.

2. Why Open Economy Macroeconomics Matters

The US as a “Closed Economy”

- **Unique Position:** US is the **most closed** major economy in the world.
 - Can describe most US economic dynamics without referencing the rest of the world.
 - Exchange rate (dollar) plays a secondary role compared to other countries.
- **Recent Lecture:** Discussed banking crisis, inflation, Fed policy without mentioning exchange rates or foreign economies.

Open Economy Imperative for Other Countries

- **Singapore Example:** Cannot analyze Singapore’s economy without considering global trade and capital flows.
- **Even Japan:** Despite being a large economy, the yen’s value is central to economic analysis.
- **General Rule:** For almost all countries, external factors (trade, capital flows, exchange rates) are **primary** drivers.

Growing US Openness

- **Historical Trend:** US has become more integrated with global economy over time.
 - Trade (exports + imports) as share of GDP has risen steadily (until recent “deglobalization” pushback).
 - Financial market integration even larger (cross-border asset holdings).
- **Future Outlook:** Uncertain due to post-COVID deglobalization trends, but openness still significant.

3. Three Dimensions of Openness

1. Openness in Goods Markets

Definition: Ability to import goods/services from abroad and export domestically-produced goods/services.

Measurement

- **Openness Ratio:** $\frac{\text{Exports} + \text{Imports}}{\text{GDP}}$
 - US: ~15% of GDP in imports, ~15% in exports (total ~30%).
 - **Underestimates True Openness:** Many non-traded goods (e.g., domestically produced cars) face international competition (prices determined by foreign competitors).
 - * Example: Ford prices reflect Toyota competition, even if Ford is produced domestically.
 - * Sectors with no international competition: haircuts, local services.

Trade Balance

- **Definition:** $NX = X - IM$ (Net Exports = Exports - Imports).
- **US Pattern:** Chronic trade deficit (~\$70 billion/month in 2023).
 - Exports less than imports persistently.
- **Political Concern:** Trade deficits often criticized as “unfair competition.”
 - **Economic Reality:** Reflects **savings-investment imbalance**, not competitiveness per se.
 - * US saves less than it invests $\Rightarrow$ Must borrow from abroad $\Rightarrow$ Trade deficit.

Top US Trade Partners (Deficits)

1. **China:** Largest deficit (~\$380 billion/year).
 - Major political tensions during “global imbalances” era (2000s-2010s).
2. **Mexico:** ~\$130 billion/year deficit.
3. **Vietnam:** Often “Chinese exports in disguise” (supply chain relocation).
4. **Germany:** Persistent surplus country.

Impediments to Goods Market Openness

- **Tariffs:** Taxes on imports (reduce trade flows).
- **Quotas:** Quantitative restrictions on import volumes.

2. Openness in Financial Markets

Definition: Ability to buy/sell foreign financial assets (stocks, bonds, etc.).

Scale

- **Far Larger than Goods Trade:** Financial transactions dwarf goods trade by an order of magnitude.
- **US Asset Holdings by Foreigners:** ~\$20+ trillion.
 - Top holders: Japan, China, UK, Canada (~\$2 trillion each).
 - **Composition:**
 - * Japan, UK, Canada: Mostly private sector (pension funds, wealth managers).
 - * China: Primarily central bank (foreign reserves, currency intervention).

US Holdings of Foreign Assets

- **Substantial but Asymmetric:** US residents hold large amounts of Canadian, European, Asian, Latin American assets.

The “Exorbitant Privilege”

- **Asymmetry in Asset Composition:**
 - **Foreigners Buy from US:** Safe assets (US Treasuries) for insurance/safety (low yield).
 - **US Buys Abroad:** Risky assets (equity, FDI, emerging market bonds) for high returns.
- **Result:** US earns **higher average return** on foreign assets than it pays on foreign holdings of US assets.
 - **Implication:** US can run persistent trade deficits funded by investment income surplus.
 - Example: Brazilian sovereign bonds yield 10-14% vs. US Treasuries at 3-4%.

Impediments to Financial Openness

- **Capital Controls:** Government restrictions on cross-border capital flows.
 - **Rare in Developed Economies:** US, Europe, Japan have fully open capital accounts.
 - **Common in Emerging Markets:** Used to prevent destabilizing capital flight or sudden inflows.
 - * Example: Limit outflows during crises; limit inflows during booms to prevent reversal.

3. Openness in Factor Markets

Definition: Mobility of labor and capital across borders.

Labor Mobility

- **Immigration:** Workers move from one country to another.
 - Critical for US labor market (as discussed earlier: 500,000/year short-fall during COVID).

Capital Mobility

- **Foreign Direct Investment (FDI):** Firms relocate production facilities.
 - Example: Japanese car manufacturers (Toyota, Honda) build plants in US/Canada to serve North American market (avoid trade costs).

Scope of This Course

- **Focus:** Goods and financial market openness.
 - **Factor Mobility:** Not covered in detail (though implicit in some discussions).
-

4. Exchange Rates: Nominal vs. Real

Nominal Exchange Rate (E)

Definition

- **Convention Used in This Course:** Price of **domestic currency** in terms of **foreign currency**.
 - If US is domestic: $\$E = \$ \text{ Yen per Dollar}$ (or Pounds per Dollar, etc.).
 - Example: $E = 130 \text{ ¥}/\$$ means 1 dollar buys 130 yen.

Appreciation vs. Depreciation

- **Appreciation** ($E \uparrow$): Domestic currency becomes more valuable.
 - Example: E rises from 100 $\text{¥}/\$$ to 150 $\text{¥}/\$ \Rightarrow$ Dollar appreciated.
 - Dollar is now more expensive (foreigners must pay more yen to get one dollar).
- **Depreciation** ($E \downarrow$): Domestic currency loses value.
 - Example: E falls from 150 $\text{¥}/\$$ to 100 $\text{¥}/\$ \Rightarrow$ Dollar depreciated.
 - Dollar is now cheaper (foreigners pay fewer yen per dollar).

Recent Example: Dollar-Yen (2022-2023)

- **Sharp Appreciation:** Dollar rose from $\sim 100 \text{ ¥}/\$$ to $\sim 150 \text{ ¥}/\$$ (massive 50% appreciation).
 - Driven by Fed tightening (raising US interest rates) while Bank of Japan kept rates near zero.

- **Intervention:** Bank of Japan intervened to slow yen depreciation (rare for major economies).

Limitations of Nominal Exchange Rate

- **Not Sufficient for Economic Decisions:** Knowing E alone doesn't tell you where to buy goods.
 - Example: $E = 130 \text{ ¥}/\$$ tells you currency conversion rate.
 - But to decide whether to buy a car in US or Japan, you need to know **prices in both countries**.

Real Exchange Rate (ϵ)

Definition

$$\epsilon = \frac{E \times P}{P^*}$$

- E : Nominal exchange rate (foreign currency per unit of domestic currency).
- P : Domestic price level (price index or price of specific good).
- P^* : Foreign price level.

Interpretation

- **Relative Price of Goods:** How many units of foreign goods can one unit of domestic goods buy?
- **Purchasing Power:** The real exchange rate determines trade competitiveness.

Example: US vs. UK Car Purchase

- **US Car Price (P):** \$10,000.
- **UK Car Price (P^*):** £8,000.
- **Exchange Rate (E):** 0.80 £/\$ (1 dollar buys 0.80 pounds).

Step 1: Convert US price to pounds.

$$\text{US Car in Pounds} = P \times E = 10,000 \times 0.80 = \text{£}8,000$$

Step 2: Compare to UK price.

$$\epsilon = \frac{E \times P}{P^*} = \frac{0.80 \times 10,000}{8,000} = \frac{8,000}{8,000} = 1$$

Interpretation: One US car trades for exactly one UK car (equal purchasing power).

Alternative Scenario

- If $P^* = £9,000$ (UK car more expensive), then:

$$\epsilon = \frac{8,000}{9,000} = 0.89 < 1$$

- US car is **cheaper** than UK car $\Rightarrow$ Expect high US exports, low US imports.
- If $P^* = £7,000$ (UK car cheaper), then:

$$\epsilon = \frac{8,000}{7,000} = 1.14 > 1$$

- US car is **more expensive** $\Rightarrow$ Expect low US exports, high US imports.

Real Appreciation vs. Real Depreciation

- **Real Appreciation** ($\epsilon \uparrow$): Domestic goods become **relatively more expensive**.
 - **Causes:**
 - * Nominal appreciation ($E \uparrow$).
 - * Domestic inflation higher than foreign inflation ($P \uparrow$ faster than $P^* \uparrow$).
 - **Effects:**
 - * Exports $\downarrow$ (foreign buyers find domestic goods expensive).
 - * Imports $\uparrow$ (domestic buyers find foreign goods cheap).
- **Real Depreciation** ($\epsilon \downarrow$): Domestic goods become **relatively cheaper**.
 - **Causes:**
 - * Nominal depreciation ($E \downarrow$).
 - * Foreign inflation higher than domestic inflation.
 - **Effects:**
 - * Exports $\uparrow$ (foreign buyers find domestic goods cheap).
 - * Imports $\downarrow$ (domestic buyers find foreign goods expensive).

Importance of Real vs. Nominal

- **Trade Flows:** Determined by **real** exchange rate, not nominal.
 - Nominal exchange rate is only one component of ϵ .
 - **Nominal Volatility Dominates:** In practice, most short-run movements in ϵ come from changes in E (prices P, P^* adjust slowly).
-

5. Global Business Cycle Synchronization

Evidence of Synchronization

- **Business Cycles Move Together:** Recessions and expansions tend to occur simultaneously across countries.
 - 2008-2009 Global Financial Crisis: Nearly universal recession.
 - 2020 COVID Recession: Synchronized global contraction.
 - Smaller US recessions (e.g., 2001 dot-com) also spread globally.

Mechanisms

1. **Trade Linkages:** US recession $\Rightarrow$ Lower US imports $\Rightarrow$ Lower foreign exports $\Rightarrow$ Foreign recession.
2. **Financial Linkages:** US financial crisis $\Rightarrow$ Global credit tightening $\Rightarrow$ Global demand contraction.
3. **Confidence Effects:** Crisis in large economy reduces global business/consumer confidence.

Asymmetry

- **US to World:** Strong spillovers (US recession hurts everyone).
- **Small Country to US:** Weak spillovers (Argentinian recession has negligible effect on US).
- **Emerging Role of China:** China's size now provides some diversification.
 - China's cycle not perfectly correlated with US $\Rightarrow$ Stabilizing force for commodity exporters.

Historical Data

- **Advanced Economies** (blue line): Track closely with global cycle.
 - **Emerging Markets** (red line): Higher average growth (catch-up) but similar cyclical fluctuations.
-

6. Rising Global Integration

Historical Trend: Globalization

- **US Trade Openness:** Steady increase in $(X + IM)/GDP$ from 1970s through 2019.
 - Exports rose from ~10% of GDP to ~15%.
 - Imports rose similarly.
- **Financial Integration:** Even more dramatic increase in cross-border asset holdings.

Size Matters

- **Smaller Countries $\Rightarrow$ More Open:**
 - Limited domestic variety of goods $\Rightarrow$ Must import.
 - Example: Chile (small economy) has much higher trade ratio than US or Japan.
- **Europe Anomaly:** Large European countries (Germany, France) have very high openness ratios.
 - **Reason:** Intra-European trade (same currency zone, minimal trade barriers).
 - If Europe treated as single entity, openness ratio similar to US.

Recent Developments (Post-2020)

- **Deglobalization Pressures:** COVID, US-China tensions, supply chain reshoring.
 - **Uncertain Future:** Trend toward openness may have reversed, but structural integration remains high.
-

7. Summary and Preview

Key Concepts Introduced

1. **Three Dimensions of Openness:**
 - Goods markets (trade).
 - Financial markets (capital flows).
 - Factor markets (migration, FDI).
2. **Exchange Rates:**
 - **Nominal (E):** Price of currency (affects financial returns).
 - **Real (ϵ):** Relative price of goods (affects trade flows).
3. **Trade Balance:** $NX = X - IM$ (influenced by ϵ, Y, Y^*).
4. **Global Linkages:** Business cycles synchronized; policies in large economies have spillovers.

Why This Matters

- **Policy Effectiveness:** Fiscal/monetary policy effects differ in open vs. closed economies.
 - Open economy: Policy “leaks” abroad through imports and capital flows.
 - Smaller multipliers, but new policy channels (exchange rate).
- **External Shocks:** Foreign demand, foreign interest rates, exchange rate movements are new sources of fluctuations.
- **Exchange Rate as Policy Tool:** Can use ϵ to stimulate demand (depreciation boosts net exports).

What's Next

- **Lecture 18-19:** Extend IS model to open economy (goods market equilibrium with trade).
- **Lecture 20-21:** Introduce financial openness (Mundell-Fleming model: IS-LM with open capital account).
- **Lecture 22+:** Expectations, asset pricing, and exchange rate dynamics.

Bottom Line: Opening the economy introduces powerful new linkages and policy tradeoffs. For most countries, these are **first-order** considerations. Even for the US, they matter increasingly over time.

Lecture 18: Review for Quiz 2

Overview

This lecture reviews the material covered in the second part of the course, which will be the subject of Quiz 2. It consolidates key concepts from the labor market, Phillips curve, IS-LM-PC model, and economic growth theory.

1. The Labor Market & Natural Rate of Unemployment

Building Blocks of the Labor Market Model

Wage Setting (WS) Equation

$$W_t = P_t^e \times F(u_t, z)$$

Dividing by expected price level:

$$\frac{W_t}{P_t^e} = F(u_t, z)$$

Key Properties: - **Decreasing in unemployment** ($\partial F / \partial u < 0$): Higher unemployment weakens worker bargaining power and makes unemployment more costly (harder to find jobs). - **Increasing in institutional parameters** ($\partial F / \partial z > 0$): Higher z represents: - Greater union bargaining power. - More generous unemployment benefits. - Stronger labor market protections. - **Depends on expected price level:** Workers negotiate nominal wages but care about real wages, so they base demands on what they expect prices to be during the contract period.

Price Setting (PS) Equation **Production Function** (simplified): $Y = N$ (one unit of labor produces one unit of output). - **Marginal Cost:** $MC = W$ (wage per unit of labor).

Pricing by Firms (markup pricing):

$$P = W \times (1 + m)$$

Where m is the markup over marginal cost, capturing: - Market power (degree of competition). - Other costs not explicitly modeled (capital depreciation, overhead, etc.).

Real Wage Offered by Firms:

$$\frac{W}{P} = \frac{1}{1+m}$$

Key Insight: Higher markup ($m \uparrow$) means lower real wage offered by firms.

Equilibrium: Natural Rate of Unemployment

Definition of Natural Rate (u_n)

- The unemployment rate at which **actual price equals expected price** ($P_t = P_t^e$).
- **Not truly “natural”** - depends on structural parameters (m, z), not biological constants.

Solving for u_n At natural rate, real wage demanded equals real wage offered:

$$F(u_n, z) = \frac{1}{1+m}$$

For a specific functional form (e.g., $F(u, z) = z - \alpha u$):

$$z - \alpha u_n = \frac{1}{1+m}$$

Solving:

$$u_n = \frac{1}{\alpha} \left(z - \frac{1}{1+m} \right)$$

Comparative Statics: What Changes u_n ?

Increase in Bargaining Power ($z \uparrow$)

- **WS Curve:** Shifts **up** (workers demand higher real wage at any unemployment rate).
- **PS Curve:** Unchanged (firms' markup unchanged).
- **Result:** $u_n \uparrow$ (natural rate of unemployment rises).
- **Intuition:** Higher wage demands inconsistent with firms' willingness to pay; only way to restore equilibrium is for unemployment to rise (weakens worker bargaining power).

Increase in Markup ($m \uparrow$)

- **WS Curve:** Unchanged.
- **PS Curve:** Shifts **down** (firms offer lower real wage).
- **Result:** $u_n \uparrow$ (natural rate of unemployment rises).
- **Intuition:** Lower real wage offered is inconsistent with worker demands at initial u_n ; unemployment must rise to lower wage demands.

2. From Labor Market to Inflation: The Phillips Curve

Deriving the Phillips Curve

Starting Point: Wage setting equation

$$W_t = P_t^e \times F(u_t, z)$$

Linearize F :

$$W_t = P_t^e(z - \alpha u_t)$$

Substitute into price setting:

$$P_t = W_t(1 + m) = P_t^e(z - \alpha u_t)(1 + m)$$

Divide by P_{t-1} :

$$\frac{P_t}{P_{t-1}} = \frac{P_t^e}{P_{t-1}}(z - \alpha u_t)(1 + m)$$

Define inflation rates: $\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} - \pi_t^e = \frac{P_t^e - P_{t-1}}{P_{t-1}}$

After approximations (assuming $(1 + \pi_t) \approx 1 + \pi_t$ for small π):

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

Modern Form: Expectations-Augmented Phillips Curve

Using Natural Rate: At u_n : $\pi = \pi^e$ (by definition).

This gives:

$$\pi^e + (m + z) - \alpha u_n = \pi^e$$

Therefore:

$$(m + z) = \alpha u_n$$

Substituting back:

$$\pi_t = \pi_t^e + \alpha u_n - \alpha u_t$$

$$\pi_t - \pi_t^e = -\alpha(u_t - u_n)$$

In terms of output gap (using Okun's Law: $u_t - u_n \approx -\beta(Y_t - Y_n)/Y_n$):

$$\pi_t - \pi_t^e = \lambda(Y_t - Y_n)$$

Where $\lambda = \alpha\beta L/Y_n$ (positive constant).

Inflation Expectations: Anchored vs. Unanchored

Anchored Expectations ($\theta = 0$)

$$\pi_t^e = \bar{\pi}$$

Where $\bar{\pi}$ is the central bank's inflation target (e.g., 2% in US).

Phillips Curve becomes:

$$\pi_t = \bar{\pi} - \alpha(u_t - u_n)$$

Interpretation: Inflation deviates from target when unemployment deviates from natural rate, but expectations remain anchored at target.

Characteristics: - Credible central bank (strong anti-inflation reputation). - Temporary inflation shocks don't affect long-run expectations. - **Example:** US currently (2023) - inflation rose to 5%, but long-run expectations remain ~2.5%.

Unanchored Expectations ($\theta = 1$)

$$\pi_t^e = \pi_{t-1}$$

Phillips Curve becomes:

$$\pi_t - \pi_{t-1} = -\alpha(u_t - u_n)$$

Interpretation: Change in inflation (acceleration/deceleration) depends on unemployment gap.

Characteristics: - Weak central bank credibility. - People extrapolate past inflation into future. - **Example:** US in 1970s - inflation spiraled as expectations became unanchored.

Implications for Policy

With Anchored Expectations

- **Positive output gap** ($Y > Y_n$): $\pi > \bar{\pi}$ (inflation above target).
- **Policy Response:** Raise interest rates to close output gap.
- **Outcome:** Inflation returns to target without persistent deviation.

With Unanchored Expectations

- **Positive output gap** ($Y > Y_n$): $\pi_t > \pi_{t-1}$ (inflation accelerating).
- **Problem:** Inflation doesn't stabilize - it keeps rising.
- **Policy Response:** Must create **negative** output gap ($Y < Y_n$) for extended period to bring inflation down.
- **Cost:** Severe recession required to disinflate (Volcker disinflation 1980-82).

The Phillips Curve in Historical Context

1960s: Apparent Stable Trade-off

- **Observation:** Downward-sloping Phillips Curve (π vs. u).
- **Interpretation:** Policymakers thought they could permanently lower unemployment by accepting higher inflation.
- **Reality:** Expectations were implicitly anchored at low inflation levels.

1970s: Breakdown

- **Observation:** No clear relationship - high inflation **and** high unemployment (stagflation).
- **Causes:**
 1. **Supply shocks** (oil price shocks): $m \uparrow$ shifts Phillips Curve up.
 2. **Unanchored expectations:** $\theta \rightarrow 1$ (people expected high inflation to persist).
- **Result:** Vertical or positively-sloped relationship in (π, u) space.

1980s-Present: Credibility Restoration

- **Volcker disinflation** (1980-82): Fed raised rates dramatically, caused recession.
- **Outcome:** Expectations re-anchored; Phillips Curve relationship restored.
- **Current:** Most advanced economies have anchored expectations ($\theta \approx 0$).

3. The IS-LM-PC Model

Structure of the Model

IS Curve (Goods Market Equilibrium)

$$Y = C(Y - T) + I(Y, r) + G$$

Key Features: - Output determined by aggregate demand. - **Decreasing in real interest rate** (r): Higher r reduces investment. - **Shifts:** Changes in G, T (fiscal policy), or exogenous demand shocks (confidence, financial conditions captured by x in $I(Y, r, x)$).

LM (Monetary Policy Rule)

$$r = \bar{r}$$

Simplification: Central bank directly sets the real interest rate. - In reality, sets nominal rate i , but with fixed π^e , can think of setting r .

Phillips Curve (Inflation Dynamics)

$$\pi_t - \pi_t^e = \lambda(Y_t - Y_n)$$

Key Features: - Links output gap to inflation deviation from expectations. - Provides constraint on how long output can deviate from potential.

Short Run vs. Medium Run

Short Run (IS-LM)

- **Time Horizon:** Months to a year.
- **Prices:** Sticky (inflation doesn't adjust).
- **Output:** Demand-determined (can deviate from Y_n).
- **Policy:** Monetary or fiscal policy can move output.

Medium Run (IS-LM-PC)

- **Time Horizon:** Several years.
- **Prices:** Adjust gradually via Phillips Curve.
- **Output:** Converges to Y_n (supply-side constraint binds).
- **Policy:** Cannot keep output above Y_n indefinitely without accelerating inflation.

Policy Analysis with IS-LM-PC

Example 1: Output Above Potential Initial Situation: - IS-LM equilibrium at $Y_1 > Y_n$. - Phillips Curve: $\pi_1 > \pi^e$ (inflation above expectations).

Problem: Inflation rising (if expectations unanchored, accelerating).

Policy Response (Central Bank): 1. **Raise real interest rate:** $r \uparrow$ (shift LM up or move along IS). 2. **Output falls:** New equilibrium at $Y_2 < Y_1$. 3. **Iterate:** Continue raising r until $Y = Y_n$.

Medium-Run Equilibrium: - Output: $Y = Y_n$ (output gap closed). - Inflation: $\pi = \pi^e$ (stable inflation). - Interest rate: $r = r_n$ (natural rate of interest).

Example 2: Fiscal Expansion from Equilibrium Shock: $\Delta G > 0$ (government spending increases).

Short Run: - IS shifts **right**. - Output increases: $Y_1 > Y_n$ (positive output gap). - Inflation: $\pi_1 > \pi^e$ (above expectations).

Medium Run: - Central bank raises r to combat inflation. - Output returns to Y_n . - Result: Fiscal expansion has **no long-run effect on output**. -

Crowding out: Higher r reduces investment; $\Delta I = -\Delta G$ in long run.

Example 3: Supply Shock (e.g., Oil Price Increase) Shock: Markup increases ($m \uparrow$), e.g., due to higher energy costs.

Effects: 1. **Natural rate of unemployment rises:** $u_n \uparrow$ (from WS-PS analysis). 2. **Natural level of output falls:** $Y_n \downarrow$ (fewer people employed). 3. **Phillips Curve shifts left:** For any Y , inflation now higher (positive supply shock).

Policy Dilemma (if central bank doesn't recognize Y_n has fallen): - **Initially:** At old output Y_0 , now $Y_0 > Y_n$ (positive output gap). - **Phillips Curve:** Inflation starts rising. - **Central bank reaction:** Must raise r to bring Y down to new (lower) Y_n . - **Outcome:** Recession is **necessary** to stabilize inflation.

Historical Example: 1970s oil shocks led to both recession and inflation (stagflation).

Zero Lower Bound (ZLB) and Deflationary Spirals

The ZLB Problem Scenario: Economy in recession, needs negative real interest rate to reach Y_n .

$$r_n < 0$$

But nominal rate constrained:

$$i \geq 0$$

Fisher Equation: $r = i - \pi^e$

Problem: If $\pi^e \approx 0$ (or negative), then $r = i - 0 \geq 0 > r_n$ (real rate too high).

Deflationary Spiral Mechanism Starting Point: $Y < Y_n$ (recession), $i = 0$ (ZLB binding).

Step 1: Phillips Curve implies $\pi < \pi^e$ (deflation).

Step 2: Deflation causes $\pi^e \downarrow$ (if expectations not anchored).

Step 3: Lower π^e raises real rate: $r = i - \pi^e = 0 - \pi^e \uparrow$.

Step 4: Higher r reduces demand further: $Y \downarrow$.

Step 5: Return to Step 1 (vicious cycle).

Result: Self-reinforcing downward spiral.

Policy Responses at ZLB Monetary Policy (Limited): - **Quantitative Easing** (QE): Buy long-term bonds to lower long-term rates. - **Forward Guidance:** Promise to keep rates low even after recovery (affects expectations). - **Negative nominal rates:** Some central banks (ECB, BoJ) went slightly negative.

Fiscal Policy (Primary Tool): - **Expansionary fiscal policy:** $\Delta G > 0$ or $\Delta T < 0$. - Shifts IS right, raises output without needing lower interest rates. - **Historical Example:** Japan used massive fiscal stimulus (1990s-2000s), leading to high public debt.

Inflation Expectations Management: - Commit to higher inflation target temporarily (raise π^e to lower real rate).

4. Economic Growth: The Solow Model

Motivation: Growth Facts

Fact 1: Convergence Among Similar Countries

- **Observation:** Countries with similar institutions but lower initial GDP/capita grow faster.
- **Evidence:** OECD countries (1950-2020) show negative relationship between initial income and growth rate.
- **Interpretation:** Convergence toward common steady state.

Fact 2: Persistent Inequality Across Countries

- **Observation:** Some countries (Sub-Saharan Africa) remain poor despite decades.
- **Evidence:** Income gap between richest and poorest countries increased from 20:1 (1950) to 40:1+ (2020).
- **Interpretation:** Different steady states due to differences in technology, institutions.

Fact 3: Growth Miracles and Disasters

- **Miracles:** South Korea, Taiwan, China (rapid growth from low levels).
- **Disasters:** Argentina (relative decline), Zimbabwe (absolute decline).

Production Function and Key Assumptions

Cobb-Douglas Production Function

$$Y = F(K, N) = K^\alpha N^{1-\alpha}$$

Properties: 1. **Constant Returns to Scale (CRS):** $F(\lambda K, \lambda N) = \lambda F(K, N)$.
 - Doubling all inputs doubles output. 2. **Diminishing Returns to Each Factor:** $\frac{\partial^2 Y}{\partial K^2} < 0$, $\frac{\partial^2 Y}{\partial N^2} < 0$. - Increasing one input (holding other fixed) yields progressively smaller output gains.

Special Case (used in lecture): $\alpha = 0.5$

$$Y = \sqrt{K} \times \sqrt{N} = \sqrt{K \times N}$$

Intensive Form (Per Worker) Dividing by N :

$$\frac{Y}{N} = F\left(\frac{K}{N}, 1\right) = f(k)$$

Where $k = K/N$ (capital per worker) and $y = Y/N$ (output per worker).

For Cobb-Douglas:

$$y = k^\alpha$$

Graphical Representation: Concave function (diminishing returns). - Slope $= f'(k) = MPK$ (marginal product of capital). - As $k \uparrow$, output rises but at decreasing rate.

Solow Model Without Technological Progress

Key Equations 1. Production Function:

$$y = f(k)$$

2. Savings = Investment:

$$I = S = sY$$

Where s is the (exogenous) savings rate ($0 < s < 1$).

3. Capital Accumulation:

$$K_{t+1} = K_t - \delta K_t + I_t$$

Where δ is the depreciation rate ($0 < \delta < 1$).

Per Worker Form:

$$k_{t+1} = k_t - \delta k_t + sf(k_t)$$

Or:

$$\Delta k = sf(k) - \delta k$$

Steady State Definition: $\Delta k = 0$ (capital per worker constant).

Condition:

$$sf(k^*) = \delta k^*$$

Interpretation: - **Investment per worker** ($sf(k)$) exactly offsets **depreciation per worker** (δk). - Capital stock neither growing nor shrinking.

Graphical Solution: - Plot $sf(k)$ (concave, increasing) and δk (linear, through origin). - Intersection determines k^* .

Dynamics: Convergence to Steady State If $k < k^*$: - Investment ($sf(k)$) $>$ Depreciation (δk). - $\Delta k > 0$ (capital per worker rising). - Economy accumulates capital, moving right toward k^* .

If $k > k^*$: - Investment ($sf(k)$) $<$ Depreciation (δk). - $\Delta k < 0$ (capital per worker falling). - Economy decumulates capital, moving left toward k^* .

At $k = k^*$: - $\Delta k = 0$ (steady state). - Output per worker constant: $\Delta y = 0$.

Growth Rates in Steady State (No Population Growth) Capital per worker: $g_k = 0$. Output per worker: $g_y = 0$. Total output: $g_Y = 0$ (if N constant).

Key Implication: No long-run growth in living standards without technological progress.

Comparative Statics: Changes in Savings Rate

Increase in Savings Rate ($s \uparrow$) Effect on Diagram: - $sf(k)$ curve shifts up (more investment at every k). - New steady state: $k^* \uparrow$ (higher capital per worker).

Dynamics (if initially at old steady state): - $sf(k)$ now exceeds δk (investment $>$ depreciation). - $\Delta k > 0$ (capital accumulates). - Economy grows **faster than steady state** during transition. - Eventually converges to new (higher) steady state.

Long-Run Effect: - **Level effect:** Higher y^* (richer in steady state). - **Growth effect:** None - still $g_y = 0$ in new steady state. - **Transitional growth:** Faster growth during transition from old to new steady state.

Policy Implication: Saving can explain growth miracles (e.g., East Asia 1960s-1980s) but only temporary acceleration, not permanent increase in growth rate.

Solow Model with Population Growth

Modification Population grows at rate g_N :

$$N_{t+1} = (1 + g_N)N_t$$

Capital Accumulation (per worker):

$$\Delta k = sf(k) - (\delta + g_N)k$$

Intuition for $g_N k$ term: - To keep k constant, must not only replace depreciation (δk). - Also need to equip new workers with capital ($g_N k$). - **Total required investment:** $(\delta + g_N)k$.

New Steady State Condition:

$$sf(k^*) = (\delta + g_N)k^*$$

Diagram: - Now plot $sf(k)$ vs. $(\delta + g_N)k$ (steeper line). - Intersection determines k^* (lower than before for same s).

Growth Rates in Steady State Capital per worker: $g_k = 0$ (by definition of steady state). **Output per worker:** $g_y = 0$ (since k constant and $y = f(k)$). **Total capital:** $g_K = g_N$ (capital grows to equip new workers). **Total output:** $g_Y = g_N$ (output grows with population).

Key Implication: Population growth is a source of GDP growth, but not growth in living standards (GDP/capita).

Solow Model with Technological Progress

Modeling Technology Labor-Augmenting Technological Progress:

$$Y = F(K, AN)$$

Where: - A = Technology level (or “efficiency of labor”). - AN = Effective labor (number of workers in efficiency units).

Growth Rate of Technology:

$$A_{t+1} = (1 + g_A)A_t$$

Effective Worker Units Redefine variables in terms of effective workers: - $\tilde{k} = \frac{K}{AN}$ (capital per effective worker). - $\tilde{y} = \frac{Y}{AN}$ (output per effective worker).

Production Function (intensive form):

$$\tilde{y} = f(\tilde{k})$$

Capital Accumulation (Per Effective Worker)

$$\Delta \tilde{k} = sf(\tilde{k}) - (\delta + g_N + g_A)\tilde{k}$$

Intuition for $g_A \tilde{k}$ term: - To keep $\tilde{k} = K/(AN)$ constant, need to: 1. Replace depreciation: $\delta \tilde{k}$. 2. Equip new workers: $g_N \tilde{k}$. 3. Equip existing workers with better technology: $g_A \tilde{k}$. - **Total required investment:** $(\delta + g_N + g_A)\tilde{k}$.

Steady State with Technological Progress Condition:

$$sf(\tilde{k}^*) = (\delta + g_N + g_A)\tilde{k}^*$$

Diagram: - Plot $sf(\tilde{k})$ vs. $(\delta + g_N + g_A)\tilde{k}$. - Intersection determines $\tilde{k}^*$.

Growth Rates in Steady State Capital per effective worker: $g_{\tilde{k}} = 0$.

Output per effective worker: $g_{\tilde{y}} = 0$.

But in terms of actual workers:

Capital per worker:

$$k = \frac{K}{N} = A\tilde{k}$$
$$g_k = g_A + g_{\tilde{k}} = g_A$$

Output per worker:

$$y = \frac{Y}{N} = A\tilde{y}$$
$$g_y = g_A + g_{\tilde{y}} = g_A$$

Total output:

$$Y = AN\tilde{y}$$
$$g_Y = g_A + g_N + g_{\tilde{y}} = g_A + g_N$$

Key Implications: 1. **Long-run growth in living standards** (GDP/capita) is **entirely driven by technological progress** (g_A). 2. Savings rate, depreciation, etc. affect **levels** but not **long-run growth rates**. 3. **Population growth** adds to GDP growth but not GDP/capita growth.

Convergence and Conditional Convergence

Unconditional Convergence (Fails Empirically) Prediction: If all countries have same (s, δ, g_N, g_A) , then: - Poor countries (low k) should grow faster than rich countries (high k). - Eventually, all converge to same steady state.

Reality: Not observed globally. - Many poor countries (e.g., Sub-Saharan Africa) remain poor. - No evidence of unconditional convergence.

Conditional Convergence (Holds Empirically) Prediction: Countries converge to **their own** steady states, which differ due to: - Different savings rates (s). - Different population growth (g_N). - Different human capital levels (H). - Different technology levels (A) and growth rates (g_A).

Evidence: - Within groups of similar countries (OECD), conditional convergence observed. - Controlling for determinants of steady state, poor countries do grow faster.

Policy Implication: To raise long-run income, need to: 1. Increase s (transitional growth). 2. Lower g_N (raises k^* and y^*). 3. Invest in human capital (H). 4. **Most important:** Improve technology (A) and institutions that foster innovation (g_A).

Growth Accounting: Solow Residual

Decomposing Growth Cobb-Douglas Production Function:

$$Y = K^\alpha (AN)^{1-\alpha}$$

Taking logs and differentiating:

$$g_Y = \alpha g_K + (1 - \alpha)(g_A + g_N)$$

Rearranging:

$$g_A = g_Y - \alpha g_K - (1 - \alpha)g_N$$

Or (including human capital H):

$$g_A = g_Y - \alpha g_K - \beta g_H - (1 - \alpha - \beta)g_N$$

Solow Residual: The part of output growth not explained by growth in measured inputs (capital, labor, human capital). - Interpreted as **Total Factor Productivity (TFP) growth**.

Findings from Growth Accounting Advanced Economies: - Roughly 50% of growth explained by capital accumulation. - Roughly 50% by TFP growth (g_A).

Cross-Country Income Differences: - Differences in K, H explain some variation. - Large residual: Differences in A (technology/efficiency). - **Interpretation:** Institutions, policies, geography affect A .

Policy Implication: Policies that improve TFP (innovation, education quality, rule of law) are crucial for long-run prosperity.

5. Key Takeaways for Quiz 2

Labor Market

- Understand derivation of natural rate of unemployment from WS-PS model.
- Know how z (bargaining power) and m (markup) affect u_n .
- Natural rate is not constant - depends on structural parameters.

Phillips Curve

- Derive Phillips Curve from wage-price dynamics.
- Distinguish anchored vs. unanchored expectations (critical for policy).
- Understand output gap form: $\pi - \pi^e = \lambda(Y - Y_n)$.

IS-LM-PC

- Analyze short-run (IS-LM) vs. medium-run (convergence to Y_n).
- Policy responses: monetary (change r) vs. fiscal (change G, T).
- Supply shocks shift Phillips Curve and change Y_n .
- ZLB creates deflationary spiral; fiscal policy becomes primary tool.

Solow Growth Model

- Derive steady state: $sf(k^*) = (\delta + g_N + g_A)k^*$.
- Understand dynamics: countries below steady state grow faster (convergence).
- Comparative statics: $s \uparrow$ raises k^*, y^* (level effect) but not long-run g_y (growth effect).
- With technological progress: $g_y = g_A$ (only source of sustained growth in living standards).
- Conditional convergence: countries converge to different steady states depending on s, g_N, A .

Lecture 19: Open Economy - Goods Market

Overview

This lecture extends the IS model to an open economy, introducing trade and the real exchange rate while keeping financial markets closed for now.

1. Key Concepts in Openness

Real Exchange Rate ()

- **Definition:** $\epsilon = \frac{E \times P}{P^*}$
 - E : Nominal exchange rate (units of domestic currency per foreign currency)
 - P : Domestic price level
 - P^* : Foreign price level
- **Real Appreciation** ($\epsilon \uparrow$): Domestic goods become more expensive relative to foreign goods.
- **Real Depreciation** ($\epsilon \downarrow$): Domestic goods become cheaper relative to foreign goods.

Uncovered Interest Parity (UIP)

- **Condition:** $(1 + i_t) = \frac{E_t}{E_{t+1}^e}(1 + i_t^*)$
- **Approximation:** $i_t \approx i_t^* + \frac{E_{t+1}^e - E_t}{E_t}$
 - Domestic interest rate = Foreign interest rate + Expected appreciation of domestic currency
- **Interpretation:** In equilibrium, expected returns on domestic and foreign bonds (in same currency) must be equal.
- **Note:** This lecture focuses on goods markets only; UIP will be integrated later in the Mundell-Fleming model.

2. Demand for Domestic Goods vs. Domestic Demand for Goods

Key Distinction (Open Economy)

- **Domestic Demand for Goods (D):** $D = C(Y - T) + I(Y, r) + G$
 - What domestic residents demand (from anywhere).
- **Demand for Domestic Goods (Z):** $Z = D + X - IM/E$
 - What is demanded of domestically-produced goods (by anyone).
 - In closed economy: $Z = D$ (they were the same).
 - In open economy: $Z \neq D$ due to imports and exports.

Exports (X)

- $X = X(Y^*, \epsilon)$
- **Increasing in Foreign Income** ($Y^* \uparrow \rightarrow X \uparrow$): Higher foreign income increases foreign demand for our goods.
- **Decreasing in Real Exchange Rate** ($\epsilon \uparrow \rightarrow X \downarrow$): Appreciation makes our goods more expensive for foreigners.

Imports (IM)

- $IM = IM(Y, \epsilon)$
- **Increasing in Domestic Income** ($Y \uparrow \rightarrow IM \uparrow$): Higher income increases consumption of all goods, including imports.
- **Increasing in Real Exchange Rate** ($\epsilon \uparrow \rightarrow IM \uparrow$): Appreciation makes foreign goods cheaper, increasing imports.

3. Equilibrium in the Goods Market (Open Economy)

Equilibrium Condition

- $Y = Z = C(Y - T) + I(Y, r) + G + X(Y^*, \epsilon) - IM(Y, \epsilon)/E$
- **ZZ Curve:** Demand for domestic goods as a function of domestic output.
- **Diagram:**
 - ZZ curve intersects 45° line to determine equilibrium output Y .

- ZZ is **flatter** than the closed economy DD curve.

Multiplier Effect (Open Economy)

- **Smaller Multiplier:** Opening reduces the multiplier because part of the demand increase “leaks” to imports.
- **Why?:** When $Y \uparrow$, $C \uparrow$, but part of ΔC goes to imports, not domestic goods.
- Result: $\frac{dY}{dG} < \frac{1}{1-c_1}$ (open) compared to closed economy multiplier.

Trade Balance

- **Net Exports (NX):** $NX = X - IM/E = Z - D$
- **Trade Balance = 0** when ZZ = DD.
- **Trade Surplus (NX > 0):** When Y is low (imports are low).
- **Trade Deficit (NX < 0):** When Y is high (imports are high).

4. Policy Experiments

Fiscal Expansion ($\Delta G > 0$)

- **Effect on Output:** ZZ shifts up $\rightarrow Y \uparrow$ (but less than in closed economy).
- **Effect on Trade Balance:** $NX \downarrow$ (trade deficit worsens).
 - Higher Y increases imports.
 - Exports unchanged (depends on Y^* , not Y).

Foreign Demand Shock ($\Delta Y^* > 0$)

- **Effect on Output:** Exports $\uparrow \rightarrow$ ZZ shifts up $\rightarrow Y \uparrow$.
- **Effect on Trade Balance:** $NX \uparrow$ (trade surplus improves).
- **Note:** Preferable to domestic fiscal expansion because:
 1. Improves trade balance (vs. worsening it).
 2. No fiscal deficit required.

Real Depreciation ($\epsilon \downarrow$)

- **Assumption:** Volume effect dominates price effect (realistic except very short run).
 - Net exports increase with depreciation: $NX \uparrow$ when $\epsilon \downarrow$.
- **Effect on Output:** ZZ shifts up $\rightarrow Y \uparrow$ (expansionary).
- **Effect on Trade Balance:** $NX \uparrow$ (improves).
- **Expenditure Switching:** Depreciation shifts expenditure (domestic and foreign) toward domestic goods.

Combined Policy: Depreciation + Fiscal Contraction

- **Goal:** Improve trade balance without changing output.
- **Mechanism:**

- Depreciation: Improves NX , increases Y (expansionary).
- Fiscal contraction ($\downarrow G$): Reduces Y (contractionary).
- **Result:** $NX \uparrow$, Y unchanged.
- **Historical Example:** China (late 1990s–2000s) accused of “mercantilist policies” (keeping currency artificially low to boost exports).

5. Summary

- **Open Economy Insight:** $Z \neq D$ due to trade.
- **New Parameters:** Foreign output (Y^*), real exchange rate (ϵ).
- **Smaller Multipliers:** Import leakage reduces fiscal policy effectiveness.
- **Trade Balance:** Determined by $Z - D = X - IM/E$.
- **Depreciation:** Improves trade balance and is expansionary (expenditure switching).
- **Cross-Country Spillovers:** Policy in one country (e.g., China) affects others through Y^* .

Next: Integrate financial openness (Mundell-Fleming model).

Lecture 20: The Mundell-Fleming Model

Overview

The Mundell-Fleming model extends IS-LM to an open economy with both goods market and financial market openness. It explains exchange rate determination and the interaction between monetary policy, fiscal policy, and exchange rates in the short run.

1. Motivation: Exchange Rate Fluctuations

Recent Dynamics (2020s)

- **Dollar Appreciation (pre-2022):** US tightened monetary policy before rest of world.
- **Dollar Depreciation (post-2022):** Markets expect US to peak in tightening while Europe/Japan continue.
- **Key Driver:** Relative interest rate expectations across countries.

Policy Rate Paths

- **US:** Markets expect limited further hikes, then cuts.
- **Europe:** Expects more prolonged hiking cycle (energy shocks, inflation).
- **Japan:** Has been at zero lower bound for decades; minimal expected changes.

2. The Mundell-Fleming Model: Structure

IS Equation (Open Economy)

$$Y = C(Y - T) + I(Y, i) + G + NX(Y, Y^*, E)$$

- **Net Exports:** $NX = X(Y^*, E) - IM(Y, E)$
 - NX decreasing in Y (higher income $\rightarrow$ more imports).
 - NX increasing in Y^* (higher foreign income $\rightarrow$ more exports).
 - NX decreasing in E (appreciation $\rightarrow$ exports fall, imports rise).

Assumptions (Short Run)

- **Fixed Prices:** $P = P^* = \text{constant}$ (no inflation).
- **Nominal = Real:**
 - Real interest rate = Nominal interest rate ($r = i$).
 - Real exchange rate = Nominal exchange rate ($\epsilon = E$).

Uncovered Interest Parity (UIP)

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e$$

- **Approximation:** $i_t \approx i_t^* + \frac{E_{t+1}^e - E_t}{E_t}$
- **Interpretation:** Domestic interest rate = Foreign interest rate + Expected depreciation of domestic currency.
- **Key Assumption (for now):** E_{t+1}^e is **fixed** (exogenous parameter).
 - Simplifies analysis; will be relaxed later.

LM (Monetary Policy)

$$i = \bar{i}$$

- Central bank sets the nominal interest rate.

3. Understanding UIP

Why Does $i \uparrow$ Cause $E \uparrow$ (Appreciation)?

Starting Point: $i = i^*$, so $E = E^e$ (no expected change).

Scenario: Domestic interest rate increases ($i \uparrow$).

- **Logic:** If $i > i^*$, domestic bonds pay higher return.
- **Arbitrage:** To maintain equal expected returns, must expect domestic currency to depreciate ($E_{t+1} < E_t$).
- **Mechanism:** Given E^e is fixed, only way to generate expected depreciation is to appreciate E_t **today**.
 - If $E_t > E^e$, then expect E to fall (depreciation) from t to $t + 1$.
 - This capital loss offsets higher interest rate differential.

Equilibrium: $i \uparrow \Rightarrow E \uparrow$ (appreciation today) $\Rightarrow$ Expect depreciation tomorrow
 $\Rightarrow$ Returns equalized.

Why Does $E^e \uparrow$ Cause $E \uparrow$ (Appreciation)?

- **Logic:** If expect dollar to be stronger in future ($E^e \uparrow$), everyone wants to buy dollars now.
- **Result:** Current exchange rate E jumps immediately (one-for-one with E^e if $i = i^*$).
- **Why?:** If E didn't jump, domestic bonds would offer same interest rate **plus** expected capital gain (violates UIP).

4. The Mundell-Fleming Diagram

Three-Panel Diagram

1. **Top Left: IS-LM** (Output Y vs. Interest Rate i)
2. **Bottom: UIP** (Exchange Rate E vs. Interest Rate i)

IS Curve (Open Economy)

- **Slope:** Flatter than closed economy (import leakage).
- **Shifts:**
 - $\uparrow G$ or $\downarrow T$: IS shifts **right** (but smaller effect than closed economy).
 - $\uparrow Y^*$: IS shifts **right** (higher exports).
 - $\uparrow E^e$: IS shifts **left** (appreciation $\rightarrow$ lower net exports).

UIP Curve

- **Slope:** Upward sloping (E vs. i).
- **Interpretation:** Higher i requires higher E (appreciation) to generate expected depreciation.
- **Special Point:** If $i = i^*$, then $E = E^e$.
- **Shifts:**
 - $\uparrow i^*$: UIP shifts **left** (depreciation).
 - $\uparrow E^e$: UIP shifts **right** (appreciation).

5. Policy Experiments

Contractionary Monetary Policy ($\uparrow i$)

- **IS-LM:** Output falls ($Y \downarrow$).
- **UIP:** Exchange rate appreciates ($E \uparrow$).
- **Two Channels** (why monetary policy more powerful in open economy):
 1. **Investment Channel** (closed economy): $i \uparrow \Rightarrow I \downarrow \Rightarrow Y \downarrow$.
 2. **Net Export Channel** (open economy): $i \uparrow \Rightarrow E \uparrow \Rightarrow NX \downarrow \Rightarrow Y \downarrow$.
- **Result:** More powerful than closed economy (especially for small open economies).

Expansionary Fiscal Policy ($\uparrow G$, accommodated by Fed)

- **Assumption:** Fed keeps i constant (LM doesn't shift).
- **IS-LM:** Output increases ($Y \uparrow$), but **less** than in closed economy (import leakage).
- **UIP:** No change in E (since i unchanged).
- **Note:** In practice, if fiscal expansion causes overheating, Fed may raise $i \Rightarrow E \uparrow$.

Increase in Expected Exchange Rate ($\uparrow E^e$)

- **IS:** Shifts **left** (current E appreciates $\Rightarrow NX \downarrow$).
- **UIP:** Shifts **right** (for same i , need higher E).
- **Result:** $E \uparrow$ (appreciation), $Y \downarrow$ (contractionary).

Foreign Output Shock ($\downarrow Y^*$)

- **IS:** Shifts **left** (exports fall).
- **UIP:** No change.
- **Result:** $Y \downarrow$, E unchanged (unless Fed reacts by cutting i).

Foreign Interest Rate Increase ($\uparrow i^*$)

- **IS:** May shift depending on how E changes (to be analyzed).
- **UIP:** Shifts **left** (depreciation).
 - For same i , now need lower E (depreciation) to expect appreciation and equalize returns.
- **Result:** $E \downarrow$ (depreciation), effect on Y depends on net export response (analyzed next lecture).

6. Key Takeaways

- **Mundell-Fleming = IS-LM + UIP + Open Economy**
- **Two Channels for Monetary Policy:** Investment + Net Exports (more powerful in open economy).
- **Fiscal Policy:** Smaller multiplier (leakage to imports); no effect on E if i unchanged.
- **Exchange Rate Determination:** UIP (arbitrage condition) determines E given i, i^*, E^e .
- **Expectations Matter:** Changes in E^e have immediate effects on current E (asset pricing).
- **Small vs. Large Economies:** Net export channel more important for small open economies (US less affected).

Next: Complete analysis of foreign shocks and extensions to flexible exchange rates.

Lecture 21: Exchange Rate Regimes

Overview

This lecture analyzes different exchange rate regimes (fixed vs. floating), their implications for policy autonomy, and the costs and benefits of each regime. It also covers speculative attacks and how expectations drive exchange rate dynamics.

1. Review: Mundell-Fleming Shocks (Floating Regime)

Increase in Expected Exchange Rate ($\uparrow E^e$)

- **UIP:** Shifts **right** (for same i , need higher E).
- **IS:** Shifts **left** (appreciation $\rightarrow NX \downarrow$).
- **Result:** $E \uparrow$ (appreciation), $Y \downarrow$ (contractionary).

Foreign Output Decline ($\downarrow Y^*$)

- **UIP:** No change.
- **IS:** Shifts **left** (exports decline).
- **Result:** $Y \downarrow$, E unchanged (unless Fed responds).

Foreign Interest Rate Increase ($\uparrow i^*$)

- **UIP:** Shifts **left** (depreciation needed to expect appreciation).
- **IS:** Shifts **right** (depreciation $\rightarrow NX \uparrow$).
- **Result:** $E \downarrow$ (depreciation), $Y \uparrow$ (expansionary if i unchanged).

Why IS Shifts Right: - $i^* \uparrow \Rightarrow$ Domestic bond less attractive $\Rightarrow E \downarrow$ (depreciation today). - Depreciation $\Rightarrow$ Domestic goods cheaper $\Rightarrow$ Exports rise, Imports fall $\Rightarrow NX \uparrow \Rightarrow$ IS shifts right.

2. Exchange Rate Regimes

Floating (Flexible) Exchange Rate

- **Definition:** Exchange rate freely adjusts to equilibrate financial markets (UIP holds).
- **Examples:** US-Euro, US-Japan, US-UK, Australia, Canada, Sweden.
- **Characteristics:**
 - Independent monetary policy.
 - Exchange rate volatility.
 - Clean floats (Euro, Yen) vs. Managed floats (emerging markets).

Fixed (Pegged) Exchange Rate

- **Definition:** Exchange rate fixed to another currency (or basket).

- **Examples:** Hong Kong (pegged to USD), Eurozone (common currency), Singapore (target zone vs. basket).
- **Implication:** If credible peg, then $E = E^e = \bar{E}$ (constant).
 - UIP $\Rightarrow i = i^*$ (no expected appreciation/depreciation).
 - **Loss of Monetary Policy Independence:** Must follow foreign central bank's policy.

Hybrid Regimes

- **Most countries** have hybrid systems between pure float and pure peg.
- **Examples:**
 - **Managed Floats:** Brazil, South Africa, Colombia (intervene frequently, inflation trends).
 - **Target Zones:** Singapore (basket peg within narrow bands, secret weights).
 - **Capital Controls:** China (semi-pegged, can deviate i from i^* due to capital controls).

3. Policy Under Different Regimes

Floating Exchange Rate

Recession Response:

- **Fiscal Policy:** Expansionary ($\uparrow G$) $\Rightarrow$ IS shifts right $\Rightarrow Y \uparrow$ (smaller multiplier than closed economy).
- **Monetary Policy:** Cut interest rate ($\downarrow i$) $\Rightarrow$ Two channels:
 1. Investment $\uparrow$ (traditional).
 2. $E \downarrow$ (depreciation) $\Rightarrow NX \uparrow$ (expenditure switching).
 - **More Powerful** than closed economy (net export channel reinforces investment channel).

Fixed Exchange Rate

Recession Response:

- **Fiscal Policy:** Same as floating (expansionary $\uparrow G \Rightarrow Y \uparrow$).
- **Monetary Policy: No independence.** Must follow foreign country's policy ($i = i^*$).
 - If recession is **country-specific** (not global), cannot use monetary policy tool.
 - **Example:** Hong Kong must follow US interest rate policy exactly.

Costs of Fixed Regime:

1. **Loss of Policy Tool:** No ability to adjust i for domestic cycle.
2. **Misalignment with Foreign Cycle:** If foreign country not in recession, their policy inappropriate for you.

4. Speculative Attacks

Mechanism

- **Scenario:** Fixed exchange rate, but markets lose confidence ($E^e \downarrow$).
- **UIP:** $E_t = \frac{1+i_t}{1+i_t^*} E_{t+1}^e$
 - If $E^e \downarrow$ (expect devaluation), then $E_t \downarrow$ (pressure to depreciate).
- **Defense:** To maintain peg, must raise $i \uparrow \uparrow$ (offset expected depreciation with interest rate differential).

Cost of Defense

- **Domestic Recession:** High interest rates contract output ($Y \downarrow$).
- **Example:** Russia (2022 invasion) raised rates from 4-5% to 20% to defend ruble.

Historical Examples

ERM Crisis (1992)

- **Context:** European Monetary System (narrow bands around exchange rates).
- **Trigger:** German reunification $\Rightarrow$ Expansionary fiscal policy $\Rightarrow$ German interest rates $\uparrow \Rightarrow$ Deutschmark appreciation.
- **Response:**
 - **France:** Raised rates, defended peg, suffered recession.
 - **UK:** Gave up, left ERM (allowed pound to depreciate).
- **Outcome:** System collapsed; later replaced by Euro (common currency, no speculative attacks possible).

Argentina & Turkey

- **Pattern:** Chronic speculative attacks due to:
 - Low credibility.
 - Insufficient reserves.
 - High inflation.
- **Cycle:** Attempt to stabilize $\rightarrow$ Attack $\rightarrow$ Raise $i \rightarrow$ Recession $\rightarrow$ Abandon peg $\rightarrow$ Repeat.

5. Volatility of Floating Rates

Excess Volatility Problem

- Exchange rates fluctuate **more than fundamentals** (productivity, demand shocks) justify.
- **Why?:** Forward-looking nature of UIP.

Recursive UIP

$$E_t = \frac{1 + i_t}{1 + i_t^*} \frac{1 + i_{t+1}^e}{1 + i_{t+1,e}^*} \frac{1 + i_{t+2}^e}{1 + i_{t+2,e}^*} \dots E_{t+N}^e$$

- **Current exchange rate** depends on:
 - Expected path of domestic interest rates.
 - Expected path of foreign interest rates.
 - Long-run expected exchange rate (E_{t+N}^e).
- **Problem:** Expectations about distant future are highly volatile and driven by “imagination” (news, sentiment, narratives).
- **Result:** Exchange rates overreact to news, creating noise unrelated to current fundamentals.

Costs of Excess Volatility

- **Transaction Costs:** Harder to plan trade and investment.
- **Financial Risk:** Currency exposure adds risk to cross-border investment.
- **Example:** Russia (2022) - ruble collapsed on invasion news, then recovered (not due to fundamentals, but massive rate hikes).

6. Choosing an Exchange Rate Regime

Case for Fixed Exchange Rate

When to Peg? 1. **Similar Shocks:** If shocks are correlated with the anchor country, foreign monetary policy appropriate for you. - **Example:** Eurozone (similar business cycles across members). 2. **High Fiscal Capacity:** Can fight recessions with fiscal policy (don't need monetary policy). - **Example:** Hong Kong (large reserves, flexible fiscal policy). 3. **Flexible Domestic Prices/Wages:** If P adjusts easily, nominal peg doesn't constrain real exchange rate. - **Example:** Hong Kong (flexible labor markets). 4. **Low Inflation Credibility:** Peg to anchor credibility of low-inflation country. - **Example:** Argentina (attempted currency board to import US credibility). 5. **High Trade Integration:** Reduces transaction costs if trade heavily with anchor country. - **Example:** Eurozone (frequent cross-border transactions).

Case for Floating Exchange Rate

When to Float? 1. **Idiosyncratic Shocks:** If shocks differ from trading partners, need independent monetary policy. 2. **Low Fiscal Capacity:** Need monetary policy as stabilization tool. 3. **Rigid Prices/Wages:** Need nominal exchange rate to adjust real exchange rate. 4. **Credible Monetary Policy:** Can achieve low inflation without anchor. 5. **Large Economy:** Less exposed to external shocks; net export channel less important (e.g., US).

7. Summary

- **Fixed Exchange Rate:** $E = \bar{E} \Rightarrow i = i^* \Rightarrow$ No monetary policy independence.
 - **Benefits:** Reduces volatility, lowers transaction costs, anchors inflation expectations.
 - **Costs:** Lose policy tool, vulnerable to speculative attacks, must suffer foreign shocks.
- **Floating Exchange Rate:** E adjusts freely $\Rightarrow$ Independent monetary policy.
 - **Benefits:** Can respond to domestic shocks, no speculative attacks on currency.
 - **Costs:** Excess volatility, complicates trade/investment decisions.
- **Speculative Attacks:** If $E^e \downarrow$, must raise i to defend peg $\Rightarrow$ Domestic recession.
- **Most Regimes Are Hybrid:** Managed floats, target zones, capital controls (e.g., China, Singapore, Brazil).
- **Expectations Matter:** Exchange rates driven by expected future policies and distant fundamentals (excess volatility).

Next: Financial markets, asset pricing, and expectations.

Lecture 22: Expectations and Asset Pricing I - Bonds

Overview

This lecture introduces the concept of expected present discounted value (EPDV) and applies it to pricing bonds. It covers discounting future cash flows, bond pricing, yield curves, and the role of risk premiums.

1. Motivation: Why Asset Prices Move

Recent Events

- **First Republic Bank:** Collapsed and sold to JPMorgan (second-largest US bank failure, \$100B deposit outflow).
- **Equity Markets:** S&P 500 volatility driven by COVID, policy support, inflation, Fed hikes.
- **Bond Markets:** Long-duration Treasury ETFs experienced 40% declines due to rising interest rates.

Key Insight

Asset prices move dramatically based on **expectations** about the future (cash flows, interest rates, risk).

2. Expected Present Discounted Value (EPDV)

The Problem

- You pay for an asset **today** (in current dollars).
- The asset pays cash flows **in the future** (future dollars).
- **Question:** How do we compare payments across time?

The Solution: Discounting

Why discount the future? - \$1 today > \$1 one year from now (can invest and earn interest). - If \$1 invested today yields $1 \times (1 + i_t)$ next year, then \$1 next year is worth $\frac{1}{1+i_t}$ today.

Key Formula: Value of \$1 received in n years:

$$\text{Value today} = \frac{1}{(1 + i_t)(1 + i_{t+1}) \cdots (1 + i_{t+n-1})}$$

General EPDV Formula (Known Future)

$$V_t = Z_t + \frac{Z_{t+1}}{1 + i_t} + \frac{Z_{t+2}}{(1 + i_t)(1 + i_{t+1})} + \cdots$$

- Z_t : Cash flow at time t (received today, no discounting).
- Z_{t+1} : Cash flow one year from now (discount by $1 + i_t$).
- Z_{t+2} : Cash flow two years from now (discount by product of two rates).

EPDV with Uncertainty (Expected Future)

$$V_t = Z_t + \frac{E_t[Z_{t+1}]}{1 + i_t} + \frac{E_t[Z_{t+2}]}{(1 + i_t)(1 + E_t[i_{t+1}])} + \cdots$$

- Replace unknown future cash flows and interest rates with **expectations** formed at time t .

3. Special Cases

Constant Interest Rate (i constant)

$$V_t = Z_t + \frac{Z_{t+1}}{1 + i} + \frac{Z_{t+2}}{(1 + i)^2} + \cdots$$

Constant Payment (Z constant)

$$V_t = Z \left(1 + \frac{1}{1 + i} + \frac{1}{(1 + i)^2} + \cdots \right)$$

Constant Rate & Payment (Finite Horizon n)

$$V_t = Z \frac{1 - (1 + i)^{-n}}{i/(1 + i)}$$

Perpetuity (Constant Payment Forever)

$$V_t = \frac{Z}{i}$$

Ex-Dividend Perpetuity (first payment at $t + 1$):

$$V_t = \frac{Z}{i}$$

Key Insight: As $i \rightarrow 0$, $V_t \rightarrow \infty$ (low interest rates inflate asset prices).

4. Bond Pricing

Bond Characteristics

- **Maturity (n):** Number of years until final payment.
- **Face Value:** Principal repaid at maturity (e.g., \$100).
- **Coupons:** Periodic interest payments (may be zero).

One-Year Bond

- Pays \$100 one year from now.
- **Price:** $P_{1t} = \frac{100}{1+i_{1t}}$
- **Inverse Relationship:** Higher $i_{1t} \Rightarrow$ Lower P_{1t} .

Two-Year Bond

- Pays \$100 two years from now (zero-coupon).
- **Price:** $P_{2t} = \frac{100}{(1+i_{1t})(1+E_t[i_{1,t+1}])}$
- **Inverse Relationship:** Higher i_{1t} or $E_t[i_{1,t+1}] \Rightarrow$ Lower P_{2t} .

Arbitrage Pricing (Alternative Method)

Compare two strategies for investing \$1 for one year: 1. **Buy one-year bond:** Return = $1 + i_{1t}$. 2. **Buy two-year bond, sell after one year:** Pay P_{2t} today, receive $E_t[P_{1,t+1}]$ next year.

Arbitrage Condition:

$$1 + i_{1t} = \frac{E_t[P_{1,t+1}]}{P_{2t}}$$

Result: Same price as EPDV method:

$$P_{2t} = \frac{E_t[P_{1,t+1}]}{1 + i_{1t}} = \frac{100}{(1 + i_{1t})(1 + E_t[i_{1,t+1}])}$$

5. Yield to Maturity

Definition

The **constant** annual interest rate that equates the bond price to its EPDV.

Two-Year Bond:

$$P_{2t} = \frac{100}{(1 + i_{2t})^2}$$

Where i_{2t} is the **two-year yield**.

Expectation Hypothesis

From $P_{2t} = \frac{100}{(1+i_{1t})(1+E_t[i_{1,t+1}])} = \frac{100}{(1+i_{2t})^2}$:

$$i_{2t} \approx \frac{i_{1t} + E_t[i_{1,t+1}]}{2}$$

Interpretation: Long-term rate = Average of expected short-term rates.

6. Yield Curve (Term Structure)

Definition

Relationship between **maturity** and **yield** (plot i_n vs. n).

Shapes

- **Upward Sloping** (steep): Markets expect rising short-term rates (normal in expansions).
 - Example: November 2000 (pre-recession, Fed expected to tighten).
- **Downward Sloping** (inverted): Markets expect falling short-term rates (common in recessions or before cuts).
 - Example: June 2001 (recession, Fed cut rates).
 - Example: Today (2023, inflation peak passed, Fed expected to ease).

Why Shapes Change

- **Fed controls short end** (overnight rate).
- **Long end** determined by expectations of future short rates.
- **Steep curve:** Fed tight today, expect loosening soon.
- **Inverted curve:** Fed tight today, expect easing later (recession signal).

7. Risk and Term Premiums

Price Risk (No Default Risk)

- **One-year bond:** No price risk (receive \$100 at maturity).
- **Two-year bond held for one year:** Price risk (don't know $P_{1,t+1}$).

Risk-Adjusted Arbitrage

$$1 + i_{1t} + x_b = \frac{E_t[P_{1,t+1}]}{P_{2t}}$$

- x_b : **Risk premium** for holding longer-maturity bond.

Term Premium in Yields

$$i_{2t} \approx \frac{i_{1t} + E_t[i_{1,t+1}]}{2} + \frac{x_b}{2}$$

- **Term premium** (x_b): Compensation for price risk.
- Usually positive (longer bonds riskier), but can be negative (hedging value).

Recent Example

- **Pre-2020**: $x_b < 0$ (long bonds hedged against crises, rates fell in downturns).
- **2022-2023**: $x_b > 0$ (inflation risk dominates, rates rise with inflation shocks).

8. Summary

- **EPDV**: Core tool for valuing future cash flows.
- **Discounting**: Interest rates convert future dollars to present value.
- **Bond Prices**: Inversely related to interest rates.
- **Yield Curve**: Reflects expectations of future short rates (+ risk premium).
- **Term Premium**: Compensation for holding longer-maturity bonds (price risk).

Next: Apply EPDV to equity pricing and understand policy effects on asset prices.

Lecture 23: Expectations and Asset Pricing II - Equity

Overview

This lecture extends the EPDV framework to equity pricing, analyzes how monetary and fiscal policy affect asset prices, and explores the role of expectations and risk premiums in financial markets.

1. Equity vs. Bonds: Key Differences

1. No Fixed Coupons

- **Bonds:** Promise fixed payments (coupons + face value).
- **Equity:** Pay dividends (variable, not guaranteed).
 - Companies set dividend **policy** (not commitment).
 - Can cut dividends to zero without default (e.g., regional banks during crises).

2. No Maturity

- **Bonds:** Have fixed maturity date (except perpetuities).
- **Equity:** Lasts as long as the company exists (in principle, forever).
 - No terminal date $\Rightarrow$ Always a “future price” term in valuation.

2. Equity Pricing via Arbitrage

Setup

Compare two one-year investment strategies: 1. **Buy one-year bond:** Return $= 1 + i_{1t}$. 2. **Buy stock, hold for one year, sell:** Pay Q_t today, receive $E_t[D_{t+1}]$ (dividend) + $E_t[Q_{t+1}]$ (sale price).

Arbitrage Condition (with Risk Premium)

$$1 + i_{1t} + x_s = \frac{E_t[D_{t+1}] + E_t[Q_{t+1}]}{Q_t}$$

- x_s : **Equity risk premium** (typically much larger than bond risk premium x_b).

Stock Price Today

$$Q_t = \frac{E_t[D_{t+1}] + E_t[Q_{t+1}]}{1 + i_{1t} + x_s}$$

3. Fundamental Value of Equity

Recursive Substitution

- At $t + 1$: $Q_{t+1} = \frac{E_{t+1}[D_{t+2}] + E_{t+1}[Q_{t+2}]}{1 + i_{1,t+1} + x_s}$
- Substitute $E_t[Q_{t+1}]$ into Q_t equation.
- Repeat infinitely.

Result: Gordon Growth Model (Simplified)

$$Q_t = \sum_{n=1}^{\infty} \frac{E_t[D_{t+n}]}{(1 + i_{1t} + x_s)(1 + E_t[i_{1,t+1}] + x_s) \cdots}$$

Interpretation: Stock price = EPDV of **all future dividends** (discounted by interest rate + risk premium).

Fundamental Value vs. Bubbles

- **Fundamental Value:** Based on expected dividends.
- **Bubbles:** Price exceeds any reasonable EPDV of dividends.
 - Example: Dot-com bubble, Bitcoin (2017), South Sea Bubble (Isaac Newton lost £20,000).
 - Newton: “I can calculate the motions of the heavenly bodies, but not the madness of people.”

4. Policy Effects on Asset Prices

A. Expansionary Monetary Policy ($\downarrow i$)

Effect on Bonds

- **Mechanism:** Lower $i \Rightarrow$ Less discounting of future payments $\Rightarrow$ Higher bond prices.
- **Result: Bond prices** $\uparrow$ (inverse relationship with interest rates).

Effect on Equity

- **Channel 1 (Discounting):** Lower $i \Rightarrow$ Less discounting of future dividends $\Rightarrow Q \uparrow$.
- **Channel 2 (Output):** Lower $i \Rightarrow$ Expansionary $\Rightarrow Y \uparrow \Rightarrow$ Sales $\uparrow \Rightarrow$ Dividends $\uparrow \Rightarrow Q \uparrow$.
- **Result: Equity prices** $\uparrow\uparrow$ (stronger effect than bonds).

Wealth Channel of Monetary Policy: Fed cuts rates $\Rightarrow$ Asset prices rise $\Rightarrow$ Households feel wealthier $\Rightarrow$ Consumption $\uparrow$.

B. Increase in Consumer Spending ($\uparrow C_0$)

Scenario 1: Fed Accommodates (Does Not React)

- **Bonds:** No change (interest rate unchanged).
- **Equity:** $C \uparrow \Rightarrow$ Sales $\uparrow \Rightarrow$ Dividends $\uparrow \Rightarrow Q \uparrow$.

Scenario 2: Fed Dislikes Overheating (Raises i)

- **Bonds:** Fed expected to hike $\Rightarrow i \uparrow$ (anticipated) $\Rightarrow$ **Bond prices** $\downarrow$.
- **Equity:** Two offsetting effects:
 1. Higher dividends (direct effect): $Q \uparrow$.
 2. Fed fully offsets demand shock $\Rightarrow$ No change in $Y \Rightarrow$ No change in dividends $\Rightarrow$ Only get discounting effect: $Q \downarrow$.
- **Result: Good news is bad news** for asset markets (common in 2022-2023).

Key Insight: Asset market response depends critically on **expected Fed reaction**.

5. Risk Premium and Market Volatility

VIX (“Fear Index”)

- Measures implied volatility from S&P 500 option prices.
- **High VIX** = High risk aversion (risk-off).
- **Low VIX** = Low risk aversion (risk-on).

Example: COVID-19 Crash (March 2020)

- **VIX spiked** $\Rightarrow x_s \uparrow\uparrow \Rightarrow$ S&P 500 fell 35%.
- **Recovery:** VIX declined $\Rightarrow x_s \downarrow \Rightarrow$ S&P 500 rallied 114% (to end-2021).
 - Driven by: Fed easing ($i \downarrow$) + Risk appetite recovery ($x_s \downarrow$).

Example: Regional Banking Crisis (May 2023)

- **First Republic collapse** $\Rightarrow$ Fear of contagion $\Rightarrow$ VIX spike $\Rightarrow$ S&P 500 intraday decline.
- **PacWest:** Circuit breakers triggered (halts trading when price falls too fast).

Key Insight: Risk premium (x_s) is a major driver of equity volatility.

6. Bubbles and Excess Volatility

Historical Bubbles

- **Nikkei 225** (Japan, 1980s): 500% gain, then collapse.
- **Bitcoin** (2017): 1,500% gain, then 80% decline.
- **Dot-com Bubble** (1990s): Extreme valuations, then crash.

Why Bubbles Occur

- Equity has **no terminal value** (unlike bonds) $\Rightarrow$ Future price always matters.
- If $E_t[Q_{t+n}]$ grows faster than discounting, price can explode (self-fulfilling).
- **Imagination** and **narratives** drive expectations.

Excess Volatility

- Equity prices move **much more** than justified by changes in expected dividends.
- Large component driven by **speculative** factors (not fundamentals).

7. Real vs. Nominal Pricing

Nominal vs. Real EPDV

- **Nominal:** Discount nominal cash flows by nominal interest rate.
- **Real:** Discount real cash flows by real interest rate ($r = i - \pi$).

Result: Same valuation (if consistent), but real terms clarify purchasing power.

8. Summary

Asset Pricing Formula

- **Bonds:** $P_t = \frac{\text{Face Value}}{(1+i)^n}$ (+ coupons if any).
- **Equity:** $Q_t = \sum_{n=1}^{\infty} \frac{E_t[D_{t+n}]}{\prod (1+i_{t+k}+x_s)}$ (EPDV of dividends).

Policy Effects

- **Expansionary Monetary Policy:** $\uparrow$ Bond prices, $\uparrow\uparrow$ Equity prices.
- **Aggregate Demand Shocks:** Effect depends on **expected Fed reaction**.
 - Fed accommodates $\Rightarrow$ Good news is good news.
 - Fed tightens $\Rightarrow$ Good news is bad news (2022-2023 environment).

Risk Premium

- **Equity risk premium** (x_s) is large and volatile.
- **VIX** measures risk appetite; spikes during crises cause asset price collapses.

Bubbles

- Equity prone to bubbles (no maturity $\Rightarrow$ future price always matters).
- Speculative component often dominates fundamentals.

Next: Integrate expectations into IS-LM model (permanent income, life-cycle theories).

Lecture 24: Expectations in IS-LM

Overview

This lecture integrates expectations into the IS-LM framework, introducing permanent income and life-cycle theories of consumption, and analyzing how forward-looking behavior affects policy effectiveness.

1. Motivation: Expectations Matter for All Agents

Why the Static IS-LM Overweights the Present

- **Consumption:** Depends on current disposable income ($Y - T$).
- **Investment:** Depends on current output (Y) and current interest rate (i).

Reality: The Future Dominates Decisions

- **Consumers:** Care about lifetime income (permanent income, human wealth).
- **Firms:** Invest based on expected future profits (not current sales).
- **Governments:** Long-term fiscal sustainability matters.
- **Foreign Investors:** Future political/economic conditions drive FDI.

Key Insight: Expectations about the future often matter **more** than current conditions.

2. Consumption with Expectations

Permanent Income Hypothesis (Milton Friedman)

- **Core Idea:** Consumption depends on **permanent income** (expected average lifetime income), not current income.
- **Implication:** Temporary income changes have **small** effects; permanent changes have **large** effects.

Life-Cycle Theory (Franco Modigliani)

- **Core Idea:** Individuals smooth consumption over their lifetime.
 - **Young:** Borrow (income < consumption).
 - **Middle-aged:** Save heavily (income > consumption, prepare for retirement).
 - **Old:** Dissave (income < consumption, live off savings).

Wealth-Based Consumption Function

$$C = c_0 + c_1 \times (\text{Total Wealth})$$

Total Wealth = Financial Wealth + Human Wealth

Financial Wealth

Financial Wealth = EPDV of asset cash flows – Debts

- Includes stocks, bonds, real estate, expected inheritances. - **Borrowing against wealth:** Rich often borrow against assets (tax advantages) rather than selling.

Human Wealth

Human Wealth = EPDV of future labor income

- For most people, **human wealth** » **financial wealth**. - Example: Students have low current income but high human wealth $\Rightarrow$ Borrow/consume more than current income.

Realistic Consumption Function

$$C = c_0 + c_1(Y - T) + c_2 \times (\text{Total Wealth})$$

- **Current income** ($Y - T$) matters: Liquidity constraints, hand-to-mouth consumers.
- **Wealth** matters: Forward-looking, captures permanent income effects.

Interpretation of c_0 in Static IS-LM: Captured consumer confidence, expectations $\Rightarrow$ Now we model this explicitly via wealth.

3. Investment with Expectations

Value of a Machine

Firm buys machine for price P_K (normalize to 1). Machine depreciates at rate δ (geometric).

EPDV of Machine:

$$V_t = \frac{E_t[\pi_{t+1}]}{1 + i_t} + \frac{(1 - \delta)E_t[\pi_{t+2}]}{(1 + i_t)(1 + E_t[i_{t+1}])} + \dots$$

- π_{t+n} : Expected profits in year n .
- $(1 - \delta)^{n-1}$: Probability machine still works in year n .

Investment Decision: Buy machine if $V_t > P_K$.

Investment Function with Expectations

$$I = I(V_t, Y_t, i_t)$$

- V_t (**EPDV of future profits**): $\uparrow V_t \Rightarrow \uparrow I$ (main driver).
- Y_t (**current output**): Relaxes financial constraints (internal funds).
- i_t (**interest rate**): Borrowing cost + discounting effect.

Key Insight: Future expected profits matter more than current conditions.

Simplification: Use Yield Curve

Firms don't need to forecast future short rates $i_{t+1}, i_{t+2}, \dots$ - Can use current term structure: $i_{2t}, i_{3t}, \dots$ (market already incorporates expectations).

4. Expectations-Augmented IS-LM

Augmented Aggregate Demand

$$Z = C(Y, T, i, \mathbf{Y}^e, \mathbf{T}^e, \mathbf{i}^e, \dots) + I(Y, i, \mathbf{Y}^e, \mathbf{i}^e, \dots) + G$$

- **Current variables:** Y, T, i, G .
- **Future expectations** (bolded): $E_t[Y_{t+1}], E_t[T_{t+1}], E_t[i_{t+1}], \dots$

Signs (same as static model): - $\frac{\partial Z}{\partial Y} > 0$, $\frac{\partial Z}{\partial Y^e} > 0$ (higher future income $\Rightarrow$ higher human wealth). - $\frac{\partial Z}{\partial T} < 0$, $\frac{\partial Z}{\partial T^e} < 0$ (higher future taxes $\Rightarrow$ lower human wealth). - $\frac{\partial Z}{\partial i} < 0$, $\frac{\partial Z}{\partial i^e} < 0$ (higher future rates $\Rightarrow$ lower EPDV).

IS Curve: Much Steeper

- **Why?:** Moving i_t alone (without changing $E_t[i_{t+1}]$) has **small** effect.
- **Movement along IS:** Small ΔY for given Δi (most of value comes from future).

Recovering Monetary Policy Power: Shift IS

- **Cut i_t AND convince markets $E_t[i_{t+1}] \downarrow \Rightarrow$ IS shifts **right**.**
- **Mechanism:**
 1. Lower future rates $\Rightarrow$ Higher EPDV (consumption + investment).
 2. Lower future rates $\Rightarrow$ Higher future $Y \Rightarrow$ Higher human wealth $\Rightarrow$ Higher consumption.

Forward Guidance: Central banks emphasize future policy path to maximize impact.

5. Policy Implications

A. Temporary vs. Permanent Changes

Temporary Tax Cut

- **Current effect:** $\downarrow T \Rightarrow \uparrow (Y - T) \Rightarrow \uparrow C$ (small).
- **Wealth effect:** No change (human wealth = EPDV of future income, which is unchanged).
- **Result: Small multiplier** (most of windfall saved).

Permanent Tax Cut

- **Current effect:** $\downarrow T \Rightarrow \uparrow (Y - T) \Rightarrow \uparrow C$ (small).
- **Wealth effect:** $\downarrow E_t[T_{t+1}] \Rightarrow \uparrow$ Human wealth $\Rightarrow \uparrow C$ (large).
- **Result: Large multiplier** (similar to static IS-LM).

Key Insight: Permanent fiscal changes have much larger effects than temporary changes.

B. Monetary Policy with Forward Guidance

Scenario 1: Cut i_t Only (No Forward Guidance)

- **Effect:** Small movement along steep IS $\Rightarrow \Delta Y$ is small.

Scenario 2: Cut i_t + Commit to Low i_{t+1}

- **Direct effect:** Movement along IS (small).
- **Expectation effect:** IS shifts right (large).
 - Lower $E_t[i_{t+1}] \Rightarrow$ Higher EPDV $\Rightarrow \uparrow I, C$.
 - Lower $E_t[i_{t+1}] \Rightarrow$ Higher $E_t[Y_{t+1}] \Rightarrow$ Higher human wealth $\Rightarrow \uparrow C$.
- **Result:** Much larger ΔY .

Modern Central Banking: Fed/ECB speeches emphasize policy path (“higher for longer”, “patient approach”) to shape expectations.

C. Fiscal Policy and Expectations

Standard Effect: Contractionary

- $\downarrow G \Rightarrow$ IS shifts left $\Rightarrow Y \downarrow$ (direct effect dominates).

Expansionary Fiscal Contraction (Rare) Can occur if fiscal consolidation dramatically improves expectations: 1. **Removes uncertainty:** Fiscal crisis risk eliminated $\Rightarrow \uparrow E_t[Y_{t+1}] \Rightarrow \uparrow$ Human wealth $\Rightarrow \uparrow C$. 2. **Enables monetary easing:** Central bank cuts rates in response $\Rightarrow E_t[i_{t+1}] \downarrow \Rightarrow$ IS shifts right.

Example: Ireland (Late 1980s) - Context: Chronic fiscal deficits, economic stagnation, dominated political discourse. - **Policy:** Aggressive fiscal consolidation (deficit reduction). - **Result:** - Y increased (GDP growth accelerated). - Unemployment rose (lagging indicator), but consumption + investment surged. - Household savings rate plummeted (optimism about future). - **Why?:** Removed fiscal uncertainty $\Rightarrow$ Expectations improved $\Rightarrow$ Offset contractionary direct effect.

Caveat: This is **rare**. Most fiscal contractions are contractionary (direct effect dominates). Requires extreme starting conditions (chronic crisis, high uncertainty).

6. The Greenspan Conundrum

Context

- **Alan Greenspan** (Fed Chair, 1987-2006): Widely regarded as successful.
- **Problem (mid-2000s):** Economy overheating, Fed raised short rates, but long rates fell.

Mechanism

- Fed hiked short rates $\Rightarrow$ But couldn't convince markets of persistent tightening.
- Long rates = Average of expected future short rates $\Rightarrow$ Markets expected Fed to reverse course.
- **Result:** Fed ineffective (couldn't cool economy via long-rate channel).

Actual Cause

- **External factor:** Massive capital inflows from China (savings glut) pushed down long rates.
- **Lesson:** Expectations matter, but sometimes external forces dominate.

7. Summary

Consumption

- Depends on **total wealth** (financial + human), not just current income.
- **Permanent changes** have much larger effects than temporary changes.

Investment

- Driven by **EPDV of future profits** (not current sales alone).
- Firms can use term structure (no need to forecast rates).

IS-LM with Expectations

- **IS is steeper:** Current policy variables have smaller effects.
- **Shift IS via expectations:** Future policy commitments restore power.

Policy Lessons

1. **Temporary policies** are weak (wealth barely changes).
2. **Permanent policies** are strong (wealth changes significantly).
3. **Forward guidance** is critical for monetary policy effectiveness.
4. **Fiscal contractions** can be expansionary if expectations improve dramatically (rare).

Central Banking = Expectations Management: Policy actions matter less than communication about future policy path.

Next: Review for Quiz 3 (open economy, asset pricing, expectations).

Lecture 25: Review for Quiz 3

Overview

This lecture reviews material for Quiz 3, covering open economy (Mundell-Fleming), asset pricing (bonds and equity), and expectations in IS-LM. ~73% of quiz relates to Mundell-Fleming model.

1. Open Economy: Goods Market Only

Demand for Domestic Goods vs. Domestic Demand

- **Domestic Demand (D):** $D = C(Y - T) + I(Y, i) + G$ (what residents buy).
- **Demand for Domestic Goods (Z):** $Z = D + X - IM$ (what domestic producers sell).

Net Exports

$$NX = X(Y^*, \epsilon) - IM(Y, \epsilon)$$

- **Exports:** $\frac{\partial X}{\partial Y^*} > 0$, $\frac{\partial X}{\partial \epsilon} < 0$ (appreciation $\Rightarrow$ less exports).
- **Imports:** $\frac{\partial IM}{\partial Y} > 0$, $\frac{\partial IM}{\partial \epsilon} > 0$ (appreciation $\Rightarrow$ more imports).
- **Assumption (Always Holds on Quiz):** $\frac{\partial NX}{\partial \epsilon} < 0$ (appreciation worsens net exports).

ZZ Curve Diagram

1. Start with $DD = C + I + G$ (domestic demand).
2. Subtract $IM(Y, \epsilon) \Rightarrow$ Shifts down + rotates (flatter, due to $\frac{\partial IM}{\partial Y} > 0$).
3. Add $X(Y^*, \epsilon) \Rightarrow$ Parallel shift up (exports not function of Y).
4. Result: ZZ curve (demand for domestic goods).

Key Point: $ZZ - DD = NX$ (vertical distance = trade balance).

Equilibrium

$$Y = Z \quad (45^\circ \text{ line intersection})$$

- **Trade Surplus:** $ZZ > DD$ (output low, imports low).
- **Trade Deficit:** $ZZ < DD$ (output high, imports high).

Multiplier

- **Open economy multiplier** $<$ **Closed economy** (import leakage: $\frac{\partial IM}{\partial Y} > 0$).

Policy Experiments

1. $\uparrow G$: IS shifts right $\Rightarrow Y \uparrow$ (less than closed), $NX \downarrow$ (trade deficit worsens).
2. $\uparrow Y^*$: $X \uparrow \Rightarrow ZZ$ shifts up $\Rightarrow Y \uparrow$, $NX \uparrow$ (trade surplus improves).
3. $\downarrow \epsilon$ (depreciation): $X \uparrow$, $IM \downarrow \Rightarrow ZZ$ shifts up $\Rightarrow Y \uparrow$, $NX \uparrow$ (expenditure switching).

2. Mundell-Fleming Model (~73% of Quiz)

Components

1. **IS (Open Economy)**: $Y = C(Y - T) + I(Y, i) + G + NX(Y, Y^*, E)$.
2. **UIP**: $E_t = \frac{1+i_t}{1+i_t^*} E_{t+1}^e$ (uncovered interest parity).
3. **LM**: $i = \bar{i}$ (central bank sets rate).

Assumptions

- **Sticky prices**: $P = \text{constant} \Rightarrow \text{Real exchange rate} = \text{Nominal exchange rate}$ ($\epsilon = E$).
- **Fixed expected exchange rate**: $E_{t+1}^e = \bar{E}^e$ (parameter, not endogenous).

UIP Condition (Critical)

$$E_t = \frac{1+i_t}{1+i_t^*} E_{t+1}^e$$

Approximation:

$$i_t \approx i_t^* + \frac{E_{t+1}^e - E_t}{E_t}$$

Interpretation: Domestic rate = Foreign rate + Expected appreciation of domestic currency.

Key Results: - $\uparrow i$ (domestic) $\Rightarrow E \uparrow$ (appreciation today $\Rightarrow$ expect depreciation tomorrow). - $\uparrow i^* \Rightarrow E \downarrow$ (depreciation today $\Rightarrow$ expect appreciation tomorrow). - $\uparrow E^e \Rightarrow E \uparrow$ (appreciation today, one-for-one if $i = i^*$).

UIP Diagram

- **Axes**: E (horizontal), i (vertical).
- **Upward sloping**: Higher i requires higher E (to generate expected depreciation).
- **Key Point**: When $i = i^*$, then $E = E^e$ (no expected change $\Rightarrow$ no interest differential).

Shifts: - $\uparrow i^*$: UIP shifts **left** (depreciation). - $\uparrow E^e$: UIP shifts **right** (appreciation).

IS Curve (Open Economy)

- **Steeper than closed economy** (less sensitive to i due to import leakage).
- **Interest rate has two effects:**
 1. **Investment channel:** $\uparrow i \Rightarrow \downarrow I \Rightarrow \downarrow Y$.
 2. **Net export channel:** $\uparrow i \Rightarrow E \uparrow$ (via UIP) $\Rightarrow NX \downarrow \Rightarrow \downarrow Y$.
- **Result:** Monetary policy **more powerful** in open economy (two contractionary channels).

Mundell-Fleming Diagram

1. **Top:** IS-LM (determines Y and i).
2. **Bottom:** UIP (determines E given i).

Policy Experiments

1. Contractionary Monetary Policy ($\uparrow i$)

- **IS-LM:** LM shifts up $\Rightarrow Y \downarrow$ (investment $\downarrow + NX \downarrow$).
- **UIP:** Move along curve $\Rightarrow E \uparrow$ (appreciation).

2. Expansionary Fiscal Policy ($\uparrow G$, Fed accommodates)

- **IS-LM:** IS shifts right $\Rightarrow Y \uparrow$ (smaller multiplier than closed economy).
- **UIP:** No change (LM unchanged $\Rightarrow i$ unchanged $\Rightarrow E$ unchanged).

3. Increase in Expected Exchange Rate ($\uparrow E^e$)

- **UIP:** Shifts **right** $\Rightarrow E \uparrow$ (appreciation).
- **IS:** Shifts **left** (appreciation $\Rightarrow NX \downarrow$).
- **Result:** $Y \downarrow$, $E \uparrow$ (contractionary).

4. Foreign Output Decline ($\downarrow Y^*$)

- **IS:** Shifts **left** (exports $\downarrow$).
- **UIP:** No change.
- **Result:** $Y \downarrow$, E unchanged (unless Fed responds).

5. Foreign Interest Rate Increase ($\uparrow i^*$)

- **UIP:** Shifts **left** $\Rightarrow E \downarrow$ (depreciation).
- **IS:** Shifts **right** (depreciation $\Rightarrow NX \uparrow$).
- **Result:** $E \downarrow$, $Y \uparrow$ (if domestic i unchanged).

3. Fixed vs. Floating Exchange Rates

Fixed Exchange Rate (Peg)

- **Definition:** $E = E^e = \bar{E}$ (constant).
- **UIP Implication:** $i = i^*$ (no expected change $\Rightarrow$ no interest differential).
- **Loss of Monetary Independence:** Central bank must follow foreign policy.

Example: Hong Kong (pegged to USD) $\Rightarrow$ Hong Kong policy rate tracks US Fed rate exactly.

Speculative Attack

- **Scenario:** Markets lose confidence, expect $E^e \downarrow$ (devaluation).
- **UIP:** To maintain peg (E constant), must raise $i \uparrow\uparrow$ (offset expected depreciation).
- **Cost: Domestic recession** (high rates contract economy).

Examples: - **ERM Crisis (1992):** German reunification $\Rightarrow$ German rates $\uparrow \Rightarrow$ UK/France forced to raise rates $\Rightarrow$ Recessions $\Rightarrow$ UK left ERM. - **Argentina/Turkey:** Chronic attacks due to low credibility, insufficient reserves.

Floating Exchange Rate

- **Benefits:** Independent monetary policy, no speculative attacks.
- **Costs: Excess volatility** (driven by expectations of distant future, narratives).

Recursive UIP: E_t depends on expected paths of i and i^* far into future $\Rightarrow$ Highly sensitive to news/sentiment.

Choosing a Regime

Fixed Exchange Rate Preferable If: 1. **Similar shocks** to anchor country (foreign policy appropriate). 2. **High fiscal capacity** (can fight recessions with fiscal policy). 3. **Flexible domestic prices/wages** (real exchange rate adjusts even if nominal fixed). 4. **Low inflation credibility** (import anchor's credibility). 5. **High trade integration** (reduces transaction costs).

Examples: Eurozone (similar cycles), Hong Kong (large reserves + flexible markets).

4. Asset Pricing

Expected Present Discounted Value (EPDV)

$$V_t = Z_t + \frac{E_t[Z_{t+1}]}{1 + i_t} + \frac{E_t[Z_{t+2}]}{(1 + i_t)(1 + E_t[i_{t+1}])} + \dots$$

Key Insight: \$1 received in n years worth $\frac{1}{\prod_{k=0}^{n-1} (1+i_{t+k})}$ today.

Bond Pricing

- **One-year bond:** $P_{1t} = \frac{100}{1+i_{1t}}$.
- **Two-year bond:** $P_{2t} = \frac{100}{(1+i_{1t})(1+E_t[i_{1,t+1}])}$.

Inverse Relationship: $\uparrow i \Rightarrow \downarrow P$ (less discounting of future payments).

Yield to Maturity

Definition: Constant rate that equates price to EPDV.

$$P_{2t} = \frac{100}{(1+i_{2t})^2} \Rightarrow i_{2t} \approx \frac{i_{1t} + E_t[i_{1,t+1}]}{2}$$

Expectation Hypothesis: Long rate = Average of expected short rates (+ term premium).

Yield Curve

- **Upward sloping:** Expect rising rates (expansions, pre-tightening).
- **Inverted:** Expect falling rates (recessions, Fed expected to ease).
- **Current (2023):** Inverted (markets expect Fed to cut after inflation peak).

Equity Pricing

$$Q_t = \sum_{n=1}^{\infty} \frac{E_t[D_{t+n}]}{\prod (1+i+x_s)}$$

- **No maturity** $\Rightarrow$ Always future price term (prone to bubbles).
- **No fixed coupons** $\Rightarrow$ Dividends risky.

5. Expectations in IS-LM

Consumption: Permanent Income

$$C = c_0 + c_1(Y - T) + c_2 \times (\text{Financial Wealth} + \text{Human Wealth})$$

- **Human Wealth:** EPDV of future labor income.
- **Permanent changes** have large effects; **temporary changes** have small effects.

Investment: EPDV of Profits

$$I = I(V_t, Y, i) \quad \text{where } V_t = \text{EPDV of future profits}$$

Augmented IS-LM

$$Z = C(Y, T, i, \mathbf{E}[Y], \mathbf{E}[T], \mathbf{E}[i]) + I(Y, i, \mathbf{E}[Y], \mathbf{E}[i]) + G$$

- **IS is steeper:** Current i has small effect (future matters more).
- **Forward Guidance:** Commit to future i path $\Rightarrow$ Shift IS $\Rightarrow$ Larger effect.

Policy Implications

1. **Temporary tax cut:** Small effect (wealth unchanged).
2. **Permanent tax cut:** Large effect (human wealth $\uparrow$).
3. **Monetary policy:** Effective only if convinces markets of persistent change.
4. **Fiscal consolidation:** Usually contractionary, but can be expansionary if dramatically improves expectations (Ireland example).

6. Key Formulas

Mundell-Fleming

- **UIP:** $i_t \approx i_t^* + \frac{E_{t+1}^e - E_t}{E_t}$
- **IS:** $Y = C(Y - T) + I(Y, i) + G + NX(Y, Y^*, E)$
- **NX:** $\frac{\partial NX}{\partial \epsilon} < 0$ (appreciation worsens trade balance)

Asset Pricing

- **EPDV:** $V_t = \sum_{n=0}^{\infty} \frac{E_t[Z_{t+n}]}{\prod_{k=0}^{n-1} (1+i_{t+k})}$
- **Bond price:** $P_{nt} = \frac{100}{(1+i_{nt})^n}$
- **Yield:** $i_{2t} \approx \frac{i_{1t} + E_t[i_{1,t+1}]}{2} + \text{term premium}$

Expectations-Augmented Consumption

- $C = c_1(Y - T) + c_2 \times (\text{EPDV of assets} + \text{EPDV of labor income})$

7. Study Tips

1. **Master Mundell-Fleming diagrams** (73% of quiz).
2. **Practice all comparative statics:** $\uparrow G, \uparrow i, \uparrow i^*, \uparrow E^e, \uparrow Y^*$.
3. **Understand UIP deeply:** Why does $\uparrow i$ cause appreciation? (Expected depreciation offsets interest differential).
4. **Know EPDV formula:** Apply to bonds, equity, human wealth.
5. **Distinguish temporary vs. permanent:** Small vs. large effects on wealth.

Good luck on Quiz 3!